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NAME : ....

PART-I

Objective Type Questions

1. Let x,y and z be real numbers such that $x + y + z = 20$ and $x + 2y + 3z = 16$, then the value of $x + 3y + 5z$ is-

(A) –4 (B) 4 (C) –12 (D) 12

2. The sum of all distinct real solutions of the equation $(x^2-3)^3-(4x+6)^3+216=18(4x+6)(3-x^2)$ is-

(A) –3 (B) 4 (C) 1 (D) –7

3. The number of integral solutions of the equation $x+2y = 2xy$ is (are)

(A) 2 (B) 1 (C) 4 (D) infinitely many

4. If $\sqrt{5-\sqrt{10}-\sqrt{15}+\sqrt{6}}=\dfrac{\sqrt{a}+\sqrt{b}-\sqrt{c}}{\sqrt{b}}$, then the value of a + b + c is (where a,b,c are coprime)

(A) 10 (B) 11 (C) 12 (D) 13

5. If $x = \sqrt[3]{\sqrt{127}+10}-\sqrt[3]{\sqrt{127}-10}$ then $x^3 + 9x$ is divisible by

(A) 3 (B) 4 (C) 5 (D) 6

6. Each set X_r contains 5 elements and each set Y_r contains 2 elements and $\bigcup_{r=1}^{20} X_r = S = \bigcup_{r=1}^{n} Y_r$. If each element of S belong to exactly 10 of the X_r's and to exactly 4 of the Y_r's, then n is

(A) 10 (B) 20 (C) 100 (D) 50

Subjective Type Questions

7. From 50 students taking examinations in Mathematics, Physics and Chemistry, each of the student has passed in at least one of the subject, 37 passed Mathematics, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics, at most 29 Mathematics and Chemistry and at most 20 Physics and Chemistry. What is the largest possible number that could have passed all three examination?

8. Consider the number N = 975X44Y (X and Y are single digit numbers)

(i) If X = 3 and N is divisible by 3 find all possible values of Y.

(ii) If X = 4 and N is divisible by 6, find all possible values of Y.

(iii) If N is divisible by 8, find all all possible values of Y.

9. Find all natural numbers 'n' for which $n^5 + n^4 + 1$ is a prime number

10. Factorize following :

(i) $6x^3 - 5x^2 - 3x + 2$

(ii) $(1 + x + x^2 + x^3)^2 - x^3$.

(iii) $(x + y - 2xy)(x + y - 2) + (1 - xy)^2$.

(iv) $(x^2 + y^2 - 2x + 1)^2 - (4y - 4xy)(x^2 - y^2 - 2x + 1)$.

(v) $16(6x - 1)(2x - 1)(3x + 1)(x - 1) + 25$.

(vi) $(6x - 1)(4x - 1)(3x - 1)(x - 1) + 9x^4$.

(vii) $(6x - 1)(2x - 1)(3x - 1)(x - 1) + x^2$.

(viii) $(x + 1)^4 + (x + 3)^4 - 272$.

(ix) $x^4+\left(2\sqrt{7}-1\right)x+\sqrt{7}-7$ into two quadratic factors

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11. Factorize : $2xyz + x^2y + y^2z + z^2x + xy^2 + yz^2 + zx^2$

12. Simplify $\sqrt{x(x+4)(x+8)(x+12)+256}$, $(x>0)$.

13. Find the value of $\sqrt{2+\sqrt{2-\sqrt{2+\sqrt{2......\infty}}}}$

14. The value of the expression $\sqrt{7-\sqrt{48}}+\sqrt{5-\sqrt{24}}+\sqrt{3-\sqrt{8}}$ is

15. If a,b,c are positive integer such that $a \geq b \geq c$ such that $a^2 - b^2 - c^2 + ab = 2011$ and $a^2 + 3b^2 + 3c^2 - 3ab - 2ac - 2bc = -1997$ find a.

16. Find the value of $(a + b)(b + c)(c + a)$ if $a + b + c = 21$ and $a^3 + b^3 + c^3 = 3213$

17. If $x^2 + 5y^2 + 10z^2 = 4xy + 6yz + 2z - 1$ find $x + y + z$

18. Positive real numbers a,b and c satisfy the equation $a^2 + b^2 + c^2 = 2703$ and $(a-1)(b-1)(c-1) = abc - 1$ compute $a + b + c$

19. Find $3x + 2y + 6z$ if x, y, and z satisfy the system of equations

$$\begin{cases} \frac{x}{2}=\frac{y}{3}=\frac{z}{5} \\ 2x+3y-z=16 \end{cases}$$

20. If x, y, z are non zero real numbers such that

$\frac{y+z}{x}=\frac{z+x}{y}=\frac{x+y}{z}$, find the value of $\frac{y+z}{x}$.

(A) –1 or 2 (B) –2 (C) –1 (D) 2 (E) 2 or 1

21. Find $x^6 + y^6$ if $x=\sqrt{5+\sqrt{5}}$ and $y=\sqrt{5-\sqrt{5}}$.

22. If $x^2 - x - 1 = 0$, then find the value of $\frac{x^{16}-1}{x^8+2x^7}$.

23. Find the remainder when $x + x^9 + x^{25} + x^{49} + x^{81}$ is divided by $x^3 - x$

24. Find number of real roots of the equation $x^8 - 4x^5 - x^4 + 2x^2 - 1 = 0$

25. Find all real solutions of the equation $\left(7+4\sqrt{3}\right)^{x^2+2x+1}+\left(7-4\sqrt{3}\right)^{x^2+2x+1}=14$

26. Solve the equation $(3x-1)\left(\sqrt{x}+3x-1\right)=2x$

27. Solve the system $1+\frac{1}{4x^2}=\frac{1}{y}, 1+\frac{1}{4y^2}=\frac{1}{z}, 1+\frac{1}{4z^2}=\frac{1}{x}$

28. Solve the system of equation $\sqrt{x+1}+\sqrt{x^2-x+1}-\sqrt{x^3+1}=1$

29. Solve for x

$$(2x^2-1)^{5x+2}=(2x^2-1)^{x^2+6}$$

30. Let a,b,c,d be real numbers which satisfy the system of equations

$$a + b + 2ab = 3$$
$$b + c + 2bc = 4$$
$$c + d + 2cd = 5$$

then the value of $d + a + 2da$ is

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31. Solve for x,y,z

$$x^3 = \frac{z}{y} - \frac{2y}{z};\ y^3 = \frac{x}{z} - \frac{2z}{x};\ z^3 = \frac{y}{x} - \frac{2x}{y}$$

32. How many ordered pair of integers (a,b) satisfy the equation $a^{2020} + 4^b = 2^{b+1}$?

33. Find all real number x such that $x = \sqrt{x - \frac{1}{x}} + \sqrt{1 - \frac{1}{x}}$

34. Solve : $16x^2(1-x)^2 = 1 + 2(1-2x)^2$

35. Solve the equation $x^{x^{8888}} = 8888$.

36. Find the product of all real solutions of x that satisfy $\frac{1}{x^2+2x-3} + \frac{18}{x^2+2x+2} = \frac{18}{x^2+2x+1}$.

37. Find the smallest and the greatest values of $9\left(\frac{x^2-x+1}{x^2+x+1}\right)^2 + 5$ if x is real.

38. Let a, b, and c denote the lengths of the sides of a triangle. Prove the following inequality $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < 2$.

39. Let set $A = \left[5 - a^2, |a| - 1\right]$ and set $B = \left[-\frac{1}{a^2}, a\right]$, $a \neq 0$. Find set of values of 'a' for following

(i) set A contains atleast 10 integers
(ii) set A contains atmost 8 integers
(iii) set B contains atmost 10 integers
(iv) $A \cap B = \phi$
(v) $A \cap B$ contains atleast 20 integers

40. Find minimum value of following

(i) $x^3 + \frac{1}{x}, x \in \mathbb{R}^+$
(ii) $x^m + \frac{1}{x^n}, m, n \in \mathbb{N}, x \in \mathbb{R}^+$
(iii) $\frac{1}{x(1-x^3)}; x \in (0,1)$

41. If $f(x)$ is polynomial of 50^{th} degree such that $f(0) = 0,\ f(1) = \frac{1}{2},\ f(2) = \frac{2}{3},\ f(3) = \frac{3}{4} \ldots\ f(50) = \frac{50}{51}$, then find

(i) $f(51)$
(ii) $f(-1)$

42. If P(x) is polynomial with minimum degree, such that $P(x) = \frac{x}{x^2+3x+2}$ for $x = 0,1,2,.....10$. Find

(i) P(–1)
(ii) P(–2)

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PART-II
Objective Type Questions

1. In a square ABCD, E is a point on diagonal BD. P and Q are the circumcentres of ΔABE and ΔADE respectively then which of the option is always correct ?

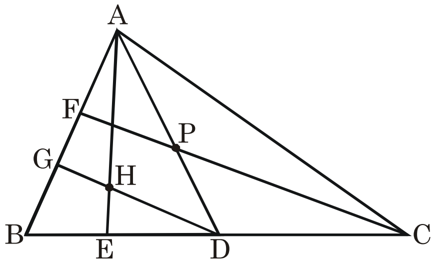
(A) APEQ is a rhombus but not always square

(B) APEQ is a parallelogram but not always square

(C) APEQ is always square

(D) None of these

2. In the figure that follows, BD = CD, BE = DE, AP = PD and DG || CF.



Then $\dfrac{\text{area of } \Delta ADH}{\text{area of } \Delta ABC}$ is equal to

(A) $\dfrac{1}{6}$ (B) $\dfrac{1}{4}$ (C) $\dfrac{1}{5}$ (D) None of these

Paragraph for Question 3 and 4

Consider a square with side length 'x' inscribed in a right angled triangle with sides of length 3cm, 4cm and 5cm so that one vertex of square coincides with right angled vertex of the triangle and another square with side length 'y' is inscribed so that one side of square lies on hypotenuse of the triangle.

3. Value of 'x' is equal to-

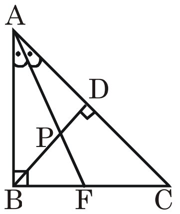
(A) $\dfrac{12}{7}$ cm (B) $\dfrac{12}{5}$ cm (C) $\dfrac{9}{7}$ cm (D) $\dfrac{9}{5}$ cm

4. The ratio $\dfrac{x}{y}$ is equal to-

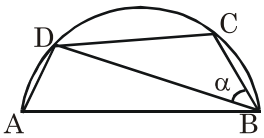
(A) $\dfrac{37}{35}$ cm (B) $\dfrac{5}{7}$ cm (C) $\dfrac{111}{140}$ cm (D) $\dfrac{111}{100}$ cm

Subjective Type Questions

5. Given ΔABC with $AB \perp BC$, BD is altitude to AC, AF bisects $\angle BAC$, AP = 12 and PF = 8. Find $\tan(\angle BAF)$ and find PD.



6. AB is diameter of circle centred at O. If $AB = \sqrt{2}DC$, then $\tan\alpha$ is equal to



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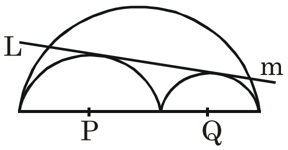
7. Let 'S' be the circle circumscribing right angled ΔABC right angled at A. Perpendicular from A meets the hypotoneous at E and circumcircle at D. If $\angle CDE = 45°$ and AB = 1. Then find length ED.

8. If radii of two smaller circles are 6 and 4, then find LM.



9. In ΔABC, a,b and c are the lengths of the sides of the triangle such that

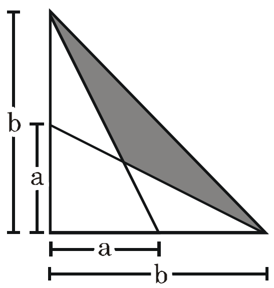
$$b^2 = \frac{2a^4}{a^4+1}, c^2 = \frac{3b^6}{b^8+b^4+1}, a^2 = \frac{4c^8}{c^{12}+c^8+c^4+1}$$. Prove that ΔABC is equilateral.

10. Let DIAL, FOR, and FRIEND be regular polygons in the plane. If ID = 1, find the product of all possible areas of OLA.

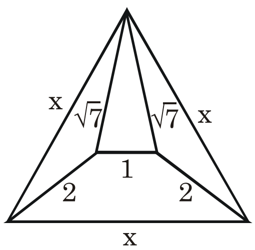
11. Let ABCDEF be a regular hexagon with side length 2. A circle with radius 3 and center at A is drawn. Find the area inside quadrilateral BCDE but outside the circle.

12. In rectangle ABCD, points E and F lie on sides AB and CD respectively such that both AF and CE are perpendicular to diagonal BD. Given that BF and DE separate ABCD into three polygons with equal area, and that EF = 1, find the length of BD.

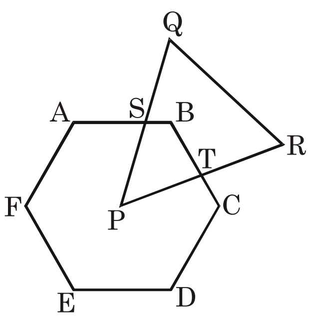
13. Find an expression in terms of a and b for the area of the hatched region in the right triangle in Figure.



14. Determine x in the equilateral triangle shown in figure



15. ABCDEF is a regular hexagon with center P and PQR is an equilateral triangle, as shown in figure. If $\overline{AB} = 3$, $\overline{SB} = 1$, and $\overline{PQ} = 6$, determine the area common to both figures



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ANSWER KEY

PART-I

1. D **2.** C **3.** A **4.** A **5.** B,C **6.** B **7.** 14

8. (i) {1,4,7} (ii) {0,6} (iii) {0,8} **9.** n = 1

10. (i) $(x-1)(3x+2)(2x-1)$ (ii) $(1+x+x^2)(1+x+x^2+x^3+x^4)$ (iii) $(x+y-xy-1)^2$

(iv) $(x^2+2xy-y^2-2x-2y+1)^2$ (v) $(24x^2-16x-3)^2$ (vi) $(9x^2-7x+1)^2$

(vii) $(6x^2-6x+1)^2$ (viii) $2(x+5)(x-1)(x^2+4x+19)$ (ix) $\left(x^2-x+\sqrt{7}\right)\left(x^2+x-\sqrt{7}+1\right)$

11. $(x+y)(y+z)(z+x)$ **12.** $x^2+12x+16$ **13.** $\frac{1+\sqrt{5}}{2}$ **14.** 1 **15.** 253

16. 2016 **17.** 10 **18.** 53 **19.** 84 **20.** –1 or 2 **21.** 400 **22.** 21

23. 5x **24.** 2 **25.** 0,–2 **26.** $\frac{1}{9},\frac{7+\sqrt{13}}{18}$ **27.** $x=y=z=\frac{1}{2}$ **28.** x = 0,1

29. $\left\{\pm1,\frac{1}{\sqrt{2}},4\right\}$ **30.** $\frac{34}{9}$ **31.** $(x,y,z)=(1,1,-1),(1,-1,1)(-1,1,1)(-1,-1,-1)$

32. 3 **33.** $\frac{1+\sqrt{5}}{2}$ **34.** $-\frac{1}{2},\frac{1}{2},\frac{3}{2}$ **35.** $x=\sqrt[8888]{8888}$ **36.** 56 **37.** 6 and 86

39. (i) $\left(-\infty,-\sqrt{12}\right]\cup\left[\sqrt{12},\infty\right)$ (ii) $\left(-\sqrt{11},\sqrt{11}\right)$ (iii) $\left(-\infty,\frac{-1}{\sqrt{11}}\right)\cup\left(\frac{1}{\sqrt{10}},10\right)$

(iv) $(-\infty,2)$ (v) $[20,\infty)$

40. (i) $\frac{4}{\sqrt[4]{27}}$ (ii) $\frac{m+n}{(mn)^{\frac{1}{m+n}}}$ (iii) $\frac{8}{3\sqrt[3]{2}}$ **41.** (i) $\frac{25}{26}$ (ii) $-\left(\frac{1}{2}+\frac{1}{3}+\dots\frac{1}{51}\right)$

42. (i) $\frac{5}{6}-\left(\frac{1}{2}+\frac{1}{3}+\dots\frac{1}{11}\right)$ (ii) $2\left(\frac{1}{3}+\frac{1}{4}+\dots+\frac{1}{12}\right)-10$

PART-II

1. C **2.** C **3.** A **4.** A **5.** $\tan(\angle BAF)=\frac{1}{2}$, $PD=\frac{24}{\sqrt{5}}$

6. 1 **7.** $\frac{1}{\sqrt{2}}$ **8.** $\frac{8\sqrt{114}}{5}$ **10.** $\frac{1}{32}$ **11.** $4\sqrt{3}-\frac{3}{2}\pi$ **12.** $\sqrt{3}$

13. $\frac{b^2}{2}\left(\frac{b-a}{a+b}\right)$ **14.** $\frac{5+\sqrt{13}}{2}$ **15.** $\frac{9\sqrt{3}}{4}$

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ASSIGNMENT # 02 QUADRATIC EQUATION MATHEMATICS

Straight Objective Type

1. If the roots of $\ell x^2+2mx+n=0$ are real & distinct, then the roots of $(\ell+n)(\ell x^2+2mx+n)=2(\ell n-m^2)(x^2+1)$ will be -

(A) real & equal
(B) real & different
(C) imaginary
(D) may be real or imaginary

2. If α and β (where $\alpha,\beta \in R$) are the roots of equation $x^2+4x+b=0$ and $|\alpha-\beta| \le 2$, then the complete range of b is-

(A) (3,4) (B) (4,5) (C) [3,4] (D) (3,5)

3. Let $f(x)=\dfrac{x^2-5x+6}{x^2-3x+2}$, then range of $f(x)$ is -

(A) $(-\infty,1)\cup(1,\infty)$
(B) $(-\infty,-1)\cup(-1,1)\cup(1,\infty)$
(C) $(-\infty,\infty)$
(D) $(-\infty,-1)\cup(-1,\infty)$

4. The smallest positive number 'k' such that there exists $\alpha,\beta \in R$ satisfying $\alpha+\beta=k$ and $\alpha\beta=k$, is

(A) 4 (B) 1 (C) 2 (D) 8

5. Quadratic equation in x, $(k^2-2k+2)x^2-2(\sqrt{k^4+4})x+(k^2+2k+2)=0\ \forall\ k\in R$ has

(A) real & distinct roots
(B) equal roots
(C) imaginary roots
(D) roots of opposite sign

6. If equation $x^2-px+p=0$ has positive integral roots α,β, then value of $\alpha^{2016}-\beta^{2016}$ is (where $p\in R$)

(A) 4 (B) 2016 (C) p^2-4 (D) 0

7. If $a+b+c>\dfrac{9c}{4}$ and quadratic equation $ax^2+2bx-5c=0$ has non-real roots, then -

(A) $a>0,\ c>0$ (B) $a>0,\ c<0$ (C) $a<0,\ c<0$ (D) $a<0,\ c>0$

8. Let $f(x)=x^3+x+1$ and P(x) be a cubic polynomial such that $P(0)=-1$ and the roots of $P(x)=0$ are the squares of the roots of $f(x)=0$, then value of P(9) is -

(A) 98 (B) 899 (C) 80 (D) 898

9. If α and β are roots of the equation $ax^2+bx+c=0$ then roots of the equation $a(2x+1)^2-b(2x+1)(3-x)+c(3-x)^2=0$ are -

(A) $\dfrac{2\alpha+1}{\alpha-3},\dfrac{2\beta+1}{\beta-3}$ (B) $\dfrac{3\alpha+1}{\alpha-2},\dfrac{3\beta+1}{\beta-2}$ (C) $\dfrac{2\alpha-1}{\alpha-2},\dfrac{2\beta+1}{\beta-2}$ (D) none of these

10. The set of values of 'a' for which the inequality $x^2-(a+2)x-(a+3)<0$ is satisfied by atleast one positive real x, is -

(A) $[-3,\infty)$ (B) $(-3,\infty)$ (C) $(-\infty,-3)$ (D) $(-\infty,3]$

11. If one root of the equations $ax^2+bx+c=0$ and $bx^2+cx+a=0$ is common, then the value of $\left(\dfrac{a^3+b^3+c^3}{abc}\right)^2$ is -

(A) 1 (B) 3 (C) 9 (D) 16

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12. Set of the values of parameter 'm' for which every solution of the inequality $x^2 - 3x + 2 \le 0$ is also a solution of the inequality is $2x^2 - (m + 1)x - (2m + 3) < 0$,is -

(A) $\left(\frac{3}{4}, \infty\right)$ (B) $\left(-\frac{2}{3}, \infty\right)$ (C) $\left(-\frac{2}{3}, \frac{3}{4}\right)$ (D) $(-\infty, \infty)$

13. Number of rational roots of the equation $x^3 - 3x + 1 = 0$

(A) 0 (B) 1 (C) 2 (D) 3

14. The integral values of k for which the roots of $kx(x+2) + k - (x+2) = 0$ are rational is given by -

(A) 5 (B) –3 (C) –6 (D) 12

15. Number of values of $x \in R$ satisfying $\left|x - \frac{1}{x}\right| + \left|x + \frac{1}{x}\right| = 2$ are -

(A) 1 (B) 2 (C) 3 (D) 4

16. If one of the roots of $ax^2 + bx + c = 0$ is greater than 1 and the other is less than –3 and if the roots of $cx^2 + bx + a = 0$ are α & β, then -

(A) $\alpha < 1$ & $\beta > -\frac{1}{3}$ (B) $\alpha < 1$ & $\beta < -\frac{1}{3}$ (C) $\alpha > 1$ & $\beta > 1$ (D) $\alpha > 1$ & $\beta < -\frac{1}{3}$

17. If $px^2 - qx + r = 0$ has imaginary roots and $q < \frac{4p + r}{2}$ then set of real values of x satisfying the equation

$\left|p\left(x^2 + \frac{6}{p}\right) - x(q+5) + r\left(1 + \frac{x^2}{r}\right)\right| = \left|\left|px^2 - qx + r\right| - \left|x^2 - 5x + 6\right|\right|$, is -

(A) $(-\infty, 2] \cup [3, \infty)$ (B) $[2, 3]$ (C) R (D) no value of x

18. The set of values of 'a' for which $f(x) = ax^2 + 2x(1 - a) - 4$ is negative for exactly three integral values of x, is -

(A) (0, 2) (B) (0, 1] (C) [1, 2) (D) $[2, \infty)$

19. The number of possible value of 'α' for which the expression $y = \frac{\alpha x^2 + 7x - 2}{\alpha + 7x - 2x^2}$ has atleast one common linear factor in numerator and denominator, is -

(A) 0 (B) 1 (C) 2 (D) 3

20. If α, β are real numbers such that $\beta > \alpha$ & $|\alpha - \beta| < 2$, then the equation $(x - \alpha)(x - \beta) + 1 = 0$ has -

(A) both roots in (α, β) (B) both roots in $(-\infty, \alpha)$ (C) both roots in (β, ∞) (D) no real roots

21. If α, β, γ are roots of equation $111x^3 - 11x - 1 = 0$, then $(\alpha\beta)^{-2} + (\beta\gamma)^{-2} + (\gamma\alpha)^{-2}$ equals

(A) 2332 (B) 1331 (C) 1210 (D) 2442

22. If $4(a-1)x^2 + 4(a-1)x + (119 + 5a - 4a^2) > 0 \ \forall \ x \in R$, then number of integral values of 'a' is -

(A) 4 (B) 5 (C) 9 (D) 10

23. Let $f(x)$ is polynomial of two degree such that $f(1) = 6$, $f(-1) = 30$ and $f(3) > 0$. If one root is twice of the other root of the equation $f(x) = 0$, then the value of $f(3)$ is -

(A) 54 (B) 78 (C) 96 (D) 110

24. If $\frac{9 - |2x+3|}{2 - |x|} < 0$, then number of integral values of x is -

(A) 3 (B) 4 (C) 5 (D) more than 5

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25. If $a_n = \sin^n\theta + \cos^n\theta\ \forall\ n \in N$ and $4a_{n+2} - 2\sqrt{6}a_{n+1} + a_n = 0\ \forall\ n \in N$, then value of $(\sin\theta + \cos\theta)$ is -

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{1}{\sqrt{2}}$

26. The number of integers 'x' satisfying the equation $(x^2 - x - 1)^{x+2} = 1$ is-

(A) 2 (B) 3 (C) 4 (D) 5

27. Complete set of values of 'm' for which $\text{cosec}^4x + 2m\,\text{cosec}^2x + m^2 - 3 > 0\ \forall\ x \in R - \{n\pi\}$, is -

(A) R (B) $\left(\sqrt{3}-1,\infty\right)$ (C) $\left(-\infty,-\sqrt{3}-1\right)$ (D) $(-1,\infty)$

28. If sum of the roots of quadratic equation $ax^2 + bx + c = 0$ is equal to sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2}$ is equal to -

(A) 2 (B) –2 (C) 1 (D) –1

29. If $\alpha = \log_3 4$ is one of the roots of equation $x^3 + ax^2 + bx + 1 = 0$, $a,b \in R$, then equation which always have a root $\beta = \log_2 48$, is-

(A) $x^3 + 2(b - 6)x^2 + 4(a - 4b + 12)x + 8(4b - 2a - 7) = 0$

(B) $x^3 + bx^2 + ax + 1 = 0$

(C) $(65 + 16a + 4b)x^3 + 2(48 + 8a + b)x^2 + 4(12 + a)x + 8 = 0$

(D) $(65 + 16a + 4b)x^3 + (48 + 8a + b)x^2 + (12 + a)x + 8 = 0$

30. If $3a + 4b = 5$, where $a, b \in R$, then minimum value of $a^2 + b^2$ is -

(A) 1 (B) 4 (C) 5 (D) 10

31. If the equation $x^3 + 2x^2 - 4x + 5 = 0$ has roots α, β, γ, then the value of $\frac{(\alpha^3+5)(\beta^3+5)(\gamma^3+5)}{8\alpha\beta\gamma}$ is

(A) 104 (B) 8 (C) 13 (D) –104

32. If $p, q, r, s \in R$, then equation $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ must has-

(A) 6 real roots (B) atleast two real roots

(C) 2 real & 4 imaginary roots (D) 4 real & 2 imaginary roots

33. If $a^2\sin^2x - 2\sin x - (2a + 1) = 0$ $(a > 0)$ has exactly one root in $\left[0, \frac{\pi}{2}\right]$, then range of a is

(A) $[2, \infty)$ (B) $[1, \infty)$ (C) $[3, \infty)$ (D) none of these

34. The number of solutions of equation $|2ax - 3| + |ax + 1| + |5 - ax| = \frac{1}{2}$ is/are -

(A) 1 (B) 0 (C) 2 (D) 3

35. If α, β are roots of $x^2 + px + q = 0$, $p \neq 0$ and also of $x^{2n} + p^nx^n + q^n = 0$ and if $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$ are roots of $x^n + 1 + (x + 1)^n = 0$, then n is

(A) an integer (B) odd integer (C) even integer (D) none of these

36. Roots of the equation $2(x - a)(x - b) - (x - a^2 - b^2 - 1)^2 = 0$ are (where $a,b \in R$)

(A) real & equal (B) real & different (C) imaginary (D) imaginary and equal

37. If α, β, γ are the roots of the equation $ax^3 + bx^2 + cx + d = 0$, then

$a\left(\frac{\alpha^3}{\beta}+\frac{\beta^3}{\gamma}+\frac{\gamma^3}{\alpha}\right)+b\left(\frac{\alpha^2}{\beta}+\frac{\beta^2}{\gamma}+\frac{\gamma^2}{\alpha}\right)+c\left(\frac{\alpha}{\beta}+\frac{\beta}{\gamma}+\frac{\gamma}{\alpha}\right)$ is equal to-

(A) a (B) b (C) c (D) d

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38. If all roots of $x^4 - 8x^3 + ax^2 - bx + 16 = 0$ are positive real numbers, then $(a + b)$ is equal to -

(A) 32 (B) 24 (C) 56 (D) 8

39. Given α, β be the roots of quadratic equation, $x^2 - 2ax + 7 = 0$ and γ, δ be the roots of the equation, $x^2 - 2bx + 1 = 0$. If $\alpha\delta = \beta\gamma$, then $\frac{a^2}{b^2}$ is equal to -

(A) 5 (B) 6 (C) 7 (D) 8

40. Let α, β be roots of $x^2 - x + b = 0$ and $S_k = \alpha^k + \beta^k$, then the value(s) of b such that S_2, S_3 and S_4 are in arithmetic progression is -

(A) 0 (B) 1

(C) $\frac{1}{5}$ (D) no real value(s) of b exists

41. Let α,β,γ be real numbers which satisfy the relation $\alpha + \frac{1}{\beta\gamma} = \frac{1}{2}, \beta + \frac{1}{\alpha\gamma} = -\frac{1}{10}, \gamma + \frac{1}{\alpha\beta} = \frac{1}{18}$, then the value $\frac{\beta - \alpha}{\alpha - \gamma}$ is equal to-

(A) $-\frac{11}{40}$ (B) $\frac{11}{40}$ (C) $-\frac{27}{20}$ (D) none of these

42. There is only one real value of λ for which the quadratic equation $\lambda x^2 - 4x + 9 = 0$ has two positive integral solutions. The sum of these two solutions is-

(A) 4 (B) 8 (C) 9 (D) 12

43. Let α,β,γ and δ be the roots (real or non real) of equations $x^4 - 3x + 1 = 0$. The value of $\alpha^3 + \beta^3 + \gamma^3 + \delta^3$ is

(A) 3 (B) 12 (C) 9 (D) –9

Multiple Correct Answers Type

44. For the equation $\log_{x^2+6x+8}\left(\log_{2x^2+2x+3}(x^2-2x)\right) = 0$ -

(A) number of solutions is two (B) sum of solutions is '–4'

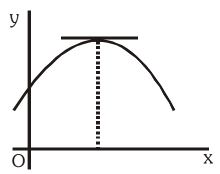
(C) number of solution is one (D) sum of solutions is '–1'

45. If eqautions $(a + 2)x^2 + bx + c = 0$ and $2x^2 + 3x + 4 = 0$ have a common root where $a, b, c \in N$ then -

(A) $b^2 - 4ac < 0$ (B) minimum value of $a + b + c$ is 16

(C) $b^2 < 4ac + 8c$ (D) minimum value of $a + b + c = 7$

46. Graph of $f(x) = px^2 - qx - r$ is shown in the adjacent figure then -



(A) $r(p - q - r) < 0$ (B) $q < 0$

(C) $\sqrt{q^2 + 4pr} > |q|$ (D) $p < 0$

47. If α, β and γ are the roots of the equation $x^3 - 3x + 1 = 0$ then -

(A) $\Pi(\alpha + 1) = -3$ (B) $\Sigma(\alpha + 1) = 0$

(C) $\Sigma(\alpha + 1)(\beta + 1) = -3$ (D) $\Sigma\alpha^2 = 6$

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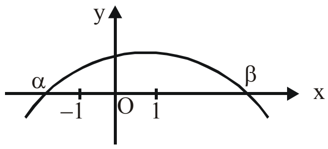
48. The values of k for which the system of equations 2x + ky = k, 3x – y = 10 has solution(s) satisfying x > 0, y > 0 are given by-

(A) $k < 0$ (B) $k < -\frac{2}{3}$ (C) $k > 0$ (D) $k > \frac{20}{3}$

49. The graph of quadratic polynomial $f(x) = ax^2 + bx + c$ is shown below.



Which of the following are correct ?

(A) $\frac{c}{a} < -1$ (B) $|\beta - \alpha| > 2$

(C) $f(x) > 0 \ \forall \ x \in (0,\beta)$ (D) $abc < 0$

50. If $x^2 + y^2 = 4$ and $E = x^2 + 2xy - y^2$, then-

(A) Maximum value of E is $4\sqrt{2}$ (B) Minimum value of E is $-4\sqrt{2}$

(C) $x \in [-2,2]$ (D) $y \in [-2,2]$

51. If the cubic polynomials $x^3 + ax^2 + 11x + 6$ and $x^3 + bx^2 + 14x + 8$ may have a common factor of the form $x^2 + px + q$, then -

(A) $a + p = b + q$ (B) $ap < bq$ (C) pq divides ab (D) p + q divides a + b

52. If the quadratic equation $x^2 - \frac{(\alpha^2+11)x}{9} + \frac{15}{4}(\alpha+\beta) + 16 = 0$ has two integral roots α & β, then -

(A) $\alpha + \beta = 20$ (B) $|\alpha - \beta| = 120$ (C) $|\alpha - \beta| = 6$ (D) $\alpha + \beta = 128$

53. If $ax^2 + bx + c$ (a > 1 & a, b, c are integers) is equal to p for two distinct integral values of x, $p \in$ prime, then $ax^2 + bx + c$ cannot be equal to 2p for -

(A) x = 0 (B) x = –1 (C) x = 1 (D) x = 3

54. If $2y^3 + \ell y^2 + my + 4 = 0$ such that $\ell + 15 < 2m$ and $2\ell + m + 10 < 0$, then atleast one root of the equation must lie in -

(A) (–1, 0) (B) (–2, –1) (C) (0, 2) (D) (1, 2)

55. If a, b, c are non zero real numbers and $(ax^2 + bx + c)(ax^2 - bx - c) = 0$, then which of the following can be true -

(A) equation has four real roots

(B) equation has two real and two imaginary roots

(C) equation has three real and one imaginary roots

(D) equation has four imaginary roots

56. If equation $(x - a)(x - 3) = -1$ has integral roots, then

(A) number of possible values of 'a' is 2. (B) number of possible values of 'a' is 4.

(C) sum of all possible values of 'a' is 6. (D) product of all possible values of 'a' is 5.

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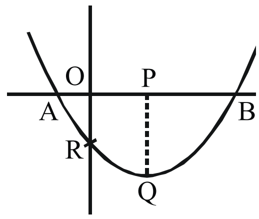
57. Let α,δ are roots of equation $x^2 + k_1x - 5 = 0$ and β, γ are roots of equation $x^2 + k_2x + k_3 = 0$. If α, β, γ & δ are in A.P., then which of the following is/are correct -

(A) $k_1 = k_2$
(B) If $k_3 = 3$, then $|\alpha + \beta + \gamma + \delta| = 8$
(C) $\alpha + \beta + \gamma + \delta = 2k_1$
(D) $\alpha + \beta + \gamma + \delta = 2k_3$

58. If equations $bx^2 + acx + b^2c = 0$ and $cx^2 + abx + b^2c = 0$ have a common root, (where a, b, c are non-zero distinct real numbers), then which of the following is/are correct -

(A) $\frac{a}{c}+\frac{b}{a}+\frac{a}{b}=0$
(B) $\frac{a}{b}+\frac{b}{c}+\frac{c}{a}=0$
(C) $\frac{1}{bc^2}+\frac{1}{a^2c}+\frac{1}{cb^2}=0$
(D) $\frac{1}{ac^2}+\frac{1}{ba^2}+\frac{1}{cb^2}=0$

59. Graph of $y = ax^2 + bx + c$ is shown in the figure and length of PQ & OR are 9 units & 5 units respectively (where Q is the vertex of the parabola) and area of ΔOBQ is $\frac{45}{4}$ $(\text{unit})^2$, then



(A) length of AB is 3
(B) length of AB is 4
(C) $\left|\frac{b}{a}\right|>1$
(D) $\left|\frac{b}{a}\right|<1$

60. If equation $x^2 + (2a + 4)x + (6a^2 + 4ab + b^2 + 8) = 0$ has real roots, then

(A) $a^b.b^a = 1$
(B) product of roots is -4
(C) product of roots is 16
(D) $(\sin a)^a + \left(\sec\left(\frac{b}{2}\right)\right)^{b/2} = 1$

61. For the equation $(x)^4 + (x + 2)^4 = 16(x + 1)^4$

(A) sum of real roots $= -2$
(B) sum of imaginary roots $= -2$
(C) product of real roots $= \frac{8}{7}$
(D) product of imaginary roots $= \frac{8}{7}$

62. If roots of the equation $\left(4^a+4^b+4^c\right)x^2-2\left(2^{a+b}+2^{b+c}+2^{c+d}\right)x+\left(4^b+4^c+4^d\right)=0$ are real, then which of the following is/are always correct-

(A) $a + b = c + d$
(B) $b=\frac{a+c}{2}$
(C) $a + d = b + c$
(D) $c=\frac{b+d}{2}$

63. If α and β are the roots of the equation $x^2 - 6x - 4 = 0$, then $\frac{\alpha^{17}+\beta^{17}-4\left(\alpha^{15}+\beta^{15}\right)}{\left(\alpha^8+\beta^8\right)^2-2^{17}}$ is a

(A) real number
(B) composite number
(C) prime number
(D) rational number

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64. Let P_1 and P_2 are possible real values of P for which expression $3x^2 + 2Pxy + 2y^2 + 4x - 4y + 1$ can be resolved into linear factors, then -

(A) $P_1 + P_2 = 8$ (B) $P_1 + P_2 = -8$

(C) P_1, P_2 both are positive (D) P_1, P_2 both are negative.

65. Let $E = (x + y)(y + z)$ & $x, y, z > 0$ also $xyz(x + y + z) = 1$, then λ = minimum value of E, then λ is

(A) Integer (B) Even

(C) Prime (D) Algebraic

66. Integral values of x for which $x^2 + 17x + 71$ is a perfect square of an integer is/are -

(A) –10 (B) –7 (C) 10 (D) 7

67. Let $P(x) = x^2 + bx + c$ where $b,c \in R$, $P(P(1)) = P(P(2)) = 0$ and $P(1) \neq P(2)$, then-

(A) $2b + 1 = 0$ (B) $2c + 3 = 0$ (C) $2b + 3 = 0$ (D) $2c + 1 = 0$

68. For the quadratic equation $x^2 + (2 - m)x - 4 = 0$ in x identify the correct statement(s)-

(A) if $m < 5$ then exactly one root is smaller than – 1

(B) if $m < 5$ then both roots are smaller than – 1

(C) if $m > 5$ then both roots are greater than –1

(D) roots are distinct real numbers for each $m \in R$

69. If the equation $x^4 + 2ax^3 + x^2 + 2ax + 1 = 0$ have at least two distinct negative roots, then the possible values of a is/are-

(A) 1 (B) 2 (C) 3 (D) 4

70. If $x^2 + 3x + 5$ is the greatest common divisor of $(x^3 + ax^2 + bx + 1)$ and $(2x^3 + 7x^2 + 13x + 5)$, then the value of $[a + b]$ is greater than or equal to-

(A) 7 (B) 8 (C) 9 (D) 10

[note [k] denotes greatest integer less than or equal to k]

71. If the equation $4x^4 + 2x^3 - 2x^2 + 3ax - a^2 = 0$ has four real roots, then a can be -

(A) $-\frac{1}{2}$ (B) 0 (C) $\frac{1}{16}$ (D) $\frac{1}{8}$

72. Suppose 'a' and 'b' are integers and $b \neq -1$. If the quadratic equation $x^2 + ax + b + 1 = 0$ has a positive integeral root, then which of the following is/are true-

(A) the other root is also a positive integer (B) the other root is an integer

(C) $a^2 + b^2$ is a prime number (D) $a^2 + b^2$ has a factor other than 1 and itself.

73. The equation $x^4 - (3m + 2)x^2 + m^2 = 0$ has four real roots in arithmetic progression. The possible value of m is/are -

(A) $-\frac{6}{19}$ (B) $\frac{6}{19}$ (C) 6 (D) –6

74. If x,y,z are three real number such that $x + y + z = 4$ and $x^2 + y^2 + z^2 = 6$, then-

(A) $x \in \left[\frac{2}{3}, 2\right]$ (B) $y \in \left[\frac{2}{3}, 2\right]$ (C) $z \in (2,3)$ (D) $y \in (2,3)$

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Linked Comprehension Type

Paragraph for Question 75 and 76

If α, β, γ are roots of the equation $x^3 + 2x^2 + 3x + 3 = 0$. Then

75. The value of $\frac{\alpha}{\alpha+1}+\frac{\beta}{\beta+1}+\frac{\gamma}{\gamma+1}$ is -

(A) 5 (B) 6 (C) –5 (D) –6

76. The value of $\left(\frac{\alpha}{\alpha+1}\right)^3+\left(\frac{\beta}{\beta+1}\right)^3+\left(\frac{\gamma}{\gamma+1}\right)^3$ is -

(A) 14 (B) 44 (C) 45 (D) 15

Paragraph for Question 77 and 78

Let a, b, c be real numbers with $a > 0$ such that the quadratic equation $ax^2 + bcx + b^3 + c^3 - 4abc = 0$ has no real roots. Let $P(x) = ax^2 + bx + c$ and $Q(x) = ax^2 + cx + b$.

77. Which one of the following is true

(A) $(c^2 - 4ab)(b^2 - 4ac) < 0$ (B) $(c^2 - 4ab)(b^2 - 4ac) > 0$

(C) $(a^2 - 4bc)(b^2 - 4ac) < 0$ (D) $(a^2 - 4bc)(b^2 - 4ac) > 0$

78. Which of the following is true for P(x) and Q(x)

(A) $P(x) > 0\ \forall\ x \in R$ and $Q(x) < 0\ \forall\ x \in R$

(B) $P(x) < 0\ \forall\ x \in R$ and $Q(x) > 0\ \forall\ x \in R$

(C) neither $P(x) > 0\ \forall\ x \in R$ nor $Q(x) > 0\ \forall\ x \in R$

(D) exactly one of P(x) or Q(x) is positive for all real x.

Paragraph for Question 79 to 80

Let α,β,γ be distinct real numbers such that

$$\left(2\sin\theta+1\right)\alpha^2+\left(\sqrt{3}\tan\theta-1\right)\alpha+\left(\cot\theta-\sqrt{3}\right)=0$$

$$\left(2\sin\theta+1\right)\beta^2+\left(\sqrt{3}\tan\theta-1\right)\beta+\left(\cot\theta-\sqrt{3}\right)=0$$

$$\left(2\sin\theta+1\right)\gamma^2+\left(\sqrt{3}\tan\theta-1\right)\gamma+\left(\cot\theta-\sqrt{3}\right)=0$$

79. For the value of θ obtained from the given information, the number of values of θ in $[-\pi, 2\pi]$ is -

(A) 2 (B) 3 (C) 4 (D) 5

80. For the value of θ obtained from the given information, the value of k for which the equations $x^2 + 2\sin\theta\, x + \cot^2\theta = 0$, $x^2 + (1 - k)x + k = 0$ has one root in common -

(A) k = 2 (B) k = 3 (C) $k = \frac{3}{2}$ (D) can't have common root

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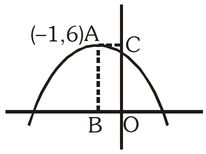
Paragraph for Question 81 to 83

Graph of $f(x) = ax^2 + bx + c$ is shown adjacently, for which A(–1,6) is the vertex and $b^2 - 4ac = 72$.



On the basis of above informations, answer the following questions :

81. The value of a + b + c is equal to -

(A) –6 (B) 6 (C) –10 (D) 10

82. The quadratic equation with rational coefficients whose one of the roots is $a + b - \sqrt{c}$, is -

(A) $x^2 - 18x + 78 = 0$ (B) $x^2 + 18x - 72 = 0$ (C) $x^2 + 18x + 78 = 0$ (D) $x^2 - 18x + 72 = 0$

83. Range of $g(x) = (a + \frac{1}{2})x^2 + (b + 2)x + c$ when $x \in [-1, 0]$ is -

(A) $\left[-\frac{3}{2}, \frac{9}{2}\right]$ (B) $\left[3, \frac{9}{2}\right]$ (C) $\left[0, \frac{23}{5}\right]$ (D) $\left[3, \frac{23}{5}\right]$

Paragraph for Question 84 to 86

A polynomial equation is given by $2x^4 + kx^3 + 22x^2 + kx + 2 = 0$.

On the basis of above information, answer the following questions :

84. Interval of k for which the equation has no imaginary roots can be -

(A) $k \in (12, 14)$ (B) $k \in (11, 12)$ (C) $[-12, -11]$ (D) $k \in [-13, -12]$

85. Interval of k for which exactly 2 roots of the equation are real can be -

(A) $k \in (-10, -8)$ (B) $k \in (-14, -12)$ (C) $k \in \phi$ (D) $k \in (13, 16)$

86. Interval of k for which there are no real roots of the equation can be -

(A) $k \in [-10, 10]$ (B) $k \in [3, 13]$ (C) $k \in (-13, -11]$ (D) $k \in [0, 13)$

Paragraph for Question 87 & 88

Let graph of quadratic expression $ax^2 + bx + c$, $a > 0$ be symmetric about $x = -\frac{1}{2}$ and $4a + 2b + c < 0$.

87. Sum of roots of the equation $ax^2 + bx + c = 0$ is -

(A) 1 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) –1

88. Value of (a + c) is -

(A) positive (B) negative (C) zero (D) can't say

Paragraph for Question 89 and 90

If α, β are the roots of equation $x^2 - 4x + 1 = 0$ and $P_n = \alpha^n + \beta^n$, $n \in N$

89. Value of $\frac{P_6 + P_4}{P_5}$ is-

(A) 1 (B) 2 (C) 4 (D) 6

90. Value of P_5 is-

(A) 724 (B) 1672 (C) 1364 (D) 1984

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Paragraph for Question 91 and 92

Let x_1,x_2,x_3 denotes three real roots of equation $x^3 - 3x - 1 = 0$ such that $x_1 < x_2 < x_3$

91. Value of $x_3^2 - x_2^2$ is equal to -

(A) $x_3 - x_1$ (B) $x_1(x_2 - x_3)$ (C) $x_1 - x_3$ (D) $x_1(x_3 - x_2)$

92. The value of expression $(2 - x_1)(2 - x_2)(2 - x_3)$ is equal to -

(A) 1 (B) -1

(C) $(1 + x_1)(1 + x_2)(1 + x_3)$ (D) $|(1 + x_1)(1 + x_2)(1 + x_3)|$

Paragraph for Question 93 and 94

Let $P : R \to R$ be a polynomial function with leading coefficient one which satisfies following properties.

(i) $P(x)$ is divisible by $x^2 + 1$.

(ii) $P(x) + 1$ is divisible by $x^3 + x^2 + 1$.

(iii) $P(x)$ of degree four.

93. If the equation $P(x) = \lambda$ (where λ is prime number) has atleast one integral solution, then sum of possible value(s) of λ is

(A) 2 (B) 5 (C) 7 (D) 12

94. The value of $P(3)$ is -

(A) 10 (B) 11 (C) 110 (D) 0

Paragraph for Question 95 and 96

Given that α, β are the roots of the equation $ax^2 + bx + c = 0$ such that $a, b, c, \alpha, \beta \in R$ and $a \neq 0$, $1 < \alpha < 3 < \beta$.

95. Roots of the equation $a(x + 3)^2 + b(x + 3) + c = 0$ are -

(A) both imaginary (B) both real and positive

(C) both negative (D) of different sign

96. Number of solutions of the equation $a(\sin x + 2)^2 + b(\sin x + 2) + c = 0$ in $(0, 2\pi]$ are -

(A) 1 (B) 2 (C) 3 (D) 4

Matrix Match Type

97. Let $f(x) = \dfrac{x^2 - 11x + 18}{x^2 - 7x + 12}$

Column-I		**Column-II**
(A)	Complete solution of $0 < f(x) < 1$	(P) $(3, 4)$
(B)	Complete solution of $f(x) < 1$	(Q) $\left(\frac{3}{2}, 2\right) \cup (9, \infty)$
(C)	Complete solution of $f(x) < 0$	(R) $(2, 3) \cup (4, 9)$
(D)	Complete solution of $f(x) > 2$	(S) $(-\infty, 2) \cup (3, 4) \cup (9, \infty)$
		(T) $\left(\frac{3}{2}, 3\right) \cup (4, \infty)$

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98.

	Column-I		Column-II
(A)	If $\|x + 2\| + \|x - 3\| = 5$, then sum of integral values of x satisfying the equation lies in the interval	(P)	$(-3, -1)$
(B)	If $\sqrt{x+\sqrt{x}}-\sqrt{x-\sqrt{x}}=\frac{3}{2}\sqrt{\frac{x}{x+\sqrt{x}}}$, then the value/s of x satisfying the equation may lie in the interval	(Q)	$[3, 5]$
		(R)	$(1, 3)$
(C)	If a, b, c are positive numbers such that $2a + 3b + 4c = 9$ then maximum value of $a^4b^2c^3$ lies in the interval	(S)	$(1, 2]$
(D)	If 6 lies between roots of the equation $x^2 + 2(m - 3)x + 9 = 0$ then 'm' may belong to the interval	(T)	$[5, 10]$

99. Match the set of values of x in column-II which satisfy the inequations in column-I.

	Column-I		Column-II
(A)	$\|x + 1\|+\|x - 2\|+\|x - 3\| \geq 6$	(P)	$x < 0$
(B)	$\|\|x\| - 1\| < 1 - x$	(Q)	$x = 3$
(C)	$\|x - 4\| < \sqrt{3-x}$	(R)	$x \leq 3$
(D)	$\sqrt{x^3-4x^2+4x-3}+\sqrt{3-x}+x^2 \geq 9$	(S)	$x \geq 4$
		(T)	$x \in \phi$

100.

	Column-I		Column-II
(A)	If x, y and z are real numbers such that $x^2 + y^2 + 2z^2 = 4x - 2z + 2yz - 5$, then the possible value of $(x - y - z)$ is	(P)	1
		(Q)	2
(B)	Value of $\left(1+\frac{1}{2}\right)\left(1+\frac{1}{2^2}\right)\left(1+\frac{1}{2^4}\right)\left(1+\frac{1}{2^8}\right)......\infty$ is	(R)	-3
(C)	If 'α' is root of $x^4 + x^2 - 1 = 0$ then value of $(\alpha^6 + 2\alpha^4)^7$ is	(S)	-1
(D)	The sum of all the real roots of the equation $(x - 1)^2 - 2\|x - 1\| - 3 = 0$ is	(T)	4

Numerical Grid

101. The equation $x^2 + bx + c = 0$ has distinct roots. If 3 is subtracted from each root, the result are reciprocal of the original roots. Then the value of $(b^2 + 6c + bc)$, is

102. The roots of a quadratic equation $ax^2 - 45x + 100 = 0$ are two consecutive natural numbers, where $a \in N$. The greater root is

103. It is given that α, β are roots of equation $x^2 + 2ax + 3 = 0$ and γ, δ are roots of equation $x^2 + 7x + 5 = 0$. If $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ is minimum, then [a] is equal to

(where [.] denotes greatest integer function)

104. If $\alpha=\sqrt[3]{45+29\sqrt{2}}+\sqrt[3]{45-29\sqrt{2}}$, then the value of $\alpha^3-21\alpha-90$ is equal to

105. Number of solutions of the equation $\left(\sqrt{3}+\sqrt{2}\right)^{x^2-15x}+\left(\sqrt{3}-\sqrt{2}\right)^{x^2-15x}=2\sqrt{3}$ is/are

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106. Number of rational values of x satisfying equation $x^{(\log_5 x)^2+5\log_5 x+2} = 5^8$ is/are

107. If the three equations $x^2 + px + 12 = 0$

$x^2 - qx + 15 = 0$

$x^2 + (p - q)x + 36 = 0$

have a common negative root then the value of $|p + q|$ is

108. The number of negative integral solutions of the equation $x^2 2^{x+1} + 2^{|x-3|+2} = x^2.2^{|x-3|+4} + 2^{x-1}$ is

109. If the equations $x^2 - 4x + 1 = 0$ and $x^2 - (4\sin\theta)x - 7 = 0$ has one root common, then number of values of θ in $(0, 2\pi)$ is

110. Solve the system of equation $(x \geq y \geq z \geq 2)$

$x^3 - 6z^2 + 12z - 8 = 0$

$y^3 - 6x^2 + 12x - 8 = 0$

$z^3 - 6y^2 + 12y - 8 = 0$

111. Solve for x satisfying $\sqrt{x+2\sqrt{x+2\sqrt{x+....+2\sqrt{x+2\sqrt{3x}}}}} = x \;\; (x > 0)$

112. Solve for x and y

$y^2 = x^3 - 3x^2 + 2x$

$x^2 = y^3 - 3y^2 + 2y \; (x \neq y)$

113. $\dfrac{(7^4+64)(15^4+64)(23^4+64)(31^4+64)(39^4+64)}{(3^4+64)(11^4+64)(19^4+64)(27^4+64)(35^4+64)} = ?$

114. Let P(x) be a monic polynomial with degree 4 and P(1) = 3, P(3) = 11, P(5) = 27 find P(–2) + 7P(6)

115. If $x + \sqrt[4]{5-x^4} = 2$, then find $x\sqrt[4]{5-x^4} \; (x \in R)$

116. Find sum of all real solutions to the equation $(x + 1)(x^2 + 1)(x^3 + 1) = 30x^3$

117. If $a + b + c = 3$

$a^2 + b^2 + c^2 = 29$

$abc = 11$

find $a^5 + b^5 + c^5$

118. Prove that sum of cubes of legs of right angled triangle is less than the cube of hypotenuse

119. Let $f(x) = x^2 - 2ax - a^2 - \dfrac{3}{4}$. Find all values of a such that $|f(x)| \leq 1 \; \forall \; x \in [0,1]$

120. Find the monic polynomial of eighth degree with rational coefficients having one root as $\sqrt{8+\sqrt{6}+4\sqrt{3}+3\sqrt{2}}$

121. Find number of real solutions to the equation $2x^4 - 3x^2 - 2x\sin x + 3 = 0$

122. If equations $2x^2 + bx + b = 0$ and $x^2 + 2x + 2b = 0$ have a common root, then number of integral values of 'b' is

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123. If α, β, γ are roots of $x^3 + 2x^2 - x + 3 = 0$, then value of $\dfrac{(\alpha^3+\alpha^2+3)(\beta^3+\beta^2+3)(\gamma^3+\gamma^2+3)}{\alpha\beta\gamma}$ is

124. Number of real solution(s) of the equation $3^{3x} - 3(3^{2x}) - 49(3^x) - 45 = 0$ is

125. The sum of all real values of 'm' such that the equation $(x^2 - 2mx - 4(m^2 + 1)).(x^2 - 4x - 2m(m^2 + 1)) = 0$ has exactly three different real roots is

126. Let α, β are roots of $x^2 - bx - b = 0$, $b > 0$ and $|\alpha|$, $|\beta|$ are roots of $x^2 + px + q = 0$. If minimum value of $(p^2 - 8q)$ is k, then value of $|k|$ is

127. For equation $(2x + 3)(3x + 4)(5x + 6)(30x + 31) = 10$. The value of greatest integer which is less than the product of real roots of this equation is

128. If $x+\dfrac{1}{x}=3$ and $x^5+\dfrac{1}{x^5}=a$, then sum of digits of 'a' is

129. If $a,b,c \in R - \{0\}$, such that both roots of the equation $ax^2 + bx + c = 0$ lies in (2,3), then both roots of the equation $a(x + 1)^2 + b(x^2 - 1) + c(x - 1)^2 = 0$ are always greater than k, then maximum integral value of k is

130. Let $P(x) = x^2 - x + 1$ and α be a root the equation $P(P(P(P(x)))) = 0$, then the absolute value of $(P(\alpha) - 1)P(\alpha)P(P(\alpha))P(P(P(\alpha)))$ is

131. Let $f(x)$ be a polynomial of degree 3 with integer coefficients such that $f(0) = 3$ and $f(1) = 11$. If $f(x) = 0$ has exactly 2 integer roots, then number of such polynomials is

132. The number of integral value(s) of $a \in (8,\infty)$ for which the expression $\log_{10}(4x^2 - ax + 3)$ is always non-negative for all $x \in [1,3]$ is

133. If each pair of $x^2 + ax + b = 0$, $x^2 + cx + d = 0$ and $x^2 + ex + f = 0$ has exactly one root in common and all equations do not have common roots, then the value of $\dfrac{(a+c+e)^2}{ac+ce+ea-b-d-f}$ is

134. If α and β are roots of $x^2 + 2x + 2 = 0$, then the value of $\alpha^3 + 2\beta$ is

135. Least positive integral value of k for which the equation $e^{2x} + (1 - 2k)e^x + k^2 - 1 = 0$ does not have any real solution, is (where e is Napier's constant)

136. The complete range of k for which the equation $x^4 - 2kx^2 + x + k^2 - k = 0$ has all real solutions is $\left[\dfrac{m}{n},\infty\right)$ where m and n are relatively prime numbers. Then m + n is

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ANSWER KEY

1.	C	**2.**	C	**3.**	B	**4.**	A	**5.**	B
6.	D	**7.**	B	**8.**	B	**9.**	B	**10.**	B
11.	C	**12.**	A	**13.**	A	**14.**	D	**15.**	B
16.	A	**17.**	B	**18.**	C	**19.**	D	**20.**	D
21.	D	**22.**	B	**23.**	D	**24.**	A	**25.**	C
26.	C	**27.**	B	**28.**	A	**29.**	A	**30.**	A
31.	C	**32.**	B	**33.**	C	**34.**	B	**35.**	C
36.	B	**37.**	C	**38.**	C	**39.**	C	**40.**	A
41.	C	**42.**	D	**43.**	C	**44.**	C,D	**45.**	B,C
46.	A,B,C,D	**47.**	A,D	**48.**	B,D	**49.**	A,B,C,D	**50.**	A,B,C,D
51.	A,C	**52.**	A,C	**53.**	A,B,C,D	**54.**	A,C	**55.**	A,B
56.	A,C,D	**57.**	A,B	**58.**	A,C	**59.**	A,C	**60.**	A,C,D
61.	A,B,D	**62.**	B,C,D	**63.**	A,B,D	**64.**	B,D	**65.**	A,B,C,D
66.	A,B	**67.**	A,B	**68.**	A,C,D	**69.**	A,B,C,D	**70.**	A,B
71.	A,B,C,D	**72.**	B,D	**73.**	A,C	**74.**	A,B	**75.**	A
76.	B	**77.**	A	**78.**	D	**79.**	A	**80.**	D
81.	A	**82.**	C	**83.**	D	**84.**	D	**85.**	D
86.	A	**87.**	D	**88.**	B	**89.**	C	**90.**	A
91.	A,B	**92.**	A,D	**93.**	C	**94.**	C	**95.**	D

96. B **97.** (A)→(Q); (B)→(T); (C)→(R); (D)→(P)

98. (A)→(Q); (B)→(R,S); (C)→(Q); (D)→(P)

99. (A) → (P,S); (B) → (P); (C) → (T); (D) → (Q)

100. (A)→(T); (B)→(Q); (C)→(P); (D)→(Q)

101.	6	**102.**	5	**103.**	2	**104.**	0	**105.**	4
106.	3	**107.**	1	**108.**	0	**109.**	2	**110.**	(2,2,2)
111.	3	**112.**	No solution	**113.**	337	**114.**	1112	**115.**	$\frac{8-\sqrt{42}}{2}$
119.	$\left[-\frac{1}{2},\frac{\sqrt{2}}{4}\right]$	**121.**	0	**122.**	1	**123.**	5	**124.**	1
125.	3	**126.**	4	**127.**	1	**128.**	6	**129.**	2
130.	1	**131.**	0	**132.**	0	**133.**	4	**134.**	0
135.	2	**136.**	7						

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One or more than one Correct Answer Type

1. If for an increasing GP, a,b,c and d are ℓ^{th}, m^{th}, n^{th} and p^{th} terms respectively and roots of equation $(a^2 + b^2 + c^2)x^2 - 2(ab + bc + cd)x + b^2 + c^2 + d^2 = 0$ are real, then-

(A) ℓ,m,n,p are in G.P

(B) ℓ,m,n,p are in A.P

(C) $b^2 + c^2 = ac + bd$

(D) $b^2 + c^2 = ad + bc$

2. The value of $\sum_{k=1}^{\infty} \frac{3k^2 + 3k + 1}{(k^2 + k)^3}$ is equal to

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

3. If the equation in x, $x^4 + px^3 + qx^2 = 16(2x - 1)$, where $p, q \in R$ has all positive roots, then

(A) $q : |p| = 3 : 2$ (B) $p > 8$ (C) $q \geq 4$ (D) $p < 0 < q < 8$

4. Least value of $\frac{x^2y^2 - 2x^2y + 2x^2 + 2xy - 2x + 1}{x^2y + x}$ is λ, then {where $x,y \in R^+$, $x^2y + x \neq 0$}

(A) $\lambda \in (0,1)$ (B) $\lambda \in [1,3)$ (C) $\lambda \in [3,4]$ (D) $\lambda \in (4,7)$

5. Value of $S = \sum_{n=1}^{\infty} \frac{\cot\left(\frac{(2n-1)\pi}{4}\right)}{3^{n+1}}$ is-

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{1}{3\sqrt{3}}$ (C) $\frac{1}{9}$ (D) $\frac{1}{12}$

6. Let $\frac{a}{b}$,ab, a – b and a + b are first 4 terms of an A.P. in order ($a,b \in R_0$), then-

(A) common difference of A.P is $-\frac{6}{5}$

(B) common difference of A.P is $-\frac{5}{6}$

(C) fifth term of A.P. will be $-\frac{117}{40}$

(D) $ab = \frac{27}{40}$

7. Let (a_n), n = 1,2,3.... be the sequence of real numbers such that $a_1 = 2$ and $a_n = \left(\frac{n+1}{n-1}\right)(a_1 + a_2 + ... + a_{n-1}), (n \geq 2)$, then-

(A) $\left[\frac{a_{2016}}{2^{2016}}\right] = 1008$

(B) $\left[\frac{a_{2016}}{2^{2016}}\right] = 2016$

(C) a_{2015} is divisible by 2^{2019}

(D) If $a_1 + a_2 + ... + a_n > 1008$, then least value of n is 8.

[where [.] denotes greatest integer function]

8. If a_1, a_2, a_3, a_4, are in G.P., then value of $(a_2 - a_3)^2 + (a_3 - a_1)^2 + (a_4 - a_2)^2$ is equal to

(A) $(a_1 + a_4)^2$ (B) $(a_1 - a_4)^2$ (C) $a_4^2 + a_1^2$ (D) $a_1^2 + 2a_4^2$

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9. The sum of infinite series $\frac{1}{3}+\frac{3}{3.7}+\frac{5}{3.7.11}+\frac{7}{3.7.11.15}+....$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$ (C) $\frac{4}{3}$ (D) $\frac{8}{3}$

10. If root of $28x^3 - 13kx^2 + 12x - 1 = 0$ are in H.P., then value of k is -

(A) –4 (B) –3 (C) 3 (D) 6

11. If a, b, c be three unequal positive quantities in H.P., then -

(A) $a^{100} + c^{100} > 2b^{100}$ (B) $a^3 + c^3 > 2b^3$ (C) $a^5 + b^5 < 2b^5$ (D) $a^2 + c^2 > 2b^2$

12. If $\sum_{r=1}^{n} r(r+1)(2r+3) = an^4 + bn^3 + cn^2 + dn + e$, then

(A) $a - b = d - c$

(B) $e = 0$

(C) $a,\ b-\frac{2}{3},\ c-1$ are in A.P.

(D) $\frac{b+d}{a}$ is an integer

13. If $2a^2, b^2, 2c^2$ are three distinct numbers in harmonic progression then-

(A) $4a^2 - b^2, b^2, 4c^2 - b^2$ are in G.P.

(B) $4a^2 - b^2, b^2, 4c^2 - b^2$ are in A.P.

(C) $\frac{ab}{c}, \frac{ac}{b}, \frac{bc}{a}$ are in A.P.

(D) $\frac{ab}{c}, \frac{2ac}{b}, \frac{bc}{a}$ are in A.P.

14. Let x,y,z be positive real numbers, then the least value of $\frac{x(1+y)+y(1+z)+z(1+x)}{\sqrt{xyz}}$ is equal to-

(A) $\frac{9}{\sqrt{2}}$ (B) 6 (C) $\frac{1}{\sqrt{6}}$ (D) 3

15. Let $S = \log_a bc + \log_b ca + \log_c ab$ where a,b,c are real numbers greater than 1, then 'S' can be equal to-

(A) 4 (B) 6 (C) 3 (D) 8

17. Value of $\frac{1}{1.7.13}+\frac{1}{7.13.19}+\frac{1}{13.19.25}+.....\infty$ is

(A) $\frac{1}{12}$ (B) $\frac{1}{7}$ (C) $\frac{1}{84}$ (D) $\frac{1}{91}$

16. Let $S_n = \sum_{r=2}^{n} \frac{3^{r-1}(2r-3)}{r(r-1)}$, then

(A) S_9 is divisible by 4

(B) S_9 is divisible by 21

(C) $10.S_{10}+3 = 3^{10}$

(D) $7(S_7 + 3), 10(S_{10} + 3), 13(S_{13}+ 3)$ are in G.P.

18. If for non-zero, distinct a, b and c, sum of roots of equation $ax^2 + bx + c = 0$ is equal to sum of squares of their reciprocals, then

(A) $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in H.P.

(B) $\frac{b}{c}, \frac{a}{b}, \frac{c}{a}$ are in H.P.

(C) $\frac{b}{c}, \frac{a}{b}, \frac{c}{a}$ are in A.P.

(D) $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are in A.P.

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19. Let x,y & z > 0 & x + 2y + 3z = 12, then-

(A) maximum value of xy^2z^3 is $\frac{32}{3}$
(B) maximum value of xy^2z^3 is 64
(C) minimum value of $\frac{1}{x}+\frac{2}{y}+\frac{3}{z}$ is 3
(D) minimum value of $\frac{1}{x}+\frac{2}{y}+\frac{3}{z}$ is 1

20. Between two numbers a and b, if A_1,A_2 are arithmetic means and H_1,H_2 are harmonic means, then

$\left[\frac{H_1H_2(A_1+A_2)}{H_1+H_2}\right]^2$ is equal to-

(A) $\frac{A_1H_2}{A_2H_1}$
(B) $A_1A_2H_1H_2$
(C) $\frac{A_2H_2}{A_1H_1}$
(D) $A_1^2H_2^2$

21. Consider a sequence $< t_n >$ such that $t_n^{\ 2} + t_n = t_{n+1}$, $\forall\ n \in N$ and $t_1 = 1$, then-

(A) $\prod_{m=1}^{99}(t_m+1)=t_{99}$
(B) $\prod_{m=1}^{99}(t_m+1)=t_{100}$
(C) $1+\sum_{m=1}^{99}t_m^2=t_{100}$
(D) $1+\sum_{m=1}^{99}t_m^2=t_{99}$

22. If a,b,c are positive real numbers, such that minimum value of $E = a^4 + 2b^4 + 4c^4 - 8abc$ is 'm' and a_0,b_0,c_0 are respective value of a,b,c for which E is minimum, then -

(A) m = 2
(B) m = –2
(C) $a_0b_0c_0 = 1$
(D) $a_0b_0c_0 = 4$

23. Let $S_p = T_1 + T_2 + ... + T_p$ & $S'_q = T_1' + T_2' + ... + T_q'$ (where $T_1,T_2.....T_p$ & $T_1', T_2',........T_q'$ both sequences are in A.P.). If $\frac{S_p}{S'_q}=\frac{p^2+p}{q^2+q}$, then $\frac{T_p}{T_q'}$ is -

(A) $\frac{p}{q}$
(B) $\frac{q}{p}$
(C) $\frac{2p+1}{2q+1}$
(D) $\frac{2p-1}{2q-1}$

24. If $\alpha,\beta,\gamma + 2$ are roots of $x^3 - \alpha x^2 + \beta x + 3\gamma = 0$ and they form an A.P. (in that order), then its common difference is

(A) 1
(B) 2
(C) –1
(D) –2

25. $\sum_{r=0}^{\infty}\frac{r-1}{2^{r+1}}$ is equal to-

(A) $\frac{1}{2}$
(B) 0
(C) 2
(D) $-\frac{1}{2}$

26. If $f(x)=\frac{(x-3)^7+3(x-3)^6+(x-3)^5+1}{(x-3)^5}, \forall x > 3$.

Then the least value of $f(x)$ is-

(A) 6
(B) 10
(C) 12
(D) 14

27. Sum of series $\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\frac{1}{20}+.....\infty$ is

(A) $\frac{1}{2}$
(B) 1
(C) $\frac{3}{2}$
(D) 2

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28. Let $a_1, a_2, a_3 \ldots\ldots a_n \ldots\ldots\ldots$ are in arithmetic progression. If $a_4 + a_8 + a_{12} = k$ $(k \neq 0)$, then sum of the first fifteen term of A.P is-

(A) 5k (B) $\frac{3}{2}k$ (C) 7k (D) $\frac{5}{3}k$

29. Which of the following cubic polynomials given below do not have their roots in arithmetic progression

(A) $x^3 + 3x^2 - 2x - 1$ (B) $x^3 + 3x^2 - 2x - 2$ (C) $x^3 + 3x^2 - 2x - 3$ (D) $x^3 + 3x^2 - 2x - 4$

30. In ΔABC if $\sin A \sin(B - C) = \sin C. \sin(A - B)$, then- (where $A \neq B \neq C$)

(A) tanA, tanB, tanC are in A.P (B) cotA, cotB, cotC are in A.P

(C) cos2A, cos2B, cos2C are in A.P (D) sin2A, sin2B, sin2C are in A.P

31. Let $a_1, a_2, a_3, \ldots$ be in A.P. and $g_1, g_2, g_3, \ldots$ be in G.P. if $a_1 = g_1 = 2$ and $a_{10} = g_{10} = 3$ then

(A) $a_7 g_{19}$ is an integers (B) $a_{19} g_7$ is an integer

(C) $a_7 g_{19} = a_{19} g_{10}$ (D) $a_7 g_{19} = a_{19} g_7$

32. If $x_1, x_2, x_3 \ldots\ldots x_n$ are 'n' distinct odd natural numbers not divisible by any prime number greater than 5, then $\left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \ldots + \frac{1}{x_n}\right)$ is always smaller than $(n \in N)$

(A) 1 (B) 3 (C) 5 (D) 7

33. If $T(k,n) = \sum_{r=1}^{n}\left(\sum_{i=1}^{r}\frac{1}{k^r}\right)$, (where $i,n,r \in N$), then which of the following is/are true -

(A) T(2,n) is always less than 2 (B) T(2,n) is always less than 3

(C) $T(3,\infty) = \frac{3}{4}$ (D) $T(3,\infty) = \frac{9}{4}$

34. Sum of the series $\left[\frac{1}{1.2}+1\right]+\left[\frac{1}{2.3}+1\right]+\left[\frac{1}{3.4}+1\right]+\left[\frac{1}{4.5}+1\right]\ldots$ upto to 30 terms-

(A) $\frac{30\times31}{32}$ (B) $\frac{31\times32}{30}$ (C) $\frac{30\times32}{31}$ (D) $\frac{31\times16}{30}$

35. If the first and $(2n - 1)^{th}$ term's of an A.P., a G.P. and a H.P. are equal and their n^{th} terms are a, b, c respectively, then -

(A) $a = b = c$ (B) $a \geq b \geq c$ (C) $a + c = b$ (D) $ac - b^2 = 0$

36. The sum of the series, $\frac{x}{1-x^2} + \frac{x^2}{1-x^4} + \frac{x^4}{1-x^8} + \ldots\ldots$ upto infinite terms. If $|x| < 1$, is

(A) $\frac{x}{1-x}$ (B) $\frac{1}{1-x}$ (C) $\frac{1+x}{1-x}$ (D) 1

37. Consider the sequence 1, 2, 2, 4, 4, 4, 4, 8, 8, 8, 8, 8, 8, 8, 8, ..., then 1025^{th} term is

(A) 2^9 (B) 2^{11} (C) 2^{10} (D) 2^{12}

38. If the inequality $\sqrt{2-x} + \sqrt{4-3x} + \sqrt{5-4x} + \sqrt{8x-7} \leq k$ holds good for all x in $\left[\frac{7}{8}, \frac{5}{4}\right]$, then possible values of k is (are)

(A) 3 (B) 7 (C) 5 (D) 9

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39. If $a_{n+1} = 2^n a_n$ and $a_1 = 1$, then $\log_2(a_{100})$ is equal to-
(A) 5050 (B) 4950 (C) 1020 (D) 5150

40. Sum of infinite terms of the series $\frac{5}{1.2}.\frac{1}{3}+\frac{7}{2.3}.\frac{1}{3^2}+\frac{9}{3.4}.\frac{1}{3^3}+\frac{11}{4.5}.\frac{1}{3^4}+....\infty$ equals to
(A) 1 (B) $\frac{5}{2}$ (C) $\frac{5}{3}$ (D) 2

41. If $a=\sum_{r=1}^{\infty}\frac{1}{r^2}$ and $b=\sum_{r=1}^{\infty}\frac{1}{(2r-1)^2}$, then $\frac{3a}{b}$
(A) 2 (B) 3 (C) 4 (D) 5

42. If equation $x^4 - 4x^3 + \lambda x^2 + \mu x + 1 = 0$ has four positive real roots, then-
(A) $|\lambda| + |\mu| = 2$ (B) $\lambda - \mu = 10$ (C) $|\lambda| + |\mu| = 10$ (D) $|\lambda| - |\mu| = 2$

43. If $T_n = (n^2 + 4n + 2)2^n$ and $S_{10}=\sum_{r=1}^{10}T_r$, then the value of $2 + S_{10}$ is equal to-
(A) $2^{10}.10^2$ (B) $2^{12}.12^2$ (C) $2^{11}.11^2$ (D) $2^{12}.11^2$

44. If x, y, z are in A.P. and x^2, y^2, z^2 are in H.P., then $(x \neq y \neq z)$
(A) $x + y = z$ (B) $x + z = y$
(C) $xy + yz + zx = 0$ (D) $(x + y + z)^2 = x^2 + y^2 + z^2$

45. In an AP, $t_1, t_2, t_3 ...... t_m$, if $t_1 + t_4 + t_8 + t_{12} + t_{16} + t_{20} + t_{24} + t_{27} = 224$, then $t_1 + t_2 + t_3 + ... + t_{27}$ is equal to -
(A) 576 (B) 448 (C) 675 (D) 756

46. From an ATM a person withdraw money by many transaction in arithmetic progression of absolute common difference ₹300. ATM has only ₹500 & ₹100 notes only (with maximum limit per transactions ₹3200). If ATM gives as long as possible maximum possible amount of any transaction in ₹500 notes, then choose correct option(s) -
(A) To get maximum amount number of transactions are 10.
(B) Maximum number of ₹500 notes he can get is 33.
(C) Maximum number of ₹500 notes he can get is 32.
(D) To get maximum amount number of transactions are 9.

47. If a, b > 0 such that $a^2 + 2b^2 = 12$, then-
(A) maximum value of $a + 2b$ is 6 (B) maximum value of ab^2 is 8
(C) maximum value of $\frac{2}{a}+\frac{4}{b}$ is 3 (D) minimum value of $\frac{2}{a}+\frac{4}{b}$ is 3

48. If $T_n=\frac{\left(n^2+n+2\right)2^n}{(n+1)(n+2)(n+3)}$, then -
(A) $\sum_{r=1}^{5}T_r=\frac{6.2^6}{7.8}-\frac{2}{2.3}$ (B) $\sum_{r=1}^{4}T_r=\frac{5.2^5}{6.7}-\frac{2}{2.3}$
(C) $\sum_{r=1}^{\infty}T_r=-\frac{2}{2.3}$ (D) $\sum_{r=1}^{\infty}T_r$ does not exist finitely

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49. The positive integers are in form of triangle as shown

$$\begin{array}{c} 1 \\ 2 \quad 3 \\ 4 \quad 5 \quad 6 \\ 7 \quad 8 \quad 9 \quad 10 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \end{array}$$

then row in which the number 100 will be

(A) 15 (B) 20 (C) 14 (D) 22

50. If a, b, c, d are four positive real numbers such that abcd = 1, then minimum value of $(1 + \Sigma a + \Sigma ab + \Sigma abc + abcd)$ is equal to -

(A) 16 (B) 8 (C) 12 (D) 14

51. $\sum_{k=1}^{\infty} \frac{6^k}{(3^{2k+1} + 2^{2k+1}) - (3^k 2^{k+1} + 2^k 3^{k+1})}$ is equal to -

(A) 3 (B) $\frac{1}{3}$ (C) $\frac{4}{3}$ (D) 2

52. If $x = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots$ upto ∞ terms

$y = \frac{1}{1^2} + \frac{3}{2^2} + \frac{1}{3^2} + \frac{3}{4^2} + \ldots\ldots$ upto ∞ terms

$z = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots\ldots$ upto ∞ terms, then

(A) x, y, z are in A.P.

(B) $\frac{y}{6}, \frac{x}{3}, \frac{z}{2}$ are in A.P.

(C) $z, \frac{x}{3}, \frac{y}{6}$ are in A.P.

(D) 6y, 3x, 2z are in A.P.

53. $\sum_{r=1}^{2n} (-1)^{r+1} r^3 + \sum_{r=1}^{n} (2r-1).(2r)$ is equal to

(A) $\frac{n(n+1)(2n+1)}{6}$ (B) $-\frac{n(2n+1)(4n+1)}{3}$ (C) $\frac{n(n+1)(2n+1)}{3}$ (D) $\frac{n(2n+1)(4n+1)}{6}$

54. If $5^{\log_2\left(\frac{x^2}{2}\right)} - 3^{2\log_4\left(\frac{x^2}{2}\right)} = \sqrt{3^{\log_{\sqrt{2}} 2x^2}} - 5^{2(\log_2 x^2 - 1)}$, then x is

(A) 2 (B) –2 (C) 1 (D) –1

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55. Let a_1,a_2,a_3,a_4 and b be real numbers such that $b+\sum_{k=1}^{4}a_k=8$, $b^2+\sum_{k=1}^{4}a_k^2=16$ then b can be-

(A) 2 (B) 3 (C) 4 (D) 5

56. $\sqrt{\underbrace{(111......1)}_{2n\text{ times}}-\underbrace{(22......2)}_{n\text{ times}}}=N$, then number of distinct digits in N is m, then m is equal to

(A) natural number (B) prime number
(C) composite number (D) algebraic number

57. If sum of n terms of the series 2.2 + 6.4 + 12.8 + 20.16 + 30.32 + ... $n(n + 1)2^n$ is expressed as $(n^2-\lambda n+\mu).2^{n+\delta}-4$, then which of the following is/are correct -

(A) $\lambda+\mu=3$ (B) $\mu+\delta=3$ (C) $\lambda+\mu=1$ (D) $\mu+\delta=-1$

58. Let $S_n=n+\frac{1}{2}\left(n-1+\frac{1}{2}\left(n-2+\frac{1}{2}\left(n-3+.......\frac{1}{2}(2)\right)\right)....\right)$. Then which of the following is/are true -

(A) $S_{2005}=4008$ (B) $S_{2005}=4010$ (C) $S_{2004}=4006$ (D) $S_{2004}=2008$

59. Value of $S=\sum_{k=1}^{n}\frac{4k}{4k^4+1}$ is -

(A) $1-\frac{1}{2n^2+2n+1}$ (B) $1-\frac{1}{2n^2+n+1}$ (C) $1-\frac{1}{n^2+n+1}$ (D) $2-\frac{1}{n^2+n+1}$

60. Let a_n be a sequence such that $a_1=2$ and $a_{n+1}=a_n+2n$ for all $n\in N$, then a_{100} is-

(A) 5050 (B) 9902 (C) 9950 (D) 10100

61. If a_i is positive or negative according as 'i' is even or odd and equation $x^4+a_1x^3+a_2x^2+a_3x+5=0$ has four positive real roots then if the minimum value of a_1a_3 is N, then the sum of the digits in 'N' is-

(A) 7 (B) 8 (C) 9 (D) 10

62. If $T_n=(n^2+1)n!$ & $S_n=T_1+T_2+T_3+......T_n$. Let $\frac{T_{10}}{S_{10}}=\frac{a}{b}$ where a & b are realtively prime natural numbers, then the value of (b – a) is

(A) 8 (B) 9 (C) 10 (D) 11

63. Given a_n is a sequence such that $a_1=1$, $a_2=2017$ and $\frac{a_{n-1}+n\,a_{n-2}}{n}=\frac{a_{n-1}^2}{a_n}$ for n > 2, then-

(A) a_5 is the greatest term of the sequence (B) a_6 is the greatest term of the sequence
(C) $a_m<a_{m-1}$ holds true for m = 6,7,8.... (D) $a_m<a_{m-1}$ holds true for m = 7,8,9....

64. If f(x) = 0 is a cubic equation with positive and distinct roots α, β, γ such that β is the harmonic mean of the roots of f '(x) = 0 and $k=\left[\frac{2\beta}{\alpha+\gamma}\right]$, (where [.] denotes greatest integer function) then the value of $\sum_{i=k}^{5}2^k$ is equal to

(A) 10 (B) 5 (C) 6 (D) 12

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65. Let S denote the set of all real values of x such that $(x^{2010}+1)\ (1+x^2+x^4+.....+x^{2008})=2010x^{2009}$, then

(A) The number of elements in S is 2

(B) The number of elements in S is 1

(C) Point (x, 2) lies inside the parabola $y=x^2-2x+2$

(D) Image of the point (x, 2) in the line mirror y = x lies on x + y = 4

66. If $S_n=\sum_{r=1}^{n}\frac{2r+1}{r^4+2r^3+r^2}$, then S_{20} is equal to $\frac{n}{n+1}$, then find $\sqrt{n+1}$

(A) 20 (B) 21 (C) 22 (D) 23

67. Consider the sequence of natural numbers $s_0, s_1, s_2, ...$ such that $s_0=3$, $s_1=3$ and $s_n=3+s_{n-1}s_{n-2}(n>1)$, then

(A) s_{1545} is odd and s_{1546} is even

(B) s_{1545} is even and s_{1546} is odd

(C) Both s_{1545} and s_{1546} are even

(D) Both s_{1545} and s_{1546} are odd

68. If x > 0, then greatest value of the expression $\frac{x^{50}}{1+x+x^2+......+x^{100}}$ is

(A) $\frac{1}{102}$ (B) $\frac{1}{101}$ (C) $\frac{1}{100}$ (D) None of these

69. The value of $\sum_{n=2}^{\infty}\frac{n}{1+n^2(n^2-2)}$ is equal to

(A) $\frac{5}{4}$ (B) 1 (C) $\frac{5}{16}$ (D) $\frac{1}{4}$

70. Let $E=x^{2017}+y^{2017}+z^{2017}-2017xyz$ (where $x, y, z \geq 0$), then the least value of E is -

(A) 0 (B) –2014 (C) –2017 (D) 2017

71. Let $T_n = 3n-2$ and $T'_m = 7m+2$ be two sequences each having 2004 terms. The number of common terms to the two sequences is-

(A) 283 (B) 284 (C) 285 (D) 286

Linked Comprehension Type

Paragraph for Question 72 & 73

Let S_i denotes the sum of first i terms of arithmetic sequence $a_1,a_2,a_3.......$, similarly S_j and S_k ($a_\alpha > 0$ where $\alpha = 1,2,3,.....$ and $i,j,k \in N$) then-

72. The value of the expression $\frac{S_i}{i}(j-k)+\frac{S_j}{j}(k-i)+\frac{S_k}{k}(i-j)$ is independent of -

(A) i

(B) j

(C) k

(D) choice of arithmetic sequence $a_1,a_2,a_3....$

73. For i = 5j, ratio $\frac{S_i}{S_j}$ does not depend on **j**, then which of the following can be true-

(A) $\frac{S_i}{S_j}=15$ (B) $\frac{S_i}{S_j}=25$ (C) $\frac{S_i}{S_j}=10$ (D) $\frac{S_i}{S_j}=5$

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Paragraph for Question 74 & 75

If $x_i \in \{-1, 0, 1, 2\}$ & $i = 1, 2, \ldots\ldots, n$ & $\sum x_i = 19$, $\sum x_i^2 = 99$.

74. If $\lambda = \sum x_i^3$ is minimum, then λ is -

(A) 19 (B) 0 (C) –19 (D) –100

75. If $\mu = \sum x_i^3$ is maximum, then

(A) $\mu = 133$
(B) if $x_i = 2$, then number of such x_i is 19.
(C) if $x_i = 1$, then number of such x_i is 2.
(D) cann't be calculated

Paragraph for Question 76 & 77

Let S_n be the sum of infinite terms of geometric series whose first term is n and the common ratio is $\frac{n+1}{2n+1}$ (where n = 1, 2, 3, ...).

76. Value of $S_1S_n + S_2S_{n-1} + \ldots\ldots + S_nS_1$ is -

(A) n
(B) 2n(n + 1)
(C) $\frac{n}{3}(2n^2+12n+13)$
(D) $\frac{n}{3}(2n+3)^2$

77. Value of $S_1^2 + S_2^2 + \ldots\ldots + S_n^2$ is -

(A) $\frac{n}{2}(2n+3)^2$
(B) $n(2n + 3)^2$
(C) $\frac{n}{3}(4n^2+12n+11)$
(D) $(2n + 1)^2$

Paragraph for Question 78 & 79

Consider the number sequence $a_n = \frac{1}{(n+1)\sqrt{n}+n\sqrt{n+1}}$, where n is a positive integer. The sum of the first n term is given by S_n.

78. How many terms in $S_1, S_2, S_3, S_4 \ldots\ldots S_{2014}$ are rational numbers -

(A) 42 (B) 43 (C) 44 (D) 41

79. Let $T_n = (1 - S_n)^{-2}$, the value of $T_1 + T_2 + T_3 + \ldots\ldots + T_{10}$ is equal to -

(A) 65 (B) 56 (C) 11 (D) 55

Paragraph for Question 80 & 81

Consider the series

$S_n = 1.2x + 2.3x^2 + 3.4x^3 + 4.5x^4 + 5.6x^5 + 6.7x^6 + 7.8x^7 + \ldots$.upto n terms.

80. If x = –1, then S_{30} is equal to-

(A) 250 (B) 500 (C) 480 (D) 450

81. If $x = \frac{1}{2}$, then S_∞ is equal to-

(A) 2 (B) 4 (C) 6 (D) 8

Paragraph for Question 82 & 83

Given $< a_n >$ is a sequence of real numbers for which $a_1 = 0$, $a_2 = 1$ and for $n \geq 3$, $a_n = a_{n-1} - a_{n-2}$, then

82. a_{2016} is equal to-

(A) 0 (B) 1 (C) –1 (D) none of these

83. Sum of first 100 terms of the sequence is-

(A) 2 (B) –2 (C) 1 (D) –1

Paragraph for Question 84 to 85

As we know that if T_n of any series can be written as $T_n = f(n) - f(n+1)$ or $f(n+1) - f(n)$, then sum of n terms can be $f(1) - f(n+1)$ or $f(n+1) - f(1)$ respectively.

If T_n(n^{th} term) of a series is $\dfrac{(n-2)2^n}{n(n+1)(n+2)}$ such that $s_n = \sum_{r=1}^{n} T_r$ and $S = \sum_{n=1}^{\infty} \dfrac{s_n + 1}{2^{n+1}}$.

On the basis of above information, answer the following questions :

84. Value of $\lim_{n \to \infty} s_n$ is -

(A) –1 (B) 1 (C) –2 (D) D.N.E.

85. Value of S is -

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) not defined

Paragraph for Question 86 & 87

Let $T_r = \dfrac{rx}{(1-x)(1-2x)(1-3x)....(1-rx)}, x \neq 0$ & $V_r = \dfrac{1}{(r^2 + 7r + 12)}$, then

86. $\sum_{r=2}^{\infty} T_r$ is-

(A) $\dfrac{1}{(1-x)}$ (B) $\dfrac{1}{(x-1)}$ (C) $\dfrac{1}{(x-1)(x-2)}$ (D) $\dfrac{1}{(x-1)(x-2)} - 2$

87. $\sum_{r=1}^{\infty} V_r$ is-

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{5}$ (D) $\frac{1}{4}$

Paragraph for Question 88 & 89

If a,b,c are in G.P and (loga – log2b), (log2b – log3c) and (log3c – loga) are in A.P then-

88. If a,b,c denotes sides of a triangle then a : b : c is-

(A) 9 : 6 : 4 (B) 4 : 2 : 1 (C) 9 : 3 : 1 (D) 16 : 4 :1

89. The value of $\dfrac{\tan\left(\frac{B+C}{2}\right)}{\tan\left(\frac{B-C}{2}\right)}$ for ΔABC with sides a,b and c is-

(A) 5 (B) $\frac{1}{5}$ (C) 6 (D) 4

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Numerical Grid Type

90. If α,β,γ are roots of equation $x^3 - 2x + 1 = 0$, then value of $\prod\left(\frac{1-\alpha}{1+\alpha}\right)-\sum\frac{1-\alpha}{1+\alpha}$ is equal to

91. If the number of three digit natural numbers whose consecutive digits are in G.P., is N, then the sum of digits of N is

92. If $a_1, a_2, a_3....$ is an arithmetic progression with common difference 1 and $a_1 + a_2 + a_3 + ... + a_{98} = 147$, then $a_2 + a_4 + a_6 + ... + a_{98} = N$, then number of digits in N is

93. If $S = \frac{3}{5} + \frac{10}{5^2} + \frac{21}{5^3} + \frac{36}{5^4} + \frac{55}{5^5} + ...\infty$ then 4S is equal to

94. Let $S = \sum_{k=0}^{\infty}\frac{2^k}{7^{2^k}+1}$ then $\frac{1}{S}$ is equal to

95. Let a,b are two integers such that $0 < a < b < 10^6$ and arithmetic mean of a and b is exactly 8 more than its geometric mean. If number of such ordered pairs is N, then N–990 is equal to

96. Let $<a_n>$ be an arithmetic sequence such that $\sum_{i=1}^{50} a_{2i-1} = 50$, then $\left|\sum_{j=1}^{50}(-1)^{\frac{j^2+j}{2}} a_{2j-1}\right|$ is equal to

97. Let x,y,z are three positive real numbers such that $xyz = 1$ then minimum value of $\frac{1+xy}{1+x}+\frac{1+yz}{1+y}+\frac{1+zx}{1+z}$ is equal to

98. If $\frac{p}{q}$ denotes the sum of the infinite series $\frac{1}{2}+\frac{1}{4}+\frac{2}{8}+\frac{3}{16}+\frac{5}{32}+\frac{8}{64}+\frac{13}{128}+\frac{21}{256}+\frac{34}{512}+.....$, where p,q are relatively prime positive integers, then $\left(\frac{p+q}{p-q}\right)$ is equal to

99. If a, b, c are three distinct positive numbers are in H.P. such that

$\left(\frac{a+b}{2a-b}\right)+\left(\frac{c+b}{2c-b}\right) > \sqrt{\lambda\sqrt{\lambda\sqrt{\lambda\sqrt{\lambda........\infty}}}}$, then maximum integral value of λ is equal to

100. Let α(a) and β(a) be the roots of the equation $\left(\sqrt[3]{1+a}-1\right)x^2+\left(1-\sqrt{1+a}\right)x+\sqrt[3]{1+a}=\sqrt[6]{1+a}$, where $a > 0$, $\alpha(a) > \beta(a)$. If minimum value of expression $f(a) = \alpha(a) + 12\beta(a) + \frac{4}{\alpha(a)\left(3\beta(a)\right)^2}, a > 0$ is m, then value of [m] is

[**Note :** [m] denotes the largest integer less than or equal to m]

101. If $\alpha = 1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{n}+\frac{1}{n+1}$ and $\beta = \frac{n+2}{2}-\left\{\frac{1}{(n+1)n}+\frac{2}{n(n-1)}+\frac{3}{(n-1)(n-2)}+....+\frac{n-1}{3.2}\right\}$.

Then the value of α – β is

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102. Consider S = {±1, ±2, ±3, ±4, ... ±10}. If T denotes sum of elements taken two at a time as product from S, then number of prime numbers involved in prime factorisation of |T| is

103. Let S_n represent sum to first n terms of a G.P. whose first term is 1 and common ratio is 3 and T_n represent the n^{th} term of another G.P. whose first term is 5 and common ratio is 5. If $\sum_{n=1}^{\infty}\frac{S_{n+1}}{T_{n+2}}=\frac{p}{q}$, where p and q are in their lowest form (p, q ∈ N), then $\left(\frac{q-10p}{6}\right)$ is equal to

104. If $\sec(2\theta-2\alpha), \frac{1}{2}\operatorname{cosec}^2\theta, \sec(2\theta+2\alpha)$ are in H.P., then the value of $\sec 2\theta \sec^2\alpha$ is

ANSWER KEY

1. B,C	**2.** D	**3.** C	**4.** A	**5.** D	**6.** A,C,D	**7.** A,C,D
8. B	**9.** A	**10.** C	**11.** A,B,D	**12.** A,B,C,D	**13.** A,D	**14.** B
15. B,D	**16.** A,B,D	**17.** C	**18.** A,C	**19.** B,C	**20.** B,D	**21.** B,C
22. B,C	**23.** A	**24.** B	**25.** B	**26.** A	**27.** B	**28.** A
29. A,B,C	**30.** B,C	**31.** A,C	**32.** B,C,D	**33.** A,B,C	**34.** C	**35.** B,D
36. A	**37.** C	**38.** B,C,D	**39.** B	**40.** A	**41.** C	**42.** B,C,D
43. C	**44.** C,D	**45.** D	**46.** B	**47.** A,B,D	**48.** A,B,D	**49.** C
50. A	**51.** D	**52.** B	**53.** B	**54.** A,B	**55.** A,B	**56.** A,D
57. A,B	**58.** A,C	**59.** A	**60.** B	**61.** B	**62.** B	**63.** B,D
64. C	**65.** B,C	**66.** B	**67.** D	**68.** B	**69.** C	**70.** B
71. D	**72.** A,B,C,D	**73.** B,D	**74.** A	**75.** A,B,C	**76.** C	**77.** C
78. B	**79.** A	**80.** C	**81.** D	**82.** C	**83.** A	**84.** D
85. B	**86.** B	**87.** D	**88.** A	**89.** A	**90.** 4	**91.** 8
92. 2	**93.** 5	**94.** 6	**95.** 5	**96.** 2	**97.** 3	**98.** 3
99. 4	**100.** 8	**101.** 0	**102.** 3	**103.** 5	**104.** 2	

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NAME : ....

Straight Objective Type

1. If $|x-5|^{x^2-3x-10}=1$, then the sum of solutions is -

(A) 13 (B) 8 (C) 15 (D) 3

2. Sum of solutions of the equation $3^{2x+1} + 2.3^x.7^x - 3.7^{2x+1} = 0$, is -

(A) 0 (B) –1 (C) 1 (D) 2

3. Values of x satisfying the equation $5^{6x^2} - 2.5^{3x^2+5x+2} + 5^{10x+4} = 0$ are -

(A) $2, \frac{2}{3}$ (B) $-2, -\frac{1}{3}$ (C) $2, -\frac{1}{3}$ (D) $-2, \frac{2}{3}$

4. Which of the following is true -

(A) $3^{150} > 2^{200} > 6^{100}$ (B) $6^{100} > 2^{200} > 3^{150}$ (C) $6^{100} > 3^{150} > 2^{200}$ (D) $3^{150} > 6^{100} > 2^{200}$

5. Product of the solutions of the equation $\left|\log_{\sqrt{3}} x - 2\right| - \left|\log_3 x - 2\right| = 2$ is

(A) –1 (B) 3 (C) 2 (D) 1

6. If $\log_2 9 + \log_9 2 = k$ $(k \in N)$ then largest integer less than k is

(A) 1 (B) 2 (C) 3 (D) 4

7. If α and β are solutions of equation $6^x - 3^{2x-1} - 2^{2x-1} + 6^{x-1} = 0$, then -

(A) $\alpha + \beta = 1$ (B) $\alpha + \beta = 0$ (C) $\alpha + \beta = \frac{1}{1-\log_2 3}$ (D) $\alpha + \beta = \log_2 3$

8. If $2^{2x+1} = 3^y$ and $3^{2y+1} = 2^x$ then $(x - y)$ is equal to-

(A) $\frac{\log_3 2 + \log_2 3}{3}$ (B) $\frac{\log_3 2 - \log_2 3}{3}$ (C) $\log_3 2 + \log_2 3$ (D) $\log_3 2 - \log_2 3$

9. The real x and y satisfy simultaneously $\log_{16} x + \log_4 y^2 = 9$ & $\log_{16} y + \log_4 x^2 = 6$, then the value of xy is equal to -

(A) 2^{12} (B) 3^{12} (C) 24 (D) 2^{-12}

10. The root of the equation $\sqrt{(\log_2 x)^2 - 4(\log_2 x) + 19} = \log_2(x^2 + 12)$, is
(where $x > 1$ and $\log_2 x$, $\log_2(x^2 + 12) \in I$)

(A) 5 (B) 2 (C) 3 (D) 4

11. The product of all real values of t satisfying the equation $\left|5^{\log_5^2 t} - 25\right| - 4t^{\log_5 t} = 0$, is -

(A) 1 (B) 5 (C) 25 (D) 125

12. Which of the following is incorrect :

(A) $\log_2 3 > \log_3 11$

(B) $\sum_{r=1}^{n} \log_{n!}(r) = 1$

(C) if $\frac{\log_8\left(\frac{8}{x^2}\right)}{(\log_8 x)^2} = 3$ then $x = \frac{1}{8}, 2$

(D) $\sqrt[3]{5^{\frac{1}{\log_7 5}} + \frac{1}{\sqrt{-\log_{10}(0.1)}}} = 2$

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13. The value of x for which $2^{\log_2^2(x-4)} + (x-4)^{\log_2(x-4)} = 32$ is -

(A) $x = 9$ (B) $x = 8$ (C) $x = 12$ (D) $x = 1$

14. If $\log_2 x = \log_7 8$, $\log_3 y = \log_5 9$, $\log_5 z = \log_{11} 25$, then the value of $x^{(\log_2 7)^2} + y^{(\log_3 5)^2} + z^{(\log_5 11)^2}$ is equal to-

(A) 389 (B) 489 (C) 589 (D) 289

15. Let α,β,γ are different real number such that $\frac{1}{(\alpha-\beta)^2} + \frac{1}{(\beta-\gamma)^2} + \frac{1}{(\gamma-\alpha)^2} = (5)^{2[\log_5 11 - \log_5 9]}$, then value of $\left(\frac{1}{(\alpha-\beta)} + \frac{1}{(\beta-\gamma)} + \frac{1}{(\gamma-\alpha)}\right)^2$ is

(A) $\frac{121}{81}$ (B) $\frac{11}{9}$ (C) $\frac{81}{121}$ (D) $\frac{9}{11}$

16. Let p,q are the roots of equation $x^2 + 2bx + c = 0$ and if

$2\log\left(\sqrt{y-p} + \sqrt{y-q}\right) = \log a + \log\left(y + b + \sqrt{y^2 + 2by + c}\right)$, (wherever defined), then a is -

(A) 1 (B) 2 (C) 0 (D) 4

17. The value of $(7)^{\log_{28} 112}.(4)^{\log_{28} 4}$ is -

(A) 12 (B) 14 (C) 28 (D) 112

18. If $\frac{1}{p} - \frac{1}{q} = \frac{1}{q} - \frac{1}{r}$, then $\frac{\log(p-r)}{\log(p+r) + \log(p-2q+r)}$ equals (Assume all terms are defined)

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) 4

19. If a,b,c are positive real numbers, then the value of $\frac{(ab)^{\log\frac{a}{b}}.(bc)^{\log\frac{b}{c}}.(ca)^{\log\frac{c}{a}}}{a^{\log\frac{b}{c}}.b^{\log\frac{c}{a}}.c^{\log\frac{a}{b}}}$ is equal to -

(A) 0 (B) –1 (C) 1 (D) none of these

20. If $x = \sqrt{\log_a b}$, $y = \sqrt{\log_b a}$, $a > 0$, $b > 0$, $a, b \neq 1$, then $x^a - y^{-b} = 0$ implies -

(A) a and b can not be equal (B) a and b must be equal

(C) a + b must be zero (D) a + b can be zero

21. If log x : log y : log z = (y – z) : (z – x) :(x – y), then-

(A) $x^y y^z z^x = 1$ (B) $x^x y^y z^z = 1$ (C) $\sqrt[x]{x}\,\sqrt[y]{y}\,\sqrt[z]{z} = 1$ (D) $xyz = 2$

22. If $\frac{\log_2 a}{2x+3y} = \frac{\log_2(b+1)}{5y+6z} = \frac{\log_2(c+2)}{3z+5x}$ (wherever defined) such that $(a-1)b(c+1) \neq 0$ and $abc + 2ab + ac + 2a - 1 = 0$, then value of $(7x + 8y + 9z)$ is equal to

(A) 0 (B) 1 (C) 24 (D) 6

23. Which of the following is greatest

(A) $\log_3 5$ (B) $\log_4 6$ (C) $\log_{15} 36$ (D) $\log_{16} 36$

24. Which of the following is smallest number -

(A) $\log_3 108$ (B) $\log_4 192$ (C) $\log_5 500$ (D) $\log_6 1080$

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25. If $x = \log_3 5$, $y = \log_{17} 25$, which one of the following is correct?

(A) $x < y$ (B) $x = y$ (C) $x > y$ (D) none of these

26. Identify the correct order -

(A) $\log_2 6 < \log_3 6 < \log_3 8 < \log_2 5$
(B) $\log_3 6 < \log_3 8 < \log_2 5 < \log_2 6$
(C) $\log_2 5 < \log_2 6 < \log_3 6 < \log_3 8$
(D) $\log_3 8 < \log_3 6 < \log_2 6 < \log_2 5$

27. Which of the following is largest ?

(A) $\log_2(\log_3(\log_4 5))$
(B) $\log_4(\log_3(\log_2 5))$
(C) $\log_2(\log_4(\log_3 5))$
(D) $\log_4(\log_3(\log_3 5))$

28. If x, y, z are positive integers such that $x^{y^z}.y^{z^x}.z^{x^y} = 5xyz$, then the value of $x+y+z$ is

(A) 2 (B) 4 (C) 6 (D) 8

29. Let (x_0, y_0) be the solution of the following equations $\left(5(x+1)\right)^{\ell n5} = (2y)^{\ell n2}$, $(x+1)^{\ell n2} = 5^{\ell ny}$, then x_0 is-

(A) $\frac{1}{5}$ (B) $-\frac{2}{5}$ (C) $-\frac{4}{5}$ (D) $\frac{4}{5}$

30. Let m denotes the number of positive integers whose logarithms to the base 3 having characteristic 4 and n denotes number of zero's after the decimal and before first significant digit in 2^{-100}, then remainder obtained when m is divided by n -

(A) 12 (B) 6 (C) 11 (D) 7

31. Number of solution of the equation $\log_{11}(x^4+5) = \log_4(1-x^6)$ (wherever defined)

(A) 0 (B) 1 (C) 4 (D) 6

32. The number of integer(s) which satisfy the inequality $\log_{x^2+3}\log_{|x|+3}\log_{|2x-1|+4}\left(\frac{x^2-1}{x^2+1}\right) \geq 0$ is

(A) infinite (B) 4 (C) 2 (D) No solution

33. The value of $\sqrt{(\log_7 9 - \log_3 5)^2} + \sqrt{(\log_7 9 + \log_3 5)^2}$ is -

(A) $2\log_7 9$ (B) $2\log_3 5$ (C) less than 4 (D) more than 2

Multiple Correct Answer Type

34. If $4^{\log_4^2 x} + x^{\log_4 x} = 512$, then x will be -

(A) 16 (B) $\frac{1}{16}$ (C) 8 (D) $\frac{1}{18}$

35. If $\log_{\frac{1}{2^n}} x^{\frac{1}{n}} + \log_{\frac{1}{2^{n-1}}} x^{\frac{1}{n-1}} + \ldots\ldots + \log_2 x = 8$, then x can be equal to -

(A) 2 (B) 4 (C) 16 (D) 256

36. Which of the following quantities are irrational ?

(A) $\sin 20°$ (B) $\log_{2+\sqrt{3}}(7+4\sqrt{3})$ (C) $\log_3 5$ (D) $\tan 5°$

37. If $0 < x < 1$, then

(A) $\log_e \pi > \log_x e > \log_x \pi$
(B) $\log_x \pi > \log_e \pi > \log_x e$
(C) $\log_x \frac{1}{\pi} > \log_{\frac{1}{x}} e > 0$
(D) $|\log_x \pi| > |\log_x e| > \log_e \frac{1}{\pi}$

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38. If $x = 2^{\log_4 3}$, $y = 2^{\log_8 4}$, $z = 2^{\log_{16} 5}$, $w = 2^{\log_{16} 8}$, then correct option(s) is/are -

(A) $x > w$ (B) $y > z$ (C) $x > z$ (D) $y > w$

39. Which of the following statement(s) is/are true ?

(A) $\log_{10}3$ lies between $\frac{1}{3}$ and $\frac{1}{2}$

(B) $\log_{\operatorname{cosec}\frac{\pi}{3}} \cos\frac{\pi}{6} = 1$

(C) $e^{\ell n(\ell n 7)}$ is greater than 1

(D) $\log_{10} 2 + \frac{1}{2}\log_{10} 5 + \log_{10}\left(2+\sqrt{5}\right) = \log_{10}\left(4\sqrt{5}+10\right)$

40. The values of x satisfying the equation $x^{\log_2^2 x - 5\log_2 x + 8} = 2^{\log_2 x^2}$ are -

(A) 1 (B) 2 (C) 4 (D) 8

41. Which of the following is correct -

(A) Number of solutions of the equation $\log_2(-2x) = 4\log_{x^2} x$ is zero.

(B) If $\log_{30}3 = a$, $\log_{30}5 = b$, then $\log_{30}8 = 3(1 - a + b)$.

(C) Number of solutions of the equation $x\log_{\sqrt{x}} x = \sqrt{x} + x$ is one.

(D) If $\log_5 2 = a$ and $\log_5 3 = b$, then $\log_{18}45 = \frac{2b+1}{2b+a}$.

42. If $\log_7 \log_{7^7} x - \log_5 \log_7 x = -1$, then-

(A) only 1 natural solution

(B) at least 1 prime solution

(C) one of the solution also satisfy the equation $\log_7\log_5 x = 0$

(D) one of the solution is twin prime with 3.

43. If $y = \log_{9-a}(2x^2 + 2x + a + 5)$ is defined for all $x \in R$, then possible values of 'a' are

(A) –5 (B) –4 (C) 4 (D) 5

44. Let $x, y, z > 1$ such that $\frac{\log_2^2 x}{\log_2 x + \log_3 y} = \frac{\log_2^2 x}{\log_2 x + \log_5 z} - 75$, $\frac{\log_3^2 y}{\log_3 y + \log_5 z} = \frac{\log_3^2 y}{\log_3 y + \log_2 x} + 36$ and

$\frac{\log_5^2 z}{\log_5 z + \log_2 x} = \frac{\log_5^2 z}{\log_5 z + \log_3 y} + P$, then which can be true -

(A) $\log_2 P > 4$ (B) $\log_3 P < 4$ (C) $P > 50$ (D) $P < 50$

45. If $2\left[\left(x+2\right)^{\log_2 y} - y\right] = xy$,where x and y are positive numbers, then which of the following can be possible ?

(A) $x > y$ (B) $x < y$ (C) $x + y = 1$ (D) $xy = 2$

46. If $\log_a x = b$, where everything is defined & $x \neq 1$, then identify correct statement(s) -

(A) $a, b \in Q$, then x can be rational

(B) $a, b \in \bar{Q}$, then x can be rational

(C) $a, b \in Q$, then x can be transcedental

(D) $a, b \in \bar{Q}$, then x can be transcedental

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Linked Comprehension Type (One or more than one correct)

Paragraph for Question 47 to 49

If the characteristic of logarithm of a positive real number N to the base 10 is

(i) n, then the number N will have n + 1 digits before the decimal.

or

(ii) –n, then the number N will have n–1 zeroes after the decimal and before first significant digit appears.

Let $\log_{10} 2 = 0.3010$ & $\log_{10} 3 = 0.4771$.

On the basis of above information, answer the following questions :

47. The number of digits in $(6)^{100}$ are -

(A) 75 (B) 76 (C) 77 (D) 78

48. The number of zeroes in $10^{100(\log_{10} 2 - \log_{10} 5)}$ after the decimal and before the first significant digits appears are -

(A) 37 (B) 38 (C) 39 (D) 40

49. The number of integers the logarithms of whose reciprocal to the base 10 have the characteristic "–n" are -

(A) 9.10^{n} (B) 9.10^{n-1} (C) 9.10^{n+1} (D) 9.10^{-n}

Paragraph for Question 50 to 51

Let $f(x) = 4^{(x + 1.5)} + 9^{(x + 0.5)} - 10 \times 6^{x}$ & $g(x) = \log_{10}(2^{x} + 1) - \log_{10}6 - x\log_{10}5$.

On the basis of above information, answer the following questions :

50. Sum of values of x satisfying the equation $f(x) = 0$, is -

(A) $\log_2 5$ (B) $\log_7 11$ (C) $\log_{2/3}\left(\frac{3}{8}\right)$ (D) $\log_{3/4}\left(\frac{5}{9}\right)$

51. Value of x satisfying $g(x) = -x$ is -

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Question 52 to 53

Let $f(x) = 4^{\log_7 x} + 3^{\log_7 x}$ and P is smallest solution(s) of the equation $f(x) = x$ and $g(x) = x^2 - (P-2)x + P - 3$.

52. Number of solution(s) of the equation $f(x) = x$ is -

(A) 1 (B) 2 (C) 3 (D) infinite

53. Number of integral value(s) of P for which roots of equation $g(x) = 0$ are imaginary, is -

(A) 0 (B) 1 (C) 2 (D) 3

Matching list type (4 × 4)

54. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	Value of $\log_2\left(\sqrt{2-\sqrt{5}+\sqrt{21-4\sqrt{5}}+\sqrt{14-6\sqrt{5}}}\right)$, is	(1)	25
(Q)	Value of $(9)^{3\log_{3\sqrt{3}}(5-\sqrt{20})-8.\log_{81}(\sqrt{5}-\sqrt{4})}$ is	(2)	3
(R)	Natural solution of the equation,	(3)	15
	$\frac{3x^4+x^2-2x-3}{3x^4-x^2+2x+3}=\frac{6x^4+2x^2-7x+3}{6x^4-2x^2+7x-3}$, is	(4)	1
(S)	Value of $3^{\log_{12}(60)}.(4)^{\log_{12}(5)}$, is		

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55. Let (x_1,y_1,z_1) and (x_2,y_2,z_2) are 2 set of solution satisfying the following system of equations

$\log_{10}(2xy) = 4 + (\log_{10}x - 1)(\log_{10}y - 2)$

$\log_{10}(2yz) = 4 + (\log_{10}y - 2)(\log_{10}z - 1)$

$\log_{10}(zx) = 2 + (\log_{10}z - 1)(\log_{10}x - 1)$ such that $x_1 > x_2$,

then match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	$\frac{y_1}{x_1}$	(1)	2
(Q)	$\frac{z_1}{x_2}$	(2)	100
(R)	$\frac{z_1x_2}{z_2}$	(3)	1000
(S)	$\frac{y_2+z_1}{x_2}$	(4)	150

Codes :

	P	Q	R	S
(A)	1	2	4	3
(B)	1	2	3	4
(C)	2	1	4	3
(D)	2	1	3	4

INTEGER TYPE

56. If the product of all real solutions of the equation $x^{2\log_2^3 x+\log_2^2 x+1} = \frac{2^{11\log_2^2 x}}{4}$ is λ, then the value of $\lambda\sqrt{2}$ is equal to

57. If $a = \left(\sqrt{4+\sqrt{15}} + \sqrt{4-\sqrt{15}} - 2\sqrt{3-\sqrt{5}}\right)$, $b = \left(\sqrt{17+4\sqrt{13}} - \sqrt{17-4\sqrt{13}}\right)$, then the value of $\log_a b$ is

58. The value of $(\log_6 2)^3 + \log_6 8.\log_6 3 + (\log_6 3)^3$ is

59. If $a = \log_{245}(175)$ and $b = \log_{1715}(875)$, then the value of $\frac{1-ab}{a-b}$ is

60. If $(\log_b a)^3 + (\log_c b)^3 + (\log_a c)^3 = 3$, where a, b, c are real numbers greater than 1, then value of $\frac{ab+bc+ca}{a^2+b^2+c^2}$ is equal to

61. If $\log_{2n}(1944) = \log_n(486\sqrt{2})$. Find the number of significant digits in n^6.
(Note : $\log_{10}2 = 0.3010$, $\log_{10}3 = 0.4771$)

62. Number of values of x satisfying $x^{\log_7 13} + (13)^{\log_{11} 17\log_7 x} = 2.(17)^{\log_{11}(13)^{\log_7 x}}$, is-

63. If $(\log_\beta \alpha)^{2/3} + (\log_\alpha \beta)^{2/3} = 3$, then the value of $\left(\frac{(\log_\beta \alpha + \log_\alpha \beta)^2}{5}\right)$ is

64. The number of solution of the equation $(x-2) + 2\log_2(2^x + 3x) = 2^x$ is

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65. $\left(\sum_{r=1}^{160}\frac{(\log_{(r+1)}3)(\log_{(r+2)}3)}{\log_{\frac{(r+2)}{(r+1)}}3}\right)(\log_3 162)(\log_3 2)$ is

66. The simplified value of the expression $2^{\log_3 5}\cdot 2^{(\log_3 5)^2}-5^{(\log_3 2)\cdot(\log_3 15)}$ is

67. $\frac{2^{\sqrt{\log_2 3}+\sqrt{\log_{10}3}}.5^{\sqrt{\log_{10}3}+\sqrt{\log_5 3}}}{3^{\sqrt{\log_3 2}+\sqrt{\log_3 10}+\sqrt{\log_3 5}}}$ is equal to

68. $\log_3\left(\sqrt[3]{1+\frac{2}{3}\sqrt{\frac{7}{3}}}+\sqrt[3]{1-\frac{2}{3}\sqrt{\frac{7}{3}}}\right)$

69. $\log_9 p = \log_{12} q = \log_{16}(p+q)$. If $\frac{p}{q}=\frac{x+\sqrt{y}}{z}$ (where x,y,z coprime integers), then x + y + z is

70. If x, y, z ∈ 2^n, where 'n' is a whole number and $\log_4 x + 2\log_4 y + 3\log_8 z = 2$, then number of possible ordered triplets (x, y, z) is

71. If x < 0, & $x^2 - x - 1 = 0$, then the characteristic of $(\log_{10}(x^4+1)(x^8+1)(x^{16}+1) - 14\log_{10}(-x))$ is

72. Number of real number(s) x for which the expression $\log_2|1+\log_2|2+\log_2|x|||$ is not defined is/are

73. Solve the equation $\ell n^2 y + \left(2^{x+1}+2^{1-x}\right)\ell n y + 2^{2x+1} + 2^{1-2x} = 0$

74. If $\sum_{r=1}^{44}\log_2(\sin 2r^\circ) - \sum_{r=1}^{89}\log_2(\cos r^\circ) = k$, then the value of $\left[\frac{k}{10}\right]$ is, where [·] denotes greatest integer function.

75. Let 1 < x < y < z such that $\log_x y + \log_y z + \log_z x = 8$, $\log_x z + \log_z y + \log_y x = \frac{25}{2}$. If $\log_y z = \frac{a+\sqrt{b}}{c}$, where a, b, c ∈ ℕ, gcd (a, b, c) = 2, b not a perfect square, then a + b + c is

76. Number of solution/s of the equation $\log_7\left(x^4+7\right) = \frac{7}{6}\log_{\frac{6}{7}}\left(x+\frac{7}{x}\right), x > 0$ is

77. Prove that there exists exactly one triplet satisfying $\begin{cases} x+\log\left(x+\sqrt{x^2+1}\right)=y \\ y+\log\left(y+\sqrt{y^2+1}\right)=z \\ z+\log\left(z+\sqrt{z^2+1}\right)=x \end{cases}$

78. Solve the system log(2xy) = logx.logy, log(yz) = logy.logz, log(2zx) = logz.logx, base of log is 10.

79. Let x,y and z be real numbers satisfying the system of equations

$$\log_2(xyz - 3 + \log_5 x) = 5$$
$$\log_3(xyz - 3 + \log_5 y) = 4$$
$$\log_4(xyz - 3 + \log_5 z) = 4$$

then the value of $|\log_5 z| - |\log_5 x| - |\log_5 y|$ is

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80. Let (x_0, y_0) be the solution of the equations

$$\log_{1024}\left(x^{\log_{1024} y}\right)+\log_{1024}^2 y+\log_{1024} x = 2\log_{1024} y \text{ and } 2\log_{32}^2 x+2\log_{32} x\log_{32} y = 3\log_{32} y$$

where $x_0 > y_0 > 1$, then the value of $\log_{64} x_0^{y_0}$ is

81. If $1+x^{\frac{3}{2}} = 3^{\log_2 x}$ then sum of all possible value(s) of x is

82. If a,b,c are distinct non-zero real numbers such that $a+\frac{1}{b}=b+\frac{1}{c}=c+\frac{1}{a}$, find $\log_2|abc|$

83. If x,y,z are positive real numbers, then the minimum value of $x^{\log\left(\frac{y}{z}\right)}+y^{\log\left(\frac{z}{x}\right)}+z^{\log\left(\frac{x}{y}\right)}$ is equal to

ANSWER KEY

1.	B	**2.**	B	**3.**	C	**4.**	C	**5.**	D
6.	C	**7.**	A	**8.**	B	**9.**	A	**10.**	B
11.	A	**12.**	A	**13.**	B	**14.**	B	**15.**	A
16.	B	**17.**	C	**18.**	B	**19.**	C	**20.**	B
21.	B	**22.**	A	**23.**	A	**24.**	B	**25.**	C
26.	B	**27.**	B	**28.**	D	**29.**	C	**30.**	A
31.	A	**32.**	D	**33.**	B,C,D	**34.**	A,B	**35.**	A,B,C,D
36.	A,C,D	**37.**	A,C,D	**38.**	A,B,C	**39.**	A,C,D	**40.**	A,C,D
41.	A,D	**42.**	A,B	**43.**	B,C,D	**44.**	A,B,D	**45.**	A,B,D
46.	A,B,D	**47.**	D	**48.**	C	**49.**	B	**50.**	C
51.	B	**52.**	A	**53.**	A	**54.**	A	**55.**	A
56.	1	**57.**	4	**58.**	1	**59.**	5	**60.**	1
61.	12	**62.**	1	**63**	4	**64.**	2	**65.**	4
66.	0	**67.**	1	**68.**	0	**69.**	6	**70.**	6
71.	2	**72.**	7	**73.**	$x = 0,\ y = e^{-2}$	**74.**	4	**75.**	$\frac{6+\sqrt{34}}{2}$, 42
76.	0	**78.**	$(200,100,100);\left(\frac{1}{2},1,1\right)$			**79.**	3	**80.**	2
81.	4	**82.**	0	**83.**	3				

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Straight Objective Type

1. sinx + cosx = y^2–y+a has no values of x for any y, if a belongs to :

(A) $\left(0,\frac{\sqrt{3}}{2}\right)$ (B) $\left(-\sqrt{3},0\right)$ (C) $\left(-\infty,-\sqrt{3}\right)$ (D) $\left(\sqrt{3},\infty\right)$

2. If $x \in [-\pi,3\pi]$, then number of solution(s) of the equation $(3 + \sin x)^2 = 4-2\cos^2 x$ is

(A) 0 (B) 1 (C) 2 (D) 4

3. If $\alpha,\beta \in \left(0,\frac{\pi}{4}\right)$, then the value(s) of k for which the equation $\sin(\alpha+\beta) = k\sin\left(2\sqrt{\alpha\beta}\right)$ has solution, is -

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}}{2}$ (D) π

4. Number of ordered pairs (θ, ϕ) satisfying the simultaneous equations $\sin\theta = \cos 2\phi$, $\sin 2\phi = \cos 2\theta$, where $\theta, \phi \in (0, \pi)$, is -

(A) 1 (B) 2 (C) 0 (D) 3

5. Number of integral values of x which satisfies equation

$\frac{\sin^3 x-\cos^3 x}{\sin x+\cos x} = \frac{\operatorname{cosec}^2 x-\cot x-2\cos^2 x}{1-\cot^2 x}, x \in [0,2\pi]$, is-

(A) 0 (B) 3 (C) 6 (D) 7

6. The least positive value of x satisfying equation $\frac{4\sin^4 x}{3+\cos^2 2x-4\sin^2 x} = \frac{1}{9}$ is $\frac{\pi}{A}$, then A is .

(A) 1 (B) 4 (C) 6 (D) 8

7. Number of solutions of $\cos^{2m+1}x - \sin^{2m+1}x = 1$ in $(-\pi,\pi]$ is equal to $(m \in N)$

(A) 2m +1 (B) 2 (C) 4m + 2 (D) 1

8. The number of values of $x \in (0°,90°)$ satisfying the equation $\tan x = \tan(x + 10°)\tan(x + 20°)\tan(x + 30°)$ is -

(A) 1 (B) 2 (C) 3 (D) 4

9. Complete set of values of x in $(0,\pi)$ satisfying the inequality $1 + \log_2(-\cos x) + \log_2 \cos 3x \geq 0$, is-

(A) $\left(\frac{\pi}{2},\frac{3\pi}{4}\right]$ (B) $\left(\frac{\pi}{2},\frac{5\pi}{6}\right]$ (C) $\left[\frac{3\pi}{4},\frac{5\pi}{6}\right]$ (D) $\left[\frac{2\pi}{3},\frac{3\pi}{4}\right]$

10. Consider system of equations in x,y,z,

$$x + y + z = 2\pi,$$
$$\tan x \tan y = 3,$$
$$\tan y \tan z = 9.$$

If total number of ordered triplet (x,y,z), where $x,y,z \in [-2\pi,2\pi]$ is N, then -

(A) $N \geq 20$ (B) $16 \leq N \leq 19$ (C) $11 \leq N \leq 15$ (D) $N \leq 10$

11. Sum of all $x \in [-2\pi,2\pi]$ satisfying $\sin^4 x + \cos^3 x = 1$, is-

(A) 0 (B) π (C) 2π (D) 3π

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12. If $a+b+c=\frac{\pi}{2}$, then least positive solution of equation $\tan(x+a)\tan(x+b)+\tan(x+b)\tan(x+c)+\tan(x+c)\tan(x+a)=1$, is -

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

13. $\cos^8x + b\cos^4x + 1 = 0$ will have a solution if b belongs to :

(A) $(-\infty,2]$ (B) $[2,\infty)$ (C) $(-\infty,-2]$ (D) None of these

14. Number of solutions of the equation $\sqrt{1-\cos x} = \sin x$, which lie in the interval $[\pi, 5\pi]$:

(A) 6 (B) 5 (C) 4 (D) 3

15. Number of solutions of equation $(\sin x + \cos x + 2)^4 = 128 \sin 2x$ in $x \in [0, \pi/2]$

(A) 0 (B) 1 (C) 2 (D) 3

16. Total number of ordered pairs (x, y) satisfying the equation $|x| + |y| = 2$ and $\sin\left(\frac{\pi x^2}{3}\right) = 1$:

(A) 2 (B) 4 (C) 6 (D) 8

17. The complete set of values of x, satisfying the inequality $\cos 2x > |\sin x|$ in $x \in \left(-\frac{\pi}{2},\pi\right)$:

(A) $\left(-\frac{\pi}{6},\frac{\pi}{6}\right)$

(B) $\left(-\frac{\pi}{2},-\frac{\pi}{6}\right)\cup\left(\frac{\pi}{6},\frac{5\pi}{6}\right)$

(C) $\left(-\frac{\pi}{2},-\frac{\pi}{6}\right)\cup\left(\frac{5\pi}{6},\pi\right)$

(D) $\left(-\frac{\pi}{6},\frac{\pi}{6}\right)\cup\left(\frac{5\pi}{6},\pi\right)$

18. If the inequality $\sin^2x + a\cos x + a^2 > 1 + \cos x$ holds for any $x \in R$ then the largest negative integral value of 'a' is :

(A) –4 (B) –3 (C) –2 (D) –1

19. Let $\theta \in [0, 4\pi]$ satisfying the equation $(\sin\theta + 2)(\sin\theta + 3)(\sin\theta + 4) = 6$. If the sum of all values of θ is of form $K\pi$ then the value of K is :

(A) 6 (B) 5 (C) 4 (D) 2

20. If $\sec\theta + \text{cosec}\theta = p$ has four solutions between $(0, 2\pi)$, then :

(A) $p^2 - 8 > 0$ (B) $p^2 + 8 > 0$ (C) $p^2 - 8 < 0$ (D) $p^2 + 8 < 0$

21. If $4\cos\theta\cos2\theta\cos3\theta = 1$, then number of values of $\theta \in [0,\pi]$ are -

(A) 9 (B) 5 (C) 6 (D) 8

Multiple Correct Answer Type

22. If solution of the equation $\frac{\sqrt{6}-\sqrt{2}}{\sin x}+\frac{\sqrt{6}+\sqrt{2}}{\cos x}=8$ in $x \in \left(0,\frac{\pi}{2}\right)$ is $\frac{m\pi}{n}$ (where m & n are coprime numbers), then (m + n) can be

(A) 23 (B) 39 (C) 47 (D) 13

23. Let $-\frac{\pi}{2}<\alpha_1<\alpha_2<\alpha_3<......<\alpha_n<2\pi$ and $0 < \beta_1 < \beta_2 < .... < \beta_m < 2\pi$ are values of α and β respectively such that $2\cos2\alpha - 3 = 2\cos(2\alpha + \beta) - 2\cos\beta$, then which of the following is(are) true ?

(A) $2n > 3m$ (B) $n > m$ (C) $\sum_{i=1}^{m}\beta_i \le 2\pi$ (D) $\sum_{i=1}^{m}\beta_i \le \sum_{i=1}^{m}\alpha_i$

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24. If $\sin^3\theta + \sin\theta\cos\theta + \cos^3\theta = 1$, then θ is equal to $(n \in I)$

(A) $2n\pi$ (B) $2n\pi + \frac{\pi}{2}$ (C) $2n\pi - \frac{\pi}{2}$ (D) $n\pi$

25. Consider equation $4\left[3\sqrt{4x-x^2}\sin^2\left(\frac{x+y}{2}\right)+2\cos(x+y)\right]=13+4\cos^2(x+y)$, $x, y \in [0, 10\pi]$. Let P is set of all different values of x and Q is set of all different values of y and N is number of ordered pair (x, y) which satisfying given equation, then

(A) sum of all elements of set P is prime number

(B) sum of all elements of set Q is less than 160

(C) N is a composite number

(D) $N > 8$

26. If $\theta_1, \theta_2, \theta_3, \theta_4$ be the four values of 'θ' which satisfy the equation $\sin(\theta + \alpha) = \lambda \sin 2\theta$, no two of which differ by a multiple of 2π, then $\theta_1 + \theta_2 + \theta_3 + \theta_4$ can be :

(A) π (B) 2π (C) 3π (D) 5π

27. If $(\sin\alpha)x^2 - 2x + b \geq 2$ for all the real values of $x \leq 1$ and $\alpha \in \left(0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\pi\right)$ then possible real values of b is/are :

(A) 2 (B) 3 (C) 4 (D) 5

28. $\cos(\sin x) < \frac{1}{\sqrt{2}}$, then x lies in the interval :

(A) $\left(\frac{\pi}{4},\frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{4},0\right)$ (C) $\left(\pi,\frac{3\pi}{2}\right)$ (D) $\left[\frac{\pi}{2},\pi\right]$

29. Which of the following values of θ may satisfy the equation $\tan^6\theta + 27\tan^2\theta = 3(1 + 11\tan^4\theta)$

(A) 20° (B) 15° (C) 60° (D) 40°

30. Consider the equation $\cos 6x + \tan^2 x + \cos 6x.\tan^2 x = 1$, where $x \in [0, 2\pi]$. If L denotes the number of solutions in given interval and M denotes the sum of the solutions in the given interval, then -

(A) $L = 7$ (B) $L = 9$ (C) $M = 7\pi$ (D) $M = 9\pi$

Linked Comprehension Type

Paragraph For Questions 31 to 33

Consider the equation $\sec\theta + \cot\theta = \frac{31}{12}$

On the basis of above, answer the following

31. If $0 < \theta < \frac{\pi}{2}$, then minimum value of $[\tan\theta]$ is equal to (where $[\cdot]$ is G.I.F.)

(A) 0 (B) 1 (C) 2 (D) 3

32. Number of values of θ where $\theta \in \left(\frac{\pi}{2},\frac{3\pi}{2}\right)$ is equal to

(A) 0 (B) 1 (C) 2 (D) 3

33. Number of values of θ where $\theta \in [0, 5\pi]$ is equal to

(A) 4 (B) 6 (C) 8 (D) 10

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Paragraph for Question 34 and 35

Let x = α and y = β satisfy equation $x^4 - 2x^2\sin^2\left(\frac{\pi}{2}xy\right) + 1 = 0$, then answer the following questions.

34. Number of ordered pairs (α,β) is -

(A) 2 (B) 0 (C) 1 (D) infinite

35. Minimum value of $|\alpha - \beta| + |\alpha + \beta|$ is-

(A) 2 (B) 1 (C) 0 (D) 3

Paragraph for Question 36 & 37

Consider system of equation in variables α, β,

$[\sin(\alpha - \beta) + \cos(\alpha + 2\beta)\sin\beta]^2 = 4\cos\alpha\sin\beta\sin(\alpha + \beta)$ and $\tan\alpha = 3\tan\beta$.

36. If sum of all possible values of $\sin^2\beta$ is $\frac{a}{b}$ (where a and b are coprime numbers), then value of 2a + b is -

(A) 17 (B) 25 (C) 13 (D) 22

37. Number of possible value(s) of β in $\beta \in [-\pi, 4\pi]$, is -

(A) 10 (B) 12 (C) 14 (D) 16

Paragraph for Question 38 & 39

Consider $\cos^2\theta - \sin^2\theta = \sqrt{\frac{5+\sqrt{5}}{8}}$; $\theta \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$.

On the basis of above information, answer the following questions :

38. Number of values of θ satisfying the given equation is -

(A) 2 (B) 3 (C) 4 (D) 5

39. The difference between greatest and least value of θ is -

(A) $\frac{\pi}{5}$ (B) $\frac{\pi}{15}$ (C) $\frac{\pi}{20}$ (D) $\frac{\pi}{10}$

Paragraph for Question 40 to 42

Consider the system of equation :

$\sin x \cos 2y = (a^2 - 1)^2 + 1$

$\cos x \sin 2y = a + 1$

40. The number of values of 'a' for which the system has a solution :

(A) 1 (B) 2 (C) 3 (D) Infinity

41. The number of values of $y \in [0, 2\pi]$ when the system has solution for permissible values of a is :

(A) 2 (B) 3 (C) 4 (D) 5

42. The number of values of $x \in [0, 2\pi]$ when the system has solution for permissible values of a is :

(A) 1 (B) 2 (C) 3 (D) 4

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Paragraph for Question 43 to 45

Let S_1 be the set of all those solutions of the equation $(1 + a)\cos\theta\cos(2\theta - b) = (1 + a\cos 2\theta)\cos(\theta - b)$, which are independent of a & b and S_2 be the set of all such solutions which are dependent on a & b then

43. The sets S_1 & S_2 are given by :

(A) $\{n\pi, n \in I)$ and $\{\frac{1}{2}(m\pi + b + (-1)^m \sin^{-1}(a\sin b))\}$, $m \in I$

(B) $\left\{\frac{n\pi}{2},\ n \in I\right\}$ and $\{m\pi + (-1)^m \sin^{-1}(a \sin b)\}$, $m \in I$

(C) $\left\{\frac{n\pi}{2},\ n \in I\right\}$ & $\left\{m\pi + (-1)^m \sin^{-1}\left[\left(\frac{a}{2}\right)(\sin b)\right]\right\}$, $m \in Z$

(D) None of these

44. Conditions that should be imposed on a & b such that S_2 is non-empty :

(A) $\left|\frac{a}{2}\sin b\right| < 1$ (B) $\left|\frac{a}{2}\sin b\right| \le 1$ (C) $|a\sin b| \le 1$ (D) none of these

45. All the permissible values of b, if a = 0 and S_2 is a subset of $(0, \pi)$:

(A) $b \in (-n\pi, 2n\pi)$; $n \in Z$
(B) $b \in (-n\pi, 2\pi - n\pi)$; $n \in Z$
(C) $b \in (-n\pi, n\pi)$; $n \in Z$
(D) None of these

Subjective Type Questions

46. If $\sin x + \sin y \ge \cos\alpha \cos x\ \forall\ x \in R$, then $\sin y + \cos\alpha$ is equal to ________

47. The number of solutions of pair of equation $\sin 2\theta + \cos\theta - 4\sin\theta = 2$ and $\cos 3\theta + 1 = 2\cos 2\theta$ in $[0,2\pi]$ is

48. Number of solution of equation, $9\cos^{12}x + \cos^2 2x + 1 = 6\cos^6 x\cos 2x + 6\cos^6 x - 2\cos 2x$ in $[0,4\pi]$ are

49. Let x and y are real numbers such that

$$(\tan x - 1)^3 + 2020(\tan x - 1) = -1 \text{ and } (1 - \cot y)^3 + 2020(1 - \cot y) = -1,$$

then number of possible values of z, which satisfying the equation $\tan x + \cot y + \sin z + \cos z = 3$, $z \in [-2\pi, 2\pi]$ is

50. Number of solution of the equation $\sin^3 x(1 + \cot x) + \cos^3 x(1 + \tan x) = \cos 2x$ in $[0, 4\pi]$ is

51. If $\operatorname{cosec}^6\theta + \sin^6\theta = 3m - m^3$; $\theta \ne n\pi$ and $\frac{m^3 - m^2 - 2m}{m^2 + m - 56} \ge 0$, then number of integral values of 'm' is

52. Number of solution(s) of equation
$x \in [0,\pi]$ $(\cos^5 x) + (\sin x)(\cos^4 x) - (\sin^4 x) + (\sin^3 x) + (\cos x)(\sin^2 x) - (\cos^2 x) = 0$ is

53. Number of solution(s) of the equation $1 + (\sec^2 2x) + \cot^2 x - 2(\sin x + \cos x. \cot x) = 0$ in $(0,2\pi)$ is/are

ANSWER KEY

1.	D	**2.**	C	**3.**	D	**4.**	B	**5.**	C
6.	C	**7.**	B	**8.**	B	**9.**	D	**10.**	B
11.	A	**12.**	C	**13.**	C	**14.**	C	**15.**	A
16.	B	**17.**	D	**18.**	B	**19.**	B	**20.**	A,B
21.	C	**22.**	C,D	**23.**	A,B,C	**24.**	A,B	**25.**	A,B,C,D
26.	A,C,D	**27.**	C,D	**28.**	A,C,D	**29.**	A,D	**30.**	A,C
31.	A	**32.**	B	**33.**	D	**34.**	D	**35.**	A
36.	D	**37.**	D	**38.**	A	**39.**	D	**40.**	A
41.	D	**42.**	B	**43.**	A	**44.**	C	**45.**	B
46.	1	**47.**	2	**48.**	012	**49.**	5	**50.**	4
51.	7	**52.**	2	**53.**	1				

One or more than one Correct Answer Type (4 Marks each, –1 for wrong answer)

1. If Δ denotes the determinant of order three whose all entries are squares of odd positive integer. Then Δ is always divisible by-

(A) 4 (B) 5 (C) 8 (D) 16

2. If α,β where $\alpha>\beta$ are roots of $x^2-4x+2=0$, then value of determinant $\begin{vmatrix} 1+\alpha-\beta & 2\sqrt{\alpha\beta} & -2\sqrt{\beta} \\ 2\sqrt{\alpha\beta} & 1-\alpha+\beta & 2\sqrt{\alpha} \\ 2\sqrt{\beta} & -2\sqrt{\alpha} & 1-\alpha-\beta \end{vmatrix}$

is-

(A) 125 (B) 64 (C) 25 (D) 16

3. Consider the system of linear equations $x + 2y + z = 1$, $2x + y + z = \alpha$, $4x + 5y + 3z = \alpha^2$ Then the system has

(A) infinitely many solutions when $\alpha = -1$ or 2. (B) infinitely many solutions when $\alpha = -2$ or 1.

(C) no solution when $\alpha \in R - \{-1, 2\}$ (D) no solution when $\alpha \in R - \{-2, 1\}$

4. If $\Delta = \begin{vmatrix} 0 & 2 & 1 & \frac{1}{2} & -3 \\ -2 & 0 & 4 & -1 & 0 \\ -1 & -4 & 0 & -\frac{3}{2} & -1 \\ -\frac{1}{2} & 1 & \frac{3}{2} & 0 & 1 \\ 3 & 0 & 1 & -1 & 0 \end{vmatrix}$, then Δ is equal to -

(A) $\frac{13}{4}$ (B) 0 (C) 1 (D) $\frac{3}{4}$

5. $f(x) = \begin{vmatrix} \sec^2 x & 1 & 1 \\ \cos^2 x & \cos^2 x & \text{cosec}^2 x \\ 1 & \cos^2 x & \cot^2 x \end{vmatrix}$, then range of $f(x)$ is -

(A) (0, 1) (B) (–1, 1) (C) [0, 1] (D) [–1, 1]

6. Consider system of equations in x, y and z

$$12x + by + cz = 0$$
$$ax + 24y + cz = 0$$
$$ax + by + 36z = 0.$$

(where a, b, c are real numbers, $a \neq 12$, $b \neq 24$, $c \neq 36$).

If system of equation has solution and $z \neq 0$, then value of $\frac{1}{a-12}+\frac{2}{b-24}+\frac{3}{c-36}$ is -

(A) $-\frac{1}{3}$ (B) $-\frac{1}{12}$ (C) $-\frac{1}{6}$ (D) $-\frac{1}{4}$

7. $\begin{vmatrix} 3x^2+19 & 2x-37 & 2 \\ x^2-29 & x+31 & 3 \\ 2x^2+17 & 3x & 5 \end{vmatrix} = ax^3 + bx^2 + cx + d$, then value of a is

(A) 2 (B) 5 (C) –5 (D) –8

8. Let a, b, c, x, y, z $\in$ R such that

ax + by + cz = 1

bx + cy + az = 2

cx + ay + bz = 3.

then the value of $(a^3 + b^3 + c^3 - 3abc)(x^3 + y^3 + z^3 - 3xyz) =$

(A) 18 (B) – 18 (C) 9 (D) – 9

9. Let us consider the determinants, defined as

$$A = \begin{vmatrix} \sin\theta & \cos\theta & \tan\frac{\pi}{11} \\ \sin\left(\frac{4\pi}{3}+\theta\right) & \cos\left(\frac{4\pi}{3}+\theta\right) & \tan\frac{2\pi}{11} \\ \sin\left(\frac{8\pi}{3}+\theta\right) & \cos\left(\frac{8\pi}{3}+\theta\right) & \tan\frac{3\pi}{11} \end{vmatrix}, B = \begin{vmatrix} \sin\theta & \cos\theta & \tan\frac{\pi}{11} \\ \sin\left(\frac{8\pi}{3}+\theta\right) & \cos\left(\frac{8\pi}{3}+\theta\right) & \tan\frac{2\pi}{11} \\ \sin\left(\frac{4\pi}{3}+\theta\right) & \cos\left(\frac{4\pi}{3}+\theta\right) & \tan\frac{3\pi}{11} \end{vmatrix}$$

$$\& \quad C = \begin{vmatrix} \sin\left(\frac{4\pi}{3}+\theta\right) & \cos\left(\frac{4\pi}{3}+\theta\right) & \tan\frac{\pi}{11} \\ \sin\theta & \cos\theta & \tan\frac{2\pi}{11} \\ \sin\left(\frac{8\pi}{3}+\theta\right) & \cos\left(\frac{8\pi}{3}+\theta\right) & \tan\frac{3\pi}{11} \end{vmatrix}$$

then

(A) det (A) + det (B) = 0 (B) det (B) + det (C) = 0

(C) det (A) + det (C) – det (B) > 0 (D) det (A) + det (B) – det (C) > 0

10. Given A, B, C are angles of a triangle and $f(x) = \lim_{n\to\infty} \sum_{r=1}^{n} \begin{vmatrix} \sin(A+rx) & \sin(B+rx) & \sin(C+rx) \\ \sin A & \sin B & \sin C \\ \cos(A+rx) & \cos(B+rx) & \cos(C+rx) \end{vmatrix}$, then

$f\left(\frac{\pi}{6}\right) + f\left(\frac{\pi}{3}\right)$ is equal to -

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) 0 (D) $-\frac{1}{\sqrt{3}}$

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11. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (\ell x + my + n).(\ell'x + m'y + n')$ and $\Delta_1 = \begin{vmatrix} \ell & \ell' & 0 \\ m & m' & 0 \\ n & n' & 0 \end{vmatrix}$ and

$\Delta_2 = \begin{vmatrix} \ell' & \ell & 1 \\ m' & m & 1 \\ n' & n & 1 \end{vmatrix}$, then the product $\Delta_1\Delta_2$ is -

(A) $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ (B) $\begin{vmatrix} h & g & f \\ a & f & c \\ b & c & h \end{vmatrix}$ (C) 1 (D) –1

12. If a, b & c are unequal natural numbers, then $\begin{vmatrix} a^2+b^2 & ab+bc & ac+ba \\ ba+cb & b^2+c^2 & bc+ca \\ ca+ab & cb+ac & c^2+a^2 \end{vmatrix}$ is equal to -

(A) $a+b+c$ (B) $a+b$ (C) $b+c$ (D) 0

13. Let $a_2, a_3 \in R$ such that $|a_2 - a_3| = 6$ and $f(x) = \begin{vmatrix} 1 & a_3 & a_2 \\ 1 & a_3 & 2a_2 - x \\ 1 & 2a_3 - x & a_2 \end{vmatrix}, x \in R$. Then the greatest value of $f(x)$ is-

(A) 36 (B) 24 (C) 12 (D) 9

14. Let k_1, k_2 be the maximum and minimum values of k for which the system of equations given by
$x + ky = 1$
$kx + y = 2$
$x + y = k$
are consistent then $k_1^2 + k_2^2$ is equal to

(A) $\frac{7-\sqrt{13}}{2}$ (B) 5 (C) $\frac{9-\sqrt{13}}{2}$ (D) 7

15. If $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = 5$, then $\begin{vmatrix} bc^2 - b^2c & a^2c - ac^2 & ab^2 - ba^2 \\ b^2 - c^2 & c^2 - a^2 & a^2 - b^2 \\ c - b & a - c & b - a \end{vmatrix}$ is equal to-

(A) 5 (B) 15 (C) 25 (D) 35

16. If $a \neq 0$, 1 and $\in R$, then the value of $\Delta = \begin{vmatrix} x+1 & x & x \\ x & x+a & x \\ x & x & x+a^2 \end{vmatrix}$ is-

(A) $a^3\left\{1 + \frac{x(a^3-1)}{a^2(a-1)}\right\}$ (B) $a\left\{1 + \frac{x(a^3-1)}{a^2}\right\}$

(C) $\left\{a^3 + \frac{xa(a^2-1)}{(a^3-1)}\right\}$ (D) $a^2\left\{1 + \frac{x(a-1)}{(a^3-1)}\right\}$

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17. If $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix} = 0$ then line ax + by + c = 0 always pass through a fixed point P then P is

(A) (1, 1) (B) (1, 2) (C) (1, –2) (D) (2, 1)

18. If A, B, C are angles of a triangle ABC, then the value of $\begin{vmatrix} -1 & \cos C & \cos B \\ \cos C & -1 & \cos A \\ \cos B & \cos A & -1 \end{vmatrix}$ is :-

(A) –1
(B) 1
(C) 1 – cosA – cosB – cosC
(D) 0

19. Let $D_n = [a_{ij}]_{n \times n}$ be a (n × n) determinant with the following conditions $a_{ij} = \begin{cases} 4 & i = j \\ 2 & |i-j| = 1 \\ 0 & \text{otherwise} \end{cases}$ Then,

(A) $D_3 = 32$
(B) $D_6 + D_4 = 2D_5$
(C) $D_6 + 4D_4 = 4D_5$
(D) $D_{12} + D_{10} = D_{11}$

20. If the system of equations

$2x - 3y + 5z = 12$

$3x + y + \lambda z = \mu$

$x - 7y + 8z = 17$

has infinite solutions then $2(\lambda + \mu)$, is divisible by

(A) 3 (B) 6 (C) 11 (D) 7

21. **Statement-1 :** If a ≠ b ≠ c and ax + by + c = 0, bx + cy + a = 0, cx + ay + b = 0 are concurrent, then $2^{a^2b^{-1}c^{-1}}.2^{b^2c^{-1}a^{-1}}.2^{c^2a^{-1}b^{-1}}$ is equal to 8.

because

Statement-2 : If $a+b+c=0 \Leftrightarrow a^3+b^3+c^3=3abc$

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

22. If $f(x) = \begin{vmatrix} (x-1)^2 & (x-2)^3 & (x+1) \\ (x-1)^3 & x^2 & (x+1)^2 \\ x+2 & x+1 & x^5 \end{vmatrix}$ then coefficient of x in $f(x)$ is-

(A) –14 (B) –16 (C) 0 (D) 25

23. Let $A = \begin{vmatrix} 4 & 6 & 8 \\ 10 & 11 & 16 \\ 15 & 14 & b \end{vmatrix}$, then minimum value of b such that A can be decomposed in 80 determinants in which each entry is a natural number is-

(A) 6 (B) 16 (C) 2 (D) 8

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24. If $D_r = \begin{vmatrix} r & 2012 & 2013 \\ 11r^2 & 3r & r \\ r & \frac{1}{r} & \frac{1}{r^2} \end{vmatrix}$, then $\sum_{r=1}^{5} D_r$ is equal to-

(A) 0 (B) 155 (C) 2013 (D) –2012

25. The numbers of integral solutions of D = 8, where $D = \begin{vmatrix} y+z & z & y \\ z & z+x & x \\ y & x & x+y \end{vmatrix}$ is-

(A) 3 (B) 6 (C) 9 (D) 12

26. If the straight lines ax + may + 1 = 0, bx + (m + 1) by + 1 = 0 and cx + (m+2)cy + 1 = 0, (where a, b, c and m are non-zero) are concurrent, then a,b,c are in-

(A) A.P. only for m = 1 (B) AP for all m
(C) GP for all m (D) HP for all m

27. For the system of equations

$$x + (\ell na)y + (\ell n^3 a)z = 0$$
$$x + (\ell nb)y + (\ell n^3 b)z = 0$$
$$x + (\ell nc)y + (\ell n^3 c)z = 0,$$

which of the following statement(s) is/are correct ? (where a,b,c are different real numbers)

(A) if x,y,z not all zero, then a,b,c cannot be integers simultaneously

(B) if x,y,z not all zero, then an ordered triplet (a,b,c) can be $\left(1, 2, \frac{1}{2}\right)$

(C) if abc = 1, then system has trivial solutions

(D) system has unique solution for $abc \neq 1$

28. If system of linear equations ax + by = cz, bx + cy = az, cx + ay = bz where c = a + b, $a \neq 0$ has non trivial solutions, then

(A) $z = x + y$ (B) $z = \frac{x+y}{2}$ (C) $z = -\frac{x+y}{2}$ (D) $x + y + z = 0$

29. The number of distinct values of 2×2 determinant whose entries are from the set {–3, 0, 3} is -

(A) 3 (B) 4 (C) 5 (D) 6

30. Value of the determinant $\begin{vmatrix} pb+qc & pc+qa & pa+qb \\ -pa-rc & -pb-ra & -pc-rb \\ -qa+rb & -qb+rc & -qc+ra \end{vmatrix}$ is equal to -

(A) 0 (B) 1
(C) (p + q + r)(a + b + c) (D) (ap + bq + cr)

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31. If $x^3 + 4x + 1 = 0$ has roots a,b,c then value of $\begin{vmatrix} bc & b^2 & c^2 \\ a^2 & ac & c^2 \\ a^2 & b^2 & ab \end{vmatrix}$ -

(A) –32 (B) –64 (C) –60 (D) –16

32. For a determinant D of 5×5 its element a_{ij} is defined as $a_{ij} = \begin{cases} (i-j); & i \neq j \\ (i^2 - j^2); & i = j \end{cases}$, then the value of D is

(A) –2 (B) –1 (C) 0 (D) 1

33. Number of real roots of the equation $\begin{vmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{vmatrix} + \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix} = 0$ is -

(A) 0 (B) 1 (C) 2 (D) 3

34. Let α, β, γ, δ are distinct imaginary roots of $z^5 = 1$, then value of $\begin{vmatrix} e^{\alpha} & e^{2\alpha} & e^{3\alpha+1} - e^{-\delta} \\ e^{\beta} & e^{2\beta} & e^{3\beta+1} - e^{-\delta} \\ e^{\gamma} & e^{2\gamma} & e^{3\gamma+1} - e^{-\delta} \end{vmatrix}$ is -

(A) 0 (B) e (C) 1 (D) e^5

35. If α, β, γ are roots of the equation $x^3 + x^2 + 2x - 4 = 0$, then value of $\begin{vmatrix} 1+\alpha^2 & 1 & 1 \\ 1 & 4-\beta^3 & 1 \\ 1 & 1 & 4-\gamma^3 \end{vmatrix}$ (where α is real) -

(A) 0 (B) 40 (C) 10 (D) –15

36. The value of $\begin{vmatrix} 3a & -a+b & -a+c \\ a-b & 3b & c-b \\ a-c & b-c & 3c \end{vmatrix}$ is-

(A) –2(a + b + c) (ab + bc + ca) (B) 3(a + b + c) (ab + bc + ca)
(C) –3(a + b + c) (ab + bc + ca) (D) (ab + bc + ca) (a + b + c)

37. The value of determinant $\Delta = \begin{vmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 \\ 1+a_2b_1 & 1+a_2b_2 & 1+a_2b_3 \\ 1+a_3b_1 & 1+a_3b_2 & 1+a_3b_3 \end{vmatrix}$ is (where $a_i, b_i \neq 0\ \forall\ i = 1,2,3$)-

(A) $a_1a_2a_3 + b_1b_2b_3 + 3$
(B) $(a_1a_2 + a_2a_3 + a_3a_1)(b_1b_2 + b_2b_3 + b_3b_1)$
(C) $(a_1a_2\, b_1b_2) + b_3b_1a_3a_2 + a_3a_1b_3b_1$
(D) $a_1a_2(b_1b_2 - b_2b_3) + b_1b_2(b_2b_3 - a_1a_2) + b_2b_3(a_1a_2 - b_1b_2)$

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38. The determinant $\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix}$ is equal to-

(A) $\begin{vmatrix} bx+ay & cx+by \\ b'x+a'y & c'x+b'y \end{vmatrix}$ (B) $\begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix}$

(C) $\begin{vmatrix} bx+cy & ax+by \\ b'x+c'y & a'x+b'y \end{vmatrix}$ (D) $\begin{vmatrix} bx+cy & b'x+c'y \\ a'x+b'y & ax+by \end{vmatrix}$

39. If $A=\begin{vmatrix} \sin(\theta+\alpha) & \sin(\theta+\beta) & \sin(\theta+\gamma) \\ \cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma) \\ 1 & 1 & 1 \end{vmatrix}$ $(\alpha \neq \beta \neq \gamma)$, then which of following is/are **not** correct ?

(A) $A = 0$ for all θ (B) A is odd function of θ
(C) $A = 0$ for all α,β,γ (D) A is dependent on θ

40. If $\begin{vmatrix} \sin 3A & \sin 2A & \sin A \\ \sin 3B & \sin 2B & \sin B \\ \sin 3C & \sin 2C & \sin C \end{vmatrix} = \lambda \sin A \sin B \sin C$

$\sin\frac{A+B}{2}\sin\frac{A+C}{2}\sin\frac{B+C}{2}\sin\frac{A-B}{2}\sin\frac{B-C}{2}\sin\frac{C-A}{2}$, then the value of λ will be-

(A) 32 (B) 64 (C) –64 (D) 128

41. The value of $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\sum_{r=1}^{\infty}\begin{vmatrix} 2^{-m} & 3^{-m} & 4^{-m} \\ 3^{-n} & 4^{-n} & 2^{-n} \\ 4^{-r} & 2^{-r} & 3^{-r} \end{vmatrix}$ is equal to-

(A) positive fraction (B) integer
(C) negative real number less than –1 (D) negative real number greater than –1

42. If $f(x)=\begin{vmatrix} (1+x)^{\lambda_1\mu_1} & (1+x)^{\lambda_1\mu_2} \\ (1+x)^{\lambda_2\mu_1} & (1+x)^{\lambda_2\mu_2} \end{vmatrix}$, then which of the following statement(s) is/are correct ?

(A) constant term of $f(x)$ is 0
(B) coefficient of x in $f(x)$ is $(\lambda_1-\lambda_2)(\mu_1-\mu_2)$
(C) coefficient of x in $f(x)$ is $(\lambda_1-\mu_2)(\lambda_2-\mu_2)$
(D) constant term of $f(x)$ is 1

43. If system of equation
$(\tan\theta)x + (\cot\theta)y + (8\cos 2\theta)z = 0$
$(\cot\theta)x + (8\cos 2\theta)y + (\tan\theta)z = 0$
$(8\cos 2\theta)x + (\tan\theta)y + (\cot\theta)z = 0$
have non trivial solution, then $\sin 4\theta$ is equal to -

(A) $-\frac{\sqrt{3}}{2}$ (B) –1 (C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

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44. If a, b, c are roots of $x^3 - 5x^2 + x + 1 = 0$, then system of linear equations $ax + y = 1$, $by + z = 1$, $cz + x = 1$ is -

(A) inconsistent (B) consistent with infinite solutions
(C) consistent with unique solution (D) consistent with non-trivial solution

Linked Comprehension Type

Paragraph for Question 45 to 46

Consider the system of equations

$$2x + py + 4z = 6$$
$$x + 2y + qz = 5$$
$$x - y + 2z = 3$$

45. If system has a unique solution (α,β,γ) and $\alpha + \beta + \gamma = 1$, then q is-

(A) –2 (B) 4 (C) 2 (D) 3

46. If system has infinite solutions & $q = -2$, then $|x| : |y - 8| : |2z - 11|$ is-

(A) 1 : 1 : 1 (B) 1 : 2 : 3 (C) 3 : 2 : 1 (D) 1 : 4 : 9

Paragraph for Question 47 & 48

Consider system 'S' of linear equations in x, y, z

$$ax + 2y - z = 1,$$
$$2x + y + z = 1,$$
$$x - 3y + bz = 4,\ a, b \in N.$$

47. Number of ordered pair (a, b) for which 'S' posses infinite solutions

(A) 0 (B) 1 (C) 2 (D) more than 2

48. Number of ordered pair (a, b) for which 'S' posses no solutions

(A) 0 (B) 1 (C) 2 (D) more than 2

Paragraph for Question 49 to 50

If $P(x) = \begin{vmatrix} a_1 & x & x \\ x & a_2 & x \\ x & x & a_3 \end{vmatrix} = xg(x) - f(x)$, then (where, $f(x) = (x - a_1)(x - a_2)(x - a_3)$)

49. g(x) is equal to -

(A) $(x - a_1)(x - a_2)(x - a_3)$
(B) $(x + a_1)(x + a_2)(x + a_3)$
(C) $(x - a_1)(x - a_2) + (x - a_2)(x - a_3) + (x - a_3)(x - a_1)$
(D) $x^3 - (a_1 + a_2 + a_3)$

50. If α,β,γ are the roots of the equation $f(x) = 0$, then $\Sigma\alpha\beta$ is-

(A) g(0) (B) P(0) (C) $f(0)$ (D) P(1)

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Paragraph for Question 51 & 52

If T_r is r^{th} term of an AP with common difference D where $T_1 > 0$ & $D > 0$ such that

$$\begin{vmatrix} \frac{1}{T_1} & \frac{1}{T_1T_2} & \frac{1}{T_2T_3} \\ \frac{1}{T_2} & \frac{1}{T_2T_3} & \frac{1}{T_3T_4} \\ \frac{1}{T_3} & \frac{1}{T_3T_4} & \frac{1}{T_4T_5} \end{vmatrix} = \frac{f(D)^g}{(T_1)^a (T_2)^b (T_3)^c (T_4)^d (T_5)^e}, \text{ where } a, b, c, d, e, f, g \in N$$

51. a, b, c, g are in

(A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

52. If $\frac{T_2}{T_3} = \frac{T_4}{T_6}$, then T_5 is equal to -

(A) $5T_1$ (B) $25T_1$ (C) $4T_1$ (D) $6T_1$

Paragraph for Question 53 & 54

For system of equations $3x - y + (\sec\theta)z = 1$, $2x + y + z = 2$, $x + 2y - (\sec\theta)z = -1$, where $\theta \neq (2n+1)\frac{\pi}{2}$.

53. Which is correct option ?

(A) Given system of equations can NOT be inconsistent

(B) Given system of equations can NOT be consistent

(C) Given system of equations can NOT have infinite solutions

(D) Given system of equations can NOT have non-trivial solutions.

54. If system of equation is inconsistent, then sum of values of θ in $[-2\pi, 2\pi]$ is -

(A) 2π (B) 0 (C) 4π (D) π

Paragraph for Question 55 & 56

Consider the system of equation $ax + y = b$, $bx + y = a$, $ax + by = ab$, where $a,b \in \{0,1,2,3,4\}$.

On the basis of above information, answer the following questions :

55. Number of ordered pairs (a,b) for which system is consistent with unique solution, is -

(A) 0 (B) 2 (C) 3 (D) 4

56. Number of ordered pairs for which system is inconsistent, is -

(A) 19 (B) 20 (C) 21 (D) 22

Numerical Grid Type

57. In a triangle ΔXYZ, let a,b,c be the length of the sides opposite to the angles X,Y and Z (where $\angle Z = c$ radian) respectively and Γ dentoes $s(s-a)(s-b)(s-c)$. If $\begin{vmatrix} -a+b & c+3b-a & b+c \\ a-b & c-3b+a & -b+c \\ a+b & a+3b-c & b-c \end{vmatrix} = 6$, $b^2 + c^2 \geq a^2$, $a^2 + b^2 \geq c^2$, $a^2 + c^2 \geq b^2$, then sum of all possible integral values of $\left(\frac{1}{\Gamma}\right)$, where $s = \frac{a+b+c}{2}$ is

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58. If the sides a,b,c of ΔABC satisfy the equation $4x^3 - 24x^2 + 47x - 30 = 0$ and $\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix} = \frac{p^2}{q}$ where p & q are coprime & s is semiperimeter of ΔABC, then $(p-q)$ is

59. If A, B, C are angles of an acute angle triangle. The minimum value of

$$\frac{1}{3^5}\begin{vmatrix} (\tan B+\tan C)^2 & \tan^2 A & \tan^2 A \\ \tan^2 B & (\tan C+\tan A)^2 & \tan^2 B \\ \tan^2 C & \tan^2 C & (\tan A+\tan B)^2 \end{vmatrix}$$ is

60. If x, y & z are different even integers and minimum value of $\begin{vmatrix} 1+x^2+x^4 & 1+xy+x^2y^2 & 1+xz+x^2z^2 \\ 1+xy+x^2y^2 & 1+y^2+y^4 & 1+yz+y^2z^2 \\ 1+xz+x^2z^2 & 1+yz+y^2z^2 & 1+z^2+z^4 \end{vmatrix}$

is N, then $\frac{N}{32}$ is

61. If system of equation

$$[a]x + [a-1]y = 1$$
$$x + y = 1$$
$$[a-1]x - [a-2]\,y = 1$$

is consistent (where [.] denotes greatest integer function) then number of integral possible values of 'a' is

62. If α,β,γ are the roots of $4x^3 - 16x^2 + 19x - 6 = 0$, then the value of the determinant

$$\begin{vmatrix} \alpha-\beta-\gamma & 2\alpha & 2\alpha \\ 2\beta & \beta-\gamma-\alpha & 2\beta \\ 2\gamma & 2\gamma & \gamma-\alpha-\beta \end{vmatrix}$$ is

63. The value of determinant $\begin{vmatrix} ax+xy & a^2+xb & ab+xc \\ bx+y^2 & ab+yb & b^2+yc \\ cx+yz & ac+zb & bc+cz \end{vmatrix} + \begin{vmatrix} 1 & -1 & 1 \\ \cos\theta & \cos\theta & \sin\theta \\ -\sin\theta & -\sin\theta & \cos\theta \end{vmatrix}$ is equal to

64. If $\alpha \in [0,2\pi]$ and system of equations

$$(\cos\alpha)\,x + (\sin\alpha)\,y + (\cos\alpha) = 0$$
$$(\sin\alpha)\,x + (\cos\alpha)\,y + (\sin\alpha) = 0$$
$$(\cos\alpha)\,x + (\sin\alpha)\,y - (\cos\alpha) = 0$$

is consistent, then number of possible value(s) of α, is

65. If $\begin{vmatrix} (a+1)^2 & ab+2a+1 & ac+2a+1 \\ ab+2b+1 & (b+1)^2 & bc+2b+1 \\ ac+2c+1 & bc+2c+1 & (c+1)^2 \end{vmatrix} = \lambda a+\mu b+\delta c$, then $3\lambda + 4\mu + 5\delta$ is equal to

66. There are two triplets $(x,y,z) = (a_1,b_1,c_1), (a_2,b_2,c_2)$, which satisfy the given system of equation $x^2 + xy + xz = 18$, $y^2 + yz + yx + 12 = 0$, $z^2 + zx + zy = 30$. Then the value of $(a_1 + b_1 + c_1 + a_2 + b_2 + c_2)$ is

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67. If $\begin{vmatrix} \sqrt{3}x - x^2 & x-1 & x-\sqrt{3} \\ \sqrt{3}x & x & \sqrt{3}-1 \\ x-1 & \sqrt{3}-x & x-1 \end{vmatrix} = ax^4 + bx^3 + cx^2 + dx + e$ and $a+c = p+\sqrt{q}$, then $(p - q)$ is equal to

68. If (x_0, y_0, z_0) is any solution of the system of linear equations $1 + x + y = 2z$, $1 + y + z = 2x$, $x + z = 2y + 2$, then $\dfrac{z_0^2 - y_0^2 + 1}{x_0}$ (where $x_0 \neq 0$) is equal to

69. If the system of equation $ax + (b - \beta)y + (c - \gamma)z = 0$, $(a - \alpha)x + by + (c - \gamma)z = 0$ and $(a - \alpha)x + (b - \beta)y + cz = 0$, (where α, β, γ are distinct) has a non-trivial solution, then the value of $\dfrac{a}{\alpha} + \dfrac{b}{\beta} + \dfrac{c}{\gamma}$ is

70. Consider a system of linear equation in x,y,z as

$$x + (m-1)y + (2m-3)z = 1$$
$$mx + (2m-2)y + 2z = 0$$
$$(m+1)x + (3m-3)y + (m^2-1)z = 0$$

Then the number of value(s) of m for which the system posses at least one solution of type $(1,\alpha,\beta)$ is (where $\alpha,\beta,m \in R$)

MATCHING LIST TYPE

71. Let $\Delta = \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix}$ $\Delta_1 = \begin{vmatrix} x+y & y+2z & z-x-y \\ y+z & z+2x & x-y-z \\ z+x & x+2y & y-z-x \end{vmatrix}$

$$\Delta_2 = \begin{vmatrix} zy-x^2 & zx-y^2 & xy-z^2 \\ zx-y^2 & xy-z^2 & zy-x^2 \\ xy-z^2 & yz-x^2 & zx-y^2 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} x & -y & z \\ -y & z & -x \\ z & -x & y \end{vmatrix}, \Delta_4 = \begin{vmatrix} x+2y & y+2z & z+2x \\ y+2z & z+2x & x+2y \\ z+2x & x+2y & y+2z \end{vmatrix}$$

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	If $\Delta = 8$, then Δ_1 is equal to	(1)	4
(Q)	If $\Delta = 2$, then Δ_2 is equal to	(2)	6
(R)	If $\Delta = 6$, then Δ_3 is equal to	(3)	8
(S)	If $\Delta = 1$, then Δ_4 is equal to	(4)	9

Codes :

	P	Q	R	S
(A)	3	4	2	1
(B)	4	1	3	4
(C)	3	1	2	4
(D)	1	3	4	2

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Matrix Match Type

72. Consider the system of equations :

$$a^2x+\left(a^2-(b-c)^2\right)y+bcz=0$$

$$b^2x+\left(b^2-(c-a)^2\right)y+caz=0$$

$$c^2x+\left(c^2-(a-b)^2\right)y+abz=0\text{, then}$$

Column – I	Column – II
(A) If a + b + c = 0 and no two out of a, b, c are equal then	(P) system of equations have unique solution
(B) If a = b = c = 0, then	(Q) System of equations have infinite number of solutions
(C) If $a=b=c$, $a+b+c\neq 0$, then	(R) $\forall x,y,z\in R$, the system of equations will be true
(D) If $a+b+c\neq 0$ and no two out of a,b,c are equal, then	(S) solutions of system of equations are in the form $(\lambda_1,\lambda_2,-\lambda_1-\lambda_2)$, where $\lambda_1,\lambda_2\in R$
	(T) solutions of system of equations are in the form $(\lambda_1,\lambda_2,\lambda_1+\lambda_2)$, where $\lambda_1,\lambda_2\in R$

ANSWER KEY

1.	A,C,D	**2.**	A	**3.**	A,C	**4.**	B	**5.**	A
6.	C	**7.**	D	**8.**	A	**9.**	A,C,D	**10.**	C
11.	A	**12.**	D	**13.**	D	**14.**	D	**15.**	C
16.	A	**17.**	C	**18.**	D	**19.**	A,C	**20.**	A,B
21.	C	**22.**	B	**23.**	C	**24.**	A	**25.**	D
26.	D	**27.**	A,B,D	**28.**	C	**29.**	C	**30.**	A
31.	B	**32.**	C	**33.**	B	**34.**	A	**35.**	B
36.	B	**37.**	D	**38.**	B	**39.**	A,B,C,D	**40.**	B
41.	D	**42.**	A,B	**43.**	C	**44.**	A	**45.**	D
46.	B	**47.**	B	**48.**	B	**49.**	C	**50.**	A
51.	A	**52.**	A	**53.**	C	**54.**	B	**55.**	D
56.	A	**57.**	9	**58.**	7	**59.**	6	**60.**	8
61.	0	**62.**	64	**63.**	2	**64.**	2	**65.**	0
66.	0	**67.**	8	**68.**	2	**69.**	2	**70.**	0
71.	C	**72.**	$(A\to q, B\to q,r, C\to q,s, D\to p)$						

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One or more than one Correct Answer Type

1. ABC is a variable triangle such that A is (1, 2), B and C lie on line $y = x + \lambda$ (where λ is a variable), then locus of the orthocenter of triangle ABC is

(A) $(x - 1)^2 + y^2 = 4$ (B) $x + y = 3$ (C) $2x - y = 0$ (D) none of these

2. The sum of the distances of the point P lying inside the triangle formed by lines $y = 0$; $4x - 3y = 0$ and $3x + 4y - 9 = 0$ is 3. Then the locus of P is-

(A) $7x + 6y - 24 = 0$ (B) $x - 2y + 24 = 0$ (C) $7x - 4y + 6 = 0$ (D) $x - 2y - 6 = 0$

3. Let area of triangle formed by any line through point (1, 3) and co-ordinate axes in xy-plane is A. then

(A) If A = 8, then number of such lines is 4 (B) If A = 6, then number of such lines is 3

(C) If A = 4, then number of such lines is 2 (D) If A = 0, then number of such lines is 1

4. In right angle ΔABC, sides AB and AC are members of family of lines $(x + 2y - 3) + \lambda(2x + y - 3) = 0$, $\lambda \in R$ and hypotenuse BC is $x + y = 0$. If $\left(\frac{\sqrt{5}+1}{3+\sqrt{5}}, \frac{\sqrt{5}-1}{3+\sqrt{5}}\right)$ is incentre of ΔABC, then centroid of ΔABC is -

(A) $\left(\frac{5}{6}, \frac{-1}{6}\right)$ (B) $\left(\frac{-5}{6}, \frac{1}{6}\right)$ (C) $\left(\frac{5}{6}, \frac{1}{6}\right)$ (D) $\left(\frac{-5}{6}, \frac{-1}{6}\right)$

5. If $\alpha, \beta, \gamma \in R$ and satisfy the relations $\alpha^2 + 2\beta^2 + 3\alpha = 1 + \gamma^2$ and $2\alpha^2 + 4\beta^2 = 2\gamma^2 + 5\beta$, then

(A) Minimum value of $\alpha^2 + \beta^2$ is $\frac{4}{61}$. (B) Minimum value of $\alpha^2 + \beta^2$ is $\frac{2}{17}$

(C) Minimum value of $\alpha^2 + \beta^2 - 2\alpha + 1$ is $\frac{16}{61}$ (D) Minimum value of $\alpha^2 + \beta^2 - 2\alpha + 1$ is $\frac{1}{17}$

6. Let the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $2y^2 = 2x + 3$ at A and B. If $P(\sqrt{3}, 0)$, then value of $|PA - PB|$ is [where PA denotes distance between points P and A] -

(A) $\frac{6+4\sqrt{3}}{3}$ (B) $\frac{2}{3}$ (C) $\frac{\sqrt{76+48\sqrt{3}}}{3}$ (D) $\frac{\sqrt{76-48\sqrt{3}}}{3}$

7. A piece of graph paper is folded once so that (0,2) is matched with (4, 0) and (7, 3) is matched with (a, b). Then 5(a + b) is

(A) 24 (B) 34 (C) 44 (D) 45

8. A triangle ABC right angled at C moves such that A & B always lie on the positive x & y axes. The locus of C is

(A) straight line (B) circle (C) parabola (D) ellipse

9. From the point of intersection (P) of lines given by $x^2 - y^2 - 2x + 2y = 0$, points A, B, C, D are taken on the lines at a distance of $2\sqrt{2}$ units to form a quadrilateral whose area is A_1 and the area of the quadrilateral formed by joining the circumcentres of ΔPAB, ΔPBC, ΔPCD, ΔPDA is A_2, then $\frac{A_1}{A_2}$ equals

(A) 2 (B) 4 (C) $\sqrt{3}$ (D) 1

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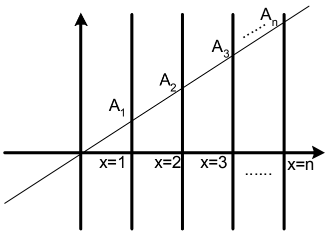
10. If the line $\sqrt{5}\,x = y$ meets the lines x = 1, x = 2… x = n, at points $A_1, A_2, \ldots A_n$ respectively then $(OA_1)^2 + (OA_2)^2 \ldots + (OA_n)^2$ is equal to



(A) $3n^2 + 3n$ (B) $2n^3 + 3n^2 + n$ (C) $3n^3 + 3n^2 + 2$ (D) $(3/2)(n^4 + 2n^3 + n^2)$

11. If a point P moves such that its distance from line $y = \sqrt{3}x - 7$ is same as its distance from $(3\sqrt{3}, 2)$, then the area of the curve, described by P, enclosed between the coordinate axes is (in square units) -

(A) $\frac{5\sqrt{3}}{2}$ (B) $5\sqrt{3}$ (C) $\frac{25\sqrt{3}}{2}$ (D) $25\sqrt{3}$

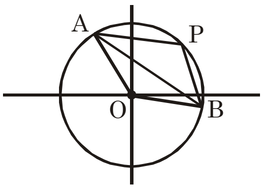
12. Let number of integral solutions of $|x| + |y| < k + 1$ be $f(k)$ (where $k \in N$), then

(A) $f(10) = 220$ (B) $f(99) = 19800$ (C) $f(10) = 211$ (D) $f(99) = 19801$

13. If $A \equiv (2, 5)$, $B \equiv (5, 2)$, $C \equiv (-1, -1)$ then inradius of Δ DEF formed by feet of altitudes drawn from vertex A, B, C to its opposite sides :

(A) $\frac{3\sqrt{5}}{2}$ (B) $\frac{2\sqrt{2}}{5}$ (C) $\frac{3\sqrt{2}}{5}$ (D) None of these

14.



Point P(3,4) lies on a circle centered at the origin (O), if straight line $3x + 4y - 7 = 0$ intersects the circle at points A and B (as shown in figure) the area of quadilateral AOBP -

(A) 20 (B) 24 (C) 26 (D) 30

15. Let the sides of a ΔABC are all integers with A as origin. If (2,–1) and (3,6) are points on line AB and AC respectively. (Line AB and AC may be extended to contain these points). If length of any two sides are prime that differs by 50. Then-

(A) smallest value of third side is 60 (B) smallest value of third side is 75

(C) minimum area is 330 (D) minimum area is 300

16. Sum of areas of all possible pentangons formed by points A(–5,0), B(5,0), C(0,2), D(4,3), E(–4,3) is-

(A) 25 sq. unit (B) 81 sq. unit (C) 17 sq. unit (D) 117 sq. unit

17. Let m_{AB} is slope of line joining the points A & B. Point $A_1(x_1, y_1), A_2(x_2, y_2), \ldots\ldots A_n(x_n, y_n)$ are such that $x_1, x_2, x_3 \ldots\ldots x_n$ are in A.P. & $y_1, y_2 \ldots\ldots y_n$ are also in A.P., then $m_{A_1A_2} + m_{A_2A_3} + \ldots\ldots + m_{A_{(n-1)}A_n}$ is equal to -

(A) 0 (B) $(n-1)m_{A_1A_n}$ (C) $(n)\,m_{A_1A_n}$ (D) $(n^2)m_{A_1A_n}$

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18. The line 3x + 2y = 6, will divide the quadrilateral formed by the line x + y = 5, y – 2x = 8, 3y + 2x = 0 and 4y – x = 0 in -

(A) two quadrilateral
(B) one pentagon and one triangle
(C) two triangle
(D) never cut the sides of quadrilateral

19. Let P(0, 3), Q(0, 5), R(x, 0), then the value of |x| for which ∠PRQ is as large as possible :-

(A) $\sqrt{15}$ (B) $\sqrt{20}$ (C) $\sqrt{35}$ (D) $\sqrt{40}$

20. If one of the line given by $2x^2 + pxy + 3y^2 = 0$ and $2x^2 + qxy - 3y^2 = 0$ is common line and other lines are perpendicular to each other, then -

(A) p = 5 (B) p = –5 (C) q = –1 (D) q = 1

21. A variable line L is drawn through (0, 0) to meet the lines y – x = 10, y – x = 20 at the point A & B respectively. If a point P is taken on line L such that OP^2 = OA.OB, then locus of P is given by-

(A) $(y - x)^2 = 100$ (B) $(x + y)^2 = 50$ (C) $(y - x)^2 = 200$ (D) $(y + x)^2 = 200$

22. The line 6x + 8y = 48 intersects the co-ordinate axes at A and B respectively. If line L intersect line segment AB and bisects the area and perimeter of ΔOAB, where O is origin, then

(A) the number of line L is 2.
(B) the number of line L is 1.
(C) y intercept of line L is $\sqrt{6}$.
(D) y intercept of line L is $2\sqrt{6}$.

23. Consider ΔABC such that AB is parallel to y-axis, BC is parallel to x-axis and centroid is at (2, 1). If median trough A is y = x – 1, then which of the following is/are true -

(A) Equation of median through C is x – 4y + 2 = 0
(B) Equation of median through C is x + 4y = 5
(C) Acute angle between median through A and C is $\tan^{-1}\left(\frac{3}{5}\right)$.
(D) Acute angle between medians through A and C is $\tan^{-1}\left(\frac{5}{3}\right)$.

24. External angle bisectors of angle B and angle C are y = x and y = –2x respectively. If the vertex A is (1, 3), then co-ordinates of incentre of ΔABC is -

(A) (1, 1) (B) $\left(-\frac{1}{2},1\right)$ (C) (0,0) (D) $\left(\frac{1}{2},\frac{3}{2}\right)$

25. The equation of sides of triangle having (3, –1) as a vertex and x – 4y + 10 = 0 and 6x + 10y – 59 = 0 as angle bisector and median respectively drawn from different vertices are -

(A) 6x + 7y – 13 = 0
(B) 2x + 9y – 65 = 0
(C) 18x + 13y – 41 = 0
(D) 6x – 7y – 25 = 0

26. Consider points P(7, 1) and O = (0, 0). If S is a point on the line y = x and T is a point on the x-axis so that P is on the line segment ST, then the minimum possible area of ΔOST is -

(A) 5 (B) 6 (C) 11 (D) 12

27. If orthocentre of a triangle is (1, 1) and equation of one of its side is x + y = 2. Also, one vertex of the triangle is (3, 3) and length of its largest side is 3 units, then :-

(A) Length of smallest side is 1 unit
(B) Area of triangle is $\sqrt{2}$ sq. unit
(C) Equation of largest side can be $(1+2\sqrt{2})x + (1-2\sqrt{2})y = 6$
(D) Equation of largest side can be $(1-2\sqrt{2})x+(1+2\sqrt{2})y = 6$

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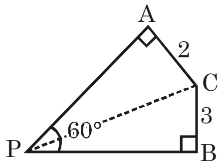
28. (1, 0), (3, 0), (3, 5) and (1, 4) are the vertices of a quadrilateral. The equation of line through the origin which divides the quadrilateral into two parts of equal area is -

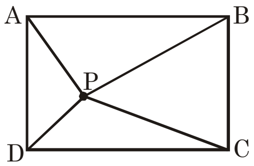
(A) $y = \frac{9}{8}x$ (B) $y = \frac{8}{9}x$ (C) $y = 8x$ (D) $y = \frac{x}{8}$

29. In the given diagram length of AC and BC are 2 and 3 respectively. If length of PC can be expressed as $\sqrt{\frac{a}{b}}$ where a, b are co-prime numbers, then a + b equals to -



(A) 81 (B) 83 (C) 78 (D) 79

30. A point (P) is in the square and its distances to the vertices A, B, C, D of square are 7, 35, 49 and x respectively, then x equals to -



(A) 36 (B) 35 (C) 34 (D) 37

31. Number of integral values of p for which the origin lies inside the triangle formed by the lines $L_1 : y + 3x = 9$, $L_2 : 2y - 3x = 18$ and $L_3 : y - px - 8p = -3$ are -

(A) 4 (B) 2 (C) 1 (D) 3

32. If lines represented by $2x^2 - 5xy + 2y^2 = 0$ intersect $x + y = 3$ at B and C and one of its angle bisector intersect $x + y = 3$ at D. E is the harmonic conjugate of 'D' with respect to BC and O is origin. Then which of the following is/are true ?

(A) $\cos(\angle BOC) = \frac{4}{5}$ (B) Point E does not exists.

(C) Area of $\Delta BOC = \frac{3}{2}$ (D) ΔBOC is isosceles.

33. If L_1 and L_2 represent equation of straight lines passing through (1, 1) and parallel to the lines represented by the equation $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$. Then which of the following holds good ?

(A) Angle between L_1 and L_2 is $\tan^{-1}\left(\frac{3}{5}\right)$.

(B) Angle between L_1 and L_2 is $\tan^{-1}\left(\frac{5}{3}\right)$.

(C) Area of parallelogram formed by L_1, L_2 and $x^2 - 5xy + 4y^2 + x + 2y - 2 = 0$ is $\frac{1}{3}$.

(D) One of L_1 and L_2 passes through origin.

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34. The vertices B and C of a triangle ABC lie on the lines 3y = 4x and y = 0 respectively and the side BC passes through the point $\left(\frac{2}{3},\frac{2}{3}\right)$. If ABOC is a rhombus, O is origin, B lies in first quadrant. Then the correct option(s) is/are -

(A) sum of coordinates of A is $\frac{12}{5}$
(B) line OA is parallel to 3y = x + 5
(C) line OA is parallel to 2y = x + 7
(D) difference of coordinates of A is $\frac{4}{5}$

35. A straight line pass through the origin O meets the parallel lines 4x + 2y = 9 and 2x + y + 6 = 0 at point P & Q respectively. The point O divides the segment PQ in the ratio-
(A) 1 : 2 (B) 3 : 4 (C) 2 : 1 (D) 4 : 3

36. The lines $y = m_1x$, $y = m_2x$ and $y = m_3x$ meet the line x + y = 1 at A, B, C respectively. If AB = BC, then $1 + m_1$, $1 + m_2$, $1 + m_3$ are in (where $m \neq m_2 \neq m_3$)
(A) arithmetic progression
(B) geometric progression
(C) harmonic progression
(D) arithmetic geometric progression

37. The distance between two parallel lines is unity. A point P lies at a distance of k unit from one of them. If Q, R two different point lies on two lines respectively such that ΔPQR is equilateral with side length $2\sqrt{\frac{7}{3}}$, then maximum value of k is -
(A) 1 (B) 2 (C) 3 (D) none of these

38. Statement-1 : Consider triangle ABC and point M is such that triangles ABM, ACM and BCM are of equal area, then point M is centroid of triangle ABC (where A, B, C and M lie in x-y plane)

Statement-2 : In triangle ABC, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = R$, then

(where symbols have their usual meaning)
(A) Statement-1 is true
(B) Statement-2 is true
(C) Statement-1 is false
(D) Statement-2 is false

39. A,B,C are respectively points (1,2), (4,2), (4,5). If T_1,T_2 are points of trisection of AC and S_1,S_2 are points of trisection of BC, (where T_1 is nearer to A and S_1 is nearer to B). Then

(A) area of quadrilateral $T_1S_1S_2T_2$ is $\frac{3}{2}$

(B) area of quadrilateral $T_1S_1S_2T_2$ is 2

(C) ratio of area of ΔCT_2S_2 to ΔABC is $\frac{1}{9}$

(D) ratio of area of ΔCT_2S_2 to ΔABC is $\frac{1}{3}$

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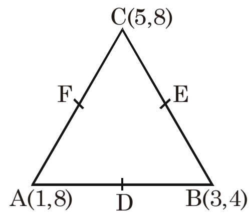
40. Consider a triangle ABC,



$$\frac{AD}{DB}=\frac{BE}{EC}=\frac{CF}{FA}=\frac{2}{1}$$

Let P,Q,R be harmonic conjugate of D,E,F respectively with respect to A and B, B and C, C and A.

(A) Centroid of $\Delta DEF=\left(3,\frac{20}{3}\right)$ (B) Centroid of $\Delta PQR=\left(3,\frac{20}{3}\right)$

(C) Centroid of $\Delta PQR=\left(-3,\frac{20}{3}\right)$ (D) Centroid of $\Delta DEF=\left(-3,\frac{20}{3}\right)$

41. Let $(a\alpha)x + y + 7 = 0$, $bx + (a\alpha)y + 3 = 0$ and $-x + y + 4 = 0$ lines are concurrent (where $\alpha \in R$ & $a \neq 0$) then-

(A) number of possible positive integral value of b is 1

(B) number of possible non-negative integral value of b is 2

(C) number of possible positive integral value of b is 3

(D) number of possible non negative integral value of b is 3

42. A rod of fixed length k has its ends sliding along the coordinate axes in Ist quadrant such that rod meets the x-axis at A(a, 0) and y-axis at B(0, b). If O is origin and $E=\left(a+\frac{1}{b}\right)^2+\left(b+\frac{1}{a}\right)^2$.

Which of the following holds good ?

(A) Minimum value of E is $k^2+\frac{4}{k^2}+2$

(B) Minimum value of E is $\left(k+\frac{2}{k}\right)^2$

(C) When E is minimum then area of ΔAOB is $\frac{k^2}{4}$

(D) When E is minimum then area of ΔAOB is $\frac{k^2}{2}$

43. A straight line through the point A(–1,3) cuts the line x + y = 4 and y = x at B and C respectively. If AB.AC = 16, and possible slopes of the line are m_1 & m_2 then $\left(\frac{m_1-m_2}{1+m_1m_2}\right)^2$ is equal to

(A) 1 (B) $\frac{3}{4}$ (C) 3 (D) $\frac{1}{3}$

44. The line $3x + 6y = k$ intersects $2x^2 + 2xy + 3y^2 - 1 = 0$ at points A and B. The circle on AB as diameter passes through origin then value of k^2 is

(A) 3 (B) 8 (C) 9 (D) 16

45. Let point 'P' be (5, 3) and a point 'R' on y = x and Q on x-axis such that PQ + QR + RP is minimum, then co-ordinates of point Q is -

(A) (17, 0) (B) $\left(\frac{17}{2},0\right)$ (C) $\left(\frac{17}{4},0\right)$ (D) $\left(\frac{23}{4},0\right)$

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46. The complete set of values the parameter 'a' so that the point $P\left(a, \frac{1}{1+a^2}\right)$ doesn't lie outside the triangle formed by the lines 15y = x + 1, 78y = 118 – 23x and y + 2 = 0 is -

(A) (0, 5) (B) [2, 5] (C) (1, 5) (D) [0, 2]

47. If the point (a, 2a) falls between the lines |x + y + 1| = 4, then complete set of values of 'a' is

(A) $\left(-\frac{5}{3},1\right)$ (B) $\left(1,\frac{5}{3}\right)$ (C) $\left(\frac{-4}{3},\frac{4}{3}\right)$ (D) $\left(-\infty,\frac{-5}{3}\right)\cup(1,\infty)$

48. The area enclosed by the curve |x + y| + |x – y| – 11 = 0 is

(A) 121 (B) 144 (C) 100 (D) 169

49. If m is the least value of $f(x)=\sqrt{x^2-2x+2}+\sqrt{x^2-4x+29}$, then [m] is equal to (where [.] denotes greatest integer function)

(A) 6 (B) 7 (C) 5 (D) 4

50. Consider ΔABC such that AB is parallel to y-axis, BC is parallel to x-axis and centroid is at (2, 1). If median trough A is y = x – 1, then which of the following is/are true -

(A) Equation of median through C is x – 4y + 2 = 0

(B) Equation of median through C is x + 4y = 5

(C) Acute angle between median through A and C is $\tan^{-1}\left(\frac{3}{5}\right)$.

(D) Acute angle between medians through A and C is $\tan^{-1}\left(\frac{5}{3}\right)$.

51. In ΔABC, let D(3,2), E and F are mid points of sides BC,CA and AB respectively. Let vertex B is (1,1) and quadrilateral BDEF is a rhombus and equation of BE is y = x, then-

(A) coordinates of A are (3,5) (B) ΔABC must be isosceles triangle

(C) centroid of ΔABC is (4,4) (D) area of ΔABC is 6 sq. units.

52. Point A lies on x-axis and point B lies on y = x. If PA.PB is minimum where P divides segment AB internally, then (given P is $(1,2-\sqrt{3})$ and 'O' is origin), -

(A) $OA=2\sqrt{2}-\sqrt{6}+\sqrt{3}-1$ (B) $OB=2\sqrt{2}-\sqrt{6}+\sqrt{3}-1$

(C) $\angle OAB=45°$ (D) $\angle OBA=22.5°$

53. Consider two isosceles triangles ΔAB_1C_1 and ΔAB_2C_2 such that $AB_1 = AC_1$, $AB_2 = AC_2$ and P(1, –2) is point of intersection of lines B_1C_1 & B_2C_2. If A, B_1, C_2 lies on 7x – y + 3 = 0 and A, B_2, C_1 lies on x + y – 3 = 0, then which of the following is/are correct ?

(A) point A lies on y-axis.

(B) product of x-coordinates of points B_1,C_1,B_2,C_2 is negative

(C) product of y-coordinates of points B_1,C_1,B_2,C_2 is positive

(D) sum of slope of lines B_1C_1 and B_2C_2 is $\frac{8}{3}$.

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54. Let $L_1, L_2, \ldots\ldots, L_n$ are lines in x-y plane which are equidistant from $(3, \pi)$, $(\pi, -4)$ and $(3 + \pi, 4)$. If $\frac{a\pi + b}{c}$, $a, b, c \in I$ is area of polygon which is formed by lines $L_1, L_2, \ldots\ldots L_n$, then which of the following is/are correct ? (where a, c are coprime numbers)

(A) $ac + b = 76$ (B) $a^2 + b^2 + c^2 = 329$

(C) $a - bc + c^2 = 251$ (D) $a + b^2 + c^3 = 667$

Linked Comprehension Type

Paragraph for Question 55 to 56

A ray of light strikes a line mirror $y = 0$ along the line $L_1 : x - 3y = 1$ and reflects along the line $L_2 = 0$. There is another line mirror $x - 4 = 0$ in the way of reflected ray. After 2nd reflection ray moves along the line $L_3 = 0$ and finally it travels in the direction of the line $L_4 = 0$ after 3rd reflection from line mirror $y + 2 = 0$.

On the basis of above informations, answer the following questions :

55. Equation of the line $L_4 = 0$ is -

(A) $x - 3y = 7$ (B) $x + 3y = 1$

(C) $x + 3y + 5 = 0$ (D) $x + 3y - 5 = 0$

56. Area of the closed figure so formed is -

(A) 6 sq. units (B) 12 sq. units

(C) 24 sq. units (D) 36 sq. units

Paragraph for Question 57 to 58

Let ABC be a triangle whose vertex A is (3, 4). $L_1 = 0$ and $L_2 = 0$ are the internal angle bisectors of angles B and C respectively, where $L_1 \equiv x + y - 2 = 0$, $L_2 \equiv 2x + y - 5 = 0$. Let A_1 is the image of point A about line L_1.

On the basis of above information answer the following :

57. Co-ordinate of point A_1 is -

(A) (2, 1) (B) (–2, –1) (C) (2, –1) (D) (–1, –2)

58. Inradius of triangle ABC is -

(A) $\sqrt{10}$ (B) $2\sqrt{10}$ (C) $\frac{3}{2}\sqrt{10}$ (D) $\frac{5}{2}\sqrt{10}$

Paragraph for Question 59 to 60

Consider in ΔABC, $A(5, -1)$, $B(\alpha, -7)$, $C(-2, \beta)$.

Let $(-6, -4)$ is image of orthocentre of ΔABC in the point mirror M which is mid point of the side BC.

59. If (p, q) is circumcentre of triangle ABC, then value of $q^2 - \frac{p}{2}$ is -

(A) $\frac{13}{2}$ (B) $\frac{11}{2}$ (C) $\frac{15}{2}$ (D) $\frac{7}{2}$

60. The value of $\beta^2 - \alpha^2 + 5\beta - \alpha$ is -

(A) 7 (B) 9 (C) 18 (D) 12

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Paragraph for Question 61 to 62

A_1 and B_1 are points on line $3x + 2y = 13$ such that ΔOA_1B_1 is equilateral triangle. Let $L_1 = 0$ and $L_2 = 0$ respresents the equation of side OA_1 and OB_1 respectively, then

61. The combined equation of $L_1 = 0$ and $L_2 = 0$ is

(A) $3x^2 - 48xy + 23y^2 = 0$ (B) $23x^2 - 48xy + 3y^2 = 0$

(C) $3x^2 + 48xy + 23y^2 = 0$ (D) $23x^2 + 48xy - 3y^2 = 0$

62. Let 'h' be the length of altitude of ΔOA_1B_1 from origin. If line through A_1 and B_1 is moved parallel towards origin through distance $\frac{h}{2}$ intersecting $L_1 = 0$ and $L_2 = 0$ in points A_2 and B_2 respectively forming ΔOA_2B_2. Again the line through A_2 and B_2 is moved parallel towards origin through distance $\frac{h}{4}$ intersecting $L_1= 0$ and $L_2= 0$ in points A_3 and B_3 respectively forming ΔOA_3B_3 and similarly ΔOA_4B_4,ΔOA_5B_5 and so on. The sum of areas of ΔOA_1B_1,ΔOA_2B_2,ΔOA_3B_3,ΔOA_4B_4 upto infinite triangles.

(A) $\frac{4\sqrt{3}}{9}$ (B) $\frac{52\sqrt{13}}{9}$ (C) $\frac{4\sqrt{13}}{9}$ (D) $\frac{52\sqrt{3}}{9}$

63. Equation of BC is -

(A) $x + y = \sqrt{3}$ (B) $x - y = \sqrt{3}$ (C) $\sqrt{3}x + y = 3$ (D) $x - \sqrt{3}y = \sqrt{3}$

64. Equation of CD is -

(A) $\sqrt{3}y + x = 4 - 3\sqrt{3}$ (B) $y + \sqrt{3}x = 4 + 3\sqrt{3}$

(C) $\sqrt{3}y + x = 4 + 3\sqrt{3}$ (D) $y + \sqrt{3}x = 4 - 3\sqrt{3}$

Paragraph for Questions 65 and 66

The line $6x + 8y = 48$ intersects, the X & Y axes at A and B respectively. A line L bisects the area as well the perimeter of triangle OAB (where O is the origin), then answer the following questions

65. The line L

(A) doesnot intersect AB (B) doesnot intersect OB

(C) doesnot intersect OA (D) passes through the origin

66. The number of such possible line(s) is -

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Question 67 and 68

A line $L_0 : 2x + 5y = 11$ rotates about a point $P(\alpha,\beta)$ on the line L_0 such $\alpha,\beta \in$ integer and $|\alpha + \beta|$is least, through an angle $(-1)^n.n^\circ$ in n^{th} second. If L_0 becomes line L_n after n seconds. Then

67. $(\alpha + \beta)$ is-

(A) –5 (B) –2 (C) 1 (D) 4

68. Perpendicular distance from (0,0) on L_{180} is-

(A) $\frac{16}{\sqrt{29}}$ (B) $\frac{11}{\sqrt{29}}$ (C) $\frac{7}{\sqrt{29}}$ (D) $\frac{4}{\sqrt{29}}$

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Paragraph for Question 69 to 71

Consider a family of lines $(4a + 3)x - (a + 1)y - (2a + 1) = 0$, where $a \in R$.

69. The locus of the foot of perpendicular from the (0, 0) on each member of this family is

(A) $(2x - 1)^2 + 4(y + 1)^2 = 5$
(B) $(2x - 1)^2 + (y + 1)^2 = 5$
(C) $(2x + 1)^2 + 4(y - 1)^2 = 5$
(D) $(2x - 1)^2 + 4(y - 1)^2 = 5$

70. An equation of member of this family with positive gredient makes an angle of $\pi/4$ with the line $3x - 4y = 2$, is -

(A) $7x - y = 5$
(B) $4x - 3y + 2 = 0$
(C) $x + 7y = 15$
(D) $5x - 3y = 4$

71. Minimum area of the triangle which a member of this family with negative greadient can make with the positive semi axes, is -

(A) 8
(B) 6
(C) 4
(D) 2

Paragraph for Question 72 to 73

Given four parallel lines L_1, L_2, L_3 and L_4. Let the distances between them be d_{12}, d_{23}, d_{34} respectively (here d_{ij} is distance between lines L_i and L_j). Let P be a point, sum of whose distances from four lines is k ($d_{12} < d_{23} < d_{34}$).

72. If $k = d_{12} + 2d_{23} + d_{34}$, then locus of 'P' is -

(A) entire region between lines L_2 and L_3.
(B) entire region between lines L_1 and L_2.
(C) entire region between lines L_3 and L_4.
(D) not possible.

73. If $k = d_{12} + 2d_{23} + d_{34} + 2\alpha$, $0 < \alpha < d_{34}$, then locus of 'P' is -

(A) entire region between lines L_2 and L_3.
(B) entire region between lines L_1 and L_4.
(C) entire region between lines L_3 and L_4.
(D) not possible.

Paragraph for Question 74 & 75

Consider set of 3 distinct lines $L_i : a_ix + b_iy + c_i = 0$ where $i \in \{1,2,3\}$, $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $a_i, b_i, c_i \in R$

74. If value of Δ is zero, then-

(A) lines must be concurrent
(B) lines cannot form a triangle
(C) lines must be parallel
(D) two of the three lines must be perpendicular

75. If equation of L_3 can be written as $\lambda L_1 + \mu L_2 = 0$, where λ and μ are non zero real numbers, then-

(A) L_3 must pass through origin
(B) $\Delta = 0$
(C) L_3 is perpendicular to both L_1 and L_2
(D) L_3 must be angle bisector of L_1 & L_2

Paragraph for Question 76 & 77

In ΔABC coordinates of A,B,C are (–1,6) (6,5) and (7,2) respectively altitudes drawn from vertices intersects at O and meet the opposite sides in D,E,F respectively, then-

76. Incentre of ΔDEF is-

(A) (8,9) (B) (10,11) (C) (–9,9) (D) (8,12)

77. Circumradius of ΔOBC is-

(A) 2 (B) $\frac{5}{2}$ (C) 3 (D) 5

Paragraph for Question 78 and 79

Consider a triangle ABC.

Let $x + y - 3 = 0$, $x - 4y + 7 = 0$ and $2x - y = 0$ respectively the equations of the altitudes BE, CF and AD on the sides AC, AB and BC.

78. If one of the vertex of triangle is of the form $(\alpha, 2\alpha)$, then the locus of centroid of triangle ABC as α varies is -

(A) $9x - 63y + 129 = 0$ (B) $9x + 63y - 129 = 0$

(C) $9x + 45y - 89 = 0$ (D) $9x + 45y - 99 = 0$

79. Let Q be any point on the line $L : x - k = 0$, where k is abscissa of orthocenter of ΔABC, P is a point (k, 0) other than point Q, QR is the acute angle bisector of angle OQP, (O is origin) meets the x-axis at piont R, then the locus of foot of perpendicular from R on the OQ is -

(A) $(x^2 + y^2)(x + 2) + x = 0$ (B) $(x^2 + y^2)(x - 1) + x = 0$

(C) $(x^2 + y^2)(x - 2) + x = 0$ (D) $(x^2 + y^2)(x + 1) + x = 0$

Numerical Grid Type

80. The equations of internal and external bisectors of $\angle B$ of a triangle are $x + y - 1 = 0$ and $x - y + 4 = 0$ and respectively. The equations of internal and external bisectors of $\angle C$ are $3x - y + 9 = 0$ and $x + 3y + 12 = 0$ respectively. If circumcentre of ΔIBC is (a, b) then $2b - a$ is .....(where I is the incentre of the triangle)

81. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and $ax^2 + 2hxy + by^2 - 2gx - 2fy + c = 0$ each represent pair of straight lines and area of parallelogram enclosed by them is $\left|\frac{\lambda |c|}{\sqrt{h^2 - ab}}\right|$, then $|\lambda|$ is

82. Let A(2, 4), B(–3, –8) and C(x, y) are points such that $\angle ACB$ is a right angle and the area of $\Delta ABC = \frac{41}{2}$ square units. Then number of such point C is

83. In ΔABC, AB = 9, BC = 10 and AC = 11. Points D, E and F lie on sides BC, AC and AB respectively so that segments AD, BE and CF all intersects at G. Also BF = CE and AG : DG = 2 : 1, then BF is equal to

84. If value of the expression

$$\left(\sqrt{(x-2)^2+(y-4)^2}-\sqrt{(x+2)^2+(y-2)^2}\right)^4+\left(\sqrt{(x-2)^2+(y-4)^2}+\sqrt{(x+2)^2+(y-2)^2}\right)^4$$

is minimum for $x = x_0$ and $y = y_0$, then $|x_0 + y_0|$ is equal to

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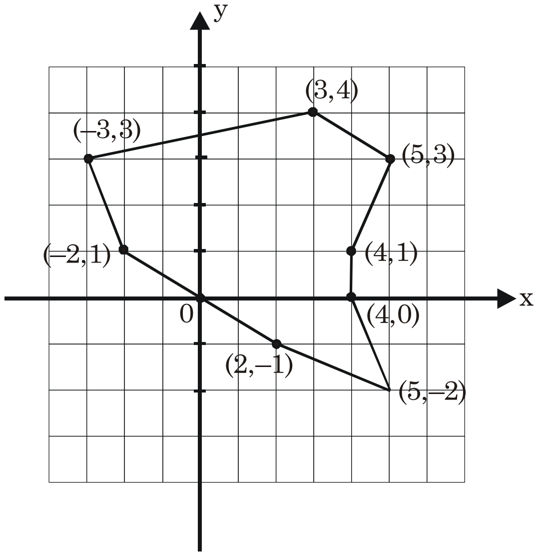
85. If area of the given figure is A sq. units. then $\frac{2A}{11}$ is equal to



86. A line passing through P(1,2) intersects the curve $(x-1)^2 = 4(y-1)$ at the points A and B, then $\frac{1}{|PA|}+\frac{1}{|PB|}$ is equal to

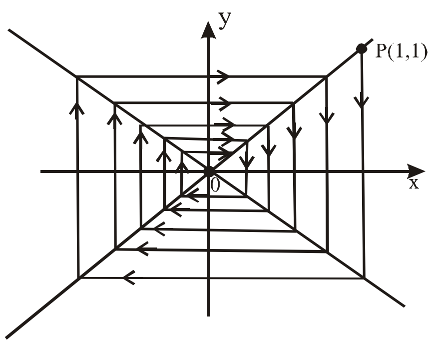
87. Let PQRS be a square such that point P lies on positive y-axis and Q lies on positive x-axis. If S is (7, 10) and the co-ordinate of point R is (a, b), then value of b is

88. A(–3,0), B(3,0) are two vertices of a triangle PAB where P is a variable point. If the angle bisector of internal angle APB divides the opposite side AB into two segments AD = 4, BD = 2, then maximum value of ordinates of the point P is

89. If a, b, c are positive numbers in H.P. and equation of line of family of lines $4acx + y(ab + bc + ca - abc) + abc = 0$ which is equally inclined to axes is given by $x - y = k$ then value of $|4k|$ is

90. Two non adjacent vertices of a rectangle are (4, 3) and (–4, –3) and the co-ordinates of other two vertices are integers, then possible number of such rectangles is

91. Consider the fixed lines $y - x = 0$ and $Ky + x = 0$, $K > 1$. A particle P starts from (1,1) to reach origin in the manner depicted as shown in figure. If the total distance travelled by the particle is 4 units, then find the value of K.



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92. Δ POR has vertices P(0,12), R(5,0) and O(0,0). There exist a line 'ℓ' cutting PR and OP at A and B respectively, such that circles can be inscribed in ΔPAB and quadrilateral ORAB. Also these circles are tangent to the line 'ℓ' at the same point. If line 'ℓ' passes through (0,8), then the area of quadrilateral ORAB is A, then find the value of $\left[\frac{A}{5}\right]$?

[**Note:** [K] denotes greatest integer less than or equal to K]

93. Consider the straight lines $L_1 : x - 7y - 8 = 0$ $L_2 : x + y + 3 = 0$ & two points $P_1(0,0)$, $P_2(\alpha^2,\alpha)$. If line segments joining P_1 & P_2 intersect the line L_1 but not L_2 then minimum positive integral value of α is

94. For a point P in the XY plane. Let $d_1(p)$, $d_2(p)$ and $d_3(p)$ be the distances of the point P from the x axis, y axis and $x - y = 0$ respectively. If A is area of the region R consisting of all points P lying in the XY plane and satisfying $d_1(P) + d_2(P) + \sqrt{2}d_3(P) = 5$ then the value of $\left[\frac{A}{3}\right]$ is

[Note [x] denotes greatest integer less than or equal to x]

95. The vertex A of ΔABC is (3, –2) and the equations to two of its medians are $5x + 3y = 11$ and $4x + 3y = 8$. If the equation of side BC is $ax + by = 7$, then $(a + b)$ is equal to

96. If the equation $x^2 + y^2 + 2xy - 2014x - 2014y - 2015 = 0$ has n solutions in (x, y) for which x, y are both positive integers and $\sum x_i$ and $\sum y_i$ represents the sum of all such possible x and y respectively, then the value of $\left[\frac{\sum x_i + \sum y_i}{400n}\right]$ ([.] is greatest integer function) is_______

97. Sachin moves from A(2,0) to B(0,b) to C(10,c) to D(9,5) along a linear path. If minimum distance travelled by Sachin is 'd', then value of d^2 is

98. A square ABCD of side length 2 cm is given, such that the points P,Q,R,S divides the sides AB, BC, CD, DA in $\alpha : 1$, $\beta : 1$, $\gamma : 1$, $\delta : 1$ respectively such that $(PQ)^2 + (QR)^2 + (RS)^2 + (SP)^2$ is minimum, then $|\alpha + \beta + \gamma + \delta|$ is equal to

99. A straight line with negative slope passing through the point (2,8) cuts the coordinate axes at A & B, then minimum value of $\frac{OA + OB}{2}$ (where O is origin) is equal to-

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ANSWER KEY

1.	**B**	**2.**	**D**	**3.**	**A,B,C,D**	**4.**	**A**	**5.**	**A,C**
6.	**B**	**7.**	**B**	**8.**	**A**	**9.**	**A**	**10.**	**B**
11.	**C**	**12.**	**D**	**13.**	**B**	**14.**	**B**	15.	A,C
16.	B	17.	B	18.	A	19.	A	20.	A,B,C,D
21.	C	22.	B,C	23.	A,C	24.	D	25.	B,C,D
26.	D	27.	A,B,C,D	28.	A	29.	D	30.	B
31.	D	32.	A,B,C,D	33.	A,C,D	34.	A,C,D	35.	B
36.	C	37.	C	38.	C,D	39.	A,C	40.	A,B
41.	A,B	42.	B,C	43.	C	44.	C	45.	C
46.	B	47.	A	48.	A	49.	A	50.	A,C
51.	A,B,D	52.	A,B	53.	A,B,C	54.	A,B,D	55.	C
56.	A	57.	B	58.	C	59.	A	60.	D
61.	A	62.	D	63.	A	64.	A	65.	C
66.	B	67.	C	68.	A	69.	D	70.	A
71.	C	72.	A	73.	C	74.	B	75.	B
76.	A	77.	D	78.	D	79.	C	80.	5
81.	2	82.	4	83.	5	84.	3	85.	5
86.	1	87.	3	88.	4	89.	7	90.	5
91.	3	92.	5	93.	8	94.	6	95.	5
96.	5	97.	194	98.	4	99.	9		

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Straight Objective Type

1. If AB is chord of contact of pair of tangents drawn from point P(5, –5) to circle $x^2 + y^2 = 5$ and (α, β) is orthocenter of ΔPAB, then $\alpha + \beta$ is -

(A) 0 (B) 2 (C) 5 (D) –2

2. Let two circles C_1 and C_2 of radii 2 and 4 be tangent at point P and tangent to a common straight line (not passing through P) at points Q and R, then value of $PQ^2 + QR^2 + RP^2$ is-

(A) 48 (B) 56 (C) 64 (D) 72

3. Let a circle S = 0 touches both the circles $x^2 + y^2 = 400$ and $x^2 + y^2 - 10x - 24y + 120 = 0$ externally and also touches x-axis. The radius of circle S = 0 is -

(A) 200 (B) 33 (C) 120 (D) 240

4. Circles are drawn by taking sides of a triangle ABC as diameters. If equations of the sides AB,BC,CA are $x - y = 0$, $3x - 4y - 17 = 0$ and $x + y - 2 = 0$, respectively, then co-ordinates of the radical centre is-

(A) (–1,2) (B) (3,–1) (C) (1,1) (D) (1,5)

5. Maximum number of points with rational co-ordinates, which can be on the circumfrence of circle centred at $(\pi,2)$, is -

(A) 0 (B) 1 (C) 2 (D) infinite

6. The sum of the slopes of the four lines tangent to both circles $(x - 2)^2 + (y - 2)^2 = 1$ and $(x - 8)^2 + (y - 2)^2 = 4$ is -

(A) 0 (B) $\sqrt{3}+\sqrt{35}$ (C) $-\sqrt{3}+\sqrt{35}$ (D) $2\sqrt{35}$

7. If the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ is inscribed in a triangle whose sides are co-ordinate axes and other side has negative slope cutting intercepts a and b on x-axis and y-axis respectively, then which of the following is/are correct.

(A) $1-\left(\frac{1}{a}+\frac{1}{b}\right)=\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}$ (B) $\frac{1}{a}+\frac{1}{b}<\frac{1}{2}$

(C) $\frac{1}{a}+\frac{1}{b}>1$ (D) $\frac{1}{a}+\frac{1}{b}-1=\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}$

8. Tangents are drawn to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ from any point A on its director circle, meets the director circle again at the points B and C, then correct option is-

(A) AB = 6, BC = 6 and equation of director circle is $x^2 + y^2 - 2x - 4y - 13 = 0$

(B) AB = 6, BC = $6\sqrt{2}$ and equation of director circle is $x^2 + y^2 = 18$

(C) $AB = 6\sqrt{2}$, $BC = 6\sqrt{2}$ and equation of direction circle is $x^2 + y^2 - 2x - 4y - 12 = 0$

(D) AB = 6, $BC = 6\sqrt{2}$ and equation of director circle is $x^2 + y^2 - 2x - 4y - 13 = 0$.

9. If the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 24x - 10y + k^2 = 0$, $k \in I$, have exactly two common tangents, then the number of possible values of k is :-

(A) 11 (B) 13 (C) 0 (D) 2

10. Let a point P is lying on x-axis, such that tangents PT_1 & PT_2 are drawn to the circle $x^2 + y^2 = 16$. If rhombus PT_1QT_2 is completed, then area of the rhombus is (Where Q lies on the circle)

(A) $24\sqrt{3}$ (B) $22\sqrt{3}$ (C) $22\sqrt{2}$ (D) $24\sqrt{5}$

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11. Given A and B are fixed points such that AB = a and point P moves on the plane in such a way that $\frac{PA}{PB} = m$ where $m \in (0,1)$. If the locus of 'P' is a curve 'C' then maximum distance between any two points of curve 'C' is -

(A) $\frac{am}{1-m^2}$ (B) $\frac{2am}{1-m^2}$ (C) $\frac{am}{2(1-m^2)}$ (D) None of these

12. Consider a circle S : $(x-11)^2 + (y-17)^2 = 1$ & a fixed point P(5,6). Two variable point Q & R lie on circle & line y = x respectively, then minimum value of PR + QR, is-

(A) 12 (B) 10 (C) 8 (D) 14

13. Let C_1 and C_2 be externally tangent circles with radius 2 and 3 respectively. Let C_1 and C_2 both touch circle C_3 internally at points A and B respectively. The tangents to C_3 at A and B meet at T and TA = 4, then radius of circle C_3 is

(A) 8 (B) 7 (C) 10 (D) 11

14. Let $C_1 : x^2 + y^2 = 1$, $C_2 : (x-10)^2 + y^2 = 1$ and $C_3 : x^2 + y^2 - 10x - 42y + 457 = 0$ are three circles. A circle C has been drawn to touch circle C_1 and C_2 externally & C_3 internally, Now circle C_1, C_2 & C_3 start rolling on circumference of C in anticlockwise direction with constant speed. The locus of the centroid of triangle formed by the centres of C_1, C_2 & C_3 as its vertices -

(A) $x^2 + y^2 - 12x - 22y + 144 = 0$ (B) $x^2 + y^2 - 10x - 24y + 144 = 0$

(C) $x^2 + y^2 - 8x - 20y + 64 = 0$ (D) $x^2 + y^2 - 4x - 2y - 4 = 0$

15. If the equation $3x^2y + 2xy^2 - 18xy = 0$ represents the sides of a triangle whose centroid is G, then the equation of circle which touches the line $x - y + 1 = 0$ and passing through G and (1, 1) is -

(A) $x^2 + y^2 - x - y = 0$ (B) $x^2 + y^2 - 9x - y + 8 = 0$

(C) $x^2 + y^2 - 4x + 2y = 0$ (D) $x^2 + y^2 + 9x + y + 8 = 0$

16. If a straight line y = mx + c touches a fixed circle such that $21m^2 - c^2 + 12m + 6c - 4mc + 16 = 0$, then radius of the circle is equal to-

(A) 2 (B) 3 (C) 5 (D) 7

17. A circle is circumscribed about a triangle ABC. A tangent is drawn to the circle through the point B to intersect the extension of the side CA beyond the point A at the point D. Let p be the perimeter of triangle ABC. Given AB + AD = AC, CD = 3 and $\angle BAC = 60°$ then the value of $(3-\sqrt{3})p$ is

(A) 3 (B) 6 (C) 9 (D) None of these

18. Let S be a circle. Tangents are drawn to the circle S from a outside point P points of contacts of tangents are Q and R. A be any point (other than Q and R) on circle S and ℓ, m, n be the length of perpendiculars from A to PQ, PR and QR respectively then

(A) $2n = \ell + m$ (B) $n^2 = \ell^2 + m^2$ (C) $n^2 = \ell.m$ (D) $2m = \ell + n$

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19. Curve $ax^2 + 2hxy + by^2 - 2gx - 2fy + c = 0$ and $a'x^2 - 2hxy + (a' + a - b)y^2 - 2g'x - 2f'y + c = 0$ intersect at 4 concyclic points A, B, C and D. If P is the point $\left(\frac{g'+g}{a'+a}, \frac{f'+f}{a'+a}\right)$ $(a+a' \neq 0)$, then the value of $\frac{PA^2 + 2PB^2 + 3PC^2}{4PD^2}$ is

(A) 5 (B) $\frac{5}{2}$ (C) $\frac{5}{4}$ (D) $\frac{3}{2}$

20. Equation of a circle touching the curve $|x-1| + |y-4| = 6$ is

(A) $x^2 + y^2 - 2x - 8y - 18 = 0$ (B) $x^2 + y^2 - 2x - 8y - 17 = 0$

(C) $x^2 + y^2 - 2x - 8y + 1 = 0$ (D) $x^2 + y^2 - 2x - 8y - 1 = 0$

21. The line $2x - y + 1 = 0$ is tangent to the circle at the point (2, 5) and the centre of the circles lies on $x - 2y = 4$. The radius of the circle is :

(A) $3\sqrt{5}$ (B) $2\sqrt{5}$ (C) $5\sqrt{3}$ (D) $5\sqrt{2}$

22. Consider the equation of circles

$S_1 : x^2 + y^2 + 24x - 10y + a = 0$

$S_2 : x^2 + y^2 = 36$ which of the following is not correct

(A) Number of non-negative integral values of 'a' such that $S_1 = 0$ represents a real circle 170

(B) If $S_1 = 0$ and $S_2 = 0$ has no point in common, then number of integral values of a is more than 49

(C) If $S_1 = 0$ and $S_2 = 0$ intersect orthogonally, then $a = 36$

(D) If $a = 0$, then number of common tangents to the circles $S_1 = 0$ and $S_2 = 0$ are 3.

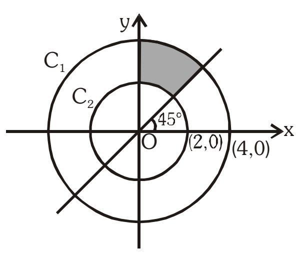
23. Consider the circles $x^2 + y^2 + 3\sqrt{2}(x + y) = 0$ and $x^2 + y^2 + 5\sqrt{2}(x + y) = 0$. A third circle is drawn internally tangent to larger circle and externally tangent to smaller circle and tangent to their common diameter. If radius of such circle is $\frac{p}{q}$ (p & q are relatively prime), then value of $p - q$ is

(A) 7 (B) 8 (C) 10 (D) –7

24. Let $A \equiv (0,4)$ and $B \equiv (3,8)$ be two points such that $\angle AXB$ is maximum where X is a point on x-axis. If x-co-ordinate of X is $a\sqrt{2} - b$, then $(a^4 + b^4)$ is

(A) 106 (B) – 602 (C) 17 (D) 706

25. As shown in figure C_1 & C_2 are two concentric circles with centres at origin and respective radii are 4, 2 units. If point (a, 3) lies inside the shaded region, then interval of a will be-



(A) $a > 0$ (B) $-\sqrt{7} < a < \sqrt{7}$

(C) $a < 3$ (D) $0 < a < \sqrt{7}$

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26. If the angle of intersection of two circles $x^2 + y^2 = a^2$ and $(x - c)^2 + y^2 = b^2$, in which the common region falls, is θ then the length of common chord of the two circles is

(A) $\dfrac{2ab\ \sin\theta}{\sqrt{a^2+b^2-2ab\cos\theta}}$ (B) $\dfrac{ab\cos\theta}{\sqrt{a^2+b^2-2ab\sin\theta}}$

(C) $\dfrac{2ab\sin\theta}{\sqrt{a^2+b^2+2ab\cos\theta}}$ (D) $\dfrac{ab\cos\theta}{\sqrt{a^2+b^2+2ab\sin\theta}}$

Multiple Correct Answer Type

27. Let f(x) is a differentiable function such that $u\ v\in R$

$$2u\,f\left(u+v^2\right)+v^3+u^4v=2v\,f\left(u^2+v\right)+u^3+uv^4$$

Consider the curve P defined by

$$P:\ f\left(x+y\right)-x-y+\frac{1}{2}=2f\left(\sqrt{xy}\right)$$

then -

(A) Area bounded by curve P with coordinate axis is $\dfrac{4-\pi}{4}$

(B) Range of the function f(x) is R^+

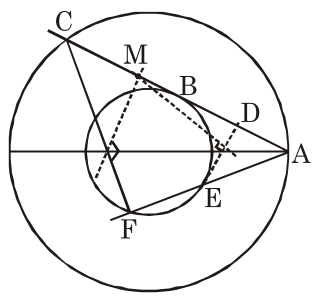
(C) Number of normals which can be drawn from point (1 ,1) to the curve P will be 4

(D) Length of tangent drawn from origin to the curve P is 1

28. If P(a,b) is a point in the first quadrant. Circles are drawn through P touching the co-ordinate axes, such that length of common chord of these circles is maximum, then possible values of $\frac{a}{b}$ is/are-

(A) $3+2\sqrt{2}$ (B) $3-2\sqrt{2}$ (C) $\sqrt{2}-1$ (D) $\sqrt{2}+1$

29. Consider circle $S_1 \equiv x^2 + y^2 - 16 = 0$ and $S_2 \equiv x^2 + y^2 - 25 = 0$. From a point A(5, 0) tangent is drawn to $S_1 = 0$ touching it at point B in 1st quadrant and meeting $S_2 = 0$ in point C. Let D be the mid-point of AB. A line through point A cuts $S_1 = 0$ in points E and F; such that perpendicular bisectors of DE and CF meets at point M on AC (as shown in figure). Now which of the following holds good.



(A) AE.AF = 9

(B) co-ordinate of point C is $\left(\dfrac{7}{5},\dfrac{24}{5}\right)$

(C) AD.AC = 18

(D) co-ordinate of point M is $\left(\dfrac{55}{20},3\right)$

30. From a point A on x-axis, 2 tangents are drawn to $x^2 + y^2 = 16$ meeting y-axis at P and Q, then

(A) the minimum value of $AP^2 + AQ^2$ is 64

(B) the minimum value of $AP^2 + AQ^2$ is 128

(C) minimum area of ΔAPQ is 32 sq. units

(D) for minimum area of triangle APQ, then co-ordinates of point A must be $(4\sqrt{2}, 0)$.

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31. An insect can go from point A(–3, 2) to $B(3\sqrt{3},2)$ by two paths such that centre of circle $x^2 + y^2 - 10y + 16 = 0$ lies in area enclosed by the paths but insect cant go inside the circle. If both paths are of smallest possible length, then which of the following option(s) is/are true -

(A) Area enclosed by both paths is $\frac{15\pi}{4}+9+3\sqrt{3}$ sq. units.

(B) Area enclosed by both paths is $\frac{15\pi}{2}+9+3\sqrt{3}$ sq. units.

(C) Absolute difference between the lengths of two paths is $\frac{5\pi}{2}$ units.

(D) Absolute difference between the lengths of two paths is $\frac{3\pi}{2}$ units.

32. Locii of a point P(x, y) satisfying the relation $x^4 + y^4 + 2x^2y^2 - 2x^2 - 6y^2 + 1 = 0$ are the two circles, these circles

(A) have centre an y-axis
(B) cut each other orthogonally
(C) are congruent
(D) have exactly two common tangents.

33. Given a circle of radius 3 and a point P outside the circle such that $\angle APB = \frac{2\pi}{3}$, where A and B are points of tangency from P to the circle. Line joining P and center O of the circle cut the circle at C and D (where C is near to point P), which of the following statement(s) is/are correct -

(A) |Area of triangle ABC - Area of triangle ABD| is $\frac{9\sqrt{3}}{2}$

(B) |Area of triangle ABC - Area of triangle ABD| is $\frac{3\sqrt{3}}{2}$

(C) Circum radius of triangle PAB is $\sqrt{3}$

(D) Circum radius of triangle PAB is $2\sqrt{3}$

34. If the equation of circle S is given by $x^2 + y^2 + 4(\lambda + 1)x - y(8 + \lambda) + 6 - 6\lambda = 0$ then which of the following option(s) is/are true.

(A) Centre of circle S = 0, always lie on a line x + 4y = 18
(B) Centre of circle S = 0, always lie on a line x + 4y = 14

(C) If S = 0 is orthogonal to $x^2 + y^2 + 2x + 4y + 1 = 0$ then $\lambda = \frac{19}{8}$

(D) If S = 0 is orthogonal to $x^2 + y^2 + 2x + 4y + 1 = 0$ then $\lambda \in \phi$

35. Two circles of radii a and 1 touch each other externally and are inscribed in the region bounded by $y = -\sqrt{4-x^2}$ and x-axis, then

(A) 100(a + 1) = 125
(B) Number of such pairs of circles is 2
(C) a = 1/2

(D) Points of contact of circle with radius a and $y = -\sqrt{4-x^2}$ are $\left(\frac{\pm 4\sqrt{2}}{3}, -\frac{2}{3}\right)$

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36. If the lines $x + y - 4 = 0$, $4x - y - 4 = 0$, $3x - y + 3 = 0$ and $mx - y = 0$ (taken in order) form a cyclic quadrilateral, then $m = \frac{a}{b}$ (where a and b are in lowest form and coprime), correct option's is/are

(A) $|a| - |b| = 1$ (B) $|a + b| = 1$ (C) $|a - b| = 13$ (D) $|a - b| = 1$

37. If $x^2 + y^2 = 4$ then

(A) maximum value of $(x + y)$ is equal to maximum value of $(y - x)$

(B) maximum value of x is equal to maximum value of $(y - x)$.

(C) minimum value of $x^2 + \frac{64}{x^2}$ is 16

(D) minimum value of $x^2 + \frac{64}{x^2}$ is 20

38. Let S_1 and S_2 be two circles touching each other externally at R. Let ℓ_1 be a line which is tangent to S_2 at P and passing through centre C_1 of circle S_1. Similarly let ℓ_2 be a line which is tangent to S_1 at Q and passing through centre C_2 of circle S_2. ℓ_1 and ℓ_2 are not parallel and intersect at K. If KP = KQ, then

(A) P and Q lie on opposite sides of the line joining C_1 and C_2

(B) P and Q lie an same side of the line joining C_1 and C_2

(C) Triangle PQR is an equilateral triangle

(D) Triangle PQR is not necessarily an equilateral triangle

39. If locus of middle points of all chords of $S_1 \equiv x^2 + y^2 - 4x - 4y - 36 = 0$ which subtends 90° at point 'P'(6,8) is a circle 'S' then

(A) Equation of circle 'S' is $x^2 + y^2 - 8x - 10y + 32 = 0$

(B) Equation of circle 'S' is $x^2 + y^2 + 8x + 10y - 32 = 0$

(C) Equation of circle of family $S_1 + \lambda S = 0$ and passes through (0, 0) is $17(x^2 + y^2) = 104x + 112y$

(D) Equation of circle with diameteric ends P(6, 8) & centre of 'S_1' is $x^2 + y^2 - 8x - 10y + 28 = 0$

40. Let point 'P' (2, 5) and point 'C' is centre of circle $x^2 + y^2 - 8x - 18y + 93 = 0$. Tangents from point 'P' touches circle at 'Q' and 'R' and mid point of chord 'QR' is 'S' and line joining 'P' to 'C' intersect circle at 'A' & 'B' then

(A) CP is arithmatic mean of AP and BP (B) PR is geometric mean of PS and PC

(C) PS is harmonic mean of PA and PB (D) Angle between PQ and PR is $\sin^{-1}\left(\frac{4}{5}\right)$

41. Let S be the set of positive value(s) of r such that the area enclosed by the tangents drawn from the point P(6,8) to the circle $x^2 + y^2 = r^2$ and the chord of contact is $\frac{75\sqrt{3}}{4}$ sq. units. Then the correct option(s) is/are -

(A) S is singleton set

(B) Equation of chord of contact is $6x + 8y = \alpha^2$, $\alpha \in S$

(C) Let A and B are point of contact of tangents then the equation of circumcircle of ΔABP is $x^2 + y^2 - 6x - 8y = 0$

(D) Number of elements in set S is 2

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42. Number of point(s) of intersection of the curves $r = \dfrac{a}{b\sin\theta + c\cos\theta}$ and $\dfrac{1}{r} = \dfrac{d}{e\sin\theta + f\cos\theta}$ can be equal to (where a,b,c,d,e,f are real numbers, $x^2 + y^2 = r^2$ and $\tan\theta = \dfrac{y}{x}$)

(A) 0 (B) 1 (C) 2 (D) 3

43. If the product of power of point (1,1) and (–1,–1) with respect to circle $x^2+y^2+2px+2qy+4=0$ is negative and the circle neither touches nor intersects co-ordinate axes, then the area of region formed by the set of ordered pair (p,q) on pq-plane is-

(A) 1 (B) 2 (C) 3 (D) 4

44. Let $S_1 = x^2 + y^2 - 1 = 0$; $S_2 \equiv x^2 + y^2 - 2x - 2y = 0$, P & Q be the points on S_1 & S_2. Now which of the following is true ?

(A) Radical axis of $S_1 = 0$ & $S_2 = 0$ is $2x + 2y = 1$

(B) The acute angle of intersection of $S_1 = 0$ & $S_2 = 0$ is $\cos^{-1}\dfrac{1}{2\sqrt{2}}$.

(C) The maximum distance between P & Q is $1+2\sqrt{2}$

(D) The minimum distance between P & Q is 1.

45. From the point of intersection of the circle $S : x^2 + y^2 - 4x + 6y + 13 = 0$ and the line $L : 2x + 5y + 11 = 0$ two tangents are drawn to the circle $x^2 + y^2 = \dfrac{121}{29}$, whose slopes are m_1 and m_2, then -

(A) $m_1 + m_2 = \dfrac{348}{5}$ (B) $m_1 . m_2 = 28$

(C) $m_1 + m_2 = -\dfrac{87}{35}$ (D) $m_1 . m_2 = -28$

46. Two tangents PA and PB are drawn to circle S = 0 where P(4,0). If line joining to point P and centre Q cuts the circle at C(2,0) and line AB at $\left(\dfrac{4}{3},0\right)$ then -

(A) radius of circle is 1 unit (B) radius of circle is 2 units

(C) $\dfrac{\text{area of }\Delta PAB}{\text{area of }\Delta AQB} = 8$ (D) $\dfrac{\text{area of }\Delta PAB}{\text{area of }\Delta AQB} = 4$

47. The equation of a circle(s) in which the chord joining the points (1,2) and (2,–1) subtends an angle of $\dfrac{\pi}{4}$ at any point on the circumference is-

(A) $x^2 + y^2 - 5 = 0$ (B) $x^2 + y^2 - 6x - 2y + 5 = 0$

(C) $x^2 + y^2 + 6x + 2y - 15 = 0$ (D) $x^2 + y^2 + 7x - 2y + 14 = 0$

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48. Let A and B are (0, 0) and (4, 0) respectively, lie on same side of a variable line 'L'. If P_1 & P_2 are perpendicular distances of point A and B from line 'L' respectively, and line 'L' touches a fixed circle, then

(A) If $P_1 + 3P_2 = 8$ then radius of circle is 2 unit and centre is (3, 0)

(B) If $P_1 + 3P_2 = 8$ then radius of circle is 3 unit and centre of circle is (1, 0)

(C) If $P_1 + 3P_2 = 4$ then radius of circle is 1 unit and centre is (3, 0)

(D) If $P_1 + P_2 = 4$ then radius of circle is 2 unit and centre is (2, 0)

49. If ax + by = k is passes through centre of circle $ax^2 + by^2 + (a + b - 4)xy - (2ax + 2by + 14) = 0$, then

(A) radius of given circle is 3 unit

(B) radius of given circle is $\sqrt{26}$ unit

(C) perpendicular distance of line ax + by = k from origin is $\sqrt{2}$

(D) length of intercept of line ax + by = k between co-ordinate axes is $2\sqrt{2}$

50. If (α, β) is a point on a circle whose center lies on the x-axis and which touches the line $x + y = 0$ at $(2, -2)$, then -

(A) least possible value of α is $6-2\sqrt{2}$

(B) least possible value of α is $4-2\sqrt{2}$

(C) largest possible value of α is $16+2\sqrt{2}$

(D) largest possible value of α is $4+2\sqrt{2}$

51. A circle C passes through the points (2, 2) and (–10, 14), then which of the following cases do not allow a unique equation of C

(A) centre of the circle C lies on y = x + 12

(B) centre of the circle C lies on y = x

(C) circle C also passes through (–3, 7)

(D) radius of the circle C is 6 units

52. Let ' C ' be the chord of a circle ' S ' of radius r which subtends angle $\frac{2\pi}{3}$ at the centre of S. If ' R ' represents the region consisting of all points inside S which are closer to C than to the circumference of S, then

(A) Area of R is $\frac{2}{\sqrt{3}}r^2$ (B) Area of R is $\frac{\sqrt{3}}{2}r^2$

(C) C divides the region R in the ratio 1 : 3 (D) C divides the region R in the ratio 1 : 4

53. The distances from origin of the centre of three circles $x^2 + y^2 - 2\lambda x = c^2$, (where c is constant, λ is variable and $\lambda > 0$) are in G.P. Let ℓ_1, ℓ_2, ℓ_3 are lengths of tangents drawn to these circles from any point on $x^2 + y^2 = c^2$, which of the following is/are not always true about ℓ_1, ℓ_2, ℓ_3 ?

(A) ℓ_1, ℓ_2, ℓ_3 are in arithmetic progression

(B) ℓ_1, ℓ_2, ℓ_3 are in geometric progression

(C) ℓ_1, ℓ_2, ℓ_3 are in harmonic progression

(D) $\ell_1 = \ell_2 = \ell_3$

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Matching list type

Answer Q.54,Q.55 and Q.56 by appropriately matching the information given in the three columns of the following table

Given 4 circles.

$S_1 : x^2 + y^2 + 4x - 4y + 4 = 0$; $S_2 : x^2 + y^2 - 2x - 2y + 1 = 0$;

$S_3 : x^2 + y^2 - 12x - 12y + 36 = 0$; $S_4 : x^2 + y^2 - 6x + 6y + 9 = 0$

Column I : Contains equation of director circle of given circles.

Column II : Contains information about number of common tangent and equation of common tangent between given circles.

Column III : Contains information about length of tangent drawn to circles from different points.

Column 1	Column 2	Column 3
(I) equation of director circle of $S_1 = 0$ is $x^2 + y^2 + 4x - 4y = 0$	(i) equation of common tangent between $S_2=0$ and $S_3=0$ is $3x+4y=12$	(P) Length of tangent drawn to $S_1 = 0$ from origin is 4.
(II) equation of director circle of $S_2 = 0$ is $x^2+y^2-2x-2y+2 = 0$	(ii) number of common tangent between $S_1=0$ and $S_2 = 0$ are 4	(Q) Length of tangent drawn to $S_2 = 0$ from (4,0) is 3.
(III) equation of director circle of $S_3 = 0$ is $x^2+y^2-12x-12y = 0$	(iii) number of common tangent between $S_2=0$ and $S_4 = 0$ are 2	(R) Length of tangent drawn to $S_3 = 0$ from (0,3) is 9.
(IV) equation of director circle of $S_4 = 0$ is $x^2 + y^2 - 6x+6y = 6$	(iv) the number of common tangents between $S_3=0$ and $S_1=0$ are 4	(S) Length of tangent drawn to $S_4=0$ from $\left(2\sqrt{2}, 2\sqrt{2}\right)$ is 5.

54. Which of the following options is the only **CORRECT** combination ?

(A) (II) (i) (Q) (B) (IV) (ii) (P) (C) (III) (iv) (S) (D) (I) (ii) (R)

55. Which of the following options is the only **INCORRECT** combination ?

(A) (I) (ii) (S) (B) (I) (i) (S) (C) (IV) (iii) (Q) (D) (III) (iv) (S)

56. Which of the following options is the only **CORRECT** combination ?

(A) (I) (ii) (Q) (B) (I) (iv) (P) (C) (III) (iii) (Q) (D) (III) (iv) (R)

Comprehension Type :

Paragraph for Question 57 to 58

Consider a circle $C : x^2 + y^2 + kx + (k + 1)y - (k + 1) = 0$ always passing through two fixed points for every $k \in R$.

57. The minimum value of the radius of circle C is

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

58. If the circles belonging to the family of circle 'C' touches y-axis, then the angle between the circles is

(A) $\tan^{-1}\left(\frac{3}{5}\right)$ (B) $\tan^{-1}\left(\frac{3}{4}\right)$ (C) $\tan^{-1}\left(\frac{4}{3}\right)$ (D) $\tan^{-1}\left(\frac{5}{3}\right)$

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Paragraph for Question 59 to 61

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two fixed points on xy plane and $R(h, k)$ is a point such that $PR : QR = k\ (k \neq 1)$ and locus of R for different values of k be curves $S_1, S_2, \ldots\ldots$ Further let $S'_1, S'_2, S'_3, \ldots\ldots$ be circles such that S_i is orthogonal to $S'_j\ \forall\ i, j$

59. Number of common tangents of S_i and S_j $(i \neq j)$ may be :

(A) 0 (B) 1 (C) 2 (D) 3

60. Number of common tangents of S'_i and S'_j $(i \neq j)$ may be :

(A) 1 (B) 2 (C) 3 (D) 4

61. If S'_k has minimum possible radius equal to that of S_i and the distance between their centres is **$(a\times PQ)$**, then a^2 equals to :

(A) 1/2 (B) 1/3 (C) 1/4 (D) 1/9

Paragraph for Question 62 and 63

Curve $x^2 + y^2 - 8x - 6y + 16 = 0$ intercepts a chord AB on the variable line $y = mx + c$, such that the circle made by taking AB as diameter always passes through origin, then

62. Locus of the centre of the circle is-

(A) $x^2 + y^2 - 6x - 4y + 4 = 0$

(B) $x^2 + y^2 - 4x - 3y + 8 = 0$

(C) $x^2 + y^2 - 4x - 4y + 4 = 0$

(D) $x^2 + y^2 - 6x - 3y + 8 = 0$

63. Relation between m & c is-

(A) $c^2 + (4m - 3)c + 8(m^2 + 1) = 0$

(B) $4c^2 + (4m - 3)c + 8(m^2 + 1) = 0$

(C) $2c^2 + (4m + 3)c + 16(m^2 + 1) = 0$

(D) $2c^2 + (4m + 3)c + 4(m^2 + 1) = 0$

Paragraph for Question 64 & 65

The line $x + 2y = a$ intersects the circle $x^2 + y^2 = 4$ at two distinct points A and B. Another line $12x - 6y - 41 = 0$ intersects the circle $x^2 + y^2 - 4x - 2y + 1 = 0$ at two another distinct points C and D.

64. The value of 'a' for which the points A,B,C and D are concyclic -

(A) 2 (B) 0 (C) –4 (D) –2

65. Equation of circle circumscribing quadrilateral ABCD is -

(A) $5x^2 + 5y^2 + 8x - 16y - 36 = 0$

(B) $5x^2 + 5y^2 - 8x - 16y - 36 = 0$

(C) $5x^2 + 5y^2 - 8x + 16y - 36 = 0$

(D) $5x^2 + 5y^2 + 8x - 16y + 36 = 0$

Paragraph for Question 66 to 67

Consider circles $C_1 : x^2 + y^2 - 4x - 6y - 3 = 0$ and $C_2 : x^2 + y^2 + 2x + 2y + 1 = 0$. Let $L_1 = 0$ represent the equation of one of the direct common tangent.

66. The equation of L_1 is

(A) $x + 2 = 0$

(B) $x - 2 = 0$

(C) $7x + 24y - 42 = 0$

(D) $7x - 24y + 42 = 0$

67. Let $C_3 = 0$ be the equation of smallest circle touching $C_1 = 0$ and $C_2 = 0$ externally and tangent to $L_1 = 0$ then radius of circle C_3 is

(A) $\frac{4}{9}$ (B) $\frac{5}{9}$ (C) $\frac{1}{3}$ (D) $\frac{2}{3}$

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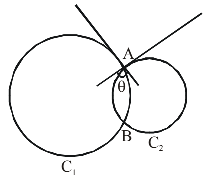
Paragraph for Question 68 and 69



If two circles $C_1 : (x-a)^2 + (y-b)^2 = r_1^2$ and $C_2 : (x-p)^2 + (y-q)^2 = r_2^2$ intersect each other at two distinct points A and B, such that angle between tangents is equal to θ (as shown in the given figure) where $\cos\theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}$ (r_1 and r_2 are radii of circles C_1 and C_2 respectively and d is the distance between centres of the given circles).

On the basis of above information, answer the following questions :

68. Value of θ is equal to-

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

69. Length of the common chord is equal to (where $r_1 \neq r_2$) -

(A) $\frac{r_1 + r_2}{2d}$ (B) $\frac{2r_1r_2}{d}$ (C) $\frac{r_1^2 - r_2^2}{2d}$ (D) $\frac{r_1^2 + r_2^2}{2d}$

Paragraph for Question 70 to 71

If the straight line $L_1 : y - 1 = k(x-1)$ and $L_2 : y - 1 = -\frac{1}{k}(x-1)$ are tangents to a variable circle such that the distance between their points of contact is $2\sqrt{2}$, then the locus of the centre of the variable circle is the curve S = 0.

On the basis of above information, answer the following questions :

70. If (x, y) satisfies the curve S = 0, then minimum value of $(x^2 + y^2 + 6)$ is -

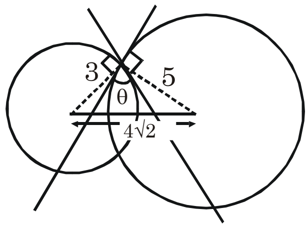
(A) $\sqrt{2} + 6$ (B) 8 (C) 14 (D) 24

71. Equation of circle which intersects the pair of lines xy = 0 orthogonally and touches the curve S = 0 internally is -

(A) $x^2 + y^2 = 4$ (B) $x^2 + y^2 = \sqrt{2}$ (C) $x^2 + y^2 = 1$ (D) $x^2 + y^2 = 2$

Integer Type

72. Let θ be the angle between tangents at the point of intersection of 2 circles having radii 3 and 5 respectively and distance between centres is $4\sqrt{2}$ units (As shown in figure) such that $\tan^2\frac{\theta}{2} = \frac{a}{b}$, then $|2a - b|$ is equal to
(where a and b are coprime numbers)



73. If P is a variable point satisfying the equation $\frac{PA}{PB} = 3$ and area of ΔPAB = 3. (where A and B are (–2,0) and (2,0) respectively), then maximum number of possible positions of point P are

74. If $\frac{p}{q}$ is the minimum value of $\left|\frac{y}{x}\right|$ for pair of real number (x,y) which satisfy the inequality $x^2 + y^2 - 50y + 400 \leq 0$, then the value of p + q is (p and q are relatively prime)

75. Let $P(x_1,y_1)$ and $Q(x_2,y_2)$ lie on $x^2 + y^2 = 1$ and $(x-1)^2 + y^2 = 1$ respectively and value of $(2x_1 - 1)^2 + (2x_2 - 1)^2 + 4(y_1^2 + y_2^2)$ lies in [a, b] then $\frac{b}{a}$ is

76. A regular hexagon is drawn circumscribing the circle with centre $\left(1, \frac{1}{\sqrt{3}}\right)$ and radius 2. $(3, \sqrt{3})$ is one vertex and other vertices are (x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4) and (x_5, y_5). If $\sum_{i=1}^{5}(x_i + y_i) = a + \sqrt{b}$ then $a + b$ is $(a, b \in N)$

77. Two circles with centres C_1 & C_2 ($\frac{15}{2}$ unit apart) are such that they have no common region of finite non-zero area. Common tangent from a point P (which divides join of C_1 and C_2 externally) touches circles at A and B respectively. If area of triangle PC_1A is $\frac{3}{2}$ unit^2 and radius of first circle is $\frac{3}{2}$ unit, then maximum area of triangle PC_2B is 3m value of m is

78. Let P and Q are two distinct points on circle $x^2 + y^2 + 2gx + 2fy + c = 0$ such that circle have minimum radius. Consider P and Q are two points from which we can draw perpendicular tangents on both circles $x^2 + y^2 - 32 = 0$ & $x^2 + y^2 - 4x - 4y - 10 = 0$ then the value of $g + f + c$ is

79. Let a point P(x, y) moving in x-y plane such that it always satisfies $x^2 + y^2 - 8|x| - 8|y| + 31 = 0$ and $\left(\left(1+x^2\right)^y - 1\right)\left(\left(1+y^2\right)^x - 1\right) < 0$. If a line passing through L(–5, 6) intersect the locus of P at A, B, C & D in order and LA . LB . LC . LD = λ then $\lambda/120$ equals

80. Let two circles are passing through a fixed point (x_1, y_1) where $x_1 > 0$, $y_1 > 0$ and both circle touch coordinate axes. If sum of their radii is equal to product of their radii, then $(x_1 + y_1) - \frac{2}{\left(\frac{1}{x_1} + \frac{1}{y_1}\right)}$ is equal to

81. Let A and B are two points on x-axis and y-axis respectively and circle passes through A,B and O (origin). Tangent at origin is at a distance of 3 unit and 5 unit from point A and B respectively, then radius of circle is

82. Tangents are drawn from point A to a circle with centre O, which touches at P and Q. Another tangent is drawn at a point R on circle, such that circle is outside the triangle formed by tangents. Perimeter of triangle is 30 cm and angle between tangents at A is 60º. If radius of circle is r then [r] equals (where [.] represent greatest integer function)

83. Consider triangle ABC. The equation of BC is y = x. The centoid of triangle is (8, 3) and circumcentre is (9, 3). If circumradius of triangle is R, then value of $\frac{R}{\sqrt{5}}$ is equal to

84. Let (α, β) be the centre of a smaller circle which touches the lines $7x - y - 5 = 0$ and $x + y + 13 = 0$ and which passes through (1,2), then the value of $|\alpha + \beta|$ is

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85. Consider circle $x^2 + y^2 + 2gx + 2fy + c = 0$, drawn with AB as diameter, where A & B are point of intersection of curve $y^2 = 4x$ and line $y = x - 1$. If perimeter of circle is $K\pi$, then value of K is equal to

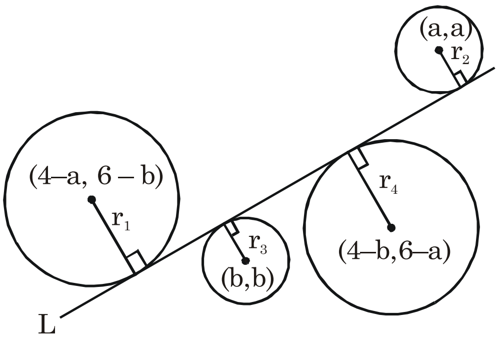
86. Let r_1 and r_2 denotes the maximum and minimum radius of circle passing through points (1,1) and (2,2) and always lie in first quadrant respectively, then $\left(\frac{r_1}{r_2}\right)^2$ is

87. If there are exactly 2 points satisfying the equations $\left|\frac{PA}{PB}\right| = \frac{2}{1}$ (where $P \equiv (x,y)$ and A and B are (0,0) and (3,0) respectively) and $(x-4)^2 + (y-5)^2 = r^2$, then number of integral values of $|r|$ is equal to

88. Let area bounded by the locus of P is A sq. units such that $\angle APB = 30^\circ$ and A,B are (0,0) and (2,0) respectively, then greatest integer less than $\left(\frac{A}{4}\right)$ is

89. Number of integral points inside the circle $x^2 + y^2 = 26$ and satisfying $\left|\sqrt{x^2+y^2} - \sqrt{(x-3)^2+(y-4)^2}\right| = 5$ is equal to

90. If four circles touch a line L (as shown in figure) such that $r_1 + r_2 = r_3 + r_4$ then lines always passes through a fixed point (h,k) where $|h + k|$ is equal to



91. The number of integral values of λ for which the variable line $3x + 4y - \lambda = 0$ lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ without having any common points with any circle is equal to 2k, then value of k is

92. A ray of light incident at the point (–3,–2) gets reflected from the tangent at (0,–2) to the circle $x^2 + y^2 = 4$. The reflected ray touches the circle. The equation of the line along which the incident ray moved is $Jx - Ky + 46 = 0$. The value of J + K is

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ANSWER KEY

1.	A	2.	C	3.	D	4.	C	5.	C
6.	A	7.	A	8.	D	9.	B	10.	A
11.	B	12.	A	13.	A	14.	B	15.	B
16.	C	17.	B	18.	C	19.	D	20.	D
21.	A	22.	D	23.	A	24.	D	25.	D
26.	C	27.	A,D	28.	A,B	29.	A,B,D	30.	B,C
31.	A,C	32.	A,B,C,D	33.	A,C	34.	B,C	35.	B,C,D
36.	A,B,C	37.	A,D	38.	B,C	39.	A,D	40.	A,B,C,D
41.	A,B,C	42.	A,B,C	43.	A	44.	A,B,C	45.	A,D
46.	A,C	47.	A,B	48.	A,C,D	49.	A,C,D	50.	B,D
51.	A,B,C,D	52.	A,C	53.	A,C,D	54.	C	55.	C
56.	A	57.	B	58.	C	59.	A	60.	B
61.	A	62.	B	63.	A	64.	D	65.	B
66.	A	67.	A	68.	D	69.	B	70.	B
71.	D	72.	9	73.	2	74.	7	75.	9
76.	6	77.	8	78.	8	79.	6	80.	2
81.	4	82.	8	83.	3	84.	3	85.	8
86.	2	87.	3	88.	5	89.	3	90.	5
91.	4	92.	7						

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ASSIGNMENT # 11 (CONIC SECTION) MATHEMATICS

Straight Objective Type

1. Pair of tangents are drawn to $x^2+4y^2+4x+3=0$ from points on the parabola $y^2=4x$. The corresponding chords of contact always touches a fixed conic section which is -

(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola

2. Let $\frac{x^2}{k}+\frac{y^2}{k-16}=1$ is equation of a conic section where $k \in (0,\infty)-\{16\}$, then which of the following is independent of k -

(A) eccentricity (B) length of latus rectum

(C) distance between vertices (D) foci

3. Four common tangents of ellipse $\frac{x^2}{4}+\frac{y^2}{1}=1$ and hyperbola $\frac{x^2}{5}-\frac{y^2}{3}=1$ make a parallelogram whose area is

(A) $4\sqrt{17}$ (B) $\frac{17}{2}$ (C) 17 (D) 34

4. An equilateral triangle is inscribed in ellipse whose equation is $x^2 + 4y^2 = 4$. One vertex of triangle is (0,1) and one altitude is contained in y-axis & length of each side is $\sqrt{\frac{m}{n}}$ (where m & n are relatively prime), then m + n is-

(A) 937 (B) 973 (C) 793 (D) 739

5. Consider right triangle ABC with hypotenuse AB and AC = 15. Altitude CH divides AB into segments AH & HB with HB = 16. Area of ΔABC is-

(A) 120 (B) 144 (C) 150 (D) 216

6. Three tangents drawn to the parabola $y^2 - 2y - 4x + 5 = 0$ such that these tangent form a triangle, then the locus of the point which cuts the line joining the circumcentre and centroid of triangle externally in 3 : 2 respectively is -

(A) $x^2 = 4y$ (B) $x = 0$ (C) $xy = 4$ (D) $x = y$

7. Let (1,0) be the vertex and $x + y - 7 = 0$ be the equation of the latus rectum of a parabola and if the equation of parabola is $x^2 + y^2 - 2xy + px + qy + r = 0$, then $(p + q + r)$ is equal to-

(A) –28 (B) –27 (C) –23 (D) –22

8. If circle whose diameter is major axis of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a > b)$ meets minor axis at point P & orthocentre of ΔPF_1F_2 lies on ellipse where F_1 & F_2 are focii of ellipse, then square of eccentricity of ellipse is-

(A) 2 sin18° (B) 2 sin15° (C) sin45° (D) sin60°

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9. A circle is described upon AA' the major axis of ellipse as diameter. Let P be any point on circle. Let AP & A'P are joined cutting the ellipse in Q and Q' respectively. If $\frac{AP}{AQ}+\frac{A'P}{A'Q'}=3$, then the value of $\frac{1}{e^2}$ is ('e' is eccentricity of ellipse) -

(A) 2 (B) 3 (C) 4 (D) 5

10. A circle of radius 'r' passes through both foci and exactly four points on the ellipse with equation $x^2 + 16y^2 = 16$. The set of all possible values of r is an interval [a,b), then a + b is equal to -

(A) $5\sqrt{2}+4$ (B) $\sqrt{17}+7$ (C) $6\sqrt{2}+3$ (D) $\sqrt{15}+8$

11 An ellipse of major and minor axis of length $\sqrt{3}$ and 1 respectively slides along the coordinate axes and always confined in the first quadrant. The locus of the centre of the ellipse will be the arc of a circle. The length of this arc is -

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) π

12. A tangent to the hyperbola $\frac{x^2}{4}-\frac{y^2}{1}=1$ meets ellipse $x^2 + 4y^2 = 4$ at two distinct points. The locus of mid points of this chord is -

(A) $(x^2 + 4y^2)^2 = 4(x^2 - 4y^2)$ (B) $(x^2 - 4y^2) = 4(x^2 + 4y^2)$
(C) $(x^2 - 4y^2)^2 = 4(4x^2 + y^2)$ (D) $(x^2 + 4y^2)^2 = 4(4x^2 - y^2)$

13. The triangle formed by tangent to the parabola $y = 4x^2$ at (a, b), (where $a \in [1,2]$) the y-axis and line y = b has the greatest area if 'a' is equal to -

(A) 1 (B) $\sqrt{2}$ (C) $\frac{3}{2}$ (D) 2

14. Given the base of a triangle and the sum of its sides, then the locus of the centre of its incircle is -

(A) parabola (B) ellipse (C) hyperbola (D) an infinite ray

15. If a circle of radius of 5 is concentric with the ellipse $3x^2 + 7y^2 = 105$. Then the common tangent can be inclined to the major axis at an angle of

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

16. Suppose the graph of quadratic polynomial $y = x^2 + px + q$ is situated so that it has arcs lying between the rays y = x and y = 2x, $x \ge 0$. These two arcs are projected on to the x axis, yielding segments S_L and S_R with S_R to the right of S_L. The difference of the lengths $\ell(S_R) - \ell(S_L) =$

(A) 0 (B) 1 (C) depends upon p & q (D) depends only on p

17. Let a second degree curve $x^2 + 4y^2 - 4 = 0$ is intersected by a variable pair of straight lines $x^2+4y^2+\lambda xy = 0$ (where 'λ' is a real parameter), at two points A and B respectively, then the locus of the point of intersection of tangents at A and B is

(A) $4x^2 + 2y^2 + 3xy = 0$
(B) $2x^2 + y^2 + 6xy = 0$
(C) $(2x - y)(2x + y) = 0$
(D) $x^2 - 4y^2 = 0$

18. Let PQ be the double ordinate of the ellipse and AA' be the major axis of this ellipse. If PA and QA' produced meets at a point R, then locus of R is a

(A) circle (B) parabola (C) ellipse (D) hyperbola

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Multiple Correct Answer Type

19. If $a^2(x^2 + y^2 - 2x - 4y + 5) = (5x - 12y + 3)^2$ is an ellipse, then the value of 'a' can be -

(A) $\sqrt{168}$ (B) 17 (C) 21 (D) $\sqrt{170}$

20. Tangents are drawn from the point P(–2,0) to $y^2 = 8x$, radius of circle (s) that would touch these tangents and the corresponding chord of contact can be equal to

(A) 4 (B) $4\left(\sqrt{2}-1\right)$ (C) $4\sqrt{2}$ (D) $4\sqrt{2}+4$

21. If the circle $x^2 + y^2 = a^2$, intersect the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then -

(A) $\sum_{i=1}^{4} x_i = 0$ (B) $\sum_{i=1}^{4} y_i = 0$ (C) $\prod_{i=1}^{4} x_i = c^4$ (D) $\prod_{i=1}^{4} y_i = c^4$

22. If a circle passes through point $\left(3, \sqrt{\frac{7}{2}}\right)$ and touches $x + y = 1$ and $x - y = 1$, then center of the circle is-

(A) (4, 0) (B) (4, 2) (C) (6, 0) (D) (7, 9)

23. Consider an ellipse $E : \frac{x^2}{16} + \frac{y^2}{4} = 1$

(A) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $4x^2 + y^2 = 9$ (where O is the origin)

(B) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $x^2 + 9y^2 = 4$ (where O is the origin)

(C) Area of quadrilateral formed by focie and end points of minor axes of the given ellipse is $8\sqrt{3}$ sq. units

(D) If a tangent is drawn at P on E to intersect the x-axis and y-axis at L and M respectively then minimum value of LM is 6 units

24. C_1 is a circle described on the line joining the positive ends of the major axis and minor axis of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$, as diameter and C_2 is the circle described on the line joining the negative ends of major axis and minor axis of the same ellipse, as diameter. If $y = mx + c$ is a common tangent to C_1 and C_2, then

(A) $m = -\frac{5}{4}, c = 0$ (B) $m = \frac{4}{5}, c = 0$ (C) $m = \frac{4}{5}, c = \frac{41}{10}$ (D) $m = \frac{4}{5}, c = \frac{-41}{10}$

Linked Comprehension Type

Paragraph for Question 25 & 26

Line x + y = 6 is tangent to a parabola 'S' at P(5,1) and M(3,–1) is foot of perpendicular from P on its directrix.

25. If coordinates of vertex of parabola 'S' are (α,β), then $\alpha - \beta$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

26. Tangents are drawn to the parabola 'S' from M which touches the parabola at the points whose coordinates are -

(A) (3,7), (12,–2) (B) (11,–1),(8,2) (C) (3,7),(7,3) (D) (3,7),(11,–1)

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Paragraph for Question 27 & 28

Consider an ellipse $E:\frac{x^2}{4}+\frac{y^2}{3}=1$, S is its focus and line L is its corresponding directrix. Curve 'C' is the locus of a variable point which moves on xy plane such that its distance from S and line L are equal.

27. Number of common tangent(s) of curve C and E is-

(A) 0 (B) 1 (C) 2 (D) 3

28. Number of normals of curve C which are tangents for ellipse E is-

(A) 0 (B) 2 (C) 4 (D) more than 4.

Paragraph for Question 29 and 30

Consider the parabola $x^2 = 4y$ and the circle $x^2 + (y - 5)^2 = r^2 (r > 0)$. Given that the circle touches the parabola at the points P and Q. Let R be the point of intersection of tangents to parabola at P and Q and S be the centre of circle.

29. Which of the following statement is/are true

(A) Centroid of ΔPQR is (0,1)

(B) Distance between chord PQ and directrix of parabola is 4

(C) Area of ΔPQR is $\frac{20}{3\sqrt{3}}$

(D) Common tangents to circle and parabola intersect at (0,–3)

30. The equation of circle which cuts intercept of length 8 units on x-axis and touches parabola $x^2 = 4y$ at point (–4, 4) is

(A) $x^2 + y^2 - 8x - 16y = 0$ (B) $x^2 + y^2 + 16x - 4y + 48 = 0$

(C) $x^2 + y^2 + 8x - 8y + 23 = 0$ (D) $x^2 + y^2 + 8x - 8y + 32 = 0$

Paragraph for Question 31 and 32

If a chord of the parabola $y^2 = 4x$ subtend a right angle at the centre of the circle $x^2 + y^2 = 16$ and the chord passes through a fixed point P and from point P three normal are drawn to intersect the parabola at A, B, C. Then

31. Area of ΔABC is

(A) $8\sqrt{2}$ (B) $4\sqrt{2}$ (C) $2\sqrt{2}$ (D) $6\sqrt{2}$

32. Out of the 3 normal chord, the length of normal chord which is minimum

(A) $6\sqrt{3}$ (B) 4 (C) $2\sqrt{3}$ (D) 2

Paragraph for Question 33 & 34

Consider the circle $S \equiv x^2 + y^2 = 4$ and a fixed point A(2, 0) on it. Let P be a variable point on circle. Chord PQ is drawn parallel to diameter of circle through A, whose mid point is R. Locus of intersection of CP and AR is the conic S^1, whose equation is $ax^2 + by^2 + 2gx + 2fy - 8 = 0$ ('C' is centre of circle). On the basis of above information, answer the following questions :

33. Length of latus rectum of the conic S^1 is -

(A) 2 (B) 4 (C) 8 (D) 16

34. If P(m, n) lies on both $y = x^2 - gx$ & $y = x - 3b$, then possible values of m are -

(A) 1 (B) 2 (C) 3 (D) 4

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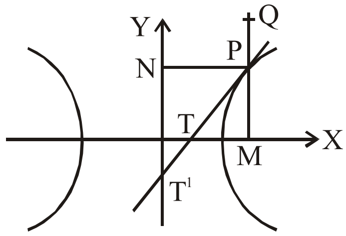
Paragraph for Question 35 & 36

Figure shows a hyperbola whose centre is C. Tangent is drawn at a variable point 'P' in Ist quadrant meeting transverse and conjugate axes at T and T^1 respectively. M and N are feet of perpendiculars from P on transverse and conjugate axes as shown. Q is a point on ordinate MP produced such that MQ = SP where S is the focus. CM.CT = 4 and CN.CT^1 = 5.



On the basis of above information, answer the following questions :

35. Eccentricity of hyperbola is -

(A) $\sqrt{2}$ (B) $\frac{3}{2}$ (C) 3 (D) 2

36. If locus of Q is straight line, then its

(A) x-intercept is $\frac{3}{4}$ (B) y-intercept is –2 (C) slope is $\frac{3}{2}$ (D) slope is $-\frac{3}{2}$

Paragraph for Question 37 & 38

Let a, b be positive real numbers, consider the circle $C_1 : (x-a)^2 + y^2 = a^2$ and ellipse $C_2 : x^2 + \frac{y^2}{b^2} = 1$.

37. The condition for which C_1 is inscribed in C_2 and having double contact with the ellipse is -

(A) $a^2 + b^2 = b^4$ (B) $a^2 - b^2 = b^4$ (C) $a^2 + b^4 = b^2$ (D) $b^4 + b^2 = a^2$

38. If $b = \frac{1}{\sqrt{3}}$, C_1 is inscribed in C_2 and having double contact with the ellipse also (p, q) are the coordinates of the point of tangency in the first quadrant for C_1 and C_2, then $(p - \sqrt{3}q)$ is equal to -

(A) 0 (B) $\sqrt{2}$ (C) 1 (D) $\frac{\sqrt{3}}{2}$

Paragraph for Question 39 & 40

Consider the curve C : xy = 4. Let any point A(t) on curve 'C' be $\left(2t, \frac{2}{t}\right)$, where $t \in R - \{0\}$.

39. If tangent at 't' point on the curve 'C' touches the curve $y^2 + 2x = 0$, then value of t is equal to-

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 3

40. If 4 points be taken on a curve C such that the chord joining any two is perpendicular to the chord joining the other two and if $\alpha, \beta, \gamma, \delta$ be the inclinations to either asymptote of the straight line joining these points to the centre, then the value of $\tan\alpha.\tan\beta.\tan\gamma.\tan\delta$ must be-

(A) 1 (B) 2 (C) $\frac{1}{2}$ (D) –1

Paragraph for Question 41 & 42

Consider a parabola $y^2 = 4x$.

41. Area of the figure formed by tangents and normals drawn at the extremities of its latus rectum is-

(A) 8 (B) 16 (C) 4 (D) 32

42. The co-ordinates of centre of a circle circumscribing a triangle formed by any tangent, normal and x-axis is-

(A) (0,0) (B) (1,0) (C) (1,1) (D) (0,1)

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Paragraph for Question 43 and 44

Let x_1, x_2, x_3, x_4 be non zero real numbers. Let A,B,C,D be distinct points on the parabola $y^2 = 4x$. Suppose that $F(x_0,y_0)$ is mid point of focal chord AB and lines BC and AE are parallel lines, where $A(x_1,y_1)$, $B(x_2,y_2)$, $C(x_3,y_3)$, $D(x_4,y_4)$ and E(2,0).

43. The value of y_3 is-

(A) $4y_0$ (B) y_0 (C) $\frac{y_0}{2}$ (D) $2y_0$

44. If $y_1y_4 = 4$, then the tangent at D and normal at A to the parabola meet at a point whose abscissa is-

(A) $\frac{1-x_0}{2}$ (B) $1 + x_0 - x_2$ (C) $1 - x_0 + x_2$ (D) $\frac{1+x_0}{2}$

Paragraph for Question 45 to 46

If the locus of the circumcentre of a variable triangle having sides y-axis, y = 2 and $\ell x + my = 1$, where (ℓ, m) lies on the parabola $y^2 = 4x$, is curve C, then

On the basis of above information, answer the following questions :

45. Vertex of curve C is -

(A) $\left(-2, \frac{3}{2}\right)$ (B) $\left(-2, -\frac{3}{2}\right)$ (C) $\left(2, \frac{3}{2}\right)$ (D) $\left(2, -\frac{3}{2}\right)$

46. Length of smallest focal chord of the curve C is -

(A) $\frac{1}{4}$ (B) $\frac{1}{12}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

Paragraph for Question 47 to 48

Consider three curves

$C_1 : x^2 - y^2 = 1 \ (x < 0)$

$C_2 : xy = 1 \ (y > 0)$

C_3 : Ellipse whose one focus is also focus of C_2 and corresponding directrix of this focus is asymptote of curve C_1.

Also length of major axis of C_3 is $\frac{8}{3}$.

On the basis of above information, answer the following questions :

47. Eccentricity of C_3 is equal to -

(A) $\frac{1}{4}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

48. If $A\left(-\frac{1}{2}, -2\right)$ and BC is double ordinate of curve C_2 then locus of orthocentre of ΔABC is a -

(A) fixed point (B) parabola (C) ellipse (D) hyperbola

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Matching List Type

49. A hyperbola 'H' intersects the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ orthogonally. The eccentricity of the hyperbola 'H' is three times the eccentricity of the ellipse. Transverse and conjugate axis of H are along x and y axis respectively. If 2a, 2b are respectively the length of transverse axis and conjugate axis, $(x_1, 0)$ is the focus and ℓ is length of latus rectum of H.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	a	(1)	1
(Q)	$\lvert x_1\rvert$	(2)	2
(R)	b^2	(3)	3
(S)	ℓ	(4)	6

Codes :

	P	Q	R	S
(A)	1	2	3	4
(B)	1	2	4	3
(C)	4	3	2	1
(D)	1	4	3	2

50. Let $C : (x - h)^2 + (y - k)^2 = r^2$ is a circle which touches hyperbola $H:\frac{x^2}{2}-y^2=1$ either at one or 2 points ($r > 0$).

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I (h,k)		List-II r
(P)	$(-1,0)$	(1)	$\sqrt{2}-1$
(Q)	$\left(\sqrt{3},0\right)$	(2)	$\frac{2}{\sqrt{3}}$
(R)	$\left(\sqrt{7},0\right)$	(3)	$\sqrt{3}-\sqrt{2}$
(S)	$(0,3)$	(4)	$2\sqrt{2}$

Codes :

	P	Q	R	S
(A)	1	3	4	2
(B)	4	3	2	1
(C)	1	4	3	2
(D)	1	3	2	4

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51. An ellipse has focus (0,0) and goes through P(4,3) and Q(–4,–3). 2a and 2b are length of its major and minor axis respectively, (x_1,y_1) is its center and (x_2,y_2) is other focus of the ellipse. Eccentricity of ellipse is $\frac{1}{\sqrt{2}}$ and y_1,y_2 are positive numbers.

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	$x_1 + y_1$	(1)	$\sqrt{2}$
(Q)	$x_2 + y_2$	(2)	$5\sqrt{2}$
(R)	b	(3)	$2\sqrt{2}$
(S)	a	(4)	10

Codes :

	P	Q	R	S
(A)	1	3	4	2
(B)	1	3	2	4
(C)	3	2	1	4
(D)	3	1	2	4

52. Let m_1, m_2, m_3 $(m_1< m_2< m_3)$ be the slopes of the distinct normals to $\frac{x^2}{16}+\frac{y^2}{12}=1$, passing through $\left(\frac{1}{2\sqrt{2}},\frac{\sqrt{2}}{2\sqrt{3}}\right)$, then

	List I		List II
(P)	$m_1 + m_2 + m_3$ is equal to	(1)	$2\sqrt{3}$
(Q)	$m_1m_2m_3$ is equal to	(2)	$-\frac{8}{3\sqrt{3}}$
(R)	$m_1 + m_3 - m_2$ is equal to	(3)	$4-\frac{2}{\sqrt{3}}$
(S)	$m_3 - m_1 - m_2$ is equal to	(4)	$4+\frac{2}{\sqrt{3}}$

	P	Q	R	S
(A)	1	2	3	4
(B)	3	2	1	4
(C)	4	2	1	3
(D)	2	2	1	3

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53. Consider an ellipse with major & minor axis having length 4 & $2\sqrt{3}$ respectively. Let P be any variable point on it. S & S' be the foci and C be the centre of ellipse

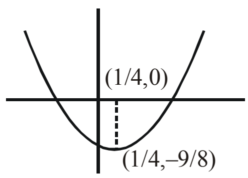
Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	Maximum value of PS.PS' is	(1)	8
(Q)	Minimum value of $(PS)^2 + (PS')^2$ is	(2)	$2+\sqrt{3}$
(R)	Minimum length of portion of tangent at P intercepted between principal axes is	(3)	4
(S)	If foot of normal from P on major axis is P' and normal at P cuts transverse axis at N, then $\frac{CN}{CP'}$ is	(4)	$\frac{1}{4}$

Codes :

	P	Q	R	S
(A)	3	1	2	4
(B)	3	1	4	2
(C)	3	4	1	2
(D)	3	2	4	1

54. Suppose a parabola has vertex $\left(\frac{1}{4}, \frac{-9}{8}\right)$ and equation $y = ax^2 + bx + c$ (where $a > 0$ and $a+b+c$ is integer). Let 'a' takes its minimum possible value under given condition, then



Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	2(latus rectum of parabola) is	(1)	9
(Q)	If equation of directrix is $4y + \lambda = 0$, then λ^2 is	(2)	81
(R)	If the value of a is $\frac{p}{q}$ (where p & q are relatively prime) then p + q is	(3)	11
(S)	If $(\alpha,0)$ lies on parabola, then $16\left(\alpha - \frac{1}{4}\right)^2$ is	(4)	36

Codes :

	P	Q	R	S
(A)	1	2	3	4
(B)	3	1	2	4
(C)	1	2	3	2
(D)	4	1	2	3

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55. Let tangent of parabola $y^2 = 4x$ at point $P(x_1, y_1)$ and $Q(x_2, y_2)$ intersect each other at $R(-15, -2)$. If S is area of ΔPQR and (x_0, y_0) is circumcentre of ΔPQR, (where $y_2 > y_1$).
Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	$x_1x_2x_0 - 2S$	(1)	208
(Q)	$y_1y_0 + \sqrt{S}x_0$	(2)	147
(R)	$\frac{x_1 + x_2 + 2S}{x_0}$	(3)	163
(S)	$\sqrt{S}x_2 + x_0$	(4)	182

Codes :

	P	Q	R	S
(A)	3	1	4	2
(B)	1	2	3	4
(C)	1	3	4	2
(D)	4	2	3	1

Matrix Match Type

56. Let S = 0 be the parabola touching x-axis at (1,0) and y = x at (1,1). Let (a,b) & (p,q) be respectively focus & vertex of parabola.

	Column-I		**Column-II**
(A)	Roots of the equation $x^2 - 5ax + 10b = 0$ are	(P)	1
(B)	$\frac{p-q}{a^2}$ is equal to	(Q)	5
(C)	Roots of the equation $x^2 - 25px + 75q = 0$ are	(R)	12
(D)	$q - 4b^2$ is equal to	(S)	2
		(T)	0

57. For parabola $y^2 - 4y - x + 3 = 0$.

	Column-I		**Column-II**
(A)	Focus of parabola is	(P)	(0,0)
(B)	Extrimity of latus rectum is	(Q)	(–1,2)
(C)	Foot of directrix is	(R)	$\left(-\frac{3}{4},2\right)$
(D)	Vertex of parabola is	(S)	$\left(-\frac{5}{4},2\right)$
		(T)	$\left(-\frac{3}{4},\frac{5}{2}\right)$

58. Consider the ellipse $x^2 + 2y^2 = 2$; with ends of major axis as A and A'. Point P lying on the ellipse is joined to A & A'. From A' perpendicular is drawn to AP & from A perpendicular is drawn to A'P. Let the locus of their point of intersection be another conic E.

	Column-I		**Column-II**
(A)	Abscissa of points of intersection of given ellipse and conic E is	(P)	$\sqrt{2}$
		(Q)	$-\sqrt{2}$
(B)	Square root of length of latus rectum of conic E is	(R)	0
(C)	The abscissa of foci of E is	(S)	$2\sqrt{2}$
(D)	If equation of directrix of E is $\lambda x + y = k$, then k is	(T)	$-2\sqrt{2}$

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Subjective Type Questions

59. Square of the area of the triangle formed by end points of a focal chord PQ of length 32 units of the parabola $y^2 = 8x$ and its vertex, is

60. Locus of a point P which divides all chords of slope $\frac{1}{2}$ of parabola $x^2 = 4y$ in the ratio 1 : 2 internally is another parabola with vertex $\left(\frac{a}{9},\frac{b}{9}\right)$ and length of latus rectum $\frac{c}{9}$, then value of abc is

61. From a point R on major axis of ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ two perpendicular normal are drawn intersecting ellipse at P & Q and area of ΔPQR is λ, then value of 25λ is

62. Let A_1 and A_2 are the vertices of the conic $C_1 : 4(x-3)^2 + 9(y-2)^2 - 36 = 0$ and a point P is moving in the plane such that $|PA_1 - PA_2| = 3\sqrt{2}$, then locus of P is another conic C_2. If D_1 denotes distance between foci of conic C_2. D_2 denotes product of the perpendiculars from the points A_1,A_2 upon any tangent drawn to conic C_2 and D_3 denotes length of the tangent drawn from any point on auxiliary circle of conic C_1 to the auxiliary circle of the conic C_2, then $\left(\frac{D_1D_2}{D_3^{\,2}}\right)^2$ is equal to

63. If a and b, (a < b) are the sides of the rectangle of the greatest area that can be inscribed in the ellipse $x^2 + 2y^2 = 8$, then b is equal to

64. If (h,k) be a point on the axis of parabola $3y^2 + 4y - 6x + 8 = 0$ from which three distinct real normals could be drawn to the parabola, then $[h]_{least}$ is equal to (where [.] denotes greatest integer function)

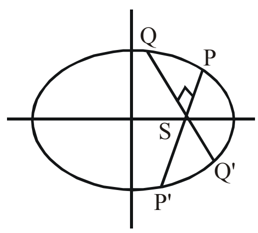
65. P(x, y) is a variable point moves in xy plane such that $x = -2t^2$, $y = t^2 - 2t + 2$, where $t \in R$, then locus of P is a conic section whose latus rectum is L. Then $\frac{5\sqrt{5}}{4}L$ is equal to

66. If P_1, P_2, P_3 are the points on ellipse $3x^2 + y^2 - 12 = 0$ and P_1', P_2', P_3' are their corresponding points on the auxillary circle, then the area of triangle $P_1' P_2' P_3'$ is λ times the area of triangle $P_1P_2P_3$, then λ^2 is

67. Let C be the centre of ellipse $\frac{x^2}{16}+\frac{y^2}{4}=1$. Let P be any point on the ellipse such that angle between tangent to ellipse at P and CP is minimum. If length of CP is $\sqrt{N}$, then $\frac{N}{2}$ is

68. If the radius of the circle touching the parabola $y^2 = 4x$ at an end of latus rectum and passing through the focus is $\sqrt{N}$, then the value of N is

69. In an ellipse $\frac{x^2}{10}+\frac{y^2}{6}=1$. If focal chords PSP' and QSQ' are at right angles to each other, then $\frac{1-e^2}{(SP)(SP')}+\frac{1-e^2}{(SQ)(SQ')}=\frac{N}{15}$. The value of [N] is (where [.] is greatest integer function)



70. The ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is such that it has least area but contains the circle $(x-1)^2 + y^2 = 1$. If eccentricity of ellipse is e, then value of $3e^2$ is equal to

71 The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ (a > 0, b > 0) and equation of its directrix is $x=\frac{1}{2}$, then eccentricity of hyperbola is

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72. From point P on hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ in first quadrant, line parallel to x-axis & y-axis are drawn cutting hyperbola at Q & R respectively. Let A be the vertex of hyperbola in 1st quadrant. If Area (ΔAPQ) = 2 Area (ΔAPR) then eccentric angle of P is $\frac{a\pi}{b}$ (where a & b are relatively prime integers). The value of a + b is

73. Two perpendicular chords are drawn through vertex 'O' of the parabola with latus rectum 4 cutting it at A an B. If the minimum area of triangle OAB is A, then the value of $\frac{A}{4}$

74. PQ and PR are two tangents to a parabola $y^2 = 4ax$ and tangent to parabola at a third point S cuts these tangents at A and B respectively then value of $\frac{PA}{PQ}+\frac{PB}{PR}$ is

75. If maximum distance of any point on the ellipse $5x^2 + 4y^2 + xy - 2 = 0$ from its centre be k and $k=\frac{a}{\sqrt{b-\sqrt{2}}}$, then (b – a) is

ANSWER KEY

1.	D	**2.**	D	**3.**	C	**4.**	A	**5.**	C
6.	B	**7.**	C	**8.**	A	**9.**	A	**10.**	D
11.	A	**12.**	A	**13.**	D	**14.**	B	**15.**	D
16.	**B**	**17.**	**D**	**18.**	**D**	**19.**	B,C,D	**20.**	B,D
21.	A,B,C,D	**22.**	A,C	**23.**	A,C,D	**24.**	A,C,D	**25.**	D
26.	D	**27.**	A	**28.**	B	**29.**	A,B,D	**30.**	A,B
31.	B	**32.**	A	**33.**	B	**34.**	B,C	**35.**	B
36.	B,C	**37.**	C	**38.**	A	**39.**	C	**40.**	A
41.	A	**42.**	B	**43.**	**D**	**44.**	**B**	**45.**	**A**
46.	**C**	**47.**	**D**	**48.**	**A**	**49.**	A	**50.**	D
51.	B	**52.**	A	**53.**	A	**54.**	C	**55.**	**A**

56. (A)→(P,S); (B)→(P); (C)→(P,R); (D)→(T)

57. (A)→(R); (B)→(T); (C)→(S); (D)→(Q)

58. (A)→(P,Q); (B)→(P); (C)→(R); (D)→(S,T)

59.	256	**60.**	064	**61.**	081 OR 150	**62.**	036	**63.**	4
64.	2	**65.**	4	**66.**	3	**67.**	5	**68.**	2
69.	4	**70.**	2	**71.**	2	**72.**	4	**73.**	4
74.	**1**	**75.**	**7**						

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One or more than one Correct Answer Type

1. Number of ways in which 19 identical gifts can be distributed to 7 girls if each gets odd number of gift is N, then choose correct option(s) -
(A) Sum of digits of N is 15. (B) Sum of digits of N is 18.
(C) Product of digit of N is 72. (D) Product of digit of N is 126.

2. The number of distinct rational numbers 'x' such that $0 < x < 1$ and $x = \frac{p}{q}$ where $p, q \in \{1, 2, 3, 6, 8, 9\}$ is -
(A) 15 (B) 13 (C) 12 (D) 11

3. Persons A,B,C,D & E are to be seated on a circular table. Number of ways they can be seated so that exactly two from A,B & C are adjacent is equal to -
(A) 6 (B) 8 (C) 12 (D) 24

4. Number of possible values of n such that n! has exactly 20 zeroes at the end is equal to -
(A) 4 (B) 5 (C) 6 (D) 7

5. Let $A = \{1,2,3,4\}$ and the subset B of A has property that if $x \in B$ and $x + 2 \in A$ then $x + 3 \in B$ (if $x + 3 \in A$), then total number of possible subsets is -
(A) 7 (B) 9 (C) 10 (D) 12

6. Number of solutions to the equation $x_1x_2x_3x_4 = y_1y_2y_3y_4$, where the variables are allowed to take values 0,1 and –1 only, is -
(A) 4323 (B) 4338 (C) 4353 (D) 4368

7. Number of ways in which letters of the word 'HONGKONG' can be arranged such that O & N never come together, is -
(A) 1440 (B) 1320 (C) 960 (D) 600

8. Number of ways in which we can choose two distinct integers from 1 to 200 such that difference between them is atmost 15, is
(A) ${}^{200}C_2 - {}^{185}C_2$ (B) ${}^{200}C_{198} - {}^{185}C_{183}$ (C) ${}^{200}C_2 - {}^{184}C_2$ (D) ${}^{200}C_{198} - {}^{184}C_{182}$

9. Five boys and four girls are to be seated in a row so that two particular girls will never sit adjacent to a particular boy and all girls are separated. If the number of ways in which they can be seated is N, then value of $\frac{N}{480}$ is-
(A) less than 38 (B) greater than 38 (C) greater than 32 (D) less than 32

10. 20 guest in a party were divided into 10 equal groups randomly and then 3 different prizes distributed to only 3 groups (each groups one prize) after some competition. Number of ways in which that can be done -
(A) $\frac{20!}{(2!)^{10}.10!} \times {}^{10}C_3$ (B) $\frac{20!}{(2!)^{10}.10!} \times {}^{10}C_3 \times 3!$
(C) $\frac{20!}{(2!)^{10}} \times {}^{10}C_3$ (D) $\frac{20!}{(2!)^{10}} \times {}^{10}C_3 \times 3!$

11. If the number of possible ordered pairs (m, n) such that $4^m + 3^n$ is divisible by 5 where m & n $\in$ N and ≤ 20, is a natural number, then sum of its digits is
(A) 3 (B) 1 (C) 5 (D) 7

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12. 8 cards and 8 envelopes are numbered 1, 2, 3, 4, 5, 6, 7, 8 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and morever the card numbered 2 & 4 always placed in envelopes numbered 1 & 3 respectively, then number of ways of placing cards in evelopes with above constraint is -

(A) 255 (B) 362 (C) 9 (D) 97

13. Define the permutation of the set {1,2,3,...n} to be sortable if upon cancelling appropriate term of such permutation remaining n–1 terms are in increasing order. If $f(n)$ is the number of sortable permutations of {1,2,3,...n}, then-

(A) $f(5) = 17$ (B) $f(5) = 16$ (C) $f(3) = 5$ (D) $f(4) = 10$

14. Consider a cube of side length 9 units. Let (x, y, z) be coordinates of points on or inside the cube such that $x, y, z \in I$ and $0 \le x, y, z \le 9$. If total number of ways of selecting two distinct points among these such that their mid-point is also having integral coordinates is N, then

(A) N is divisible by 30 (B) N is divisible by 31

(C) N is divisible by 32 (D) Number of factors of N is 40

15. The number of 4 digit natural numbers which contains

(A) exactly 2 distinct digits is 567 (B) atleast 2 distinct digits is 8090

(C) atmost 2 distinct digits is 577 (D) atleast 3 identical digits is 333

16. A line divides a plane into 2 regions. Two lines divide the plane into maximum 4 regions. If L_n is the maximum number of regions divided by n lines then which of the following is/are true :

(A) $L_{20} = 211$ (B) $L_{10} = 56$ (C) $L_{15} = 121$ (D) $L_{25} = 326$

17. Number of necklaces can be formed with 4 identical beads and two distinct jewels is ‘k’, then k divides

(A) 15 (B) 35 (C) 21 (D) 105

18. Total number of distinct 4 letters words that can be formed from the letters of the word 'NANDINI', so that at most 2 alike letters are together, is -

(A) 104 (B) 106 (C) 108 (D) 114

19. Let X and Y be subsets of S = {1,2,3,4,5} such that for each $x \in X$, $x > n(Y)$ and for each $y \in Y$, $y > n(X)$. Then number of ordered subsets (X,Y) of S is-

[Note : n(A) denotes number of elements in set A]

(A) 144 (B) 121 (C) 136 (D) none of these

20. The number of divisors of the form (4k + 2), $k \ge 0$, $k \in$ integer of the number $2^8 5^{10} 7^3 13^2$ is -

(A) 1056 (B) 924 (C) 132 (D) 32

21. All six digit numbers containing each of the digits 1,2,3,4,5,6 exactly once, and not divisible by 5, are arranged in decreasing order. Then sum of two middle digits in 200^{th} number in list is

(A) 8 (B) 7 (C) 10 (D) 11

22. Consider the number $N = 2^4 3^4 5^2 7^2 11^2$, then the sum of all the odd divisors which are divisible by '5' but not by '7' is $11^2(133\lambda)$ then λ is equal to

(1) 10 (2) 30 (3) 50 (4) 330

23. Number of squares that can be formed in 8 × 8 chessboard so that each square is surrounded by four square of same dimensions, are

(A) 36 (B) 42 (C) 45 (D) 48

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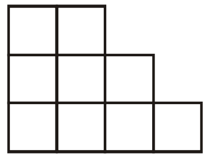
24. The number of ways in which 5 letters of word ALLEN can be placed in the squares of the figure so that no row remains empty is -



(A) 98 (B) 5880
(C) 7560 (D) none of these

25. If x,y,z,w, $\in$ N, such that each of them gives remainder 3 when divided by 5, then number of solutions of the equation $x + y + z + w = 62$, is equal to-

(A) ${}^{13}C_3$ (B) ${}^{9}C_3$ (C) ${}^{12}C_3$ (D) ${}^{12}C_2$

26. The number of ordered quadruples (a_1, a_2, a_3, a_4) of positive odd integers that satisfy $a_1 + a_2 + a_3 + a_4 = 32$ is equal to -

(A) 286 (B) 4495 (C) 680 (D) 930

27. A person wants to climb a n-step staircase using one step or two steps. Let C_n denotes the number of ways of climbing the n-step staircase. Then $C_{18} + C_{19}$ equals

(A) C_{20} (B) C_{21} (C) greater than C_{21} (D) less than C_{20}

28. The number of positive integral solutions of the equation $x + y + z + w = 15$ where $x > 1$, $y > 2$, $z > 3$ and $w < 4$ is

(A) 46 (B) 56 (C) 64 (D) None of these

29. Three digit numbers xyz are formed with digits 0, 1, 2, ... 9. Such that $x \le y \le z$ then number of such number is

(A) 150 (B) 504 (C) 165 (D) 425

30. Consider the set of all triangle OPQ where 'O' is origin & P, Q are distinct points in the plane with non-negative integral co-ordinates (x,y) such that $5x + y = 99$, then the number of such distinct triangle whose area is positive integer, are

(A) 45 (B) 15 (C) 90 (D) 120

31. Number of nine digit even natural number formed by 0 or 1 and no consecutive digits in them are 0, is

(A) 18 (B) 35 (C) 21 (D) 36

32. Number of natural solutions of the equation $xyz = 2^5 \times 3^2 \times 5^2$ is equal to-

(A) 756 (B) 520 (C) 720 (D) 120

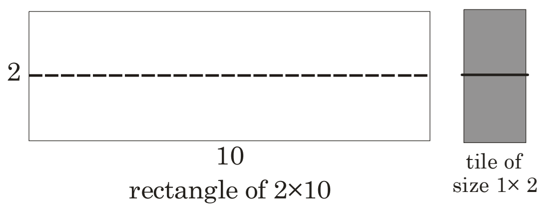
33. Four digit numbers are formed using the digits 1,2,3,4 (repetition is allowed). The number of such four digit numbers divisible by 11 is-

(A) 22 (B) 36 (C) 44 (D) 52

34. The number of ways in which 3 children can distribute 10 tickets out of 15 consecutively numbered tickets themselves such that they get consecutive blocks of 5,3 and 2 tickets is

(A) ${}^{8}C_5$ (B) ${}^{8}C_5.3!$ (C) ${}^{8}C_5.(3!)^2$ (D) ${}^{15}C_{10}.3!$

35. A 2×10 rectangle needs to be tiled with tiles of size 1×2 (domino) as shown in figure. In how many ways can this be done?



(A) 34 (B) 55 (C) 89 (D) None of these

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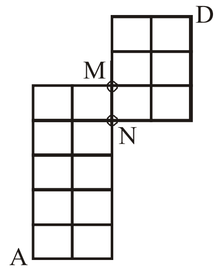
36. The number of shortest paths from point A to D (as shown in figure)



(A) 276
(B) 186
(C) 150
(D) 126

37. Number of ways in which 4A, 1B, 1C and 1D can be distributed among 4 persons such that no one gets all 4A and every one gets at least one letter is-

(A) 832 (B) 2432 (C) 2077 (D) 608

38. Total number of words which can be formed by taking all the letters of the word 'ROORKEE' such that among the alike letters only 'RR' is always together

(A) 60 (B) 84 (C) 96 (D) none of these

39. There are 10 straight lines in a plane, such that no 3 are concurrent and no 2 are parallel to each other. If points of intersection of above lines are joined, then maximum number of lines thus formed are (including old lines) -

(A) 610 (B) 620 (C) 630 (D) 640

40. If $A = \{1,2,3,4\}$, $B = \{1,2,3,4,5,6\}$ and $f : A \rightarrow B$ is an injective mapping satisfying $f(i) \neq i$, then number of such mappings are -

(A) 182 (B) 181 (C) 183 (D) none of these

41. The number of ways in which 3 identical red balls and 3 identical white balls can be distributed in between 3 students such that each student receive atleast one ball, is-

(A) 54 (B) 55 (C) 56 (D) 65

42. 3 married couples are to seated around a circular table. Let x denotes the number of ways where atmost two couples sit together and let y denotes the number of ways where two particular couples sit together, then identify the correct option(s) :-

(A) $x - 8y = 60$ (B) $x + y = 112$ (C) $x = 13y$ (D) $x = 12y$

43. Number of permutations of the word PANCHKULA where A & U are separated (Note that PAANCHKUL is one such word) is equal to -

(A) $15\underline{|7}$ (B) $21\underline{|7}$ (C) $24\underline{|7}$ (D) $36\underline{|7}$

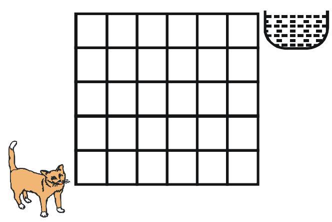
44. If r, s, t are different prime numbers and a, b, c are the positive integers such that LCM of a, b, c is $r^2s^4t^2$ and HCF of a, b, c is rs^2t, then the number of ordered triplets (a, b, c) is -

(A) 248 (B) 432 (C) 648 (D) 931

45. Sum of all the numbers greater than one lac, that can be formed by using the digits 1,2,3 is (every digit is used exactly twice) -

(A) 79999920 (B) 79999920 (C) 1999980 (D) 19999980

46. A cat moves two steps forward or one step upward in a grid having milk at one corner as shown in adjacent diagram. If cat starts from diagonally opposite corner then number of ways in which it can reach to the milk is -



(A) ${}^{11}C_6$ (B) ${}^{8}C_6$
(C) ${}^{8}C_3$ (D) ${}^{8}C_3.2!.2!2!$

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47. Let A = {6, 10, 14, ..., 1002} and B is set of divisors of the integer 360, then S(A $\cap$ B) is [S(A) denotes sum of elements in set A]

(A) 148 (B) 154 (C) 156 (D) 162

48. Consider a sequence of 10A's and 8 B's placed in a row. By a run we mean single or more letters of the same type placed side by side. For example following is an arrangement consisting of 4 runs of A & 4 runs of B : AAABBABBBAABAAAABB. The number of arrangements of 10 A's and 8 B's in a row so that there are -

(A) 4 runs of A and 4 runs of B is $2.{}^9C_3.{}^7C_3$ (B) 4 runs of A and 5 runs of B is $3.{}^9C_4.{}^7C_4$

(C) 3 runs of A and 3 runs of B is $2.{}^9C_2.{}^7C_2$ (D) 6 runs of A and 6 runs of B is $4.{}^9C_5.{}^7C_5$

49. Number of words that can be formed using all the letters of the word 'H I P H I P H U R R A Y' in which all H's lies somewhere between R's is

(A) $(198)7!$ (B) $(99)7!$ (C) $(99)8!$ (D) $(198)8!$

50. Number of ways of selecting n things out of 3 n things out of which n are of one kind (identical) and n are of IInd kind (identical) and rest unlike is :

(A) $(n+2)\,2^{n-1}$ (B) ${}^{3n}C_n$ (C) ${}^{3n}C_n - {}^{2n}C_n$ (D) None of these

51. Let there are n planes in R^3 such that any three have exactly one point in common and no four of them have a point in common. If $f(n)$ represent the number of parts in which these n planes will divide the space, then-

(A) $f(3) = 8$ (B) $f(4) = 15$ (C) $f(4) = 12$ (D) $f(10) = 176$

52. Eight cards and eight envelopes are numbered 1,1,2,3,4,5,6,7 and cards are to be placed in envelops so that each envelope contain exactly one card and no card is placed in the envelope bearing the same number. Then the number of ways it can be done is-

[Note : cards and envelopes of the same number are identical]

(A) 4230 (B) 2115 (C) 2715 (D) none of these

53. Number of points with integral co-ordinates lying inside the trapezium formed by lines x = 0, y = 0, x + y = 15 and x + y = 5 is-

(A) 95 (B) 81 (C) 99 (D) 105

54. In the many ways we can select three numbers from the set {10, 11, 12, ..., 100}. Such that they form a G.P. with integral common ratio greater than 1, is -

(A) 15 (B) 16 (C) 17 (D) 18

55. Number of seven digit numbers, which can be formed with non-zero distinct digits such that it starts and end with a prime digit and middle digit is an even digit, is-

(A) 17280 (B) 15120 (C) 12960 (D) 21600

56. There are 7 seats in a row. The number of ways three persons take seats so that middle seat is always occupied and no two persons are consecutive is equal to n!, then which of the following statements is(are) true -

(A) Number of ways in which n points are selected from 12 distinct collinear points so that no two of them are consecutive is ${}^9C_5$

(B) The number of (n + 1) digit numbers in which the digit in any place is greater than the digit to the left of it is ${}^9C_5$

(C) The number of odd proper divisors of 20412 is 3n + 1

(D) $\frac{1}{2} + \log_6 n + \log_8 6 \geq 3^{2/3}$

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57. If $A_1, A_2, ..., A_n$, (where $n \geq 6$) be n people sitting on a circle. Numbers of ways 3 people can be selected if no two of them are consecutive is (are) -

(A) ${}^{n}C_3 - (n^2 - 3n)$ (B) ${}^{n-2}C_3 - {}^{n-4}C_1$

(C) $\dfrac{{}^{n}C_1 . {}^{n-4}C_2}{3}$ (D) ${}^{n}C_1 . {}^{n-4}C_2$

58. Number of ways to choose 3 positive integers $a < b < c$ so that $abc = 30030$

(A) 121 (B) 729 (C) 726 (D) 216

59. 5 British and 4 American men checked in a Hotel such that each occupy exactly one room. The hotel has 5 rooms, 3 rooms, 3 rooms and 3 rooms at Ist floor, 2nd floor, 3rd floor and 4th floor respectively. Number of ways in which the rooms can be alloted so that no floor remains empty -

(A) $1836 \times 9!$ (B) $2002 \times 9!$ (C) $1835 \times 9!$ (D) $2001 \times 9!$

60. Let x be a set containing 8 elements and P(x) be it's power set. A and B are picked from P(x). If $n(A) = n(B)$ and $A \neq B$, then total number of ordered pair (A,B) is

[Note : n(A) represents number of elements in set A]

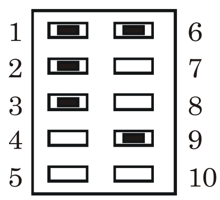
(A) 12870 (B) 13420 (C) 13124 (D) 12614

61. A code is a sequence of 0's and 1's that does not have three consecutive 0's. The possible number of codes that have exactly 11 digits is -

(A) 927 (B) 9×2^8 (C) 504 (D) $2^{10} - 9$

62. One commercially available Tenbutton lock can be opened by pressing the correct button(s) in any order. Number of possible ways that these locks are designed so that sets of as many as nine and as few as one button could serve as combination. (Assume that in every combination a button can not be pressed more than once)

{The sample shown has 1,2,3,6,9 as its combination}



(A) $2^{10} - 2$ (B) $2^{10} - 1$ (C) $\left(\sum_{r=0}^{10} ({}^{10}C_r)\right) - 1$ (D) $10! - 2$

63. Let $5 < n_1 < n_2 < n_3 < n_4$ be integers such that $n_1 + n_2 + n_3 + n_4 = 35$. The number of such distinct arrangements (n_1,n_2,n_3,n_4) is-

(A) ${}^{38}C_3$ (B) ${}^{8}C_3$ (C) 5 (D) 6

64. Consider the set $S = \{0,1,2,3,.........,9\}$, then which of the following is/are always true ?

(A) Number of ways the two integers a,b with replacement chosen from S so that $|a - b| > 6$ is 12

(B) Number of ways the two integers a,b with replacement chosen from S so that $|a - b| > 6$ is 20

(C) Number of 10-digit numbers that can be formed using each and every digit of S, which are divisible by 2 but not by 3 is 0

(D) Number of 10-digit numbers that can be formed using each and every digit of S which are divisible by 2 but not by 3 is 10

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Comprehension Type :

Paragraph for Question 65 to 66

Consider two system of parallel lines mutually perpendicular in a plane. Containing n lines in each system and distance between pair of adjacent lines is 1 unit in each system, then

65. For n = 6, total number of ways of selecting four squares of one unit side length from grid formed by above lines, such that exactly three pairs of squares have exactly one vertex in common and other three pairs has no vertex in common, is

(A) 225 (B) 12650 (C) 124 (D) 228

66. For n = 3, number of ways in which squares of 1 units side lengths can be coloured with three different colours with 2 different shades of each, such that no two squares with common sides has same colour of any shade, is -

(A) 384 (B) 288 (C) 576 (D) 216

Paragraph for Question 67 & 68

A cricket team of eleven players is to be selected out of 6 batsman, 5 bowlers, 2 all rounders and 2 wicket keepers. Consider an allrounder as a player who is a batsman as well as a bowler.

67. The number of selections containing exactly one wicket keeper and atleast one all rounder is -

(A) ${}^2C_1(2.{}^{11}C_9+{}^{11}C_8)$ (B) ${}^2C_1.{}^2C_9.{}^{11}C_9$

(C) ${}^2C_1.({}^{12}C_3+{}^{11}C_3)$ (D) ${}^4C_2.{}^{11}C_9$

68. Number of selections when a particular wicket keeper do not want to play if a particular batsman and a particular bowler are both playing, is (where it is not necessary that a wicket keeper must be a part of playing eleven)

(A) ${}^{13}C_{11}+{}^{13}C_{10}+{}^{12}C_9$ (B) ${}^{15}C_{11}-2.{}^{13}C_3$

(C) ${}^{13}C_{11}+2.{}^{13}C_{10}+{}^{12}C_9$ (D) ${}^{13}C_{11}+{}^{13}C_{10}+{}^{12}C_8$

Paragraph for Question 69 to 70

Let x & y be 4 digit numbers in each of which digits do not repeat. A pair of such numbers (x, y) is called a good pair if no digit in numbers x & y are common. Whereas pair (x, y) is called bad pair if at least one digit in numbers x and y is common.

69. The number of ordered pairs (x, y) which are good is

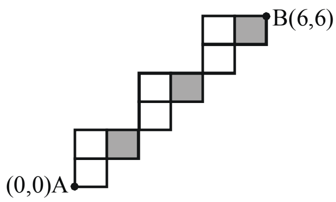
(A) $4\underline{|9}$ (B) $3\underline{|9}$ (C) $\underline{|9}$ (D) None of these

70. The number of ordered pairs (x, y) which are bad is

(A) $\frac{527\underline{|9}}{10}$ (B) $\frac{527\underline{|9}}{5}$ (C) $\underline{|9}$ (D) None of these

Paragraph for Question 71 to 73

Consider the grid path as shown in the figure.



On the basis of above information, answer the following questions :

71. The number of ways a person can move, from A to B along the grid lines via shortest path is -

(A) 125 (B) 64 (C) 27 (D) 216

72. The letters of the word TENDULKAR are filled in each box of the grid, then number of ways such that only consonants occupy shaded box of the grid is k.6!, where k is equal to -

(A) 36 (B) 72 (C) 120 (D) 180

73. The number of ways in which six X's can be filled in the grid so that no horizontal row of the grid remains empty, is -

(A) 81 (B) 64 (C) 27 (D) 8

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Paragraph for Question 74 & 75

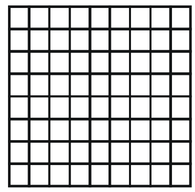
A function is defined as $f(\alpha) = (-1)^{\alpha_1} + (-1)^{\alpha_2} + (-1)^{\alpha_3} + ... + (-1)^{\alpha_k}$, where $\alpha \in N$ and $\alpha_1, \alpha_2, \alpha_3, ..., \alpha_k$ are all divisors of α including 1 and α itself.
For example, let $\alpha = 84$, then $\alpha = 2^2.3^1.7^1$ has divisors $\alpha_1, \alpha_2, \alpha_3 ... \alpha_{12}$ of which 8 are even and 4 are odd. Hence $f(\alpha) = 4$.
On the basis of above information, answer the following questions :

74. If $f(\alpha) = 4$ and $\alpha < 60$, then number of possible values of α are -
(A) 2 (B) 4 (C) 6 (D) 8

75. If $f(\alpha)$ is a negative odd number, then α can be -
(A) 2 (B) 4 (C) 5 (D) 10

Paragraph for Question 76 to 77

If 10 vertical equispaced (1 cm) lines and 9 horizontal equispaced lines (1 cm) are drawn in a plane as shown in the given figure.
On the basis of above information, answer the following questions :



76. Total number of rectangles with one side odd & one side even are given by-
(A) 600 (B) 700 (C) 800 (D) 900

77. If squares of odd side length are selected from the above grid, then sum of their areas is equal to-

(A) $\sum_{r=1}^{4}(11-2r)(10-2r)(2r-1)^2 \text{ cm}^2$ (B) $\sum_{r=1}^{8}(9-2r)(7-2r)(2r+1)^2 \text{ cm}^2$

(C) $\sum_{r=1}^{8}(11-2r)(9-2r)(2r+1)^2 \text{ cm}^2$ (D) $\sum_{r=1}^{5}(11+2r)(9+2r)(2r-1)^2 \text{ cm}^2$

Paragraph for Question 78 and 79

Let Z = DEPARTMENT.
On the basis of above information, answer the following questions :

78. Number of selection of 2 vowels and 2 consonants from the letters of Z, is-
(A) 15 (B) 32 (C) 45 (D) 105

79. If number of arrangements of letters of word Z taken all at a time when no two T's are together and no two E's are together is $K\underline{|7}$, then value of K is-
(A) 108 (B) 116 (C) 120 (D) 144

Paragraph for Question 80 to 82

Consider seven different points $P_1, P_2, P_3, P_4, Q_1, Q_2, Q_3$ in a plane such that P_1,P_2,P_3,P_4 are on straight line 'ℓ'; Q_1,Q_2,Q_3 are non-collinear points and none of them lies on straight line 'ℓ' (Q_1,Q_2,Q_3 are on same side of ℓ).

80. The maximum possible number of quadrilaterals formed by using seven given points -
(A) 34 (B) 22 (C) 37 (D) 38

81. The maximum number of possible points of intersection of straight lines formed by joining any two of the given seven points (Excluding the given points) -
(A) 51 (B) 30 (C) 61 (D) 58

82. From each of the three points Q_1,Q_2,Q_3 perpendicular lines are drawn to the straight lines formed by joining any two of the given points (excluding the point from which perpendicular line is drawn). The maximum possible number of points of intersection of perpendicular lines (excluding points Q_1, Q_2, Q_3) is-
(A) 283 (B) 286 (C) 284 (D) 289

Integer Type

83. If natural numbers from 1 to 10000 are written in a row then digit '9' appears 'k' times. Then the value of $\frac{k}{1000}$ is

84. The number of ways to select two distinct numbers 'a' and 'b' from the set $\{5^1, 5^2, 5^3 \ldots.. 5^{25}\}$ so that $\log_a b$ is an integer is denoted by k, then sum of the digits of 'k' is

85. There are four pairs of shoes of four persons with different sizes in a Shoe Rack. A person can distribute these shoes to above four persons such that each receives any two shoes and no one get his own complete pair in 'm' number of ways then largest digit in m is

86. If A.M. of smallest elements in all triplets of distinct element of set A = {1, 2, 3, 4, 5, 6} is $\frac{a}{b}$ where 'a' and 'b' are relatively prime then (a – b) is

87. If L.C.M. of A, B, C is $r^2 t^2 s^2$, where r, s, t are distinct prime numbers and A, B, C are positive integers, the number of ordered triplets (A, B, C) is K, then value of $\left[\frac{K}{1000}\right]$ is (where [.] denotes greatest integer function)

88. Let k denotes the number of integers formed using the digits 0 to 9 (digits used at most once), greater than 999 and less than 10^6 which are divisible by 125, then number of proper factors of k are

89. Let $A_1A_2A_3 \ldots\ldots A_{11}$ is a regular polygon. If number of quadrilaterals that can be formed with the help of vertices of polygon, such that no side of quadrilateral is a side of polygon, is (11p) then value of p is

90. There are 3 married couples and an unmarried couple want to sit in a row, such that no husband sit next to his own wife, then total number of ways are given by the number N then $\frac{N}{2220}$ is equal to

91. Using only 0, 1, 2, 3, 4, if number of all four digit numbers where sum of digit is 12, is N, then digit at unit place of N, is

92. Let P_n denotes the number of ways of choosing three points out of n distinct points in a line such that no two of them are consecutive. If Q_n denotes the number of ways of choosing similar points on circle then $\frac{Q_6}{P_6} + \frac{Q_7}{P_7} + \frac{Q_8}{P_8}$ is

93. A committee of four members is to be formed from 5 men and 5 women. If atleast one of Miss X and Miss Y is in the committee then Mr. A & Mr. B must serve together and if both the above ladies are not in the committee then both the above mentioned men will not serve the committee. If committee spend ₹300 per women and ₹200 per men, then number of committies with minimum expenditure is

94. Let A is set of first 7 natural numbers and B is set of first 6 even natural numbers. Consider function $f : A \to B$ such that $f(x) \neq 2x \ \forall \ x \in A$ and $f(x)$ is an onto function. If total number of such functions is $181 \times N$, then sum of digits of N is

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95. Let $N = \alpha\alpha\alpha\alpha\alpha\alpha$ be a 6 digit number (all digit repeated) and N is divisible by 924 and let α, β be the roots of the equation $x^2 - 11x + \lambda = 0$, if product of all possible values of λ is 168K then the value of K is

96. Let there be 15 letters $\left(L, M, N, O, P, \underbrace{Q, Q, Q, \ldots.Q}_{10\,Q's}\right)$. If the arrangement of these letters in a line so that there is atleast two alike letters (Q) between two distinct letters is $\frac{k!}{2}$, then 'k' is equal to

97. During the time of demonetization, two persons went to Bank to get exchange their old money of gross value 9000 and 5500 respectively and N is total number of ways in which Cashier can exchange their money if he had 5 two thousand rupee notes, 6 five hundred rupee notes and 20 hundred rupee notes, then $\frac{N}{2}$ is (Given that notes can differ only by their value)

98. If the sides of a triangle determined by throwing a dice (with faces marked 1, 2, 3, 4, 5, 6) thrice, then what is the maximum number of the obtuse angled isosceless triangle that can be formed.

99. The number of ways in which two integers chosen from 0 to 9 with replacement so that their product ends with digit 0, 1, 5 or 6 is N. Then the digit in unit place of N is

Matrix Match Type

100. **Column-I** / **Column-II**

	Column-I		Column-II
(A)	The number of 5 digit numbers of the form xyzyx in which $x < y$ is μ then $\frac{\mu}{36}$ is	(P)	0
(B)	From a point A(–3,4) two tangents are drawn to the circle $x^2 + y^2 = 4$. Line $x = 1$, cuts these tangents at B & C. Let length BC is ℓ then $\frac{20\ell}{\sqrt{5376}}$	(Q)	7
(C)	A circle with centre C(2,2) passes through origin and intersect x-axis at A(4,0) and y-axis at B. Then the area of part of the circle that lies in first quadrant is Δ then $\frac{\Delta - 4\pi}{4}$ is	(R)	10
(D)	If $x + y = a$ and $x + 2y = 2a$ are the adjacent sides of a rhombus whose diagonals intersect at (1,2). If the sum of all possible values of a is λ then $(3\lambda - 10)$ is	(S)	4
		(T)	2

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101. Consider the word "CHULBUL BULBUL".

	Column-I		**Column-II**
(A)	The number of words formed using all letters in which U's are separated, are	(P)	$\frac{9!}{3!}.2^4$
(B)	The number of words formed using all the letters in which U, L are always together, are	(Q)	$\frac{9!}{4!4!}.{}^{10}C_3$
(C)	The number of ways of arranging all the letters on a circle such that no two B's are together	(R)	$\frac{9!}{4!3!}.{}^{10}C_6$
(D)	The number of words which can be formed using all the letters without changing the relative order of vowels & consonants, are	(S)	$\frac{9!}{3!}$
		(T)	$\frac{9!}{4!3!}$

ANSWER KEY

1.	A,C	2.	D	3.	C	4.	B	5.	D
6.	C	7.	B	8.	A,B	9.	A,C	10.	B
11.	B	12.	B	13.	A,C,D	14.	B,D	15.	A,D
16.	ABCD	17.	A,C,D	18.	C	19.	A	20.	C
21.	C	22.	B	23.	C	24.	B	25.	A
26.	C	27.	A	28.	A	29.	C	30.	C
31.	C	32.	A	33.	C	34.	C	35.	C
36.	B	37.	A	38.	B	39.	D	40.	B
41.	B	42.	B,C	43.	B	44.	B	45.	D
46.	C	47.	B	48.	A,C	49.	A	50.	A
51.	A,B,D	52.	C	53.	B	54.	D	55.	B
56.	A,B,C,D	57.	A,B,C	58.	A	59.	A	60.	D
61.	A	62.	A	63.	4	64.	A,C	65.	C
66.	B	67.	A	68.	C	69.	A	70.	A
71.	A	72.	C	73.	D	74.	B	75.	C
76.	C	77.	A	78.	B	79.	B	80.	B
81.	A	10.	A	83.	4	84.	8	85.	9
86.	3	87.	6	88.	6	89.	5	90.	8
91.	4	92.	2	93.	9	94.	3	95.	4
96.	7	97.	7	98.	3	99.	2		

100. A→R; B→S; C→T; D→P 101. A→R; B→P; C→Q; D→T

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One or more than one Correct Answer Type

1. If $(1 + 5x + 6x^2)^{10} = a_0 + a_1x + \ldots\ldots\ldots + a_{20}x^{20}$, then numerically greatest coefficient is -

(A) ${}^{10}C_7.{}^{10}C_8.2^3.3^2$ (B) ${}^{10}C_7.{}^{10}C_8.2^7.3^8$ (C) ${}^{10}C_8.{}^{10}C_9.2^8.3^9$ (D) ${}^{10}C_5.{}^{10}C_5.2^5.3^5$

2. Number $(106)^{85} - (85)^{106}$ is divisible by -

(A) 5 (B) 7 (C) 9 (D) 13

3. Consider the sequence $\{a_n : n \in N\}$ defined by $a_n = \dfrac{n!}{100^n}$ then -

(A) a_{99} is smallest (B) a_{100} is greatest

(C) the sequence is increasing (D) the seqeunce is decreasing

4. If I is integral part of the number $N = \left(2+\sqrt{3}\right)^3$, then the value of I is-

(A) 50 (B) 51 (C) 52 (D) 53

5. The least term in the expansion of $(3 - 4x)^{15}$ (where $x = \dfrac{1}{4}$) is-

(A) sixteenth term only (B) fourth & fifth terms

(C) fifth term only (D) fourth term only

6. If n is a very large integer then value of the series $\displaystyle\sum_{r=1}^{n} \frac{r}{\underline{|r+1}}$ is-

(A) less than unity (B) greater than unity (C) less than 2 (D) greater than 2

7. The coefficient of x^7 in the expansion of $(1 - x - x^3 + x^4)^8$ is equal to -

(A) –648 (B) 792 (C) –792 (D) 648

8. If $(x + 1)(x + 2)(x + 3)\ldots\ldots(x + n) = A_0 + A_1x + A_2x^2 + \ldots\ldots + A_nx^n$, then $(A_0 + 2A_1 + 3A_2 + \ldots.+ (n+1)A_n)$ is equal to $(n \in N)$ -

(A) $\underline{|n}\left(1+\dfrac{1}{2}+\dfrac{1}{3}+\ldots.+\dfrac{1}{n+1}\right)$ (B) $\left(1+\dfrac{1}{2}+\dfrac{1}{3}+\ldots.+\dfrac{1}{n+1}\right)$

(C) $\underline{|n+1}\left(1+\dfrac{1}{2}+\dfrac{1}{3}+\ldots.+\dfrac{1}{n+1}\right)$ (D) $\underline{|n-1}\left(1+\dfrac{1}{2}+\dfrac{1}{3}+\ldots.+\dfrac{1}{n+1}\right)$

9. If $(1+x)^n = c_0 + c_1x + c_2x^2 + c_3x^3 + \ldots\ldots + c_nx^n$, then the value of $c_0 - 3c_1 + 5c_2 - \ldots\ldots\ldots + (-1)^n(2n+1)c_n$ is :

(A) $(n-1).2^n$ (B) 0 (C) $(1-2n).2^{n-1}$ (D) $(1-n).2^n$

10. Let $f(x) = a_0 + a_1x + a_2x^2 + \ldots.+a_nx^n + \ldots\ldots$ and $\dfrac{f(x)}{1-x} = b_0 + b_1x + b_2x^2 + \ldots\ldots\ldots + b_nx^n + \ldots\ldots\ldots\ldots$

also $a_0, a_1, a_2 \ldots\ldots a_n$ are in G.P. with $a_0 = 1$, $b_1 = 3$, then-

(A) $b_{n+1} - b_n = a_n$ (B) $b_n - b_{n-1} = a_n$ (C) $\displaystyle\sum_{r=0}^{10} a_r = 2^{11} - 1$ (D) $b_5 = 63$

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11. Let $f(n)=\sum_{k=0}^{n}{}^{n}C_k\cos\left(\frac{2k\pi}{n}\right)$ and $g(n)=\sum_{k=0}^{n-1}{}^{n-1}C_k\cos\left(\frac{2k\pi}{n}\right)$, then

(A) f(6) is a rational number
(B) g(8) is an irrational number
(C) f(n) = 2g(n + 2)
(D) f(n) = 2g(n)

12. If $\sum_{r=1}^{5}{}^{20}C_{2r-1}=k$, then the remainder when k^6 is divisible by 11

(A) 2 (B) 3 (C) 5 (D) 9

13. If P_n denotes the product of all the coefficient of $(1 + x)^n$ and $9!\,P_{n+1} = 10^9\,P_n$, then n is equal to

(A) 10 (B) 9 (C) 19 (D) None of these

14. If n is an odd positive integer and x, y, z are distinct then the number of distinct terms in the expansion of $(x + y + z)^n + (x - y - z)^n$ is

(A) $\frac{(n+1)(n+2)}{2}$ (B) $\frac{(n+2)^2}{4}$ (C) $\left(\frac{n+1}{2}\right)^2$ (D) $\left(\frac{n(n+1)^2}{2}\right)$

15. If coefficient of x^{18} in $(1 + x + x^2..........x^9)^3$ is $(|x| < 1)$ L, then $\frac{L}{11}$ is -

(A) 2 (B) 3 (C) 5 (D) 7

16. Coefficient of x^{70} in the expansion of $(x - 1)(x^2 - 2)(x^3 - 3)......(x^{12} - 12)$ is-

(A) 4 (B) 8 (C) 12 (D) 24

17. Numerically greatest term is the expansion of $(3 + 7x)^{25}$ is

(A) 19^{th} term if x = 1
(B) 18^{th} term if x = 1
(C) 19^{th} term if x = –1
(D) seventh term if $x=\frac{1}{7}$

18. Let $f(n,r)$ be arithmetic mean of minimums of all the 'r' element subset of set {1,2,3,....,n}, then

(A) $f(100,20)=\frac{101}{21}$
(B) $f(40,10)=\frac{40}{11}$
(C) $f(39,19)=2$
(D) $f(80,26)=\frac{80}{27}$

19. If $\sum_{r=0}^{n}\frac{r}{{}^{n}C_r}=\sum_{r=0}^{n}\frac{n^2-4n+6}{2.{}^{n}C_r}$, then -

(A) n = 1 (B) n = 2 (C) n = 3 (D) n = 4

20. If $(1 + x + x^2 + x^3)^n=\sum_{r=0}^{3n}b_r x^r$, then :-

(A) $\sum_{0}^{3n}b_r=2^{2n}$ (B) $\sum_{r=0}^{3n}rb_r=3n2^{2n-1}$ (C) $b_0 = 1$ (D) $\sum_{r=0}^{3n}rb_r=3n4^n$

21. $f(n)=\sum_{r=1}^{n}\left[r^2\left({}^{n}C_r-{}^{n}C_{r-1}\right)+(2r+1)\left({}^{n}C_r\right)\right]$, then

(A) f(30) = 960 (B) f(21) = 483 (C) f(16) = 64 (D) f(11) = 44

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22. If $(x+1)(x+2)\ ......(x+99) = A_0 + A_1\cdot x + A_2\cdot x^2 + ... + A_{99}\cdot x^{99}$ and

$S = \dfrac{\sum_{k=1}^{99}(k.A_k)}{100!} = \sum_{i=1}^{99}\left(\dfrac{1}{r_i}\right), \left(r_i < r_j, \forall i < j\right)$ then value of $\sum_{k=1}^{99}\dfrac{1}{\sqrt{(r_k-1)}}$ is lying between

(A) 15 and 19 (B) 15 and 17 (C) 14 and 17 (D) 17 and 19

23. Let $(2+x)(1+(1+x)^2)(1+(1+x)^4)(1+(1+x)^8)(1+(1+x)^{16}) = a_0 + a_1x + ... + a_nx^n$, then

(A) $a_r = a_{n-r}$, for each $r = 0,1,2,....n$ (B) $a_{14} = a_{16}$

(C) $\sum_{r=0}^{n} a_r = 2^{32} - 1$ (D) $a_0 + a_n = 2$

24. If $\left(\dfrac{x^2+2x+2}{x+1}\right)^{2018} = a_{2018}x^{2018} + a_{2017}x^{2017} + ... + a_1x + a_0 + \dfrac{b_1}{x+1} + \dfrac{b_2}{(x+1)^2} + ... + \dfrac{b_{2018}}{(x+1)^{2018}}\ \forall x \in R - \{-1\}$,

then (a_i, $b_i \in$ Constant)

(A) $\sum_{i=0}^{2018} a_i + \sum_{i=1}^{2018}\dfrac{b_i}{2^i} = \left(\dfrac{5}{2}\right)^{2018}$ (B) $\sum_{i=1}^{2018} b_i = 2^{2018} - \binom{2018}{1009}$

(C) $\sum_{i=1}^{2018} b_i = 2^{2018} + \binom{2018}{1009}$ (D) $a_0 = \binom{2018}{1009}$

25. If $\sum_{i=0}^{2n}\lambda_i(a-2)^i = \sum_{i=0}^{2n}\mu_i(a-3)^i$ and $\lambda_i = 1\ \forall\ i \ge n$ then :-

(A) $\mu_n = {}^{2n+1}C_n$ (B) $\mu_n = {}^{2n+1}C_3$

(C) $\mu_n = {}^{2n+1}C_n + {}^{2n+1}C_{2n}$ (D) $\mu_n = \sum_{i=n+1}^{2n}\lambda_i$

26. In the expansion of $(2x^2y + 3y^2z + 4z^2x)^9$, the coefficient of $x^8y^8z^{11}$ is $\dfrac{\underline{|a+b+c}.\,a^ab^bc^c}{\underline{|a}\ \underline{|b}\ \underline{|c}}$, then $(a+b+c)$ is equal to-

(A) 9 (B) 18 (C) 27 (D) 36

27. In the expansion of $(x-1)^{{}^nC_1}.(x-2)^{{}^nC_2}.(x-3)^{{}^nC_3}.....(x-n)^{{}^nC_n}$, the coefficient of $x^{(2^n-2)}$ is equal to-

(A) $-n2^{n-1}$ (B) $-2n.2^n$ (C) $-n.3^{n-1}$ (D) $-3n2^{n-1}$

28. The value of ${}^{404}C_4 - {}^4C_1.{}^{303}C_4 + {}^4C_2.{}^{202}C_4 - {}^4C_3.{}^{101}C_4$ is-

(A) $(201)^4$ (B) $(101)^4$ (C) 0 (D) $(301)^2$

29. Let K be the coefficient of x^4 in the expansion of $(1 + x + ax^2)^{10}$. What is the value of 'a' that minimizes K ?

(A) 4 (B) – 4 (C) – 7 (D) 7

30. The coefficient of x^5 in the expansion of $(x^2 - x - 2)^5$ is :

(A) – 83 (B) – 82 (C) – 81 (D) 0

31. The remainder when 3^{37} is divided by 80, is

(A) 0 (B) 1 (C) 3 (D) 9

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32. Greatest binomial coefficient in the expansion of $(1+x)^{40+m}$ is $\sum_{i=0}^{m} {}^{m}C_i . {}^{40}C_{m-i}$, then the value of m is (where m is an even natural number)-

(A) 10 (B) 20 (C) 30 (D) 40

33. The value of $\sum_{k=5}^{2014} {}^{2013}C_{k-1} {}^{2015}C_{k-5}$ is -

(A) ${}^{4028}C_{2019}$ (B) ${}^{4029}C_{2019}$ (C) ${}^{4029}C_{2010}$ (D) ${}^{4028}C_{2009}$

34. Let n be a positive integer and $\left(3\sqrt{3}+5\right)^{2n+1} = \alpha+\beta$, where α is integer and $0<\beta<1$, then $(n \in N)$-

(A) α is an even integer (B) $\alpha\beta+\beta^2 = 2^{2n+1}$

(C) $(\alpha+\beta)^2$ is divisible by 2^{2n+1} (D) α is an odd integer

35. The sum of the series $(1^2+1)1!+(2^2+1)2!+(3^2+1)3!+.....+(n^2+1)n!$ is :

(A) $(n+1)((n+2)!)$ (B) $n(n+1)!$ (C) $(n+1)(n+1)!$ (D) none of these

36. The value of ${}^{3n}C_n - {}^{n}C_1 . {}^{3n-3}C_n + {}^{n}C_2 . {}^{3n-6}C_n - {}^{n}C_3 . {}^{3n-9}C_n + ...$ is equal to :-

(A) 1 (B) 2^n (C) 3^n (D) 3^n-2^n

37. $\sum_{0\le i<j\le n}\sum (i+j)(C_i^2+C_j^2+C_iC_j)$ is equal to :-

(A) $\frac{n}{2}\{(2n-1)\,{}^{2n}C_n+4^n\}$ (B) $\frac{n}{2}\{(2n+1).\,{}^{2n}C_n+4^n\}$

(C) $\frac{n}{2}\{4^n-(2n-1).{}^{2n}C_n\}$ (D) $\frac{n}{2}\{4^n-(2n+1).\,{}^{2n}C_n\}$

Matching list type

Answer Q.38,Q.39 and Q.40 by appropriately matching the information given in the three columns of the following table

	Column 1		Column 2		Column 3
(I)	$f(x)=(x-1)(x-2)(x-3)...(x-100)$	(i)	coefficient of x^{99} in $f(x)=5050$	(P)	Number of zeroes in end of $f(1)$ is 24
(II)	$f(x)=(x+1)(x+2)(x+3)...(x+100)$	(ii)	coefficient of x^{49} in $f(x)=-2500$	(Q)	$f(101)=100!$
(III)	$f(x)=(x-1)(x-3)(x-5)...(x-99)$	(iii)	coefficient of x^{99} in $f(x)=-5050$	(R)	Number of zeroes in end of $f(0)$ is 0
(IV)	$f(x)=(x-2)(x-4)(x-6)...(x-100)$	(iv)	coefficient of x^{49} in $f(x)=-2550$	(S)	$f(101)$ is an odd number

38. Which of the following options is the only **CORRECT** combination ?

(A) (I) (ii) (P) (B) (III) (ii) (R) (C) (II) (iii) (Q) (D) (IV) (iii) (P)

39. Which of the following options is the only **CORRECT** combination ?

(A) (II) (ii) (P) (B) (IV) (iv) (S) (C) (III) (i) (R) (D) (IV) (i) (S)

40. Which of the following options is the only **CORRECT** combination ?

(A) (I) (iii) (Q) (B) (III) (iv) (P) (C) (II) (iii) (R) (D) (IV) (i) (R)

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Comprehension Type :

Paragraph for Question 41 to 43

Consider the series $\sum_{p=1}^{n}\left[(1+x^{n-p+1})\left(\sum_{r=0}^{p}{}^{p}C_r.{}^{p}C_{p-r}x^r\right)\right]$

and on the series of above answer the following questions :

41. Sum of the series if $x=-1$ and n is a even natural number

(A) 0 (B) ${}^{2n}C_n$ (C) $(-1)^{n/2}\left({}^{n}C_{n/2}\right)^2$ (D) $(-1)^{n/2}\left(\frac{{}^{n}C_n+{}^{n}C_{n-1}}{2}\right)$

42. If given series can be expanded as $a_0x^0 + a_1x^1 + a_2x^2 + \ldots\ldots + a_{n+1}x^{n+1}$, then value of $\frac{a_{n+1}}{a_0}+\frac{a_n}{a_1}+\frac{a_{n-1}}{a_2}+\ldots\ldots$ upto $\left(\frac{n}{2}+1\right)$ terms, where n is a even natural number, is equal to

(A) $\frac{n}{2}+1$ (B) 0 (C) $2^n\left(\frac{n}{2}+1\right)$ (D) $2n.(3n-1).2^n$

43. If x = 1, then sum of the series is -

(A) ${}^{4n}C_{2n}$ (B) ${}^{2n}C_n$ (C) $2\sum_{r=1}^{n}{}^{2r}C_r$ (D) $2\sum_{r=0}^{n}{}^{2n}C_r$

Paragraph for Question 44 to 46

Consider the series $(1-x)(1+x+x^2)^{100} = a_0x^0 + a_1x^1 + a_2x^2 + \ldots\ldots + a_{201}x^{201}$.

On the basis of above information, answer the following questions :

44. $\sum_{r=0}^{100}a_{2r}$ is equal to-

(A) $2+\sum_{r=0}^{100}a_{2r+1}$ (B) 0 (C) $\sum_{r=0}^{201}(-1)^r a_r$ (D) $\sum_{r=0}^{100}a_{2r+1}$

45. $\sum_{r=0}^{100}a_{2r}^2$ is equal to-

(A) $\sum_{r=0}^{100}a_{2r+1}^2$ (B) ${}^{200}C_{100}$ (C) 0 (D) a_{101}^2

46. $(a_0a_1 - a_1a_2 + a_2a_3 - \ldots\ldots + a_{200}a_{201})$ is equal to-

(A) a_{50} (B) a_{51} (C) a_{101} (D) 0

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Paragraph for Question 47 & 48

Consider the function $f(x,n)=\dfrac{\sum\limits_{r=0}^{\left[\frac{n}{2}\right]} {}^{n}C_{2r}x^{r}}{\sum\limits_{r=0}^{\left[\frac{n-1}{2}\right]} {}^{n}C_{2r+1}x^{r}}$ (where [.] denotes greatest integral function).

47. The value of $f(9, 10)$ is

(A) $3\left(\dfrac{2^{10}+1}{2^{10}-1}\right)$ (B) $\dfrac{1}{2}\left(\dfrac{2^{10}+1}{2^{10}-1}\right)$ (C) $3\left(\dfrac{4^{70}+1}{4^{10}-1}\right)$ (D) $\dfrac{1}{3}\left(\dfrac{4^{10}+1}{4^{10}-1}\right)$

48. If $f(100,101)=\dfrac{f(100,100)+100}{f(100,100)+\lambda}$, then the value of λ is -

(A) –1 (B) 1 (C) 0 (D) 4

Integer Type

49. If Q(x) is the quotient when $P(x) = 1111x^{1111} - 111x^{111} + 11x^{11} - 1011$ is divided by $x - 1$, then sum of the digits in the sum of coefficients of Q(x) is

50. If a,b,c be last three digits of the number $(6! + 1)^{6!}$, then $(a + b + c)$ is

51. The number of disimilar terms in the expansion of $\left(1-x+x^{2}+\dfrac{1}{x^{2}}\right)^{50}$ is equal to

52. If coefficient of x^{14} in $(1 + 2x + 3x^2 + \ldots\ldots.. + 16x^{15})^2$ is k, then ten's digit in k is

53. If number of distinct terms in the expansion of $\left(x+\dfrac{1}{x}+y+\dfrac{1}{y}\right)^{14}$ is $\alpha\beta\gamma$, where $\alpha\beta\gamma$ is a three digits number, then $(\alpha + \beta + \gamma)$ is

54. Let $(2x^2 + 3x + 4)^{10} = \sum\limits_{r=0}^{20} a_r x^r$, then the value of $\dfrac{a_7}{a_{13}}$ is

55. If $T=\sum\limits_{r=1}^{18}\dfrac{(-1)^{r-1}.r.\,{}^{18}C_{r}}{(r+1)}$ then $\left(\dfrac{1}{T}-10\right)$ is equal to

56. The remainder when $31^{\left(31^{31}\right)}$ is divided by 7 is.....

57. If the number $N=1+6\left(\dfrac{1}{2}\right)+21\left(\dfrac{1}{2}\right)^{2}+56\left(\dfrac{1}{2}\right)^{3}+\ldots..$ upto infinite terms.

If $N=a_n3^n+a_{n-1}3^{n-1}+\ldots+a_13+a_0$, where $a_0, a_1, a_2, \ldots.. a_n \in \{0, 1, 2\}$.

Then value of $a_0 + a_1 + a_2 + \ldots.. + a_n$ equals

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58. Coefficient of x^{17} in the expansion of $(1 + x^5 + x^7)^{20}$ is λ, then $\frac{\lambda}{380}$ is equal to

59. Let $(1 + x + x^2)^{10} = a_0 + a_1x + a_2x^2 + ... + a_{20}x^{20}$. If the sum $a_0^2 + a_1a_2 + a_2a_4 + a_3a_6 + ... + a_{10}a_{20}$ = coefficient of x^n in $(1 + x + x^2)^{20}(1 - x + x^2)^{10}$, then $\frac{n}{4}$ is

60. Let $(1 + x)^{15} = a_0x^{P_0} + a_1x^{P_1} + a_2x^{P_2} + ... + a_{15}x^{P_{15}}$ where $a_0 \geq a_1 \geq a_2 \geq a_3 ..... \geq a_{15} > 0$ and $\begin{cases} P_{i+1} > P_i, \text{if } a_i = a_{i+1} = 2n+1, n \in I, i \in \{0,1,2,.....,14\} \\ P_{i+1} < P_i, \text{if } a_i = a_{i+1} = 2n, n \in I, i \in \{0,1,2,.....,14\} \end{cases}$, then the value of $\frac{a_5 + P_5 + a_{11} + P_{11} + 869}{a_9 + P_9 + P_{14} + a_{14} + 32}$ is

61. If $(1 - x + x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, $n \neq 3k$, $k \in I$, then the value of $\sum_{r=0}^{n}\left({}^nC_r.a_{2n-r}\right)$ is

62. Coefficient of x^9 in expansion of $(1 + x + x^2 + x^3 + ... + x^{20})^5$ is N then the value of $\frac{N}{143}$ is

63. If $(x+8)^8 - {}^8C_1(x+7)^8 + {}^8C_2(x+6)^8 .... + (-1)^8 x^8 = x!$, then value of x is

64. The number of terms with rational coefficients in the expansion of $\left(\sqrt[3]{5}x + \sqrt[2]{3}y + z\right)^6$ is

65. If $a_n = \left(\frac{2n+1}{n+1}\right){}^8C_n ; n = \{0,1,2,.....,7,8\}$

$b_n = \left(\frac{3n+1}{n+1}\right){}^7C_n ; n = \{0,1,2,.....,7\}$

If $a_p = \max\limits_{0 \leq n \leq 8}(a_n)$ and $b_q = \min\limits_{0 \leq n \leq 7}(b_n)$, where $n \in W$

Find $(p + q)$

66. If n be the number of terms in the expansion of $(1 + x)^{101}(1 + x^2 - x)^{100}$ and if unit place and ten's place digit in 9^n are α and β respectively, then $(\beta - \alpha)$ is

67. Number of values of r, $(0 \leq r \leq 50)$, such that ${}^{50}C_r$ is divisible by 13, is

68. The value of the series $\frac{\sum_{r=0}^{100}(-1)^r\,{}^{100}C_r\,(r)^{50}}{(50)^{100}}$ is equal to

69. If $n > 3$, then $\binom{n-1}{1}(a-1)^2 - \binom{n-1}{2}(a-2)^2 + \binom{n-1}{3}(a-3)^2 - ... + (-1)^{n-2}\binom{n-1}{n-1}(a-n+1)^2 = k$ then number of positive divisors of k,(given 'a' is a prime number)

70. If $\sum_{j=0}^{2015}\sum_{i=0}^{2015}{}^iC_j = k$, then the value of $(k+2)^{\frac{1}{2016}}$ is (where ${}^nC_r = 0 \ \forall \ n < r$ & ${}^0C_0 = 0$)

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71. Consider the sequence of 51 terms as $\frac{^{50}C_0}{1.2.3}, \frac{^{50}C_1}{2.3.4}$,......... If m^{th} and n^{th} terms are the greatest terms of the sequence, then m + n is equal to :

72. If the ratio of 2^{nd} to 3^{rd} term in the expansion of $(x + y)^n$ is equal to the ratio of 3^{rd} to 4^{th} term in the expansion of $(x + y)^{n+3}$, then n is equal to

73. Let $S = \sum_{n=1}^{\infty}\left(\frac{n^2}{\sum_{r=0}^{n} r\left({}^{n}C_r\right)}\right)$, then the value of S is

74. The remainder when $\sum_{r=1}^{10}\left({}^{10}C_r\right)(r-5)^3$ is divided by 10, is

75. If $n \in N$, then the remainder when $(57)^{n+3} + (79)^{n+1} + (46)^n$ is divided by 11 is -

ANSWER KEY

1.	B	**2.**	B	**3.**	A	**4.**	B	**5.**	D
6.	A,C	**7.**	C	**8.**	C	**9.**	B	**10.**	B,C,D
11.	A,B,D	**12.**	B	**13.**	B	**14.**	C	**15.**	C
16.	A	**17.**	A,C,D	**18.**	A,C	**19.**	B,C	**20.**	A,B,C
21.	A,B	**22.**	A,D	**23.**	B,C	**24.**	A	**25.**	A
26.	A	**27.**	A	**28.**	B	**29.**	B	**30.**	C
31.	C	**32.**	**D**	**33.**	**A,D**	**34.**	**A,B**	**35.**	**B**
36.	**C**	**37.**	**A**	**38.**	B	**39.**	B	**40.**	A
41.	A	**42.**	A	**43.**	C	**44.**	A	**45.**	A
46.	C	**47.**	A	**48.**	B	**49.**	11	**50.**	5
51.	200	**52.**	8	**53.**	9	**54.**	8	**55.**	9
56.	3	**57.**	4	**58.**	9	**59.**	5	**60.**	8
61.	0	**62.**	5	**63.**	8	**64.**	7	**65.**	4
66.	7	**67.**	3	**68.**	0	**69.**	3	**70.**	2
71.	49	**72.**	**5**	**73.**	**4**	**74.**	**5**	**75.**	**0**

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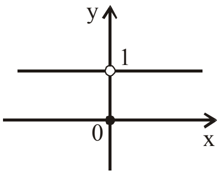
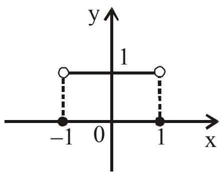
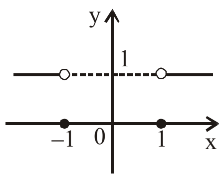
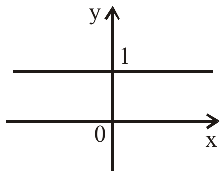
NAME : ....

ASSIGNMENT # 01 (FUNCTIONS) MATHEMATICS

One or more than one correct :

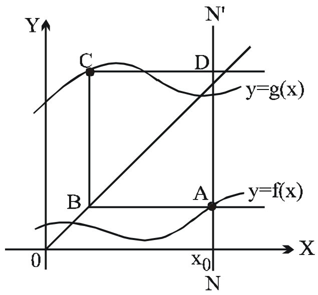
1. Which of the following best suits the graph of $f(x)=\sqrt{\operatorname{Sgn}|x|}$

(A)  (B)  (C)  (D) 

2. If $f(x)=\begin{cases} x+1 & x\in[-1,0] \\ x^2+1 & x\in(0,1] \end{cases}$, then the value of $\dfrac{f^{-1}(0)+f^{-1}(1)+f^{-1}(2)}{f(-1)+f(0)+f(1)}$ is -

(A) 0 (B) 1 (C) 2 (D) $\frac{1}{3}$

3. Given the graphs of the two functions, $y=f(x)$ and $y=g(x)$. In the adjacent figure from point A on the graph of the function $y=f(x)$ corresponding to the given value of the independent variable (sayfl x_0), a straight line is drawn parallel to the X-axis to intersect the bisector of the first and the third quadrants at point B . From the point B a straight line parallel to the Y-axis is drawn to intersect the graph of the function yfl= g(x) at C. Again a straight line is drawn from the point C parallel to the X-axis, to intersect the line NN′ at D. If the straight line NN′ is parallel to Y-axis, then the co-ordinates of the point D are



(A) $(f(x_0), g(f(x_0)))$
(B) $(x_0, g(x_0))$
(C) $(x_0, g(f(x_0)))$
(D) $(f(x_0), f(g(x_0)))$

4. Which of the following function is surjective but not injective

(A) $f:R\to R$ $f(x)=x^4+2x^3-x^2+1$
(B) $f:R\to R$ $f(x)=x^3+x+1$
(C) $f:R\to R^+$ $f(x)=\sqrt{1+x^2}$
(D) $f:R\to R$ $f(x)=x^3+2x^2-x+1$

5. If $f:R^+\to R$ is a function such that $\left(f(x^2+1)\right)^{\frac{1}{x^{1/4}}}=8$ & $x>0$, then the value $\left(f\left(\frac{256}{x^2}+1\right)\right)^{x^{1/4}}$ is equal to -

(A) 4 B) 8 (C) 32 (D) 64

6. $f:R\to R$ such that $f(x)=\ell n\left(x+\sqrt{x^2+1}\right)$. Another function g(x) is defined such that $gof(x)=x\ \forall\ x\in R$. Then g(2) is -

(A) $\frac{e^2+e^{-2}}{2}$ (B) e^2 (C) $\frac{e^2-e^{-2}}{2}$ (D) e^{-2}

7. Let $f:R\to R$ be a real valued function such that $f(10+x)=f(10-x)\ \forall\ x\in R$ and $f(20+x)=-f(20-x)\forall\ x\in R$. Then which of the following statements is true -

(A) $f(x)$ is odd and periodic
(B) $f(x)$ is odd and aperiodic
(C) $f(x)$ is even and periodic
(D) $f(x)$ is even and aperiodic

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8. Let $f(x) = \frac{x}{1+x}$ and let $g(x) = \frac{rx}{1-x}$. Let S be the set of all real numbers r such that $f(g(x)) = g(f(x))$ for infinitely many real number x. The number of elements in set S is-

(A) 1 (B) 2 (C) 3 (D) 5

9. Let $f(x) = \sin^2 x + \cos^4 x + 2$ and $g(x) = \cos(\cos x) + \cos(\sin x)$. Also let fundamental period of f (x) and gfl(x) be T_1 and T_2 respectively then

(A) $T_1 = 2T_2$ (B) $2T_1 = T_2$ (C) $T_1 = T_2$ (D) $T_1 = 4T_2$

10. If the domain of $f(x)$ is [–1,2] then the domain of the function $f([x] - x^2 + 4)$ is (where [.] denotes greatest integer function) -

(A) $\left[-1,\sqrt{7}\right]$ (B) $\left[-\sqrt{3},-1\right]\cup\left[\sqrt{3},\sqrt{7}\right]$

(C) $\left[-\sqrt{3},\sqrt{7}\right]-(-1,2)$ (D) $\left[\sqrt{13},\sqrt{15}\right]$

11. Suppose $f(x)$ is a real function satisfying $f\left(\frac{x+1}{x-1}\right) = 2f(x) + \frac{1}{x-1}\ \forall x \in R \sim \{1\}$, then $f(0)$ is equal to -

(A) 0 (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{5}{6}$

12. If $1 < x^2 < 9$, then number of solutions of the equation $\{|x|\} = \left\{\frac{1}{x^2}\right\}$ (where {.} denote fractional part function) is -

(A) 0 (B) 2 (C) 3 (D) 4

13. Period of function $f(x) = \min\{\sin x, |x|\} + \frac{x}{\pi} - \left[\frac{x}{\pi}\right]$ (where [.] denotes greatest integer function) is -

(A) $\pi/2$ (B) π (C) 2π (D) 4π

14. If $f: R \to R$ & $f(x) = \frac{\sin([x]\pi)}{x^2+2x+3} + 2x - 1 + \sqrt{x(x-1)+\frac{1}{4}}$ (where [x] denotes integral part of x), then f(x) is -

(A) one-one but not onto (B) one-one & onto

(C) onto but not one-one (D) neither one-one nor onto

15. If $f(x) = x^5 + \sin x + \left[\frac{|\cos x|}{\alpha}\right]$ is an odd function, then which is true for parameter α (where [.] denotes greatest integer function) is -

(A) $0 < \alpha < 1$ (B) $\alpha = 1$ (C) $\alpha > 1$ (D) $0 < \alpha < 2$

16. The number of solutions of the equation $[x^2 + 2x + [x^2 + 2x]] = 2\sin x$, (where [.] denotes the greatest integer function) is -

(A) 0 (B) 1 (C) 2 (D) 3

17. The set of all real values of a so that the range of the function $y = \frac{x^2+a}{x+1}$ is R, is-

(A) $[1,\infty)$ (B) $(-\infty,-1)$ (C) $(1,\infty)$ (D) $(-\infty,-1]$

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18. Let $f(x) = \max.\{\sin t : 0 \le t \le x\}$

$g(x) = \min.\{\sin t : 0 \le t \le x\}$

and $h(x) = [f(x) - g(x)]$

where [ ] denotes greatest integer function, then the range of h(x) is-

(A) $\{0,1\}$ (B) $\{1,2\}$ (C) $\{0,1,2\}$ (D) $\{-3,-2,-1,0,1,2,3\}$

19. If function $f(x) = \dfrac{x^2-(b+1)x+b}{x^2-(a+1)x+a}$ $(a \ne b$ & $a, b \in R \sim \{1\})$ can take all values except two values α & β such that $\alpha + \beta = 0$, then $\left|\dfrac{a^3+b^3-8}{ab}\right|$ equal to -

(A) 0 (B) 1 (C) 3 (D) 6

20. Consider the function $f(x) = \dfrac{1}{2\{-x\}} - \{x\}$ where { x } denotes the fractional part of x and x is not an integer.

Statement-1: The minimum value of $f(x)$ is $\sqrt{2}-1$

Statement-2: If the product of two positive numbers is a constant then minimum value of their sum is 2fltimes the square root of their product.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

21. Which of the following inequalities are not equivalent ?

(A) $|f(x)| \ge |g(x)|$ and $\sqrt{f^2(x)} \ge \sqrt{g^2(x)}$

(B) $\dfrac{x-3}{x^2-5x+6} < 2$ and $2x^2 - 11x + 15 > 0$

(C) $x+3-\dfrac{1}{x-1} > -x+2-\dfrac{1}{x-1}$ and $x + 3 > -x + 2$

(D) $\log_2(x^2 - 4) > \log_2(4x - 7)$ and $x^2 - 4x + 3 > 0$

22. Let $f(x) = |x| + |x - 1| + |x + 1|$ and $g(x) = f(x + 1)$, then

(A) If k = 2, the equation g(x) = k, has only one solution

(B) If k > 2, the equation g(x) = k has more than one solutions

(C) If $k \in [0, 2)$, the equation g(x) = k has no solution

(D) If $k \ge 2$, the equation g(x) = k has only two solutions

23. In which of the following cases there does not exist any function -

(A) $f^2(x^2) + f(2^x) + 1 = 0 \ \forall \ x \in R$

(B) $f^2(x^2 - x) - 4f(2x - 2) + x^2 + x = 0 \ \forall \ x \in R$

(C) $f(x) + f\left(\dfrac{1}{x}\right) = 2x \ \forall \ x \in R - \{0\}$

(D) $f(\sin x) + f(\cos x) = x \ \forall \ x \in R$

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24. If f(x) = max {1 + sinx, 3– cosx, 2} & g(x)= min {x^2, 1 –x, 4x – 3}, then -

(A) fog (0) = 2 (B) fog (0) > 2 (C) gof(0) = –1 (D) gof(0) < – 1

25. Which of the following is/are bijective function(s) -

(A) $f:(-\infty,0]\to(0,1]$, $f(x)=\dfrac{e^x+e^{-|x|}}{e^x+e^{|x|}}$ (B) $f:R\to R, f(x)=x^3-3x^2+2x$

(C) $f:(0,\infty)\to R, f(x)=\ell n\left(x+\dfrac{1}{x}\right)$ (D) $f:R\to R, f(x)=\log_2\log_3\log_4(4+e^x)$

26. For the function $f(x)=\sqrt{\log_{\left(\frac{x-[x]}{1+x-[x]}\right)}\left([x]^2-3[x]+3\right)}$, where [.] denotes greatest integer function -

(A) Domain of f(x) is (1,3) ~ {2} (B) Domain of f(x) is [0,4]

(C) Range of f(x) is [0,∞) (D) Range of f(x) is a singleton set

27. If $f(x) = x^3 + x^2 + 2x$, $g(x) = e^x$ (where g(x) is invertible function), then which of the following statement(s) is/are correct ?

(A) $gof(x) \le g(x)$ for $x \in (-\infty, 0]$ (B) $g^{-1}of(x) \ge g^{-1}(x)$ for $x \in (0, \infty)$

(C) $g^{-1}of(x) \le g^{-1}(x)$ for $x \in (0, \infty)$ (D) $g^{-1}of(x) \ge g^{-1}(x)$ for $x \in (-\infty,0]$

28. Let $f : \{1,2,3,4,5\} \to \{1,2,3,4,5\}$ is such that $f(x)$ is a one-one function satisfying following condition $f(x) = x + 1$ **if and only if** x is even (i.e. $f(3) \ne 4$, $f(4) = 5$ etc). Then $f^{-1}(2)$ can be-

(A) 1 (B) 3 (C) 5 (D) 2

29. If A and B are respectively the sum and product of all integral values of x satisfying the equation $|2[x] - x| = 3$, then (where [.] denotes greatest integer function)

(A) A = 0 (B) B = –18 (C) A = 6 (D) B = –9

30. The domain of the function $f(x)=\log_3\left(\log_3|\cos x|-\log_{x^2-4x+6}|\cos x|\right)$ contains which of the following intervals-

(A) $\left(1,\dfrac{\pi}{2}\right)$ (B) $\left(\dfrac{\pi}{2},3\right)$ (C) (1,3) (D) (0,3)

31. Let us consider a function $f(x)=\sqrt{a\{x\}^2+b\{x\}}$ (where {x} denotes fractional part of x), then which of the following statement(s) is/are correct ?

(A) If a > 0 > b & |b| > |a|, then domain of $f(x)$ is I.

(B) If a < 0 < b & |b| > |a|, then domain of $f(x)$ is R.

(C) If a > 0 & b > 0, then domain of $f(x)$ is R

(D) If a < 0 & b < 0, then $f(x)$ is not defined for any real value of x.

32. A function $f(x)$ satisfies $2f(x)-80=100x-3f\left(\dfrac{2x+29}{x-2}\right), x\ne 2$. If $f(x)$ can be expressed as $f(x)=\ell-mx+\dfrac{5n}{x-2}$, then

(A) $2000-\dfrac{m}{2}=5n$ (B) $17\ell = 3m$ (C) $99\ell = 34n$ (D) $5\ell = 17m$

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33. Let $f(x) = [x]^2 + [x+1] - 3$, where $[x]$ denotes greatest integer less than or equal to x, then which of the following statement(s) is/are **CORRECT** ?

(A) f(x) is many one function
(B) f(x) vanishes for atleast three values of x.
(C) f(x) is neither even nor odd function.
(D) f(x) is aperiodic.

34. Which of the following statement(s) is(are) correct ?

(A) If f is a one-one mapping from set A to A, then f is onto.
(B) If f is an onto mapping from set A to A, then f is one-one
(C) Let f and g be two functions defined from $R \to R$ such that gof is injective, then f must be injective.
(D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

35. Consider a function defined by $f(2^x) = x + 1$, then identify the correct statement(s) -

(A) $f(x) = 3$ has no solution
(B) If $f(x_1) = 3$ & $f(x_2) = 4$, then $x_1 + x_2 = 11$
(C) If $f(x_1) = 3$ & $f(x_2) = 4$, then $\frac{x_2}{x_1} = 2$
(D) $f^{-1}(1) + f^{-1}(2) = 3$

36. Let $f:\left(-\frac{\pi}{2}, a\right] \to (-\infty, b]$ be an invertible function with longest domain defined as $f(x) = \tan x + \cot x - 2$ then -

(A) Ordered pair (a,b) is $\left(-\frac{\pi}{3}, -\frac{\left(4+2\sqrt{3}\right)}{\sqrt{3}}\right)$
(B) Ordered pair (a,b) is $\left(-\frac{\pi}{4}, -4\right)$
(C) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2+\sqrt{x^2+4x}}{2}\right)$
(D) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2-\sqrt{x^2+4x}}{2}\right)$

37. If a function $f : A \to B$, $y = f(x)$ is defined by $y = \frac{ax + a^2 - 1}{x - 1}$, then which of the following statement(s) is/are correct ?

(A) $A \subseteq \{x : x \in R, x \neq 1\}$
(B) if set B is $R - \{2\}$, then for $a = 2$, $y = f(x)$ is onto
(C) if set B is R, then $\forall$ $a \in R$, $y = f(x)$ is onto
(D) $y = f(x)$ is injective function when $a \neq \frac{-1 \pm \sqrt{5}}{2}$

Comprehension Type :

Paragraph for Question 38 to 40

The function $f : R \to R$ is defined as $f(x) = \frac{e^x - e^{-x}}{2}$ and function $g : R \to [1, \infty)$ is defined as $g(x) = \frac{e^x + e^{-x}}{2}$.

On the basis of above information, answer the following questions :

38. g(2x) is not equal to -

(A) $2g^2(x) + 1$
(B) $1 + 2f^2(x)$
(C) $2g^2(x) - 1$
(D) $f^2(x) + g^2(x)$

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39. Let $h(x)=\dfrac{f(x)}{g(x)}$, then $\dfrac{2h(x)}{1+h^2(x)}$ is equal to

(A) 1 (B) 2 (C) h(2x) (D) $h(x)f(x)$

40. Number of integers contained in the range of the function $k(x)=\left[\dfrac{f(x)}{g(x)}\right]$ (where [.] denotes G.I.F.) is

(A) 0 (B) 1 (C) 2 (D) more than 2

Paragraph for Question 41 to 43

If f is a continuous real valued function such that $f(x+y)=f(x)\,f(y)$ for all real x & y, $f(5)=32$ and

$$g(x)=\begin{cases} x^2+x+3 & ; \quad x\ge 2 \\ x^2-4x+14 & ; \quad x<2 \end{cases}.$$

On the basis of above information, answer the following questions :

41. The value of $f(8)$ is -
(A) 16 (B) 256 (C) 64 (D) 128

42. Minimum value of g(x) is -
(A) 6 (B) 10 (C) 8 (D) 9

43. Minimum value of g(g(x)) is -
(A) 81 (B) 74 (C) 93 (D) 105

Paragraph for Question 44 & 45

Consider a function $f:(-1,1)\to R$, $f(x)=\ell n\left(\dfrac{1-x}{1+x}\right)$.

On the basis of above information, answer the following questions :

44. Domain of $g(x)=f(2x)$ is given by -

(A) (–1, 1) (B) (0, 1) (C) (–2, 2) (D) $\left(-\dfrac{1}{2},\dfrac{1}{2}\right)$

45. Let the solution of $f^{-1}(x)=e^x$ be $x=\ell n(\tan\theta)$, then θ is equal to -

(A) $\dfrac{\pi}{8}$ (B) $\dfrac{\pi}{6}$ (C) $\dfrac{\pi}{4}$ (D) $\dfrac{\pi}{3}$

Paragraph for Question 46 & 47

Let $f(x)=\begin{cases} x & ; \quad x<0 \\ 1-x & ; \quad x\ge 0 \end{cases}$ & $g(x)=\begin{cases} x^2 & ; \quad x<-1 \\ 2x+3 & ; \quad -1\le x\le 1 \\ x & ; \quad x>1 \end{cases}$

On the basis of above information, answer the following questions :

46. Range of $f(x)$ is -
(A) $(-\infty,1]$ (B) $(-\infty,\infty)$ (C) $(-\infty,0]$ (D) $(-\infty,2]$

47. Range of $g(f(x))$ is -
(A) $(-\infty,\infty)$ (B) $[1,3)\cup(3,\infty)$ (C) $[1,\infty)$ (D) $[0,\infty)$

Paragraph for question nos. 48 to 51

Let $f(x) = x^2 - 2x - 1 \ \forall \ x \in R$. Let $f : (-\infty, a] \to [b, \infty)$, where 'a' is the largest real number for which f (x) is bijective.

48. The value of $(a + b)$ is equal to

(A) – 2 (B) – 1 (C) 0 (D) 1

49. Let $f : R \to R$, $g(x) = f(x) + 3x - 1$, then the least value of function $y = g(|x|)$ is

(A) – 9/4 (B) – 5/4 (C) – 2 (D) – 1

50. Let $f : [a, \infty) \to [b, \infty)$, then $f^{-1}(x)$ is given by

(A) $1 + \sqrt{x+2}$ (B) $1 - \sqrt{x+3}$ (C) $1 - \sqrt{x+2}$ (D) $1 + \sqrt{x+3}$

51. Let $f : R \to R$, then range of values of k for which equation $f(|x|) = k$ has 4 distinct real roots is -

(A) (–2,–1) (B) (–2,0) (C) (–1,0) (D) (0,1)

Paragraph for question nos. 52 to 54

Let $f : [1,\infty) \to [a,\infty)$ defined by $f(x) = 2^{2x^2-4x}$ and $g : \left[\frac{\pi}{2}, \pi\right] \to [c,d]$ defined by $g(x) = \frac{\cos x + 3}{\cos x + 2}$ are two invertible functions.

52. The value of a is equal to -

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) 1

53. Range of the function $h(x) = 4ax^2 + 2cx + d$ is equal to-

(A) $\left[\frac{-1}{4}, \infty\right)$ (B) $\left[\frac{1}{4}, \infty\right)$ (C) $\left[\frac{2}{9}, \infty\right)$ (D) $\left[-\frac{2}{9}, \infty\right)$

54. $f^{-1}(x)$ is equal to -

(A) $\frac{4+\sqrt{16+8\log_2 x}}{4}$ (B) $\frac{4-\sqrt{16+8\log_2 x}}{4}$ (C) $\frac{4+\sqrt{4+\log_2 x}}{2}$ (D) $\frac{4-\sqrt{4+\log_2 x}}{2}$

Paragraph for Question 55 to 56

Consider a continuous function such that each image has atmost three preimage & atleast one image has exactly three preimages. This type of function is to be called as three-one function.

On the basis of above information, answer the following questions :

55. Which of the following function is a three-one function ?

(A) $|\ell n|x||$ (B) $e^{|x|}$

(C) $x^3 + 3x^2 - 7x + 6$ (D) $\cos(\cos^{-1}x)$

56. If $f(x)$ is a three-one function such that $f(a) = f(b)$ (where $a \neq b$), then number of maximum possible values of b is -

(A) 1 (B) 2 (C) 3 (D) 4

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Paragraph for Question 57 to 59

Let $f : A \to B$ be defined by $f(x) = 2x^2 - 6x + 5$ and A be the longest interval which is subset of $[-100, 100]$, so that $f(x)$ is invertible. Let g(x) be inverse of $f(x)$.

On the basis of above information, answer the following questions :

57. A and B are the set -

(A) $A : \left[\frac{3}{2}, 100\right]; B : \left[\frac{1}{2}, \infty\right)$ (B) $A : \left[-100, \frac{3}{2}\right]; B : \left[\frac{1}{2}, 20605\right]$

(C) $A : \left[-100, \frac{3}{2}\right]; B : \left[\frac{1}{2}, 19405\right]$ (D) $A : \left[\frac{3}{2}, 100\right]; B : \left[\frac{1}{2}, 19405\right]$

58. g(x) is the function defined by -

(A) $\frac{3}{2} + \sqrt{\frac{x}{2} - \frac{1}{4}},\ x \ge \frac{1}{2}$ and increasing (B) $\frac{3}{2} - \sqrt{\frac{x}{2} - \frac{1}{4}},\ x \ge \frac{1}{2}$ and decreasing

(C) $\frac{3}{2} + \sqrt{\frac{x}{2} - \frac{1}{4}},\ x \ge \frac{1}{2}$ and decreasing (D) $\frac{3}{2} - \sqrt{\frac{x}{2} - \frac{1}{4}},\ x \ge \frac{1}{2}$ and increasing

59. Number of distinct intersection points of $f(x)$ & g(x) is -

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Question 60 to 62

Consider the expression $f(x) = \sin^2 x + 2(1 - a)\sin x - 3a$, where $x \in [0, \pi)$ and 'a' is a real parameter.

On the basis of above information, answer the following questions :

60. If $a < 1$, minimum value of $f(x)$ is p and maximum value of $f(x)$ is q, then minimum integral value of (p + q) is -

(A) –5 (B) –4 (C) –3 (D) 3

61. If the equation $f(x) = 0$, $(a \ne 0)$ has only one real solution, then the value of 'a' is -

(A) $\frac{5}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{5}$ (D) $\frac{3}{2}$

62. If the equation $f(x) = 0$ has four real solutions, then the set of values of 'a' is -

(A) $\left(\frac{2}{3}, \frac{7}{3}\right)$ (B) $\left(-\frac{5}{3}, 0\right)$ (C) $\left(0, \frac{5}{3}\right)$ (D) ϕ

Paragraph for Question 63 & 64

Let $f(x)$ be a function defined as, $f(x) = \dfrac{[x]\sin\dfrac{\pi}{[x+1]} + \sin\pi[x+1]}{1+[x]}$, where [.] denotes greatest integer function.

On the basis of above information, answer the following questions :

63. The domain of the function is :

(A) $(-\infty, -1) \cup [0, \infty)$ (B) $(-\infty, -1] \cup [0, \infty)$ (C) $(-\infty, -1) \cup (0, \infty)$ (D) $(-\infty, -1) \cup [1, \infty)$

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64. The range of the function if $x \in [0,5)$ is-

(A) $\left\{0, \frac{1}{2}, \frac{1}{3}, \frac{3}{4\sqrt{2}}, \frac{\sqrt{5}-1}{5}\right\}$ (B) $\left\{0, \frac{1}{2}, \frac{1}{\sqrt{3}}, \frac{3}{4\sqrt{2}}, \frac{\sqrt{10-2\sqrt{5}}}{5}\right\}$

(C) $\left\{0, \frac{1}{2}, \frac{1}{\sqrt{3}}, \frac{3}{4\sqrt{2}}, \frac{\sqrt{5}+1}{5}\right\}$ (D) $\left\{0, \frac{1}{2}, \frac{1}{4}, \frac{3}{4\sqrt{2}}, \frac{\sqrt{5}+1}{5}\right\}$

Paragraph for Question 65 to 67

$f(x) = \frac{x^2+1}{ax}$ $(a \neq 0)$; $g(x) = 3\sec x$; $h(x) = \frac{x+3}{x-4}$.

On the basis of above information, answer the following questions :

65. If range of $f(x)$ and $g(x)$ are equal sets then 'a' is equal to -

(A) 3 (B) –2/3 (C) 3/2 (D) –3/2

66. $f(x)$ is one-one if-

(A) $x \in (0,\infty)$ (B) $x \in (-\infty,0)$ (C) $x \in (1,\infty)$ (D) $x \in (-\infty,1) - \{0\}$

67. Which the following is always false ?

(A) h(x) is one-one

(B) $f(x)$ is one-one if $x > 10$

(C) g(x) is many-one if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, 3\right)$

(D) The values of k for which $f(x) = k$ has exactly one solution is $k = 2$ or $k = -2$

Match the column Type :

68. **Column-I** contains functions and **column II** contains their properties. One or more entries of **Column-II** can be matches with one entry of **column-I.**

Column-I	**Column-II**
(A) $f(x) = \sin^2 2x - 2\cos^2 x$	(P) Many one but not even function
(B) $f(x) = \sqrt{\log_{0.5} \sec(\sin^2 x)}$	(Q) Both many one and even function
(C) $f(x) = \text{sgn}\left(\frac{1-x^2}{1+x^2}\right) + \cos 2\pi\left[x^2 - 5x + 6\right]$	(R) Periodic but not odd function
where sgn x and [x] denote signum function of x & greatest integer less than or equal to x respectively	(S) Range contains atleast one integer but not more than three integers.
(D) $f(x) = \frac{\cos x}{\lvert\cos x\rvert} + \cos\left(\frac{x^2 - x}{x-1}\right)$	(T) Bounded function.

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69. **Column-I** | **Column-II**

(A) Let $f : R \to R$ defined as $f(x) = e^{\operatorname{sgn} x} + e^{x^2}$, where sgn x denotes signum function of x, then f(x) is

(B) Let $f : (-1,1) \to R$ defined as
$$f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$$
where [x] denotes greatest integer less than or equal to x, then f(x) is

(C) Let $f : R \to R$ defined as
$$f(x) = \frac{x(x+1)(x^4+1)+2x^4+x^2+2}{x^2+x+1}$$ then f(x) is

(D) Let $f : R \to R$ defined as $f(x) = x + 3x^3 + 5x^5 + \ldots\ldots. + 101x^{101}$ then f(x) is

(P) Odd
(Q) Even
(R) Neither odd nor even
(S) One-one
(T) Many-One

70. **Column-I** | **Column-II**

(A) $f : R^+ \to R\ ;\ f(x) = e^{\ell nx}$ is

(B) $f : R \to R;\ f(x) = x^{1000} + 1000x + 1$ is

(C) $f : R^+ \to \left(\frac{1}{3},\infty\right);\ \frac{2}{x}f(x) + f\left(\frac{1}{x}\right) = 1 + \frac{1}{x}$ is given, then $f(x)$ is

(D) $f : R \to R,\ f(x) = |x|\operatorname{sgn}(x)$ is (where sgn(x) represents signum function)

(P) one-one
(Q) many one
(R) onto
(S) into
(T) invertible

71. **Column-I** | **Column-II**

(A) Minimum value of $|x-1| + |x-5|$ is

(B) Let $f(x) = 2 + 2\sqrt{2+2\sqrt{2+2\sqrt{2+\ldots\ldots\ldots}}}$ and $g(x) = x + 1$, then the value of $g(f(x))$ is greater than

(C) If the period of $\left|\sin\left(2x+\frac{\pi}{4}\right)\right| - \left|\cos\left(2x+\frac{\pi}{4}\right)\right|$ is $\frac{k\pi}{2}$, then the value of k is

(D) Let $f(x) = \left[\frac{1}{3}+\frac{x}{33}\right]$, then the value of $\frac{1}{27}\sum_{x=1}^{99} f(x)$

Where [.] denotes greatest integer function

(P) 1
(Q) 4
(R) 5
(S) 7
(T) 8

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72. Let $f(x) = \sqrt{[\sqrt{n^2+4}]-[n+\sqrt{x}]}$ where [.] denotes greatest integer function and $n \in N$.

	Column-I		Column-II
(A)	Number of integers in domain of $f(x)$ can be	(P)	0
(B)	Number of integral value(s) of x which satisfy $f(f(x)) = 0$ can be	(Q)	1
(C)	Number of integers in range of $f(f(x))$ can be	(R)	2
(D)	Number of points of discontinuity in domain of $f(x)$ can be	(S)	more than 2
		(T)	less than 2

Subjective :

73. Let $f(x)$ be a real valued function satisfying $xyf(xy) = (x + y)\, f(x+y)$, $\forall\, x,y \in R$. If $f(2) = \frac{1}{2}$, then the value of $\frac{1}{f(3)}$ is equal to

74. Given $f(x)$ is a function satisfying the functional equation $x^2f(x) + f(1 - x) = 2x - x^4$, then the value $|f(3)| + f(2)$ is

75. Let $f : R - \left\{\frac{1}{2}\right\} \to R - \left\{\frac{1}{2}\right\}$, $f(x) = \frac{x-2}{2x-1}$ be a function such that x = m is the solution of $f(x) + 2f^{-1}(x) + 2 = f(f(x))$, then m is equal to

76. The number of solutions of the equation $4|x^2 - 5|x| + 6| = k$ (where $k \in (0, 1)$), is

77. Let $f(x)$ be an even function satisfying $f(x) + f(x + 2) = f(x + 1)\ \forall\ x \in R$, where $f(-13) = 4$, then $f(-5) + f(1)$ is equal to

78. If $[2\sin x] + [2 + \cos x] = -1$, then number of integers in the range of the function $f(x) = \sin x + \sqrt{3}\cos x$ in $[0,2\pi]$ is (where [.] denotes greatest integer function)

79. Let $f : R \to R$ & 'k' be largest natural number for which $f(x) = x^3 + 2kx^2 + (k^2 + 12)x - 12$ is a bijective function, then $f(1) + f^{-1}(49)$ is equal to

80. The number of solutions of $3^{|x|}\,(||x|^2 - 4|x| + 3|) = 1$ is

81. Let $f(x) = \frac{-x^3+15}{7}$ and α be the number of real roots of $f(x) = f^{-1}(x)$, β be the number of real roots of $f(x) = x$, then $|\alpha - \beta|$ is

82. If $f(x + 1) = (-1)^{x+1}x - 2f(x)$ for $x \in N$ and $f(1) = f(1986)$, then the sum of digits of the number $(f(1) + f(2) + ... + f(1985))$ is

83. If $f(x)$ is a polynomial function such that $f(x + 1) - f(x) = 3x + 1$ and $f(0) = 3$, then the value of $\left(64f\left(\frac{3}{4}\right)\right)$, is

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84. Let A = {1,2,3,4,5} & B = {a,b,c,d}. A function is defined from A to B. 'p' be the number of one-one functions from A to B and 'q' be the number of onto function from A to B. Then (p + q) equals

85. Let $f(x)$ be a polynomial of degree 2005 with leading coefficient 2006 such that for n = 1, 2, 3, ..., 2005; $f(n) = n$. The number of zeros at the end of $f(2006) - 2006$ is

86. Let $f : R \to R$ be a function satisfying $f(10 - x) = f(x)$ and $f(2 - x) = f(2+x)\ \forall\ x \in R$. If $f(0) = \lambda$ (where $\lambda \in R$). Then the minimum possible number of values of x satisfying $f(x) = \lambda$ in the interval [0,38], is

87. Let $f : A \to \left[-\frac{1}{2}, 0\right]$ is such that $f^3(x) - 3xf(x) + x^3 + 1 = 0$ and $f(x)$ is an invertible function, then number of solution(s) of the equation $(f(f(x)))^2 = \frac{3}{2}f^{-1}(x) + 1$ is

88. If $2 < x^2 < 3$, then the number of positive roots of $\{x^2\} = \left\{\frac{1}{x}\right\}$ is (where {.} denotes fractional part function)

89. The number of solutions of the equation $[x + 1]^2 = 2x + 3\{x\}$ where [.] and {.} denotes the greatest integer function and fractional part function, respectively, is

90. For $x \in [0,1]$, $f(x) = \begin{cases} 2x, & 0 \le x \le \frac{1}{2} \\ 2 - 2x, & \frac{1}{2} < x \le 1 \end{cases}$

Let $f^2(x)$ denotes $f(f(x))$ and $f^{n+1}(x)$ denotes $f^n(f(x))\ \forall\ n \ge 2, n \in I$

if a^b (where a is prime number) is the number of values of x is [0,1] for which $f^{500}(x) = \frac{1}{2}$ then the value of (a + b) is

91. If $f(x) = \dfrac{x^2 - 2\sqrt{2}}{(2x + 2\sqrt{2}\cos\alpha)}$ (where $0 \le \alpha \le \pi$) has the range R, then the number of integral values of α is

92. If for all real values of u & v, $2f(u)\cos v = f(u + v) + f(u - v)$, prove that, for all real values of x.
(i) $f(x) + f(-x) = 2a\cos x$ (ii) $f(\pi - x) + f(-x) = 0$
(iii) $f(\pi - x) + f(x) = -2b\sin x$. Deduce that $f(x) = a\cos x - b\sin x$, a, b are arbitrary constants.

93. Let [x] = the greatest integer less than or equal to x. If all the values of x such that product $\left[x - \frac{1}{2}\right]\left[x + \frac{1}{2}\right]$ is prime, belongs to the set $[x_1, x_2) \cup [x_3, x_4)$, find the value of $x_1^2 + x_2^2 + x_3^2 + x_4^2$.

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94. Prove that the function defined as, $f(x) = \begin{cases} e^{-\sqrt{|\ell n\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ell n\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise, then} \end{cases}$

f(x) is odd as well as even. (where $\{x\}$ denotes the fractional part function).

95. Find the number of real values of x satisfying the equation $x^9 + \frac{9}{8}x^6 + \frac{27}{64}x^3 - x + \frac{219}{512} = 0$

ANSWER KEY

1.	A	2.	A	3.	C	4.	D	5.	D
6.	C	7.	A	8.	B	9.	C	10.	B
11.	D	12.	D	13.	C	14.	B	15.	C
16.	C	17.	B	18.	C	19.	D	20.	A
21.	B,C,D	22.	A,B,C	23.	A,B,C,D	24.	B,C	25.	A,D
26.	A,D	27.	A,B	28.	B,C	29.	A,D	30.	A,B
31.	A,B,C	32.	A,C,D	33.	A,B,C,D	34.	C,D	35.	C,D
36.	B,D	37.	A,B,D	38.	A	39.	C	40.	C
41.	B	42.	D	43.	C	44.	D	45.	A
46.	A	47.	C	48.	B	49.	C	50.	A
51.	A	52.	B	53.	A	54.	A	55.	C
56.	B	57.	B	58.	B	59.	B	60.	B
61.	C	62.	D	63.	A	64.	B	65.	B
66.	C	67.	C						

68. (A) $\rightarrow$ (Q,R,S,T); (B) $\rightarrow$ (Q,S,T); (C) $\rightarrow$ (Q,S,T); (D) $\rightarrow$ (P,S,T)

69. (A) $\rightarrow$ (R,T); (B) $\rightarrow$ (Q,T); (C) $\rightarrow$ (R,T); (D) $\rightarrow$ (P,S)

70. (A)$\rightarrow$(P,S); (B)$\rightarrow$(Q,S); (C)$\rightarrow$(P,R,T); (D)$\rightarrow$(P,R,T)

71. (A)$\rightarrow$(Q); (B)$\rightarrow$(P,Q,R,S,T); (C) $\rightarrow$ (P); (D)$\rightarrow$(R)

72. (A)$\rightarrow$(Q,S,T);(B)$\rightarrow$(Q,T); (C)$\rightarrow$(Q,R,T); (D)$\rightarrow$(P,Q,T)

73.	3	74.	5	75.	2	76.	8	77.	8
78.	0	79.	50	80.	8	81.	4	82.	7
83.	222	84.	240	85.	500	86.	13	87.	1
88.	1	89.	3	90.	502	91.	4	93.	11
95.	3								

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One or more than one correct :

1. The number of solution(s) of the equation $\cos^{-1}\left(\frac{1+x^2}{2x}\right)=\frac{\pi}{2}e^{(x^2-1)}$ is -

(A) 1 (B) 0 (C) 3 (D) 2

2. The value of the angle $\tan^{-1}(\tan 65° - 2\tan 40°)$ in degrees is equal to -

(A) –20° (B) 20° (C) 25° (D) 40°

3. The range of the function, $f(x) = (1 + \sec^{-1}x)(1 + \cos^{-1}x)$ is -

(A) $(-\infty,\infty)$ (B) $(-\infty,0] \cup [4,\infty)$ (C) $\{1,(1+\pi)^2\}$ (D) $[0,(1+\pi)^2]$

4. The value of $\tan^{-1}\left(\frac{1}{2}\tan 2A\right)+\tan^{-1}(\cot A)+\tan^{-1}(\cot^3 A)$ for $0 < A < (\pi/4)$ is

(A) $4\tan^{-1}(1)$ (B) $2\tan^{-1}(2)$ (C) 0 (D) none

5. Number of solution(s) of the equation $\cos^{-1}\sqrt{x}-\sin^{-1}\sqrt{x-1}+\cos^{-1}\sqrt{1-x}-\sin^{-1}\frac{1}{\sqrt{x}}=\frac{\pi}{2}$ is -

(A) 0 (B) 1 (C) 2 (D) 4

6. The value of $\sin\left(\frac{1}{2}\cos^{-1}\left(\frac{1}{2}\right)\right)+\sin\left(2\cos^{-1}\left(\frac{5}{13}\right)\right)$ is equal to -

(A) $-\frac{409}{338}$ (B) $-\frac{338}{409}$ (C) $\frac{338}{409}$ (D) $\frac{409}{338}$

7. Complete solution set of $\sin^{-1}x > (\sin^{-1}x)^2$ is given by -

(A) (–sin1, 0) (B) (–1, 0) (C) (0, 1) (D) (0, sin1)

8. The value of $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)+\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)-\cos^{-1}\left(\frac{1}{\sqrt{10}}\right)+\tan^{-1}\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$ is -

(A) $-\frac{\pi}{2}$ (B) $\frac{\pi}{2}$ (C) 0 (D) $-\frac{\pi}{4}$

9. If $f(x) = |\sin^{-1}x + \tan^{-1}x| + |\cos^{-1}x + \cot^{-1}x| + |\operatorname{cosec}^{-1}x + \sec^{-1}x|$, then which of following is correct about $f(x)$ -

(A) Domain of $f(x)$ is $[-1,1]$

(B) Range of $f(x)$ is $\left\{\frac{3\pi}{2}\right\}$

(C) Range of $f(x)$ is $\left[\frac{3\pi}{2},3\pi\right]$

(D) Range of $f(x)$ is $\left\{\frac{3\pi}{2},3\pi\right\}$

10. The value of x satisfying the equation $\sin(\tan^{-1}x)=\cos\left(\cot^{-1}(x+1)\right)$ is

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\sqrt{2}-1$ (D) no finite value

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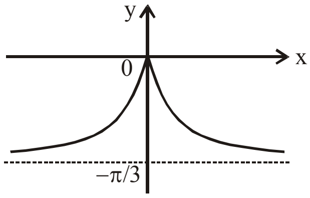
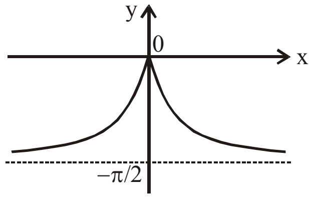
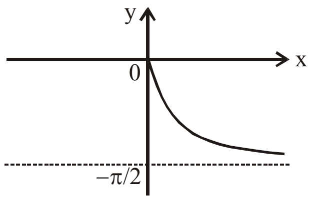
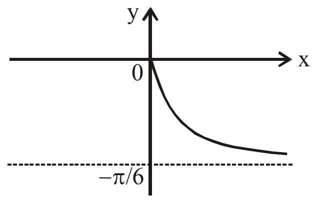
11. The graph of the functions $h(x) = -|\tan^{-1}(3x)|$, is

(A) 

(B) 

(C) 

(D) 

12. The range of values of p for which the equation $\sin\cos^{-1}(\cos(\tan^{-1}x)) = p$ has a solution is :

(A) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$ (B) $[0,1)$ (C) $\left[\frac{1}{\sqrt{2}}, 1\right)$ (D) $(-1,1)$

13. If $\sin\theta$ & $\cos\theta$ are the roots of quadratic $x^2 + ax + b = 0$, then the value of $\sin^{-1}(a^2 - 2b)$ is equal to -

(A) $-\frac{\pi}{2}$ (B) $-\pi$ (C) $\frac{\pi}{2}$ (D) π

14. If $f(x) = ax + b$ $(a < 0)$ is an onto function defined as $f : [0, 1] \to [0, 2]$, then $\cot\left(\cos^{-1}\left(\frac{e^x + e^{-x}}{2}\right) + \sin^{-1}\left(\frac{-e^x - e^{-x}}{2}\right)\right)$ is equal to

(A) $f\left(\frac{3}{4}\right)$ (B) $f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)$ (C) $f(1)$ (D) $f\left(\frac{1}{2}\right)$

15. Number of solution(s) of $\sin[\cos^{-1}x] = \sin^{-1}x$ is (where [.] denotes greatest integer function)-

(A) 0 (B) 1 (C) 2 (D) infinite

16. If $x = \cot^{-1}\sqrt{\tan\frac{\pi}{8}} - \tan^{-1}\sqrt{\tan\frac{\pi}{8}}$, then sinx is equal to -

(A) 0 (B) $1-\sqrt{2}$ (C) $\sqrt{2}-1$ (D) $\sin\frac{\pi}{8}$

17. The number of solution(s) of the equation $\sin^{-1}(2x) = \tan^{-1}\left(\frac{3-2x}{\sqrt{1-4x^2}}\right)$, is-

(A) 0 (B) 1 (C) 2 (D) 3

18. Number of solution of the equation $\sin^{-1}(4\sin^2\theta + \sin\theta) + \cos^{-1}(6\sin\theta - 1) = \frac{\pi}{2}$ lying in the interval $[0,5\pi]$, is-

(A) 9 (B) 7 (C) 6 (D) 5

19. The number of solution(s) of $\log_7\log_5\left(\sqrt{\tan^{-1}(\tan x)+5} + \sqrt{\tan^{-1}(\tan x)}\right) = 0$ in $[0,3\pi]$ is-

(A) 0 (B) 1 (C) 2 (C) 3

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20. If $\tan^{-1}(x+1)+\cot^{-1}(x-1)=\sin^{-1}\frac{4}{5}+\cos^{-1}\frac{3}{5}$, then x has the value-

(A) $4\sqrt{\frac{3}{7}}$ (B) $4\sqrt{\frac{7}{3}}$ (C) $14\sqrt{3}$ (D) $6\sqrt{7}$

21. $x=\operatorname{cosec}^{-1}\left(\alpha+\frac{1}{\alpha}\right)+\sec^{-1}\left(\beta+\frac{1}{\beta}\right)$ where $\alpha\beta<0$, then number of possible integral values of x is -

(A) 1 (B) 2 (C) 4 (D) 5

22. If $2\le a<3$, then the value of $\cos^{-1}\cos[a]+\operatorname{cosec}^{-1}\operatorname{cosec}[a]+\cot^{-1}\cot[a]$, (where [.] denotes greatest integer less than equal to x) is equal to

(A) $2-\pi$ (B) $2+\pi$ (C) π (D) 6

23. The number of solutions of the equation $|\tan^{-1}|x||=\sqrt{(x^2+1)^2-4x^2}$ is -

(A) 1 (B) 2 (C) 3 (D) 4

24. For which of the following values of 'm' the line y = mx cuts the graph of $y=\sin^{-1}(\sin x)$ in exactly 5 points-

(A) $\frac{1}{5}$ (B) $-\frac{1}{4}$ (C) $-\frac{1}{8}$ (D) $-\frac{1}{9}$

25. The number of solutions of the equation $\tan^{-1}\left(\frac{x}{1-x^2}\right)+\tan^{-1}\left(\frac{1}{x^3}\right)=\frac{3\pi}{4}$ is -

(A) 0 (B) 1 (C) 2 (D) infinite

26. If $\sum_{r=1}^{\infty}\cot^{-1}\left(\frac{(r+1)^2}{2}\right)=\tan^{-1}x$, then x is equal to -

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) 3 (D) $\frac{3}{2}$

27. The sum of the infinite terms of the series $\cot^{-1}\left(1^2+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+.....$ is equal to:

(A) $\tan^{-1}(1)$ (B) $\tan^{-1}(2)$ (C) $\tan^{-1}(3)$ (D) $\tan^{-1}(4)$

28. If $f(x)=x^{11}+x^9-x^7+x^3+1$ and $f(\sin^{-1}(\sin 8))=\alpha$, α is constant, then $f(\tan^{-1}(\tan 8))$ is equal to-

(A) α (B) $\alpha-2$ (C) $\alpha+2$ (D) $2-\alpha$

29. The solution set of the equation $\sin^{-1}\sqrt{1-x^2}+\cos^{-1}x=\cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)-\sin^{-1}x$ is -

(A) $[-1,1]-\{0\}$ (B) $(0,1]\cup\{-1\}$ (C) $[-1,0)\cup\{1\}$ (D) $[-1,1]$

30. The value of x satisfying the equation $\sin(\tan^{-1}x)=\cos(\cot^{-1}(x+1))$ is -

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $\sqrt{2}-1$ (D) no finite value

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31. The sum $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{4n}{n^4-2n^2+2}\right)$ is equal to -

(A) $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{2}{3}$ (B) $4\tan^{-1}1$ (C) $\frac{\pi}{2}$ (D) $\sec^{-1}\left(-\sqrt{2}\right)$

32. Range of the function $f(x)=\tan^{-1}\sqrt{[x]+[-x]}+\sqrt{2-|x|}+\frac{1}{x^2}$ is where [*] is the greatest integer function.

(A) $\left[\frac{1}{4},\infty\right)$ (B) $\left\{\frac{1}{4}\right\}\cup[2,\infty)$ (C) $\left\{\frac{1}{4},2\right\}$ (D) $\left[\frac{1}{4},2\right]$

33. The value of $\sum_{n=1}^{\infty}\left(\tan^{-1}\left(\frac{n}{n+2}\right)-\tan^{-1}\left(\frac{n-1}{n+1}\right)\right)$ is equal to-

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{3\pi}{4}$

34. The complete set of values of x for which $|\tan^{-1}x|+|\cot^{-1}x|=\frac{\pi}{2}$ is valid -

(A) $(-3,3)$ (B) $(-1,1)$ (C) $(-\infty,0]$ (D) $[0,\infty)$

35. Sum of values of x for which $\cot^{-1}\left(\frac{3x^2+1}{x}\right)=\cot^{-1}\left(\frac{1-3x^2}{x}\right)-\tan^{-1}6x$ is -

(A) zero (B) $-2\sqrt{3}$ (C) $2\sqrt{3}$ (D) $\frac{2}{\sqrt{3}}$

36. If $\sum_{r=1}^{\infty}\tan^{-1}\left(\frac{4r+4}{4r^3+4r^2+3r+3}\right)=-\frac{\pi}{4}+\cot^{-1}m$, then m is -

(A) $-\frac{1}{3}$ (B) $\frac{1}{3}$ (C) 3 (D) -3

37. Value of $\lim_{n\to\infty}\sum_{x=0}^{n}\tan^{-1}\left(\frac{\left(x^2+3x+1\right)\underline{|x}}{\underline{|x}\,\underline{|x+2}+1}\right)$ is-

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) Does not exist

38. $f(x)=\tan^{-1}\left(\frac{3x}{1+4x^2}\right)+\tan^{-1}\left(\frac{2+5x}{5-2x}\right)$ & $g(x)=\tan^{-1}(4x)+\tan^{-1}\left(\frac{2}{5}\right)$, then -

(A) $f(x)=g(x)$ for $x\in R-\left\{\frac{5}{2}\right\}$ (B) $f(x)=g(x)$ for $x>\frac{5}{2}$

(C) $f(x)=g(x)$ for $x<\frac{5}{2}$ (D) $f(x)$ can never be equal to $g(x)$

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39. If $\sin^{-1}\left(\frac{\sqrt{x}}{2}\right)+\sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right)+\tan^{-1}(y)=\frac{2\pi}{3}$, then

(A) Maximum value of $x^2 + y^2$ is $\frac{49}{3}$

(B) Maximum value of $x^2 + y^2$ is 4

(C) Minimum value of $x^2 + y^2$ is $\frac{1}{3}$

(D) Minimum value of $x^2 + y^2$ is 3

40. If $f(x) = 2\tan^{-1}x + \cos^{-1}\frac{1-x^2}{1+x^2}$, then the value of $\frac{2+f(-10)+f(-5)}{1+f(-7.5)+f(-3)}$ is-

(A) an even number

(B) an even prime number

(C) an odd prime number

(D) an odd number

41. Consider the equation $\sin^{-1}(\sin 2x) = |\cos x|$, $x \in [-\pi, \pi)$, then identify the correct statement(s)

(A) Number of roots of this equation is 2

(B) Number of roots of this equation is 4

(C) Exactly one root lies in $\left(\frac{\pi}{8}, \frac{\pi}{6}\right)$

(D) If $x \in [0, n\pi]$, $n \in N$, then number of roots will never be odd

42. If the mapping $f(x) = ax + b\ (a > 0)$ maps $[-1,1]$ onto $[0,2]$ then $\tan\left(\tan^{-1}\frac{1}{7}+\cot^{-1}8+\cot^{-1}18\right)$ cannot be equal to-

(A) $f\left(\frac{1}{3}\right)$

(B) $f\left(\frac{2}{3}\right)$

(C) $f\left(-\frac{2}{3}\right)$

(D) $f\left(-\frac{1}{3}\right)$

43. Consider the equation $\tan^{-1}a\sqrt{x}-\tan^{-1}\frac{\sqrt{x}}{a}=\tan^{-1}\frac{b}{\sqrt{x}}$. Identify the correct statement -

(A) if $a = 2$, $b = 1$ then equation has unique solution.

(B) if $a = 1$, $b = 1$ then equation has infinite solutions.

(C) if $a = 1$, $b \neq 0$ then equation does not have a solution

(D) if $b = 0$ then equation has infinite solutions for $a \in \{1, -1\}$

44. If $\alpha = 2\tan^{-1}(\sqrt{3-2\sqrt{2}})+\sin^{-1}\left(\frac{1}{\sqrt{6}-\sqrt{2}}\right)$, $\beta=\cot^{-1}\left(\sqrt{3}-2\right)+\frac{1}{8}\sec^{-1}(-2)$

& $\gamma=\tan^{-1}\frac{1}{\sqrt{2}}+\cos^{-1}\frac{1}{\sqrt{3}}$, then

(A) $\alpha = \beta$

(B) $\alpha + \beta = 3\gamma$

(C) $4(\beta - \gamma) = \alpha$

(D) $\beta = \gamma$

45. If $\sec^{-1}\left(\sqrt{1+x^2}\right)+\operatorname{cosec}^{-1}\left(\frac{\sqrt{1+y^2}}{y}\right)+\cot^{-1}\left(\frac{1}{z}\right)=\pi\ (x,y,z>0)$ then $x + y + z$ is always -

(A) equal to xyz

(B) greater than or equal to $3(xyz)^{1/3}$

(C) greater than or equal to $4(xyz)^{1/3}$

(D) greater than or equal to $5(xyz)^{1/3}$

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46. Which of the following functions are identical functions -

(A) $f(x) = \tan^{-1}x + \tan^{-1}(-x)$ & $g(x) = 2\tan^{-1}x$
(B) $f(x) = \cos^{-1}x + \cos^{-1}(-x)$ & $g(x) = 2\cos^{-1}x$
(C) $f(x) = \cos x + \cos|x|$ & $g(x) = 2\cos x$
(D) $f(x) = \cos^{-1}x + |\cos^{-1}x|$ & $g(x) = 2\cos^{-1}x$

47. Which of the following statement is/are true -

(A) $\cos^{-1}(\cos(1+|\sin x|)) = 1 + |\sin x|$ for all real x
(B) $\sin^{-1}\left(\sin\left((\cos^2 x - \sin^2 x)\right)\right) = \cos^2 x - \sin^2 x \quad \forall x \in R$
(C) $\tan^{-1}\left(\tan\left(x^2+2x+2\right)\right) = x^2+2x+2 \quad \forall x \in R$
(D) $\tan^{-1}(\tan(3\sin x)) = 3\sin x \quad \forall\ x \in R$

48. Let x_1, x_2, x_3, x_4 be four non zero numbers satisfying the equation

$\tan^{-1}\frac{a}{x} + \tan^{-1}\frac{b}{x} + \tan^{-1}\frac{c}{x} + \tan^{-1}\frac{d}{x} = \frac{\pi}{2}$ then which of the following relation(s) hold good ?

(A) $\sum_{i=1}^{4} x_i = a+b+c+d$
(B) $\sum_{i=1}^{4} \frac{1}{x_i} = 0$
(C) $\prod_{i=1}^{4} x_i = abcd$
(D) $(x_1+x_2+x_3)(x_2+x_3+x_4)(x_3+x_4+x_1)(x_4+x_1+x_2) = abcd$

49. If the mapping $f(x) = a^2x^2 + x + c$ maps [0, 2] onto [–1, 2], and if $\cos^{-1}\left(\frac{n}{4\pi}\right) > \frac{\pi}{3}$ such that minimum & maximum values of integer n are m & M respectively, then $\left|\frac{m+M}{3}\right|$ is equal to

(A) $f(0)$
(B) $8f(1)$
(C) 2
(D) 4

Comprehension Type :

Paragraph for Question 50 to 52

Let ${}^{50}C_{15} + 5\,{}^{50}C_{14} + 10\,{}^{50}C_{13} + 10\,{}^{50}C_{12} + 5\,{}^{50}C_{11} + {}^{50}C_{10} = {}^{55}C_p$
and ${}^{5}C_5 + {}^{6}C_5 + {}^{7}C_5 + \ldots\ldots\ldots\ldots + {}^{54}C_5 = {}^{55}C_q$, p, q < 30.

On the basis of above information, answer the following questions :

50. $\sin^{-1}\sin\left((p+q)\frac{\pi}{4}\right) + \cos^{-1}\cos(p) + \tan^{-1}\tan(q)$ equals

(A) $\frac{13\pi}{4} - 9$
(B) $21 - \frac{25\pi}{4}$
(C) $\frac{11\pi}{4} - 9$
(D) $\frac{27\pi}{4} - 21$

51. If $\sin^{-1}(x - x^2 + x^3 - x^4 + \ldots\ldots\ \infty) + \cos^{-1}(x^2 + x^3 + x^4 + \ldots\ldots\ \infty) = (p-2q-1)\frac{\pi}{4}$, $(|x| < 1)$ then number of possible real value(s) of x is

(A) 1
(B) 2
(C) 3
(D) more than 3

52. If $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{q\pi}{2(p-3)}$, then sum of all possible real value(s) of x is

(A) –1
(B) $-\frac{5}{6}$
(C) 0
(D) $\frac{1}{6}$

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Paragraph for Question 53 to 55

Let $\alpha = 2\tan^{-1}\frac{1}{2} + \sin^{-1}\frac{3}{5}$ and $\beta = \sin^{-1}\frac{12}{13} + \cos^{-1}\frac{4}{5} + \cot^{-1}\frac{16}{63}$ such that $2\sin\alpha$ and $\cos\beta$ are roots of the equation $x^2 - ax + b = 0$.

On the basis of above information, answer the following questions :

53. The value of $\tan^{-1}\left(\left(\sec\left(\cos^{-1}\left(\sin\frac{\alpha}{2}\right)\right)\right) - 1\right)$ is equal to-

(A) $\frac{\alpha}{4}$ (B) $\frac{\alpha}{2}$ (C) $\frac{\beta}{2}$ (D) $\frac{\beta}{4}$

54. The range of $f(x) = \cot^{-1}(x^2 + bx)$ is -

(A) $\left(0, \frac{\pi}{4}\right]$ (B) $(0,\pi)$ (C) $\left(0, \frac{3\pi}{4}\right]$ (D) $\left[\frac{3\pi}{4}, \pi\right)$

55. The number of solution(s) of the equation $|b| \sin^{-1}x = (a - b)x$ is -

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Question 56 to 58

Let roots of equation $x^3 - ax^2 + \left(1 + \left(\frac{\pi}{4} + \frac{4}{\pi}\right)\tan^{-1}2\right)x - \tan^{-1}2 = 0$ are $\frac{4}{\pi}$, $\tan^{-1}\alpha$ & $\tan^{-1}\beta$ where α & β are positive integers.

On the basis of above information, answer the following questions :

56. $\tan\left(2a - \frac{8}{\pi} - \frac{\pi}{2}\right)$ is equal to -

(A) $-\frac{3}{4}$ (B) -1 (C) $-\frac{2}{3}$ (D) $-\frac{4}{3}$

57. $\tan^{-1}(\alpha) + \tan^{-1}(\beta) + \tan^{-1}(\alpha + \beta)$ is equal to-

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) $\frac{3\pi}{4}$

58. If $\alpha > \beta$, then number of solutions of equation $\sin^{-1}(x + \beta) + \sin^{-1}(\alpha x) = \frac{\pi}{2}$, is -

(A) 0 (B) 1 (C) 2 (D) more than two

Paragraph for Question 59 to 61

Consider $f(x) = \sin^{-1}x + \tan^{-1}x + \sqrt{x+1}$ & let $g(x)$ be its inverse.

On the basis of above information, answer the following questions :

59. Number of integral values of k such that $f(x) = k$ has atleast one real solution is equal to -

(A) 3 (B) 4 (C) 5 (D) 6

60. If $g(0) = \ell$, then -

(A) $-1 < \ell < -\frac{1}{2}$ (B) $-\frac{1}{2} < \ell < 0$ (C) $0 < \ell < \frac{1}{2}$ (D) $-1 < \ell < -\frac{3}{4}$

61. Number of solutions of the equation $f(|x|) = 2$ is equal to -

(A) 0 (B) 1 (C) 2 (D) 4

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Match the column Type :

62. **Column-I** **Column-II**

Column-I		Column-II	
(A)	If $\tan^{-1}\left(\frac{3+x}{3}\right)+\tan^{-1}\left(\frac{3-x}{3}\right)=\frac{\pi}{6}$, then $\frac{x^2}{\sqrt{3}}$ is	(P)	0
(B)	The number of values of x, satisfying $\sin^{-1}(-x)\geq\frac{\pi}{2}$ is/are	(Q)	1
(C)	If a = cos2θ + cos4θ – cos6θ & b = 4sinθsin2θ cos3θ, then 2(a – b) is equal to	(R)	2
(D)	Let $f:R\to\left[\frac{\pi}{6},\frac{\pi}{2}\right)$, $f(x)=\sin^{-1}\left(\frac{x^2-2a}{x^2+2}\right)$ is an onto function, then the number of integral values of 'a' is	(S)	18
		(T)	36

Subjective :

63. If $m, n>0$ & $S=\tan^{-1}\frac{m}{n}-\tan^{-1}\left(\frac{m-n\sqrt{3}}{m\sqrt{3}+n}\right)$, then [S] is (where [.] denotes greatest integer function)

64. Number of integral solutions of the equation $\operatorname{sgn}\left(\sin^{-1}\left[\frac{\pi x}{6}\right]\right)=1$,

where [x] denotes the greatest integer less than or equal to x and sgn x denotes signum function of x.

65. Number of solution of the equation $2\tan^{-1}|x|.\ \ell n|x| = 1$ is

66. Sum of all values of x satisfying $\sin^{-1}|\sin x|=\sqrt{\sin^{-1}|\sin x|}$ in $x\in\left[-\pi,\frac{5\pi}{2}\right]$ is $(a\pi + b)$, where $a,b \in I$, then $(a + b)$ is equal to

67. If $\tan^{-1}x + \tan^{-1}y = \cot^{-1}z$ is such that $x + y + z = \sqrt{3}$, then value of $\frac{1}{xyz}$ can be expressed as $k\sqrt{k}$, where k is

68. If $\tan^{-1}\frac{2}{1^2}+\tan^{-1}\frac{2}{2^2}+\tan^{-1}\frac{2}{3^2}+\tan^{-1}\frac{2}{4^2}+.........\tan^{-1}\frac{2}{17^2}=\pi-\tan^{-1}\frac{p}{q}$, where p & q are co-prime, then (p – q) is

69. If $\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}+\tan^{-1}\frac{1}{21}+\tan^{-1}\frac{1}{31}=\tan^{-1}\left(\frac{a}{b}\right)$, where a & b are relatively prime numbers, then (b – a) is

70. If $\cos^{-1}\frac{p}{a}+\cos^{-1}\frac{q}{b}=\alpha$ and $\frac{p^2}{a^2}-\frac{2pq}{ab}\cos\alpha+\frac{q^2}{b^2}=f(\alpha)$, then number of integers in the range of $f(\alpha)$ is

71. Let $x_1=\sin\left(\frac{\sin^{-1}\frac{1}{\sqrt{3}}}{3}\right)$ & $x_2=\cos\left(\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)-\sin^{-1}\left(\frac{2}{\sqrt{5}}\right)\right)$, then $\frac{x_2^2}{\left(3x_1-4x_1^3\right)^2}$ is

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72. The equation $\sin \cot^{-1}\cos\tan^{-1}x = a$ has real solution iff complete range of a is [p,q). Value of $\left[q+\frac{1}{p}\right]$ is (where [.] denotes greatest integer function)

73. Let $f(x)=\left|\sqrt{3}\sin x+\cos x\right|+\pi$ and $\cos^{-1}(\cos(f(x))) = \lambda\pi - f(x)$, then 100λ is

74. Consider the curve $y = \sin^{-1}(x-1)$ & line $y = mx$ (where $m > 0$). If both the curves intersect each other at unique point, where α is maximum value of m, then the value of $\frac{8\alpha}{\pi}$ is

75. If $\tan^{-1}x+\tan^{-1}\frac{\sqrt{1-y^2}}{y}=\frac{\pi}{3}$ & $\sin^{-1}y-\cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)=\frac{\pi}{6}$, then $\frac{5\sin^{-1}x}{\sin^{-1}y}$ is

76. If 'α' is a solution of $\tan^{-1}\sqrt{x}+\cot^{-1}\frac{1}{x}=\frac{\pi}{4}$, then $\left[\frac{1}{\sqrt{\alpha}}\right]$ is equal to (where [.] denotes greatest integer function)

77. Let x_1,x_2,x_3 be the solution of $\tan^{-1}\left(\frac{2x+1}{x+1}\right)+\tan^{-1}\left(\frac{2x-1}{x-1}\right) = 2\tan^{-1}(x+1)$ where $x_1< x_2<x_3$ then $2x_1 + x_2 + x_3^2$ is equal to

78. Consider $f(x) = \sin^{-1}[2x] + \cos^{-1}([x]-1)$ (where [.] denotes greatest integer function.) If domain of $f(x)$ is [a,b) & the range of $f(x)$ is {c,d} then $a + b + \frac{2d}{c}$ is equal to (where $c < d$)

79. If $\frac{f(x)}{\sin^2 x}=-\cos^{-1}\left(\frac{2\sqrt{2}x}{\pi}\right)-|f(x)|$, then the value of $\frac{f(\pi/4)}{(-\pi/32)}$ is

80. Let $f: R\to R$, $y = f(x)$ is a function and $g(x)=\sin^{-1}f(x)$, $h(x)=\sqrt{f(x)-1}$ & $f^2(x)+f(x)-2\geq 0$, then the number of elements in the range of $f(x)$ is

81. Let $f(x) = \min(\tan^{-1}x, \cot^{-1}x)$ & $h(x) = f(x + 2) - \pi/3$. Let x_1,x_2 (where $x_1 < x_2$) be the integers in the range of h(x), then the value of $(\cos^{-1}(\cos x_1) + \sin^{-1}(\sin x_2))$ is equal to

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82. If x_1 & x_2 $(x_1 < x_2)$ are two solutions of equation $\sin^{-1}\left(\tan\left(\cos^{-1}\frac{\sqrt{x}}{\sqrt{x^2-x+1}}\right)\right)=\frac{\pi}{4}$, then $16^{x_1}+25^{x_2}$ is equal to

ANSWER KEY

1.	B	**2.**	C	**3.**	C	**4.**	A	**5.**	A
6.	D	**7.**	D	**8.**	C	**9.**	D	**10.**	D
11.	B	**12.**	B	**13.**	C	**14.**	C	**15.**	A
16.	C	**17.**	A	**18.**	C	**19.**	A	**20.**	A
21.	B	**22.**	B	**23.**	D	**24.**	A,B	**25.**	A
26.	C	**27.**	B	**28.**	D	**29.**	C	**30.**	D
31.	D	**32.**	C	**33.**	A	**34.**	D	**35.**	A
36.	A	**37.**	B	**38.**	C	**39.**	A,C	**40.**	A,B
41.	B,C,D	**42.**	A,B,D	**43.**	A,C,D	**44.**	A,C	**45.**	A,B,D
46.	C,D	**47.**	A,B	**48.**	B,C,D	**49.**	B,C	**50.**	B
51.	B	**52.**	D	**53.**	A	**54.**	C	**55.**	D
56.	D	**57.**	C	**58.**	B	**59.**	D	**60.**	B
61.	C	**62.**	(A)→(S); (B)→(Q); (C)→(R); (D)→(P)					**63.**	1
64.	2	**65.**	2	**66.**	8	**67.**	3	**68.**	7
69.	2	**70.**	2	**71.**	3	**72.**	2	**73.**	200
74.	2	**75.**	6	**76.**	1	**77.**	1	**78.**	4
79.	8	**80.**	1	**81.**	1	**82.**	629		

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HOME ASSIGNMENT # 03 MATHEMATICS

Straight Objective Type

1. $f(x)=\lim_{n\to\infty}\sqrt[n]{\sin^{2n}x+\cos^{2n}x}$. Find the number of points of its non-derivability in $[-2\pi, 2\pi]$

(A) 1 (B) 2 (C) 4 (D) 8

2. Let $f(x)=\dfrac{\ln(x^2+e^x)}{\ln(x^4+e^{2x})}$, if $\lim_{x\to\infty}f(x)=L_1$ and $\lim_{x\to-\infty}f(x)=L_2$ then $\dfrac{L_2}{L_1}$ is equal to -

(A) 1 (B) 2 (C) 0 (D) ∞

3. $\lim_{n\to\infty}\left(\dfrac{1+\sqrt[n]{4}}{2}\right)^n$ is equal to -

(A) 0 (B) –1 (C) 2 (D) 4

4. $\lim_{n\to\infty}\left[\sum_{r=1}^{n}\dfrac{1}{2^r}\right]$ (where [.] denotes greatest integer function) is -

(A) 0 (B) –1 (C) 2 (D) 4

5. If $f(x+y) = f(x) + f(y) + 2xy - 1\ \forall\ x,y \in R$ and f is polynomial function and $f'(0) = 3$, then number of points of non-differentiability for $y=||f(x)-1|-1|$ is -

(A) 2 (B) 3 (C) 5 (D) 6

6. $\lim_{x\to 0}\dfrac{\tan x.\cos(\sin x)-\sin(\sin x)}{\cos(\tan x).\tan x-\sin(\tan x)}$ is -

(A) $-\dfrac{1}{2}$ (B) –1 (C) $\dfrac{1}{2}$ (D) 1

7. If $f(x) = x + 1$, $g^{-1}(x) = x^3 + x + 1$ and $g(f(x)) = h(x)$, then $\lim_{x\to 1}\dfrac{h^{-1}(x)-h^{-1}(1)}{x-1}$ is equal to -

(A) 2 (B) 4 (C) 1 (D) 8

8. $\lim_{x\to\frac{\pi}{2}}\dfrac{1-\sin x.\sin 3x.\sin 5x.\sin 7x}{\left(\frac{\pi}{2}-x\right)^2}$ is K then $\dfrac{K}{6}$ is equal to -

(A) 6 (B) 7 (C) 8 (D) 9

9. $\lim_{n\to\infty}\left(1+\dfrac{1}{a_1}\right)\left(1+\dfrac{1}{a_2}\right)......\left(1+\dfrac{1}{a_n}\right)$ where $a_1 = 1$ and $a_n = n(1+a_{n-1})\ \forall\ n \geq 2$, is equal to

(A) e^2 (B) 2 (C) e (D) 1

10. $\lim_{n\to\infty}\dfrac{a^n}{(n!)}$, where $a \in (1, \infty)$, is

(A) 1 (B) 0 (C) 2 (D) a

11. Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$ then $265\left[\lim_{h\to 0}\frac{h^4+3h^2}{(f(1-h)-f(1))\sin(5h)}\right]$ equals

(A) 1 (B) 2 (C) 3 (D) –3

12. If $\lim_{x\to a} f(x) = \text{exist} \neq f(a)$ such that $|f(x)|$ is continuous at x = a then -

(A) $\lim_{x\to a} f(x)$ does not exist (B) $\lim_{x\to a} f(x) + f(a) = 0$

(C) $f(a) = 0$ (D) None

13. Given a function g(x) which has derivative g'(x) for all x satisfying g'(0) = 2 and $g(x+y) = e^y . g(x) + e^x g(y)$ for all x, y $\in$ R. Then the value of $g'(5) - 2e^5$ is -

(A) 5 (B) 25 (C) g(5) (D) None

14. For a > 0, let f : [–4a, 4a] $\to$ R be an even function such that f(x) = f(4a–x) $\forall$ x $\in$ [2a, 4a] and $\lim_{n\to 0^+}\frac{f(2a+n)-f(2a)}{n} = 4$ then $\lim_{n\to 0^+}\frac{f(n-2a)-f(-2a)}{2n}$ will be equal to -

(A) 2 (B) 4 (C) 3 (D) 5

15. Let f be differentiable function such that $f'(2) = \frac{1}{4}$ then $\lim_{n\to 0} 2\left[\frac{f(2+3n^4)-f(2-5n^4)}{n^4}\right]$ equal -

(A) 2 (B) 4 (C) 6 (D) 3

16. If $L = \lim_{x\to 0}\frac{(\tan x)^{\frac{3}{2}} - (\sin x)^{\frac{3}{2}}}{x^3\sqrt{x}}$, then value of πL equals

(A) $\frac{\pi}{8}$ (B) $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

17. Value of $\lim_{x\to 0^+}\frac{e^{(x^x-1)} - x^x}{((x^2)^x - 1)^2}$ is

(A) 1 (B) $\frac{1}{8}$ (C) $\frac{3}{2}$ (D) $\frac{1}{4}$

18. Let $f(x) = \max.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ and $g(x) = \min.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ then :

(A) both f(x) and g(x) are non differentiable at 5 points

(B) f(x) is not differentiable at 5 points where as g(x) is non differentiable at 7 points

(C) number of point of non differentiable for f(x) and g(x) are 7 and 5 respectively

(D) both f(x) and g(x) are non differentiable at 3 and 5 points respectively

19. If f(x) be a cubic polynomial and $\lim_{x\to 0}\frac{\sin^2 x}{f(x)} = \frac{1}{3}$, then f(1) can not be equal to -

(A) 0 (B) –5 (C) 3 (D) –2

20. $\lim_{x\to 0}\frac{\sin x^4 - x^4\cos x^4 + x^{20}}{x^4(e^{2x^4} - 1 - 2x^4)}$ is equal to

(A) 0 (B) $-\frac{1}{6}$ (C) $\frac{1}{6}$ (D) does not exist

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21. $\lim\limits_{x\to 2}\frac{(10-x)^{1/3}-2}{x-2}$ is equal to-

(A) $\frac{1}{12}$ (B) $-\frac{1}{12}$ (C) $-\frac{1}{10}$ (D) $\frac{1}{10}$

22. If $\lim\limits_{x\to 0}\frac{a\sin x-bx+cx^2+x^3}{2x^2\ln(1+x)-2x^3+x^4}$ exists and finite, then it is equal to-

(A) $\frac{1}{20}$ (B) $\frac{2}{45}$ (C) $\frac{3}{40}$ (D) None of these

23. Let $f(t)=|t|+|t-1|\ \forall\ t\in R$ and $g(x)=\begin{cases}\max.(f(t)), & x-1\le t\le x,\quad 0\le x\le 1\\ 3-x, & 1<x\le 2\end{cases}$

Number of points where g(x) is non-derivable in [0, 2] is :

(A) 0 (B) 1 (C) 2 (D) 3

24. The integer n for which the $\lim\limits_{x\to 0}\frac{\cos^2 x-\cos x-e^x\cos x+e^x-\frac{x^3}{2}}{x^n}$ is a finite non-zero number is-

(A) 2 (B) 3 (C) 4 (D) 5

25. The graph of the function y = f(x) has a unique tangent at $(e^a, 0)$ through which the graph passes then $\lim\limits_{x\to e^a}\frac{\ln(1+7f(x))-\sin(f(x))}{3f(x)}$ equals to-

(A) 1 (B) 2 (C) 7 (D) None of these

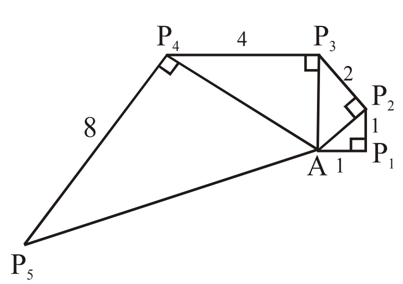
26. If $f(x)=\lim\limits_{n\to\infty}\sum_{r=0}^{n}\frac{\tan\frac{x}{2^{r+1}}+\tan^3\frac{x}{2^{r+1}}}{1-\tan^2\frac{x}{2^{r+1}}}$ then $\lim\limits_{x\to 0}\frac{f(x)}{x}$ is-

(A) 1 (B) 0 (C) –1 (D) None of these

27. Let R $\to$ R be a differentiable function at x = 0 satisfying f(0) = 0 and f '(0) = 1, then the value of $\lim\limits_{x\to 0}\frac{1}{x}\sum_{n=1}^{\infty}(-1)^n f\left(\frac{x}{n}\right)=$

(A) 0 (B) –ln2 (C) 1 (D) e

28. Let $\{P_n\}$ be a sequence of point determined as in the figure. Thus $|AP_1| = 1$, $|P_nP_{n+1}| = 2^{n-1}$ and angle AP_nP_{n+1} is a right angle $\lim\limits_{n\to\infty}\angle P_nAP_{n+1}$ equals :



(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

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29. Let O be the centre of a circle of radius 1. P , Q are the points on the circle such that $\theta = \angle POQ$ is an acute angle and R is a point outside the circle such that OPRQ is a parallelogram. If the area of the part of the parallelogram that is outside the circle is $f(\theta)$, then $\lim\limits_{\theta\to 0}\dfrac{f(\theta)}{\theta}$ is equal to

(A) $\dfrac{3}{\pi}$ (B) $\dfrac{2}{\pi}$ (C) $\dfrac{1}{2}$ (D) 1

30. The value of $\lim\limits_{x\to 0}\dfrac{\ln(\sec(ex)\sec(e^2x).....\sec(e^{50}x))}{e^2-e^{2\cos x}}$ is equal to

(A) $\dfrac{e^{100}-1}{(e^2-1)}$ (B) $\dfrac{e^{100}-1}{2(e^2-1)}$ (C) $\dfrac{2\left(e^{50}-1\right)}{e^2-1}$ (D) $\dfrac{e^2\left(e^{100}-1\right)}{2\left(e^2-1\right)}$

31. Let f(x) , f : R → R be a non-constant continuous function such that $(e^x-1)f(2x) = (e^{2x}-1)f(x)$. If $f'(0) = 1$, then $\lim\limits_{x\to 0}\left(\dfrac{f(x)}{x}\right)^{\frac{1}{x}}$ equal to

(A) e (B) $e^{1/2}$ (C) e^2 (D) e^{-2}

32. If $f(x)=\begin{cases}\ln(1+|x|+(b-2)|x+1|)+a\tan^{-1}x+2 & , \quad -2<x<0\\ b & , \quad x=0\\ \dfrac{(c+2)\sin^{-1}\sqrt{\{x\}}+e^{2x}-e^{\ln\left(4\left\{\frac{x}{2}\right\}\right)}-1}{x^2} & , \quad 0<x<2\end{cases}$ is derivable in (–2, 2) then find the value of $(9a^2+b^2+c^2)$

[Note : {y} denotes fractional part of y]

(A) 60 (B) 40 (C) 57 (D) 58

One or more than one Correct Answer Type

33. Let f : R → R be a functions such that $f\left(\dfrac{x+y}{3}\right)=\dfrac{f(x)+f(y)}{3}$, f(0) = 0 and f'(0) = 3, then -

(A) $h(x)=\begin{cases}\dfrac{f(x)}{x^2} & ; \quad x\neq 0\\ 0 & ; \quad x=0\end{cases}$ is differentiable in R (B) f(x) is continuous but not differentiable in R

(C) f(x) is continuous in R (D) f(x) is bounded in R

34. Let f : R → R be a function defined by f(x) = min (x+1, |x| + 1) then which of the following is true

(A) $f(x) \geq 1 \ \forall\ x \in R$ (B) f(x) is not differentiable at x = 0

(C) f(x) is not differentiable at x = 1 (D) f(x) is differentiable everywhere

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35. Let $f : R \to R$ be defined as

$$f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{[x]}{1+n^2}+\dfrac{3[x]}{2+n^2}+\dfrac{5[x]}{3+n^2}+\ldots+\dfrac{(2n-1)[x]}{n+n^2}\right), & x\neq\dfrac{\pi}{2}\\ 1, & x=\dfrac{\pi}{2}\end{cases}$$

where [y] denotes largest integer $\le y$, then which of the following is (are) correct ?

(A) f(x) is injective but not surjective

(B) f(x) is non-derivable at $x=\dfrac{\pi}{2}$

(C) f(x) is discontinuous at all integers and continuous at $x=\dfrac{\pi}{2}$

(D) f(x) is unbounded function

36. The possible value(s) of K for which $\lim\limits_{x\to\infty}\dfrac{2x^3-(\tan^{-1}x)^3}{\frac{8}{\pi}x^3\cot^{-1}|kx|+k^2x^6\sin\frac{1}{x^3}-3kx^3}=\dfrac{1}{2}$ is

(A) 0 (B) –1 (C) 2 (D) 4

37. Let $f(x)=\begin{cases}\dfrac{\ln(1+2x)}{x}, & \dfrac{-1}{2}<x<0\\ 2\cos x, & x=0\\ \dfrac{e^{2x}-1}{x}, & 0<x<1\\ e^2-1, & x\ge 1\end{cases}$

then -

(A) f(x) is continuous at x = 0

(B) f(x) is not differentiable at x = 0

(C) f(x) is continuous at x = 1

(D) $\lim\limits_{x\to 0^+}[f(x)]=1$

[Note : [k] denotes greatest integer less than or equal to k]

38. Let P(x) be a polynomial of n degree and $f(x)=\begin{cases}P\left(\dfrac{1}{x^3}\right)e^{-\frac{1}{x^4}}, & x\neq 0\\ 0, & x=0\end{cases}$, then

(A) f(x) is discontinuous at x = 0

(B) f(x) is continuous at x = 0

(C) f(x) is non differentiable at x = 0

(D) $f'(0)=\lim\limits_{x\to 0}f(x)$

39. If $\lim\limits_{x\to\infty}f(x^2)=a$ (a finite number), then which of the following is/are true

(A) $\lim\limits_{x\to\infty}x^2f'(x^2)=0$

(B) $\lim\limits_{x\to\infty}x^2f'(x^2)=2a$

(C) $\lim\limits_{x\to\infty}x^4f''(x^2)=0$

(D) None of these

40. $\lim_{n\to\infty}\left\{\left(\sqrt{2}+1\right)^n\right\}(-1)^{\left[\left(\sqrt{2}+1\right)^n\right]}$, $n \in N$, (where [.] is greatest integer function, {.} fractional part of x), is

(A) –1 when n is even (B) 0 when n is odd
(C) 0 when n is even (D) –1 when n is odd

Linked Comprehension Type

Passage for Q.41 to Q.42

Consider two functions $f(x)=\lim_{n\to\infty}\left(\cos\frac{x}{\sqrt{n}}\right)^n$, $g(x) = -x^{4b}$ where $b=\lim_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right)$.

41. f(x) is -

(A) e^{-x^2} (B) $e^{-\frac{x^2}{2}}$ (C) e^{x^2} (D) $e^{x^2/2}$

42. No. of solutions of f(x) + g(x) = 0 is -

(A) 0 (B) 1 (C) 2 (D) 4

Passage for Q.43 to Q.44

Let $f(x)=\begin{cases}\dfrac{1+a\cos 2x+b\cos 4x}{x^2\sin^2 x} & \text{if } x\neq 0\\ c & \text{if } x=0\end{cases}$ if f(x) is continuous at x = 0.

43. Then b + c =

(A) –3 (B) 3 (C) 0 (D) None

44. $\lim_{n\to\infty}\left(\cos^{2n} n!a\pi\right)=$

(A) 1 (B) –1 (C) 0 (D) None

Passage for Q.45 to Q.46

Let f(x) be a polynomial satisfying $\lim_{x\to\infty}\frac{x^4 f(x)}{x^8+1}=3$

f(2) = 5, f(3) = 10, f(–1) = 2, f(–6) = 37

45. Then value of $\lim_{x\to -6}\frac{f(x)-x^2-1}{3(x+6)}$ equals to -

(A) –6! (B) 6! (C) $\frac{6!}{2}$ (D) $\frac{-6!}{2}$

46. The number of points of discontinuity of $g(x)=\frac{1}{x^2+1-f(x)}$ in $\left[\frac{-15}{2},\frac{5}{2}\right]\cap D_g$ equals

(A) 4 (B) 3 (C) 1 (D) 0

Passage for Q.47 to Q.49

$$\text{Let } f(x)=\begin{cases}\lim\limits_{n\to\infty}\left(\dfrac{Px}{n}\sum\limits_{r=1}^{n}\dfrac{\left[r^2-e^{-x}+r-1\right]}{r(r+1)}\right)+\lambda & x>0\\ q & x=0\\ \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{\left\{r^2+r+e^{x}-1\right\}}{r(r+1)} & x<0\end{cases}$$

be differentiable in R :

Where [.], {.} denots greatest integer function, and fractional part function.

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47. The value of P+q+λ is equal to

(A) –2 (B) 2 (C) 1 (D) 3

48. The value of $f'(\ln 2) + f'\left(\ln\frac{1}{2}\right) + f'\left(\ln\frac{3}{2^2}\right) + f'\left(\ln\frac{5}{2^3}\right) + \ldots\ldots\ldots\infty$ is equal to

(A) 3 (B) 4 (C) 6 (D) 2

49. If g is the inverse of f then $g'\left(\frac{1}{2}\right)$ is equal to

(A) ln2 (B) – ln2 (C) 2 (D) $\frac{1}{2}$

Passage for Q.50 to Q.52

Let $f(x) = \lim_{n\to\infty}\left(\cos\sqrt{\frac{x}{n}}\right)^n, g(x) = \lim_{n\to\infty}\left(1 - x + x\sqrt[n]{e}\right)^n$. Now, consider the function y = h(x), where $h(x) = \tan^{-1}(g^{-1}f^{-1}(x))$.

50. $\lim_{x\to0^+}\frac{\ln(f(x))}{\ln(g(x))}$ is equal to

(A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) 0 (D) 1

51. Domain of the function y = h(x) is -

(A) (0, ∞) (B) R (C) (0, 1) (D) [0, 1]

52. Range of the function y = h(x) is -

(A) $\left(0,\frac{\pi}{2}\right)$ (B) $\left(-\frac{\pi}{2},0\right)$ (C) R (D) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

Passage for Q.53 to Q.55

Let f(x) = max{a, b, c} where

$$a = \lim_{\alpha\to1^+}\lim_{n\to\infty}\frac{\alpha^n|\sin x| + \alpha^{-n}|\cos x|}{\alpha^n + \alpha^{-n}}$$

$$b = \lim_{\alpha\to1^-}\lim_{n\to\infty}\frac{\alpha^n|\sin x| + \alpha^{-n}|\cos x|}{\alpha^n + \alpha^{-n}}$$

$$c = \lim_{n\to\infty}\frac{\pi}{4n}\left[1 + \cos\frac{\pi}{2n} + \cos\frac{2\pi}{2n} + \ldots\ldots + \cos\frac{(n-1)\pi}{2n}\right]. \text{ Then}$$

53. The value of a is

(A) 2|sin x| (B) |cos x| (C) |sin x| (D) $\frac{1}{2}$

54. Range of f(x) is

(A) [0, 1] (B) $\left[\frac{1}{2},1\right]$ (C) $\left[\frac{1}{\sqrt{2}},1\right]$ (D) $\left[\frac{1}{2},2\right]$

55. The value of $b + c - \frac{1}{2}$ is

(A) |cosx| (B) 2|cosx| – 1 (C) |sinx| + 1 (D) |sinx| + |cosx|

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Passage for Q.56 to Q.58

In the neighbourhood of 0 we have

$$\sqrt{1+x}=1+\frac{x}{2}-\frac{x^2}{8}+O(x^3)$$

$$\sqrt{1-x}=1-\frac{x}{2}-\frac{x^2}{8}+O(x^3)$$

$$\sqrt[3]{1+x}=1+\frac{x}{3}+O(x^2)$$

Where $O(x^3)$ means the terms containing x^3 or heigher powers and these terms are negligible in the neighbourhood of 0.

56. $\lim\limits_{x\to0}\frac{1}{x}\left[\sqrt[3]{\frac{1-\sqrt{1-x}}{\sqrt{1+x}-1}}-1\right]=$

(A) $\frac{1}{6}$ (B) $-\frac{1}{6}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

57. $\lim\limits_{x\to0}\frac{1+\sqrt{x+1}-\sqrt{1-x}-\sqrt{1-x^2}-x-\frac{x^2}{2}}{x^n}$ is finite and non-zero then n equals

(A) 1 (B) 2 (C) 3 (D) 4

58. $\lim\limits_{n\to\infty}\left(\sqrt[n]{n}+\frac{1}{n}\right)^{\frac{n}{\ell nn}}=$

(A) 1 (B) e (C) ∞ (D) 0

Passage for Q.59 to Q.60

$$L=\lim_{x\to0}\frac{(1+\sin x+\sin^2 x)^{\frac{1}{x}}-(1+\sin x)^{\frac{1}{x}}}{x}$$

$$=\lim_{x\to0}(1+\sin x)^{\frac{1}{x}}g(x)$$

59. $\lim\limits_{x\to0}g(x)$ equals

(A) e (B) –1 (C) 1 (D) $\frac{1}{e}$

60. The value of L is

(A) e^2 (B) –1 (C) 1 (D) e

Passage for Q.61 to Q.63

Let $a_n=1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+.....+\frac{1}{n!}$, $n\in N$

We have $\lim\limits_{n\to\infty}a_n=e=a_n+r_n$

Where $r_n=\frac{1}{(n+1)!}+\frac{1}{(n+2)!}+.....\infty$

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61. Which of the following is/are true ?

(A) $0 < e - a_n < \frac{1}{n(n!)}$ (B) $0 < e - a_n < 1$

(C) $e - a_n > \frac{1}{n(n!)}$ (D) None of these

62. $\lim\limits_{n\to\infty} n\sin(2\pi n!e)$ equals

(A) 0 (B) does not exist (C) 2π (D) π

63. $\lim\limits_{n\to\infty}(n+1)n!r_n$ equals

(A) ∞ (B) 1 (C) 0 (D) None of these

Passage for Q.64 to Q.65

Let $a_n = 3 - \sum\limits_{K=1}^{n} \frac{1}{K(K+1)(K+1)!}$, $n \in N$

64. $\lim\limits_{n\to\infty} a_n$ equals

(A) 0 (B) 1 (C) e (D) ∞

65. $\lim\limits_{n\to\infty}(n+1)!(a_n - e)$ equals

(A) 0 (B) 1 (C) e (D) ∞

Passage for Q.66 to Q.67

The harmonic-geometric arithmetic inequally has great applications in all branches of mathematics. Let $a_1, a_2, \ldots\ldots a_n \in R^+$, then $\frac{n}{\sum\limits_{i=1}^{n}\frac{1}{a_i}} \le \sqrt[n]{\prod\limits_{i=1}^{n} a_i} \le \frac{\sum\limits_{i=1}^{n} a_i}{n}$. Sometimes it is very useful in evaluation of some difficult limits with the help of squcezeplay theorem. Suppose

(i) $1 + \frac{x}{2+x} = \frac{2}{\frac{1}{1+x}+1} \le \sqrt{(1+x).1} \le \frac{1+x+1}{2} = 1 + \frac{x}{2}$, $x > -1$

(ii) $1 + \frac{x}{3+2x} < \frac{3}{\frac{1}{1+x}+1+1} \le \sqrt[3]{(1+x).1.1} \le \frac{1+x+1+1}{3} = 1 + \frac{x}{3}$, $x > -1$

66. $\lim\limits_{n\to\infty}\left(\sum\limits_{K=1}^{n}\sqrt{1+\frac{K}{n^2}} - n\right)$ equals

(A) $\frac{1}{2}$ (B) $\frac{1}{9}$ (C) $\frac{1}{4}$ (D) 0

67. $\lim\limits_{n\to\infty}\sum\limits_{K=1}^{n}\left(\sqrt[3]{1+\frac{K^2}{n^3}} - 1\right)$ equals

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{9}$ (D) 0

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Matrix Match Type

68. **Column-I** **Column-II**

Column-I	Column-II
(A) $\lim\limits_{n\to\infty}\cos^2\left(\pi\left(\sqrt[3]{n^3+n^2+2n}\right)\right)$ where n is an integer, equals	(P) $\frac{1}{2}$
(B) $\lim\limits_{n\to\infty} n\sin\left(2\pi\sqrt{1+n^2}\right)$ $(n \in N)$ equals	(Q) $\frac{1}{4}$
(C) $\lim\limits_{n\to\infty}(-1)^n\sin\left(\pi\sqrt{n^2+0.5n+1}\right)\left(\sin\frac{(n+1)\pi}{4n}\right)$ is (where $n \in N$)	(R) π
(D) If $\lim\limits_{x\to\infty}\left(\frac{x+a}{x-a}\right)^x = e$ where 'a' is some real constant then the value of 'a' is equal to	(S) non existent

Numerical Grid Type

69. Find the value of $\lim\limits_{n\to\infty}\frac{2}{\pi}(n+1)\cos^{-1}\left(\frac{1}{n}\right)-n$

70. If $f(x) + f(y) = f\left(\frac{x+y}{1-xy}\right)$ for all $x,\ y \in \mathbb{R}$ and $xy < 1$, $\lim\limits_{x\to 0}\frac{f(x)}{x}=2$, then value of $\frac{3f(\sqrt{3})}{\pi f'(-2)}$ is

71. Suppose f is real-valued continuous function defined on [0, 1] such that $(2x - 1)(2f(x) - x) > 0\ \forall$ $x \in R-\left\{\frac{1}{2}\right\}$. If $f'\left(\frac{1}{2}\right)$ exists, then find least possible value of $f'\left(\frac{1}{2}\right)$.

72. Let $f(x)\begin{cases}\dfrac{a(1-x\sin x)+b\cos x+5}{x^2} & x<0\\ 3 & x=0\\ \left[1+\dfrac{cx+dx^3}{x^2}\right]^{\frac{1}{x}} & x>0\end{cases}$, if f is continuous at x = 0 then find out the value of a, b, c, d.

73. If the value of limit $\lim\limits_{x\to 0}\frac{(1+x)^{\frac{1}{x}}-e+\frac{ex}{2}}{x^2}=\frac{Ae}{B}$, where A and B are co-prime natural numbers, then find the value of (A+B) :-

74. If $\lim\limits_{x\to\infty}\left[\frac{\left(1+\frac{1}{x}\right)^x}{\left(1-\frac{1}{x}\right)^x}-e^2\right]x^2=\frac{ae^2}{b}$. Find the minimum value of a + b where (a, b are natural numbers).

75. If $\lim\limits_{x\to\infty}\left(\sqrt{x^4+ax^3+3x^2+bx+2}-\sqrt{x^4+2x^3-cx^2+3x-d}\right)=4$, then $(a+c)$ is equal to

76. Let $P=\dfrac{\left(1^4+\frac{1}{4}\right)\left(3^4+\frac{1}{4}\right)\left(5^4+\frac{1}{4}\right).....\left((2n-1)^4+\frac{1}{4}\right)}{\left(2^4+\frac{1}{4}\right)\left(4^4+\frac{1}{4}\right)\left(6^4+\frac{1}{4}\right).....\left((2n)^4+\frac{1}{4}\right)}$ and $\lim\limits_{n\to\infty}(n^{\alpha}P)$ exists and is non-zero, then find α

77. Evaluate $\lim\limits_{x\to0}\sin^{-1}\left(\dfrac{\cos^{-1}x+\cos^{-1}x^2}{\pi}\right)$

78. Suppose P(x) is a polynomial function

and $f:R\to R$, $\quad f(x)=\begin{cases}\dfrac{P(x)e^{-\frac{1}{x^2}}}{x^n} & , x\neq0\ ; n\in N\\ K & , x=0\end{cases}$

If f(x) is derivable at x = 0 find K

79. $f(x)=\lim\limits_{n\to\infty}\sqrt[n]{4^x+x^{2n}+\dfrac{1}{x^{2n}}};n\in N, x\geq2$

Find number of points of non derivability in its domain

80. Find the number of points where f(x) is non-derivable

$$f(x)=\begin{cases}\tan^{-1}x & , |x|\leq1\\ \dfrac{\pi}{4}\operatorname{sgn}x+\dfrac{|x|-1}{2} & , |x|>1\end{cases}$$

81. Prove that a function $f:R\to R$ is differentiable at (α) if and only if there is a function $g:R\to R$ which is continuous at α and satisfying $f(x)+\alpha g(x)=xg(x)+f(\alpha)$ for all $x\in R$

82. Let $f(n)=\lim\limits_{x\to0}\dfrac{x-n\tan\left(\dfrac{\tan^{-1}x}{n}\right)}{n\sin\left(\dfrac{\tan^{-1}x}{n}\right)-x}$ $\qquad n\in N$

Find the value of $\sum\limits_{n=1}^{10}\dfrac{2-f(n)}{1+f(n)}$

83. Find the number of points of discontinuity $f(x)=[|x|]-|[x]|$ in $[-3, 100]$

84. If $f(x)=(\cos x)^{1/x}$ is derivable at $x=0$, $f'(0)$ is equal to

85. Find the reciprocal of sum of values of K for which $f(x)=(K+|x|)^2\ e^{(5-|x|)^2}$ is derivable for all real x.

86. Let $f:(-1,1)\to R$, be continuous function with property $f(x)=f(x^4)$ and $f\left(\dfrac{1}{2}\right)=9$ then $f\left(\dfrac{1}{9}\right)$ is -

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87. Let $f(x)=\begin{cases}\dfrac{1+\cos\pi x}{\pi^2(x+1)^2} & , \quad x<-1\\ ax+b & , \quad -1\le x\le 1\\ \dfrac{x-1}{\ln\sqrt{x}} & , \quad x>1\end{cases}$, f is continuous $\forall$ $x\in R$, then value of $3a-b$ is

88. The value of the limit $\lim\limits_{x\to0^+}4\left(\dfrac{1-\cos(x^x-1)}{x^2(\ell nx)^2}\right)$ equals

89. Let $a_1, a_2, \ldots, a_n$ be sequence of real numbers, with $a_{n+1}=a_n+\sqrt{1+a_n^2}$ and $a_0=0$ if $\lim\limits_{n\to\infty}\dfrac{a_n}{2^{n-1}}$ is $\dfrac{k}{\pi}$, then k is equal to

90. Suppose $x_1=\tan^{-1}2>x_2>x_3>\ldots$ are positive real numbers satisfying $\sin(x_{n+1}-x_n)+2^{-(n+1)}\sin x_n\sin x_{n+1}=0$ for $n\ge1$ and $\ell=\lim\limits_{n\to\infty}x_n$. Then the value of (4ℓ) where (x) denotes the least integer $\ge x$ is equal to

91. $f(x)=\lim\limits_{n\to\infty}\sin^4x+\dfrac{1}{4}\sin^4 2x+\ldots+\dfrac{1}{4^n}\sin^4(2^nx)$ and g(x) is a differentiable function satisfying $g(x)+f(x)=1$, then the maximum value of $\left(\sqrt{f(x)}+\sqrt{g(x)}\right)^4$ is

ANSWER KEY

1. D	2. A	3. C	4. A	5. D	6. A	7. B	8. B
9. C	10. B	11. C	12. B	13. C	14. A	15. B	16. B
17. B	18. B	19. C	20. C	21. B	22. C	23. B	24. C
25. B	26. A	27. B	28. C	29. C	30. B	31. B	32. C
33. C	34. D	35. C,D	36. A,B,D	37. A,B,C	38. B,D	39. D	40. A,B
41. B	42. C	43. B	44. A	45. D	46. D	47. D	48. B
49. C	50. B	51. C	52. D	53. C	54. C	55. A	56. A
57. C	58. B	59. C	60. D	61. A,B	62. C	63. B	64. C

65. A   66. C   67. C   68. A-Q; B-R; C-P; D-P   69. $1-\dfrac{2}{\pi}$ 70. 5   71. $\dfrac{1}{2}$

72. $a=-1, b=-4, c=0$ and $d=\ell n3$ 73. 35   74. 5   75. 7   76. 2

77. $\dfrac{\pi}{2}$ 78. 0   79. 0   80. 1   82. 770   83. 4

84. $-\dfrac{1}{2}$ 85. 5   86. 9   87. 1   88. 2   89. 4   90. 4   91. 4

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NAME : ....

Straight Objective Type

1. $f(x) = 2x^3 - 3(a + 1)x^2 + 6ax - 12$ has local maximum at x_1 and local minimum at x_2 and if $2x_1 = x_2$ the possible value of a is :-

(A) 1 (B) –2 (C) –1 (D) 2

2. The lateral edge of a regular hexagonal pyramid is 1 cm if the volume is maximum then the height must be $\frac{M}{\sqrt{3}}$ find the value of M :-

(A) 2 (B) 3 (C) 4 (D) 1

3. If $f''(x) = -f(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and given $F(5) = 5$, then $F(10)$ is equal to :-

(A) 4 (B) 5 (C) 6 (D) 10

4. Let $f(x) = \begin{cases} (x+1)^3 & -2 < x \le -1 \\ x^{2/3} - 1 & -1 < x \le 1 \\ -(x-1)^2 & 1 < x < 2 \end{cases}$, then find the total number of points in (–2,2) at which f attains either a local maxima or a local minima :-

(A) 1 (B) 2 (C) 3 (D) 4

5. A conical vessel is to be prepared out of a circular sheet of gold of unit radius. How much sectorial area is to be removed from the sheet so that the vessel has maximum volume :-

(A) $\frac{\pi}{\sqrt{3}}$ (B) $\frac{\pi\sqrt{2}}{\sqrt{3}}$ (C) $\frac{2\pi}{\sqrt{3}}$ (D) $\frac{2\pi}{3}$

6. Consider curve $C_1 : x^2 + \frac{y^2}{3} = 1$

and curve $C_2 : ax^2 + y^2 = 1$

If curves C_1 and C_2 intersect orthogonally each other then value of a is :-

(A) –3 (B) 3 (C) 1 (D) –1

7. Let f(x) be a monotonic polynomial of $2m - 1$ degree where $m \in N$ then the equation $f(x) + f(3x) + f(5x) + \ldots\ldots + f((2m - 1)x) = 2m - 1$ has :-

(A) atleast one real root (B) $(2m - 1)$ real roots

(C) exactly one real root (D) none of these

8. Given that $f'(x) > g'(x)$ for all real x. and $f(0) = g(0)$, then $f(x) < g(x)$ for all x belonging to :-

(A) $(0,\infty)$ (B) $(-\infty,0)$ (C) $(-\infty,\infty)$ (D) none of these

9. The function $f(x) = |ax - b| + c|x|\ \forall x \in (-\infty,\infty)$, where $a > 0$, $b > 0$, $c > 0$, assumes it's minimum only at one point if :-

(A) $a \ne b$ (B) $a \ne c$ (C) $b \ne c$ (D) $a = b = c$

10. Let domain and range of f(x) and g(x) are $[0, \infty)$. If f(x) be an increasing and g(x) be a decreasing function. Also $h(x) = f(g(x))$, $h(0) = 0$ and $p(x) = h(x^3 - 2x^2 + 2x) - h(4)$ then for all x belonging to (0, 2) :-

(A) $P(x) \in (0, -h(4))$ (B) $p(x) \in (-h(4), 0)$

(C) $p(x) \in (-h(4), h(4))$ (D) $p(x) \in (-h(4), 2h(4))$

11. If $f(-3) = f(2) = 0$ & $f'(-3) < 0$, then the largest integral value of c is (where $f(x) = x^3 + ax^2 + bx + c$) :-

(A) – 18 (B) – 19 (C) – 12 (D) – 6

12. Tangent of acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their points of intersection is :-

(A) 0 (B) $\frac{4\sqrt{2}}{3}$ (C) $\frac{4\sqrt{2}}{7}$ (D) $2\sqrt{2}$

13. If y = f(x) is twice differentiable function such that f(a) = f(b) = 0, and $f(x) > 0\ \forall\ x \in (a, b)$, then:-

(A) $f''(c) < 0$ for some $c \in (a, b)$ (B) $f''(c) > 0\ \forall\ c \in (a, b)$

(C) $f(c) = 0$ for some $c \in (a, b)$ (D) none of these

14. If $f(x) = \frac{a\sin x + b\cos x}{c\sin x + d\cos x}$ is monotonically increasing, then :-

(A) ad = bc (B) ad < bc (C) $ad \le bc$ (D) ad > bc

15. If $f(x) = \begin{cases} -x^2 & x \ge 0 \\ (a^2 - 2a) + (b^2 - 2)x & x < 0 \end{cases}$ and x = 0 is point of maxima , then :-

(A) $a \in (0, 2)$ for $b \in \left(-\sqrt{2}, \sqrt{2}\right)$ (B) $a \in [0, 2]$ for $b \in R - \left[-\sqrt{2}, \sqrt{2}\right]$

(C) $a \in \{0, 2\}$ for $b \in \left(-\sqrt{2}, \sqrt{2}\right)$ (D) $a \in R - [0, 2]$ for $b \in R - \left(-\sqrt{2}, \sqrt{2}\right)$

16. Let f, g and h are differentiable functions such that $g(x) = f(x) - x$ and $h(x) = f(x) - x^3$ are both strictly increasing functions, then the function $F(x) = f(x) - \frac{\sqrt{3}x^2}{2}$ is :-

(A) strictly increasing $\forall\ x \in R$

(B) strictly decreasing $\forall\ x \in R$

(C) strictly decreasing on $\left(-\infty, \frac{1}{\sqrt{3}}\right)$ and stricity increasing on $\left(\frac{1}{\sqrt{3}}, \infty\right)$

(D) strictly increasing on $\left(-\infty, \frac{1}{\sqrt{3}}\right)$ and stricity decreasing on $\left(\frac{1}{\sqrt{3}}, \infty\right)$

17. $f : [0,4] \to R$ is a differentiable function. Then for some $a, b \in (0, 4)$, $f^2(4) - f^2(0) =$:-

(A) 8f '(a). f(b) (B) 4f ' (b) f(a) (C) 2f ' (b) f(a) (D) f '(b) f(a)

18. If $f(x) = \int_0^1 e^{|t-x|} dt$ where $(0 \le x \le 1)$, then maximum value of f(x) is :-

(A) e – 2 (B) e – 3 (C) e – 1 (D) $2(\sqrt{e} - 1)$

19. Let $f(x) = 2x^3 - 3(2 + p)x^2 + 12px + \ln(16 - p^2)$. If f(x) has exactly one local maxima and one local minima, then the number of integral values of p is :-

(A) 4 (B) 5 (C) 6 (D) 7

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20. The equation $x^3 - 3x + [a] = 0$, where [.] denotes the greatest integer function, will have three real and distinct roots if :-

(A) $a \in (-\infty, 2)$ (B) $a \in (0,2)$ (C) $a \in (\infty,-2) \cup (0,\infty)$ (D) $a \in [-1,2)$

21. Consider function $f(x) = \dfrac{ax+b}{(x-4)(x-1)}$ has extrema at (2, –1) then value of a + b is :-

(A) 1 (B) 0 (C) 2 (D) 3

22. Let f(x) is a polynomial of 6 degree satisfies $f(x) = f(2-x)\ \forall\ x \in R$, if f(x) = 0 has 4 distinct real and two equal real roots then sum of roots of equation f(x) = 0 is :-

(A) 4 (B) 6 (C) 2 (D) can't say

23. Let f(x) be defined as :-

$$f(x) = \begin{cases} \tan^{-1}\alpha - 5x^2, & 0 < x < 1 \\ -6x, & x \geq 1 \end{cases}$$

f(x) can have a local maximum at x = 1 if value of α is

(A) 0 (B) –1 (C) –tan 1 (D) –2

24. A truck is to be driven 300 km on a highway at a constant speed of x kmph. Speed rules of the highway required that $30 \leq x \leq 60$. The fuel costs Rs. 10 per litre and is consumed at the rate of $2 + \dfrac{x^2}{600}$ liters per hour. The wages of the driver are Rs. 200 per hour. The most economical speed to drive the truck, in kmph, is :-

(A) 30 (B) 60 (C) $30\sqrt{3.3}$ (D) $20\sqrt{3.3}$

25. Slope of tangent to the curve $y = 2e^x \sin\left(\dfrac{\pi}{4} - \dfrac{x}{2}\right)\cos\left(\dfrac{\pi}{4} - \dfrac{x}{2}\right)$, where $0 \leq x \leq 2\pi$ is minimum at x = :-

(A) 0 (B) π (C) 2π (D) none of these

26. The set of values of k, for which equation $x^3 - 3x + k = 0$ has two distinct roots in (0, 1) is :-

(A) (1, 4) (B) $(0, \infty)$ (C) (0, 1) (D) ϕ

27. Which of the following is true :-

(A) $|\tan^{-1}x - \tan^{-1}y| \leq |x - y|\ \forall\ x, y \in R$ (B) $|\tan^{-1}x - \tan^{-1}y| \geq |x - y|\ \forall\ x, y \in R$

(C) $|\sin x - \sin y| \geq |x - y|\ \forall\ x, y \in R$ (D) none of these

28. Let $0 < \alpha < \theta < \beta < \dfrac{\pi}{2}$, then $\dfrac{\sin\alpha - \sin\beta}{\cos\alpha - \cos\beta}$ is equal to :-

(A) tan θ (B) –tan θ (C) cot θ (D) – cot θ

29. Find the minimum value of $(x_1 - x_2)^2 + \left(\sqrt{2 - x_1^2} - \dfrac{9}{x_2}\right)^2$ where $x_1 \in \left(0, \sqrt{2}\right)$ and $x_2 \in R^+$:-

(A) 6 (B) 7 (C) 8 (D) 9

30. A cone is made from a circular sheet of radius $\sqrt{3}$ by cutting out a sector and keeping the cut edges of the remaining piece together. The maximum volume attainable for the cone is $\dfrac{\lambda\pi}{3}$, then find λ :-

(A) 1 (B) 2 (C) 3 (D) 4

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31. Let θ be the acute angle between the curves, $y = x^x \ln x$ and $y = \dfrac{2^x - 2}{\ln 2}$ at their point of intersection on the line $y = 0$. The value of $\tan\theta$ is equal to :-

(A) $\dfrac{1}{2}$ (B) 2 (C) $\dfrac{1}{3}$ (D) 3

32. If the possible values of 'a' such that the inequality $3 - x^2 > |x - a|$ has atleast one negative solution in $a \in \left(-\dfrac{13}{4}, \lambda\right)$, then find λ :-

(A) 3 (B) 4 (C) 5 (D) 6

33. Let $f(x) = \begin{cases} xe^{ax} & , x \le 0 \\ x + ax^2 - x^3 & , x > 0 \end{cases}$ where a is positive constant and the interval in which f' (x) is increasing is $\left(-\dfrac{\lambda_1}{a}, \dfrac{a}{\lambda_2}\right)$, then find $(\lambda_1 + \lambda_2)$:-

(A) 2 (B) 3 (C) 4 (D) 5

34. If the equation $x^3 - 3x + 1 = 0$ has three real roots x_1, x_2, x_3, where $x_1 < x_2 < x_3$, then the value of $(\{x_1\} + \{x_2\} + \{x_3\})$ is equal to :-
[Note: $\{x\}$ denotes the fractional part of x.]

(A) $\dfrac{3}{2}$ (B) 1 (C) 2 (D) $\dfrac{5}{2}$

35. If $f(x) = (\sqrt{4-x^2} - 3)^2 + (\sqrt{4-x^2} + 1)^3$, then the maximum value of f(x) is :-

(A) 25 (B) 28 (C) 36 (D) 40

36. If the curves $\dfrac{x^2}{a^2} + \dfrac{y^2}{4} = 1$ and $y^2 = 16x$ intersects at right angles, then :-

(A) $a = \pm 1$ (B) $a = \pm\sqrt{3}$ (C) $a = \pm\dfrac{1}{\sqrt{3}}$ (D) $a = \pm\sqrt{2}$

37. The function $f(x) = \sin^3 x - m \sin x$ is defined on open interval $\left(\dfrac{-\pi}{2}, \dfrac{\pi}{2}\right)$ and if assumes only 1 maximum value and only 1 minimum value on this interval. then, which one of the following must be correct:-

(A) $0 < m < 3$ (B) $-3 < m < 0$ (C) $m > 3$ (D) $m < -3$

38. number of stationary points in $[0, \pi]$ for the function $f(x) = \sin x + \tan x - 2x$ is :-

(A) 0 (B) 1 (C) 2 (D) 3

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39. For which of the following function's langrange's mean value theorum is not applicable in [1, 2]:-

(A) $f(x) = \begin{cases} \frac{3}{2} - x & x < \frac{3}{2} \\ \left(\frac{3}{2} - x\right)^2 & x \geq \frac{3}{2} \end{cases}$

(B) $f(x) = \begin{cases} \frac{\sin(x-1)}{x-1} & x \neq 1 \\ 1 & x = 1 \end{cases}$

(C) $f(x) = (x - 1)\,|x - 1|$

(D) $f(x) = |x - 1|$

40. Let $x_1 = \sqrt[7]{3} + \sqrt[7]{9}$ and $x_2 = \sqrt[7]{5} + \sqrt[7]{7}$, then :-

(A) $x_1 = x_2$ (B) $x_1 > x_2$ (C) $x_1 < x_2$ (D) none of these

Multiple Correct Answer Type

41. Let f is twice differentiable function defined in [–4, 4] such that f '(–4) = 10, f '(4) = –14 and f "(x) ≥ –3 ∀ x ∈ [–4, 4], if $g(x) = \int_0^x f(t)\,dt$ and f(0) = 0 :-

(A) g(x) increases is (0, 4)

(B) g(x) decreases is (0, 4)

(C) g(x) is always concave down is (0, 4)

(D) f(x) has maximum value at x = 2

42. Let f : R – {a, b} → R, $f(x) = \frac{(x-1)(x-3)}{(x-a)(x-b)}$, then which of the following is / are correct :-

(A) f(x) = K has 3 distinct solution ∀ K ∈ (0, ∞) and ∀ a, b ∈ (3, ∞) and (a < b)

(B) f(x) = K has two solutions distinct ∀ K ∈ (0, ∞) if a ∈ (1, 3) and b ∈ (3, ∞)

(C) range of f(x) can-not be R if a, b ∈ (3, ∞) and (a < b)

(D) f(x) is strictly increasing in the interval (–∞, a) if a, b ∈ (3, ∞) and (a < b)

43. If $|f(x) - 1| \leq |g(x)|$ and $g(x) = x^3 - 12x$, than for a cubic polynomial y = f(x) the correct statements is/are:-

(A) f '(x) = 0 has one root between $\left(-2\sqrt{3}, 0\right)$

(B) f (x) = 1 has three roots common with g(x) = 0

(C) f '(x) = 0 has no root between $\left(-2\sqrt{3}, 2\sqrt{3}\right)$

(D) $f'(x)^2 + f(x)\,f''(x) = 0$ has exactly four roots between $\left(-2\sqrt{3}, 2\sqrt{3}\right)$

44. Let $f(x) = x^3 + ax^2 + bx + 5\sin^2 x$ be an increasing function in the set of real numbers R. Then a & b satisfy the condition :-

(A) $a^2 - 3b - 15 \geq 0$

(B) $a^2 - 3b + 15 \leq 0$

(C) $a^2 - 3b - 15 \leq 0$

(D) a > 0 & b > 0

45. Let $f(x) = \dfrac{(x-1)^2.e^x}{(1+x^2)^2}$, then which of the following statement(s) is(are) correct?

(A) f(x) is strictly increasing in $(-\infty, -1)$
(B) f(x) is strictly decreasing in $(1, \infty)$
(C) f(x) has two points of local extremum
(D) f(x) has a point of local minimum at some $x \in (-1, 0)$

46. If $f(x) = \int\limits_{\sin x}^{\cos x} \dfrac{dt}{e^t\sqrt{1-t^2}}; x \in \left[0, \dfrac{\pi}{2}\right]$:-

(A) f(x) is decreasing in the given interval
(B) f(x) is increasing in the given interval
(C) f(x) is nonmonotonic in the given interval
(D) If g is inverse of f then g"(0) is 0

47. Let f : R → R be defined by $f(x) = \begin{cases} 2x^2 \ln|x| - 5x^2, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Which of the following statement(s) is/are correct?

(A) f(x) has exactly one local maximum and two local minimum points
(B) f(x) is strictly increasing in $(10, \infty)$
(C) Absolute minimum value of f(x) exist but absolute maximum value of f(x) does not exist
(D) In the interval $x \in (0, \infty)$, g(x) = k has two distinct roots

48. Let us suppose that f is a twice - differentiable function satisfying f(0) = –2 ; f ' (0) = 11 ; f(2) = 5 ; f ' (2) = –3 and also suppose x = 2 to be a critical point for the function $g(x) = \dfrac{1}{x}\int\limits_0^x f(t)dt$, then :-

(A) x = 2 is a point of local maximum
(B) x = 2 is a point of local minimum
(C) $\int\limits_0^2 f(t)dt = -4$
(D) $\int\limits_0^2 f(t)dt = 10$

49. If f(x) is double differentiable function such that $|f''(x)| \le 5$ for each x in the interval [0, 4] and f takes its largest value at an interior point of this interval , then value of $|f'(0)| + |f'(4)|$ cannot be :-

(A) 10 (B) 15 (C) 21 (D) 28

50. Let $f(x) = (x-1)^4 (x-2)^n$, $n \in N$ then f(x) has :-

(A) local minima at x = 2 ; if n is even
(B) local minima at x = 1 ; if n is odd
(C) local maxima at x = 1 if n is odd
(D) local minima at x = 1 if n is even

51. Which of the following is/are correct ?

(A) $3^\pi > \pi^3$
(B) $101^{202} > 202^{101}$
(C) $\left(\dfrac{4}{3}\right)^{9/4} > \left(\dfrac{9}{4}\right)^{4/3}$
(D) $\left(1+\sin\dfrac{\pi}{3}\right)^{1+\cos\frac{\pi}{3}} > \left(1+\cos\dfrac{\pi}{3}\right)^{1+\sin\frac{\pi}{3}}$

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52. let 'f' be a continous function in [0, 3] and differentiable in (0,3) and $f(x) = 0$ has to distinct roots in (0, 3) than there exists some $\alpha \in (0, 3)$ such that f :-

(A) $\alpha f'(\alpha) + f(\alpha) = 0$
(B) $\alpha f'(\alpha) + 3f(\alpha) = 0$
(C) $\alpha f'(\alpha) + 2f(\alpha) = 0$
(D) $\alpha f'(\alpha) + 4f(\alpha) = 0$

53. Let f(x) be increasing function defined on $(0, \infty)$. if $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$ then the possible integral value of 'a' is/are :-

(A) 1
(B) 2
(C) 3
(D) 4

54. Which of the following statement's is/are correct for :-

$$f(x) = x^{1/3}(x - 1)$$

(A) has two points of inflection
(B) strictly increasing for $x > \frac{1}{4}$
(C) is concave down in $\left(-\frac{1}{2}, 0\right)$
(D) minimum value of f(x) is $\frac{-3}{4^{4/3}}$

55. Let f(x) be continuous and strictly increasing function of x and suppose $F(x) = 1 + f(x) + (f(x))^2 + (f(x))^3$ if F(x) has a positive root then :-

(A) F(x) is a decreasing function of x
(B) $f(0) < -1$
(C) $|f(0)| < 1$
(D) $F(0) < 0$

56. The function $f(x) = a(x^2 - 1)(ax + b)$, $a \neq 0$ has :-

(A) function f(x) has no point of local maxima
(B) A point of local minima at certain $x \in R^+$
(C) A point of local maxima at certain x
(D) A point of local minima at x = 1 if a + b = 0

57. If the equation $x^5 - 10a^3x^2 + b^4x + c^5 = 0$ has three equal roots, then which of the following is is/are always true :-

(A) $2b^3 - 10a^3b^2 + c^5 = 0$
(B) $6a^5 + c^5 = 0$
(C) $2c^5 - 10a^3b^2 + b^4c^5 = 0$
(D) $b^4 = 15a^4$

58. The equation $8x^3 - ax^2 + bx - 1 = 0$ has three real roots in G.P. If $\lambda_1 \le a \le \lambda_2$, then ordered pair (λ_1, λ_2) can be :-

(A) (–2, 2)
(B) (24, 29)
(C) (–10, –8)
(D) (5, 9)

59. If a, b > 0, let $f(a, b) = \frac{a^3b}{(a+b)^4}$, then :-

(A) absolute maximum value of f(a, b) is $\frac{81}{512}$

(B) absolute maximum value of f(a, b) is $\frac{27}{256}$

(C) The maximum value when a = b is smaller than the maximum value when $a \neq b$.

(D) Maximum value is same for all a and b

60. If $f(x) = \int_0^x \frac{\sin t}{t} dt$, x > 0, then :-

(A) f(x) has local maxima at $x = n\pi$ ($n = 2k$, $k \in I^+$)
(B) f(x) has local minima at $x = n\pi$ ($n = 2k$, $k \in I^+$)
(C) f(x) has neither maxima nor minima at $x = n\pi$ ($n \in I^+$)
(D) f(x) has local maxima at $x = n\pi$ ($n = 2k - 1$, $k \in I^+$)

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61. The set of values of parameter 'a' for which the point of minimum of the function $f(x) = 1 + a^2x - x^3$ satisfies the inequality $\frac{x^2+x+2}{x^2+5x+6} < 0$ is :-

(A) $\left(-3\sqrt{3},-2\sqrt{3}\right)\cup\left(2\sqrt{3},3\sqrt{3}\right)$ (B) $\left(-3\sqrt{3},2\sqrt{3}\right)\cup\left(2\sqrt{3},3\sqrt{3}\right)$

(C) $\left(-3\sqrt{3},-2\sqrt{3}\right)\cup\left(2\sqrt{3},5\sqrt{3}\right)$ (D) $\left(-3\sqrt{2},2\sqrt{3}\right)$

62. The real values of λ for which equation $x^3 + x^2 - x + \lambda = 0$ has three distinct real roots, lies in interval:-

(A) $[-2, 2]$ (B) $[-1, 0]$ (C) $(-1, 0)$ (D) $\left[-1,\frac{1}{2}\right]$

63. If α, β, γ are roots of $x^3 + (a^4 + 4a^2 + 1)x = x^2 + a^2$ then minimum value of $\sum\left\{\frac{\alpha}{\beta}+\left(\frac{\alpha}{\beta}\right)^{-1}\right\}$ is :-

(A) $\log_2 8$ (B) $\log_2 16$ (C) $\log_3 27$ (D) $\log_4 256$

64. If $f(x) = x^3 - 3x^2 + 2x$ & $f(x) = \lambda$ has exactly two solutions which are of opposite signs, then λ equals:-

(A) $\frac{-2\sqrt{3}}{9}$ (B) $\frac{-2}{3\sqrt{3}}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{2\sqrt{3}}{9}$

65. For function $f(x) = \frac{\ln x}{x}$, which of the following statements are true :-

(A) f(x) has horizontal tangent at $x = e$ (B) f(x) cuts the x-axis only at one point

(C) f(x) is many-one function (D) f(x) has one vertical tangent

66. Let $g'(x) > 0$ and $f'(x) < 0$, $\forall x \in R$, then :-

(A) $g(f(x+1)) > g(f(x-1))$ (B) $f(g(x-1)) > f(g(x+1))$

(C) $g(f(x+1)) < g(f(x-1))$ (D) $g(g(x+1)) < g(g(x-1))$

67. If the derivative of an odd cubic polnomial vanishes at two different values of 'x' then :-

(A) coefficient of x^3 & x in the polynomial must be same in sign

(B) coefficient of x^3 & x in the polynomial must be different in sign

(C) the values of 'x' where derivative vanishes are closer to origin as compared to the respective roots on either side of origin

(D) the values of 'x' where derivative vanishes are far from origin as compared to the respective roots on either side of origin.

68. Let $f(x) = \cos^2 x \,.e^{\tan x}$, $x \in \left(\frac{-\pi}{2},\frac{\pi}{2}\right)$:-

(A) f'(x) has a point of local minima at $x = \frac{\pi}{4}$

(B) f'(x) has a point of local maxima in $\left(\frac{-\pi}{4},0\right)$

(C) f'(x) has exactly two points of local maxima/ minima in $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$

(D) $f''(x) = 0$ has no root in $\left(\frac{-\pi}{2},\frac{\pi}{2}\right)$

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69. If f(x) is a differentiable real valued function satisfying $f''(x) - 3f'(x) > 3\ \forall x \geq 0$ and $f'(0) = -1$ $f(0) = 0$ then $g(x) = f(x) + x\ \forall x > 0$ is :-

(A) Positive
(B) Negative
(C) Positive as well as negative
(D) nothing can be said

70. If $(f(x) - 1)(x^2 + x + 1)^2 - (f(x) + 1)(x^4 + x^2 + 1) = 0\ \forall x \in R - \{0\}$, then which of the following statement's is/are correct :-

(A) $|f(x)| \geq 2\ \forall x \in R - \{0\}$
(B) f(x) has a local maximum at x = –1
(C) f(x) has a local minima at x = 1
(D) $\int_{-\pi}^{\pi} (\cos x) f(x)\, dx = 0$

71. if a,b,c,d $\in$ R such that $\frac{a+2c}{b+3d} + \frac{4}{3} = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has :-

(A) atleast one root is (–1,1)
(B) at least one root in (0,1)
(C) no root in (–1 , 1)
(D) no root in (0, 2)

72. a, b are real numbers such that $|a - 1| + |b - 1| = |a| + |b| = |a + 1| + |b + 1|$, then value of $|a - b|$ can be :-

(A) 1
(B) $\frac{3}{2}$
(C) 3
(D) $\frac{5}{2}$

Linked Comprehension Type

Paragraph for Q.73 & Q.74

$\int_0^x 2xf^2(t)dt = \left(\int_0^x 2f(x-t)dt\right)^2$ for $f(1) = 1$ and f(x) is continuous and differentiable function for $x > 0$ and f(x) is increasing function and $\{a_n\}$ is a sequence such that $a_{n+1} = a_n + \sqrt{1 + a_n^2}$ for $a_0 = 0$.

73. $\lim_{n\to\infty} \frac{a_k}{2^{n-1}}$, where $K = f\left(n^{\sqrt{2}-1}\right)$, is equal to :-

(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{8}$
(C) $\frac{4}{\pi}$
(D) $\frac{8}{\pi}$

74. $\tan^{-1}\left(\frac{1}{f'(1)}\right)$ is equal to (here f ' means derivative of f) :-

(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{8}$
(C) $\frac{4}{\pi}$
(D) $\frac{6}{\pi}$

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Paragraph for Q.75 & Q.76

A twice differentiable function g(x) satisfies the following conditions.

(i) $g(x) < 0\ \forall\ x \in (-\infty, 0)$ and $g(x) > 0\ \forall\ x \in (0, \infty)$

(ii) $g'(x) < 0\ \forall\ x \in (-\infty, -1)$ and $g'(x) > 0\ \forall\ x \in (-1, \infty)$

(iii) $g(0) = 0$

(iv) $\lim\limits_{x\to-\infty} g(x) = 0, \lim\limits_{x\to\infty} g(x) = \infty$

75. If $g''(x) > 0\ \forall\ x \in (-1, \infty)$ and $g'(0) = 1$ then number of solutions of equations $g(x) = x$ in $(-1, \infty)$ is :-

(A) 1 (B) 2 (C) 3 (D) 4

76. The minimum number of points where $g''(x)$ is zero is :-

(A) 3 (B) 2 (C) 1 (D) 4

Paragraph for Q.77 to Q.79

If $f(x) = |x|$; $g(x) = \begin{cases} 4-(x+4)^2 & ; \quad x \in (-5,-1) \\ x^3 - x^2 & ; \quad x \in [-1,1] \end{cases}$:-

77. The range of function g(x) is

(A) $(-5, 4]$ (B) $[0, 4]$ (C) $[3, 4]$ (D) $[-4, 4]$

78. Let m be the number of points of local extreme and n be the number of points of non-differentiability of f(g(x)) then m + n is equal to :-

(A) 3 (B) 5 (C) 6 (D) 8

79. Number of values of k satisfying the equation $f(g(f(g(x)))) = 0$; is equal to :-

(A) 3 (B) 5 (C) 6 (D) 2

Paragraph for Q.80 & 81

$f(x) = \sin 2\pi x + \{x\}$; $x \in (0,10)$ & { } denotes fractional part function

80. Number of points where f achieves local maximum is :-

(A) 10 (B) 11 (C) 20 (D) none

81. Number of points where f achieves local minima is :-

(A) 10 (B) 11 (C) 15 (D) none

Paragraph for Q.82 & 83

Consider $f(x) = \begin{cases} |x+1| & x \le 0 \\ x & x > 0 \end{cases}$

$g(x) = \begin{cases} |x|+1 & x \le 1 \\ -|x-2| & x > 1 \end{cases}$ and $h(x) = f(x) + g(x)$

82. Which one of the following is/are true :-

(A) h(x) increases in $(-\infty,1)$ and decreases in $(1,\infty)$

(B) h(x) decreases in $(-\infty,-1)$ and decreases in $(1, \infty)$

(C) h(x) has local minimum at x = 1

(D) h (x) has local maximum at x = 1

83. Which of the following is true :-

(A) h (x) is not differentiable at 4 points $\forall\ x \in R$

(B) h (x) is discontinous at 2 points $\forall\ x \in R$

(C) h(x) has no global minimum value $\forall x \in R$

(D) h(x) is non-differentiable at 3 points $\forall x \in R$

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Paragraph for Q.84 & 85

If $f(x) = (x-\alpha)^n g(x)$, then $f(\alpha) = f'(\alpha) = f''(\alpha) = \ldots = f^{n-1}(\alpha) = 0$ where f(x) and g(x) are polynomials. For a polynomial f(x) with rational coefficients, answer the following questions.

84. If f(x) touches x-axis at only one point, then the point of touching is :-

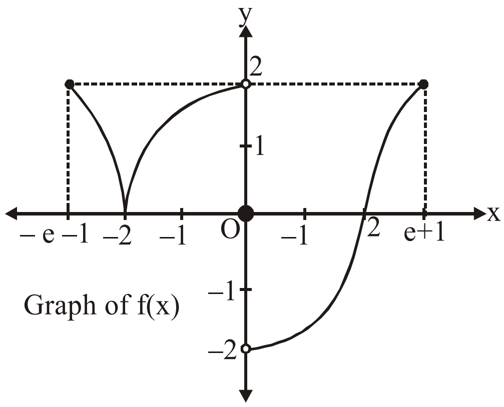
(A) always a rational number
(B) may or may not be a rational number
(C) never a rational number
(D) none of these

85. If f(x) is of degree 3 and touches x-axis, then :-

(A) all the roots of f(x) are rational
(B) only one root is rational
(C) both (A) and (B) may be possible
(D) none of these

Paragraph for Q.86 to 88

$$\text{Let } f(x) = \begin{cases} 2\ln(-x-1), & -e-1 \le x \le -2 \\ \dfrac{1}{2}(4-x^2), & -2 < x < 0 \\ 0, & x = 0 \\ \dfrac{1}{2}(x^2-4), & 0 < x < 2 \\ 2\ln(x-1), & 2 \le x \le e+1 \end{cases}$$



Graph of f(x)

and graph of f(x) is as shwon.

$$\text{Also, } g(x) = \begin{cases} \min\{f(t) : -e-1 \le t \le x\}, & -e-1 \le x < 0 \\ \max\{f(t) : 0 \le t \le x\}, & 0 \le x \le e+1 \end{cases}$$

86. Which one of the following statement does not hold good?

(A) Range of g(x) is [0, 2]
(B) g(x) is non-monotonic in [–2, 2]
(C) g(x) is a continuous function
(D) g(x) is an odd function

87. If x = α is the point of non-differentiability of g(x) in (–e –1, e + 1) then the values of α is :-

(A) –2, 1
(B) –2, –1
(C) –2, 2
(D) –2, 0

88. If the equation g(x) = k has exactly two distinct solutions in [–e –1, e + 1] then the sum of all possible integral values of k is :-

(A) 0
(B) 1
(C) 2
(D) 3

Paragraph for Q.89 to 91

Let a(t) is a function of t such that $\dfrac{da}{dt} = 2$ for all values of t and a = 0 when t = 0. Further y = m(t) x + c(t) is tangent to the curve $y = x^2 - 2ax + a^2 + a$ at the point whose abscissa is 0. Then

89. If the rate of change of distance of vertex of $y = x^2 - 2ax + a^2 + a$ from the origin with respect to t is k, then k = :-

(A) 2
(B) $2\sqrt{2}$
(C) $\sqrt{2}$
(D) $4\sqrt{2}$

90. If the rate of change of c(t) with respect to t, when t = k, is ℓ, then :-

(A) $16\sqrt{2}-2$
(B) $8\sqrt{2}+2$
(C) $10\sqrt{2}+2$
(D) $16\sqrt{2}+2$

91. The rate of change of m(t), with respect to t, at t = ℓ is :-

(A) –2
(B) 2
(C) –4
(D) 4

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Paragraph for Q.92 to 94

A function f(x) having the following properties;

(i) f(x) is continuous except at $x = 3$

(ii) f(x) is differentiable except at $x = -2$ and $x = 3$

(iii) $f(0) = 0$, $\lim\limits_{x\to3} f(x) \to -\infty$, $\lim\limits_{x\to-\infty} f(x) = 3$, $\lim\limits_{x\to\infty} f(x) = 0$

(iv) $f'(x) > 0\ \forall x \in (-\infty,-2)\cup(3,\infty)$ and $f'(x) \le 0 \forall x \in (-2, 3)$

(v) $f''(x) > 0\ \ \forall x \in (-\infty,-2)\cup(-2,0)$ and $f''(x) < 0\ \ \forall x \in (0,3)\cup(3,\infty)$

then answer the following questions

92. Maximum possible number of solutions of $f(x) = |x|$ is :-

(A) 2 (B) 1 (C) 3 (D) 4

93. Graph of function $y = f(-|x|)$ is :-

(A) differentiable for all x, if $f'(0) = 0$

(B) continuous but not differentiable at two points, if $f'(0) = 0$

(C) continuous but not differentiable at one point, if $f'(0) = 0$

(D) discontinuous at two points, if $f'(0) = 0$

94. $f(x) + 3x = 0$ has five solutions if :-

(A) $f(-2) > 6$ (B) $f'(0) < -3$ and $f(-2) > 6$

(C) $f'(0) > -3$ (D) $f'(0) > -3$ and $f(-2) > 6$

Subjective Type Questions

95. Let $g(\alpha) = \int\limits_0^{\pi/2} |\sin 2x - \alpha\cos x|\,dx,\ 0 \le \alpha \le 1$. If m and M is minimum and maximum value of $g(\alpha)\forall$ $\alpha \in [0, 1]$ then find the value of $2(M + m)$.

96. Which normal to the curve $y = x^2$ forms the shortes chord? :-

97. Prove that $1 + x < e^x < 1 + xe^x\ \forall\ x > 0$:-

98. Investigate the behaviour of the following functions for monotonicity in the given intervals, :-

(i) $f(x) = -\sin^3 x + 3\sin^2 x + 5,\ x \in [-\pi/2, \pi/2]$

(ii) $f(x) = \sec x - \operatorname{cosec} x,\ x \in (0, \pi/2)$

99. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 5, & x \le 1 \\ -2x + \log_2(b^2 - 2), & x > 1 \end{cases}$. Find all possible real values of b such that f(x) has greatest value at $x = 1$:-

100. If f(x) is continuous in [a, b], a, b > 0 and differentiable in (a, b), prove that there exists at least one $c \in [a,b]$ such that $\frac{f(b)-f(a)}{b-a} \ge \left(\frac{ab}{c^2}\right)^{\frac{n-1}{2}} f'(c)$:-

101. If the sides and angles of a plane triangle vary in such a way that its circumradius remains constant, prove that $\frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C} = 0$ where da, db and dc are small increments in the sides a, b, c respectively

102. Prove that $\frac{\sin \pi x}{x(1-x)} \le 4$, when $0 < x < 1$:-

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103. Find the absolute maximum value of $f(x) = \frac{1}{|x-4|+1} + \frac{1}{|x+8|+1}$:-

104. A point P (x, y) moves on the curve $x^{2/3} + y^{2/3} = a^{2/3}$, $a > 0$, For each position (x, y) of P, perpendiculars are drawn from origin upon the tangent and normal at P, the length (absolute value) of them being $p_1(x)$ and $p_2(x)$ respectively. Prove that $\frac{dp_1}{dx}.\frac{dp_2}{dx} \le 0$:-

105. The values of 'a' for which the function $f(x) = \sin x - a \sin 2x - \frac{1}{3} \sin 3x + 2ax$ increases throughout the number line is $[\lambda, \infty)$, then find λ :-

106. let $f(x) = ax + \cos 2x + \sin x + \cos x$ is defined for $\forall x \in R$ and $a \in R$ and is strictly increcasing fonction. if the range of a is $\left[\frac{m}{n}, \infty\right)$, ther find the value of (m –n), where m,n are coprime number :-

107. If $a^2 + b^2 = 1$ and u is the minimum value of $\frac{b+1}{a+b-2}$ then find the value of u^2.

108. Let A (1, –1), B (4, –2) and C (9, 3) be the vertices of the triangle ABC. A paralleogram AFDE is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Find the maximum area of parallelogram AFDE.

109. Let $f : [0, \infty) \to R$ be a continuous, strictly increasing function such that $f^3(x) = \int_0^x tf^2(t)\, dt$. If a normal is drawn to the curve $y = f(x)$ with gradient $\frac{-1}{2}$, then find the intercept made by it on the y-axis.

110. Let a_n $(n \ge 1)$ be the value of x for which $\int_x^{2x} e^{-t^n} dt$ $(x > 0)$ is maximum. If $L = \lim_{n\to\infty} \ln(a_n)$ then find the value of e^{-L}.

111. Let $P(x_0, y_0)$ be a point on the curve C : $(x^2 - 11)(y + 1) + 4 = 0$ where $x_0, y_0 \in N$. If area of the triangle formed by the normal drawn to the curve 'C' at P and the co-ordinate axes is $\left(\frac{a}{b}\right)$, a, b $\in$ N then find the least value of (a – 6b).

112. Let f be twice differentiable function defined in [–3, 3] such that f(0) = –4, f'(3) = 0, f'(–3) = 12 and $f''(x) \ge -2\ \forall\ x \in [-3, 3]$. If $g(x) = \int_0^x f(t)\, dt$ then find maximum value of g(x).

113. If $f(x) = a|\cos x| + b|\sin x|$ (a, b $\in$ R), has a local minimum at $x = \frac{-\pi}{3}$ and satisfies $\int_{-\pi/2}^{\pi/2} (f(x))^2 dx = 2$. Find the values of a and b and hence find $\frac{b^2}{a^2}$.

114. For $\alpha > 0$, find the minimum value of the integral $\int_0^{1/\alpha} (\alpha^2 + 4x - \alpha^5 x^2) e^{\alpha x} dx$.

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ANSWER KEY

1. (D)	2. (D)	3. (B)	4. (C)	5. (B)	6. (D)	7. (C)	8. (B)
9. (B)	10. (A)	11. (B)	12. (C)	13. (A)	14. (D)	15. (B)	16. (A)
17. (A)	18. (C)	19. (C)	20. (D)	21. (A)	22. (B)	23. (D)	24. (B)
25. (B)	26. (D)	27. (A)	28. (D)	29. (C)	30. (B)	31. (C)	32. (A)
33. (D)	34. (B)	35. (B)	36. (D)	37. (A)	38. (C)	39. (A)	40. (C)
41. (B,C)	42. (C)	43. (A,B)	44. (B,C)	45. (A,C)	46. (A,D)	47. (A,B,C)	

48. (A,D)	49. (C,D)	50. (A,C,D)	51. (A,B,D)	52. (A,B,C,D)	53. (B,C,D)
54. (A,B,C,D)	55. (B,D)	56. (B,C,D)	57. (B,D)	58. (B,C)	59. (B,C)
60. (B,D)	61. (A)	62. (A,D)	63. (A,C,D)	64. (A,B)	65. (A,B,C)
66. (B,C)	67. (B,C)	68. (A,C)	69. (A)	70. (D)	71. (A,B)

72. (C,D)	73. (C)	74. (B)	75. (A)	76. (C)	77. (A)	78. (C)	79. (C)
80. (A)	81. (D)	82. (D)	83. (A,B,C)		84. (A)	85. (A)	86. (D)
87. (C)	88. (D)	89. (B)	90. (D)	91. (C)	92. (C)	93. (B)	94. (D)

95. (3) 96. $\left(y-\frac{1}{2}\right)=\frac{-1}{\sqrt{2}}\left(x-\frac{1}{\sqrt{2}}\right),\ \left(y-\frac{1}{2}\right)=\frac{1}{\sqrt{2}}\left(x+\frac{1}{\sqrt{2}}\right)$

98. (i) $\left(0,\frac{\pi}{2}\right)$, increasing $\left(-\frac{\pi}{2},0\right)$, decreasing ; (ii) increases in $\left(0,\frac{\pi}{2}\right)$

99. $b \in [-\sqrt{130},-\sqrt{2})\cup(\sqrt{2},\sqrt{130}]$ 103. 14/13 105. $\lambda = 1$ 106. (9) 107. (9)

108. (5) 109. (9) 110.(2) 111.(3) 112. (48) 113.(3) 114.(4)

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ASSIGNMENT # 06 INDEFINITE INTEGRATION MATHEMATICS

Straight Objective Type

1. Let $F(x)=\int e^{\sin^{-1}x}\left(1-\dfrac{x}{\sqrt{1-x^2}}\right)dx$ and F(0) = 1. If $F\left(\dfrac{1}{2}\right)=\dfrac{k\sqrt{3}e^{\frac{\pi}{6}}}{\pi}$, then k is equal to -

(A) 2π (B) π (C) $3\pi/2$ (D) $\pi/2$

2. $\int\left(2x+\dfrac{1}{x}\right)\cos\dfrac{1}{x}dx$ is equal to-

(A) $x^2\cos\dfrac{1}{x}-x\sin\dfrac{1}{x}+C$ (B) $x^2\sin\dfrac{1}{x}-x\cos\dfrac{1}{x}+C$

(C) $x^2\cos\dfrac{1}{x}+x\sin\dfrac{1}{x}+C$ (D) $x^2\sin\dfrac{1}{x}+x\cos\dfrac{1}{x}+C$

3. If antiderivative of $\dfrac{x^3}{\sqrt{1+2x^2}}$ which passes through (1, 2) is $\dfrac{1}{m}(1+2x^2)^{1/2}(x^2-1)+n$. Then value of m+n is equal to -

(A) 8 (B) 5 (C) 6 (D) 7

4. $\int\tan(x-1)\tan 2x\tan(x+1)dx$ is equal to (where 'C' is the integration constant)

(A) $\log\dfrac{\cos(x+1)\cos(x-1)}{\sqrt{\cos 2x}}+C$ (B) $\log\dfrac{\sec(x+1)\sec(x-1)}{\sqrt{\sec 2x}}+C$

(C) $\log\dfrac{\sin(x+1)\sin(x-1)}{\sqrt{\sin 2x}}+C$ (D) $\log\dfrac{\mathrm{cosec}(x+1)\mathrm{cosec}(x-1)}{\sqrt{\mathrm{cosec}2x}}+C$

5. $\int\left(\sin^4 x+\cos^4 x-\sin^4 x\cos^2 x-\cos^4 x\sin^2 x\right)dx$ is-

(A) $\dfrac{\sin^7 x}{7}+\dfrac{\cos^7 x}{7}+C$ (B) $\dfrac{5x}{8}-\dfrac{3}{32}\sin 4x+C$ (C) $\dfrac{5x}{8}+\dfrac{3}{32}\sin 4x+C$ (D) $\dfrac{\sin^7 x}{7}-\dfrac{\cos^7 x}{7}+C$

(where 'C' is constant of integration)

6. $\int\dfrac{d\left(\sin^2\theta\right)}{\left(1-\sin^2\theta\right)\cos(3\theta)}$ is equal to-

(A) $\dfrac{2}{3}\ell n\left|\left(\dfrac{\cos\theta+\cos\frac{\pi}{6}}{\cos\theta-\cos\frac{\pi}{6}}\right)^{\tan\frac{\pi}{6}}e^{\sec\theta}\right|+C$ (B) $\dfrac{2}{3}\ell n\left|\left(\dfrac{\cos\theta+\cos\frac{\pi}{6}}{\cos\theta-\cos\frac{\pi}{6}}\right)^{\tan\frac{\pi}{6}}e^{\cos\theta}\right|+C$

(C) $\dfrac{2}{3}\ell n\left|\left(\dfrac{\cos\theta+\cos\frac{\pi}{6}}{\cos\theta-\cos\frac{\pi}{6}}\right)^{\tan\frac{\pi}{6}}e^{\sec(\pi-\theta)}\right|+C$ (D) $\dfrac{2}{3}\ell n\left|\left(\dfrac{\cos\theta+\cos\frac{\pi}{6}}{\cos\theta-\cos\frac{\pi}{6}}\right)^{\tan\frac{\pi}{6}}e^{\cos(\pi-\theta)}\right|+C$

(where C is constant of integration)

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7. If $\int \frac{2x^2\sec^2 x dx}{(x\sec^2 x-\tan x)^2}=f(x)+\cot x+x+C$, where C is constant of integration, then value of $f\left(\frac{\pi}{4}\right)-f\left(-\frac{\pi}{4}\right)$ is equal to

(A) $\frac{\pi}{2+\pi}$ (B) $\frac{\pi}{2-\pi}$ (C) $\frac{\pi}{4-\pi}$ (D) $\frac{\pi}{4+\pi}$

8. If $\int g(x)dx=g(x)$ then $\int g(x)\left(\frac{1+\sin x}{1+\cos x}\right)dx$ is equal to-

(A) $\frac{g(x)}{1-\cos x}+C$ (B) $\frac{1}{2}g(x)\sec^2\frac{x}{2}+C$ (C) $g(x)\tan\frac{x}{2}+C$ (D) $g(x)\cot\frac{x}{2}+C$

9. If $\int(\log_{ex} e)(\log_{e^2x} e)(\log_{e^3x} e)\frac{dx}{x}=$ A log|1 + logx| + B log|2 + logx| + C log|3 + logx|+ D then A – B + C is equal to -

(A) –1 (B) 0 (C) 2 (D) 5

10. If $3f\left(\frac{2x+3}{2x-3}\right)=2x-3$, then $\int f(x)dx$ is equal to -

(A) $\ell n(x-1)^2+c$ (B) $\ell n|(x-1)^3|+c$ (C) $\ell n(x-1)^6+c$ (D) $\ell n|(x-1)|+c$

11. $I=\int\frac{2}{(2-x)^2}\sqrt[3]{\frac{2-x}{2+x}}dx=\frac{3}{4}\sqrt[3]{f^2(x)}+c$, then $f'(1)$ is equal to -

(A) 2 (B) 1 (C) 4 (D) 3

12. If $f(x)=\int\left(\tan(\ell nx)+\frac{1}{2}\right)^2 dx$ & $f(1)=0$, then $f(e^{\pi/4})$ is -

(A) $\frac{3-e^{\pi/4}}{4}$ (B) $-\frac{3-e^{\pi/4}}{4}$ (C) $-\frac{(3+e^{\pi/4})}{4}$ (D) $\frac{e^{\pi/4}+3}{4}$

13. The value of $\int\sin x\log\left(\cot\frac{x}{2}\right)dx$ is -

(A) $-\cos x\log\cot\frac{x}{2}-\log\sin\frac{x}{2}+\log\sec\frac{x}{2}+c$

(B) $-\sin x\log\cot\frac{x}{2}+\log\tan\frac{x}{2}+c$

(C) $-\cos x\log\tan\frac{x}{2}-\log\sin\frac{x}{2}+\log\sec\frac{x}{2}+c$

(D) $-\sin x\log\left(\cot\frac{x}{2}\right)+\cos x\log\left(\tan\frac{x}{2}\right)+c$

14. $\int\frac{\operatorname{cosec}^2x-2005}{\cos^{2005}x}dx$ is equal to -

(A) $-\frac{\cot x}{(\cos x)^{2005}}+c$ (B) $\frac{\tan x}{(\cos x)^{2005}}+c$ (C) $\frac{-(\tan x)}{(\cos x)^{2005}}+c$ (D) None of these

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15. The value of integral $\int e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{1-2x^2}{\sqrt{(1+x^2)^5}}\right)dx$ is equal to -

(A) $e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{x}{\sqrt{(1+x^2)^3}}\right)+c$

(B) $e^x\left(\frac{1}{\sqrt{1+x^2}}-\frac{x}{\sqrt{(1+x^2)^3}}\right)+c$

(C) $e^x\left(\frac{1}{\sqrt{1+x^2}}+\frac{x}{\sqrt{(1+x^2)^5}}\right)+c$

(D) None of these

16. The value of $\int\frac{\sin 2x+2\tan x}{(\cos^6 x+6\cos^2 x+4)}dx$ is equal to -

(A) $2\sqrt{\frac{1+\cos^2 x}{\cos^7 x}}$

(B) $\frac{1}{12}\log\left(\frac{\cos^6 x+6\cos^2 x+4}{\cos^6 x}\right)$

(C) $\frac{1}{12}\ln\left(\frac{1+\cos^2 x}{\cos^7 x}\right)+c$

(D) $\frac{1}{12}\log\left(\frac{\cos^6 x}{\cos^6 x+6\cos^2 x+4}\right)$

17. $I=\int\frac{x^4-1}{(x^4-x^2+1)^{\frac{3}{2}}}dx$. Then I is equal to :

(A) $\frac{-x}{\sqrt{x^4-x^2+1}}+C$ (B) $\frac{x}{\sqrt{x^4-x^2+1}}+C$ (C) $\frac{1}{\sqrt{x^4-x^2+1}}+C$ (D) $\frac{-1}{\sqrt{x^4-x^2+1}}+C$

18. If $y=\cfrac{x}{x^2+\cfrac{x}{x^2+\cfrac{x}{x^2+\ldots\infty}}}$ and $\int\frac{(y-x^2)dx}{(x^2+y)(x+y^2)}=f(y)+c$ then $\frac{d}{dy}(f(y))$ for x = 1 is equal to -

(A) 1 (B) $\frac{\sqrt{5}-1}{2}$ (C) $\frac{\sqrt{5}+1}{2}$ (D) $\frac{2-\sqrt{5}}{2}$

19. If $\int\frac{dx}{x^4(1+x^3)^2}=a\ln\left|\frac{1+x^3}{x^3}\right|+\frac{b}{x^3}+\frac{c}{1+x^3}+d$, then (where d is arbitrary constant) -

(A) $a=\frac{1}{3},b=\frac{1}{3},c=\frac{1}{3}$

(B) $a=\frac{2}{3},b=\frac{-1}{3},c=\frac{1}{3}$

(C) $a=\frac{2}{3},b=\frac{-1}{3},c=\frac{-1}{3}$

(D) $a=\frac{2}{3},b=\frac{1}{3},c=\frac{-1}{3}$

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20. $\int \frac{\sqrt{\sec^5 x}}{\sqrt{\sin^3 x}} dx$ equal to -

(A) $(\tan x)^{\frac{3}{2}} - \sqrt{\tan x} + c$

(B) $2\left(\frac{1}{3}(\tan x)^{\left(\frac{3}{2}\right)} - \frac{1}{\sqrt{\tan x}}\right) + c$

(C) $\frac{1}{3}(\tan x)^{\left(\frac{3}{2}\right)} - \sqrt{\tan x} + c$

(D) $\sqrt{\sin x} + \sqrt{\cos x} + c$

21. Evaluate : $\int \frac{x^3 + 3x^2 + x + 9}{(x^2+1)(x^2+3)} dx$

(A) $\ln|x^2+3| + 3\tan^{-1} x + c$

(B) $\frac{1}{2}\ln|x^2+3| + \tan^{-1} x + c$

(C) $\frac{1}{2}\ln|x^2+3| + 3\tan^{-1} x + c$

(D) $\ln|x^2+3| - \tan^{-1} x + c$

22. Let α,β,γ and δ be 4 distinct roots of the equation $x^4-4x+3=x(x^3-f'(1)x^2 + f''(1)x-4)+f(1)$ and f(x) is a monic polynomial of degree 3. If $\int \frac{dx}{\frac{f(x)-3}{x-1}+4} = \frac{1}{2}g(x) + C$, where C is constant of integration and $g(3) = \frac{\pi}{4}$ then the value of $g(5) + g(7)$ is :-

(A) $\frac{\pi}{2}$

(B) $\frac{3\pi}{4}$

(C) π

(D) $\frac{5\pi}{4}$

23. $\int \frac{x dx}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}$ is equal to :-

(A) $\frac{1}{2}\ell n\left(1+\sqrt{1+x^2}\right) + C$

(B) $2\sqrt{1+\sqrt{1+x^2}} + C$

(C) $2\left(1+\sqrt{1+x^2}\right) + C$ (D) None of these

24. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = C - \frac{1}{f(x)}$ where f(x) is of the form of $ax^2 + bx + c$ then (a+b+c) equals :

(A) 4 (B) 5 (C) 6 (D) None of these

25. The value of integral $\int e^x \left[\frac{2\tan x}{1+\tan x} + \cot^2\left(x+\frac{\pi}{4}\right)\right] dx$ is equal to -

(A) $e^x \tan\left(\frac{\pi}{4} - x\right) + C$ (B) $e^x \tan\left(x - \frac{\pi}{4}\right) + C$ (C) $e^x \tan\left(\frac{3\pi}{4} - x\right) + C$ (D) $e^x \tan\left(x - \frac{3\pi}{4}\right) + C$

26. For any natural number m, $\int \left(x^{7m} + x^{2m} + x^m\right)\left(2x^{6m} + 7x^m + 14\right)^{1/m} dx$ (where $x > 0$) equal :

(A) $\frac{\left(7x^{7m} + 2x^{2m} + 14x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + C$

(B) $\frac{\left(2x^{7m} + 14x^{2m} + 7x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + C$

(C) $\frac{\left(2x^{7m} + 7x^{2m} + 14x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + C$

(D) $\frac{\left(7x^{7m} + 2x^{2m} + x^m\right)^{\frac{m+1}{m}}}{14(m+1)} + C$

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27. $\int P(x).e^{kx}dx = Q(x)e^{4x} + C$, where P(x) is polynomial of degree n and Q(x) is polynomial of degree 7. Then the value of n + 7 + k + $\lim\limits_{x\to\infty}\frac{P(x)}{Q(x)}$ is :

(A) 18 (B) 19 (C) 20 (D) 22

Multiple Correct Answer Type

28. If $I_n = \int_0^{\pi/4}(\tan^n(x-[x]) + \tan^{n-2}(x-[x]))dx$ and $n \in N$; $n \geq 2$, where [.] denotes greatest integer function, then -

(A) $\frac{1}{I_{100}} - \frac{1}{I_{99}} = 1$

(B) I_2, I_3, I_4...... are in H.P.

(C) I_2, I_3, I_4...... are in A.P.

(D) $\sum_{i=2}^{n}\frac{1}{I_i} = \frac{n(n-1)}{2}$

29. If $I_n = \int_0^{\pi/4}(\tan^n(x-[x]) + \tan^{n-2}(x-[x]))dx$ and $n \in N$; $n \geq 2$, where [.] denotes greatest integer function, then -

(A) $\frac{1}{I_{100}} - \frac{1}{I_{99}} = 1$

(B) I_2, I_3, I_4...... are in H.P.

(C) I_2, I_3, I_4...... are in A.P.

(D) $\sum_{i=2}^{n}\frac{1}{I_i} = \frac{n(n-1)}{2}$

30. Let $f'(x) = 3x^2.\sin\frac{1}{x} - x\cos\frac{1}{x}$, $x \neq 0$, $f(0) = 0$, $f\left(\frac{1}{\pi}\right) = 0$, then which of the following is/are not correct.

(A) f(x) is continuous at x = 0

(B) f(x) is non-differentiable at x = 0

(C) f'(x) is discontinuous at x = 0

(D) f'(x) is differentiable at x = 0

31. $\int \frac{\sin 2x + 2}{\sin 2x + \cos 2x + 1}dx = f(x) + C$, where C is integration constant and $f(0) = 0$, then -

(A) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{8} + \frac{3}{2}\ell n2$

(B) $f\left(\frac{\pi}{4}\right) = \frac{\pi}{8} + \frac{3}{4}\ell n2$

(C) $\lim\limits_{x\to\frac{3\pi}{4}} f(x)$ is does not exist

(D) $\lim\limits_{x\to\frac{3\pi}{4}} f(x) = \frac{3\pi}{8} + \ell n3$

32. If $I = \int\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$, then I equals to -

(A) $(\sqrt{x}-2)\sqrt{1-x} + \cos^{-1}\sqrt{x} + C$

(B) $(\sqrt{x}+2)\sqrt{1-x} + \sin^{-1}\sqrt{x} + C$

(C) $(\sqrt{x}-2)\sqrt{1-x} + 2\cos^{-1}\sqrt{x} + \sin^{-1}\sqrt{x} + C$

(D) $(\sqrt{x}-2)\sqrt{1-x} - \sin^{-1}\sqrt{x} + C$

(where C is integration constant)

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33. $I=\int e^{(x\sin x+\cos x)}\left(\frac{x^4\cos^3 x-x\sin x+\cos x}{x^2\cos^2 x}\right)dx$, then I equals -

(A) $e^{(x\sin x+\cos x)}\left(x-\frac{\sec x}{x}\right)+c$ (B) $e^{x\sin x}+\cos x\left(x\sin x-\frac{\cos x}{x}\right)+c$

(C) $e^{x\sin x}+\cos x\left(\frac{x}{\tan x}-\frac{\sec x}{x}\right)+c$ (D) $xe^{x\sin x+\cos x}-e^{x\sin x+\cos x}.\frac{1}{x\cos x}+c$

34. Let, $f(x)=1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}$ & $I=\int\frac{x^4}{f(x)}dx=\lambda(x-\ln|f(x)|)+C$ then the value of λ is divisible by -

(A) 8 (B) 12 (C) 24 (D) 36

35. $\int\frac{\cos x}{x}\left(x^2\ln x+1\right)dx=f(x)+c$; $f(1)=\cos 1$ and $\int f(x)\sec^2 x\,dx=g(x)+c^1$; $g(1)=0$ and $L=\lim\limits_{x\to 0^+}g(x)$, then -

(A) $f(x) = \cos x(1 + \ln x) - x \sin x \ln x$ (B) $g(x)=\frac{x\ln x}{\cos x}$

(C) $L = 0$ (D) $\lim\limits_{x\to\infty}\frac{g(x)}{x^2}=0$

36. The value of the integral $\int e^{\sin^2 x}\left(\cos x+\cos^3 x\right)\sin x\,dx$ is -

(A) $\frac{1}{2}e^{\sin^2 x}\left(3-\sin^2 x\right)+c$ (B) $e^{\sin^2 x}\left(1+\frac{1}{2}\cos^2 x\right)+c$

(C) $e^{\sin^2 x}\left(3\cos^2 x+2\sin^2 x\right)+c$ (D) $e^{\sin^2 x}\left(2\cos^2 x+3\sin^2 x\right)+c$

Linked Comprehension Type

Paragraph for Question 37 to 39

Let $f(x)=\sqrt{4\cos^2 x-3}$ then

On the basis of above information, answer the following questions :

37. The range of $f(x)$ is-

(A) $\left[0,\frac{1}{2}\right]$ (B) $\left[\frac{1}{2},1\right]$ (C) $[0,1]$ (D) $[0,2]$

38. If $\int\frac{f(x)}{\cos x}dx$ is written as $\lambda\int\frac{\cos x}{\sqrt{1-4\sin^2 x}}dx-\mu\int\frac{\cos x\,dx}{(1-\sin^2 x)\sqrt{1-4\sin^2 x}}$, then the values of λ & μ respectively are-

(A) 4,3 (B) 4,–3 (C) –4,3 (D) –4,–3

39. A primitive of $\frac{f(x)}{\cos x}$ is-

(A) $2\sin^{-1}(2\sin x)+\sqrt{3}\tan^{-1}\left(\frac{f(x)}{\sqrt{3}\sin x}\right)$ (B) $2\cos^{-1}(2\sin x)+\sqrt{3}\cot^{-1}\left(\frac{f(x)}{\sqrt{3}\sin x}\right)$

(C) $2\sin^{-1}(2\sin x)+\sqrt{3}\cot^{-1}\left(\frac{f(x)}{\sqrt{3}\sin x}\right)$ (D) $2\cos^{-1}(2\sin x)+\sqrt{3}\tan^{-1}\left(\frac{f(x)}{\sqrt{3}\sin x}\right)$

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Passage for Q.40 to 41

Let function f : R → R and g : R → R satisfy f(x + g(y)) = 2x + y + 5 and g(0) = 0

40. The value of g(x+f(y)) when x = 1 and y = 2 is -

(A) 5 (B) 4 (C) 2 (D) 1

41. $\int e^x\left(\frac{f(x)(x^2+3x+2)-2x^2+10x+11}{(x^2+3x+2)^2}\right)dx$ is equal to -

(A) $\frac{e^x(2x-1)}{x^2+3x+2}+c$ (B) $\frac{e^x(x^2+2)}{x^2+3x+2}+c$ (C) $\frac{e^x(3x+2)}{x^2+3x+2}+c$ (D) $\frac{e^x(2x+5)}{x^2+3x+2}+c$

Passage for Q.42 to Q.44

Let us consider the integrals of the following forms $f\left(x,\sqrt{mx^2+nx+p}\right)^{1/2}$.

Case-1 : If m > 0, then put $\sqrt{mx^2+nx+p}=u\pm x\sqrt{m}$

Case-2 : If p > 0, then put $\sqrt{mx^2+nx+p}=ux\pm\sqrt{p}$

Case-3 : If quadratic equation $mx^2 + nx + p = 0$ has real roots α and β there put

$$\sqrt{mx^2+nx+p}=(x-\alpha)u \text{ or } (x-\beta)u$$

Answer the following questions :

42. If $I=\int\frac{dx}{x-\sqrt{9x^2+4x+6}}$ to evaluate I, one of the most proper substitution could be

(A) $\sqrt{9x^2+4x+6}=u\pm 3x$ (B) $\sqrt{9x^2+4x+6}=3u\pm x$

(C) $x=\frac{1}{t}$ (D) $9x^2+4x+6=\frac{1}{t}$

43. $\int\frac{\left(x+\sqrt{1+x^2}\right)^{15}}{\sqrt{1+x^2}}dx$ is equal to -

(A) $\frac{\left(x+\sqrt{1+x^2}\right)^{16}}{10}+c$ (B) $\frac{1}{15\left(\sqrt{1+x^2}+x\right)}+c$ (C) $\frac{15}{\left(\sqrt{1+x^2}-x\right)}+c$ (D) $\frac{\left(x+\sqrt{1+x^2}\right)^{15}}{15}+c$

44. To evaluate $\int\frac{dx}{(x-1)\sqrt{-x^2+3x-2}}$ one of the most suitable substitution could be-

(A) $\sqrt{-x^2+3x-2}=u$ (B) $\sqrt{-x^2+3x-2}=(ux\sqrt{2})$

(C) $\sqrt{-x^2+3x-2}=u(1-u)$ (D) $\sqrt{-x^2+3x-2}=u(x-2)$

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Subjective Type Questions

45. Evaluate $\int \frac{1}{x^4+1+5x^2}dx$.

46. Evaluate $\int \frac{x^4-3x^2-3x-2}{x^3-x^2-2x}dx$.

47. Evaluate $\int x^{-11}\left(1+x^4\right)^{-1/2}dx$.

48. If $\int \frac{1}{((x-1)^3(x+2)^5)^{1/4}}dx = \frac{p}{q}\left(\frac{x-1}{x+2}\right)^{\frac{1}{4}}+c$. Then p + q is (p and q are co-prime no's) -

49. If $\int \frac{1+x^2}{3-4\sqrt{2}x+4x^4}dx = \frac{a}{x+b}$ + ctan^{-1}(x + d) + e, then a + b + c + d is equal to -

50. $\int \frac{\cos^4 x\,dx}{\sin^3 x(\sin^5 x+\cos^5 x)^{\frac{3}{5}}} = \frac{-1}{2}\left(1+\cot^A x\right)^B + C$, then value of A.B equals to

51. Evaluate : $\int \sqrt{4+\tan^2 x}\,dx$

$\int \sqrt{4+\tan^2 x}\,dx$ का मान ज्ञात कीजिये।

52. Let $f(x)$ be a differentiable function satisfying the equation $\frac{f'(x)}{2} = \frac{x}{e^{f(x)}}\ \forall x \in R$. If $f'(1) = 1$, then the number of solution(s) of the equation $f'(x) = f(x)$ is

ANSWER KEY

1. D	**2.** A	**3.** A	**4.** A	**5.** C	**6.** C	**7.** B
8. C	**9.** C	**10.** A	**11.** C	**12.** D	**13.** A	**14.** A
15. A	**16.** B	**17.** A	**18.** C	**19.** C	**20.** B	**21.** C
22. B	**23.** B	**24.** B	**25.** B	**26.** C	**27.** D	**28.** A,B,D
29. A,B,D	**30.** B,C,D	**31.** B,C	**32.** A,C,D	**33.** A,D	**34.** A,B,C	**35.** B,C
36. A,B	**37.** C	**38.** A	**39.** A	**40.** A	**41.** D	**42.** A

43. D **44.** D **45.** $\frac{1}{2}\left[\frac{1}{\sqrt{7}}\tan^{-1}\frac{x^2-1}{\sqrt{7}x}\right]-\frac{1}{2}\left[\frac{1}{\sqrt{3}}\tan^{-1}\frac{x^2+1}{\sqrt{3}x}\right]$

46. $\frac{x^2}{2}+x+\ln|x|-\frac{2}{3}\ln|x-2|-\frac{1}{3}\ln|x+1|+c$ **47.** $-\frac{r^5}{10}+\frac{r^3}{3}-\frac{r}{2}+c \quad r=\sqrt{1+\frac{1}{x^4}}$ **48.** 7

49. 0 **50.** 2 **51.** $\sqrt{3}\sin^{-1}\left(\frac{\sqrt{3}\sin x}{2}\right)$ **52.** 2

NAME : ....

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ASSIGNMENT # 07 DEFINITE INTEGRATION MATHEMATICS

Straight Objective Type

1. The value of the definite integral $\int_{t+2\pi}^{t+\frac{5\pi}{2}} (\sin^{-1}(\cos x)+\cos^{-1}(\cos x))dx$ is equal to -

(A) $\frac{\pi^2}{2}$ (B) $\frac{\pi^2}{8}$ (C) $\frac{\pi^2}{4}$ (D) None of these

2. If $k \in N$ and $I_k = \int_{-2k\pi}^{2k\pi} |\sin x|[\sin x]dx$, (where [.] denotes the greatest integer function) then $\sum_{k=1}^{100} I_k$ equal to -

(A) –10100 (B) –40400 (C) –20200 (D) None of these

3. Given that $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{\log(n^2+r^2)-2\log n}{n} = \log 2+\frac{\pi}{2}-2$, then

$\lim_{n\to\infty}\frac{1}{n^{2m}}\left[\left(n^2+1^2\right)^m\left(n^2+2^2\right)^m...(2n^2)^m\right]^{1/n}$ is equal to-

(A) $2^m e^{m(\pi/2-2)}$ (B) $2^m e^{m(2-\pi/2)}$ (C) $e^{m(\pi/2-2)}$ (D) $e^{2m(\pi/2-2)}$

4. The value of $\int_{-6}^{6} \max(|2-|x||, 4-|x|, 3)dx$ is -

(A) 37 (B) 38 (C) 39 (D) 40

5. If $I_n = \int_0^{\pi} e^x(\sin x)^n dx$, then $\frac{I_3}{I_1}$ is equal to -

(A) 3/5 (B) 1/5 (C) 1 (D) 2/5

6. If $F(x) = \frac{1}{x^2}\int_4^x (4t^2-2F'(t))dt$ then F'(4) equals to -

(A) $\frac{32}{9}$ (B) $\frac{64}{9}$ (C) $\frac{F(8)}{28}$ (D) $\frac{11}{30}F(8)$

7. $I_1 = \int_0^1 e^{-x}\cos^2 x\,dx$, $I_2 = \int_0^1 e^{-x^2}\cos^2 x\,dx$, $I_3 = \int_0^1 e^{-\frac{x^2}{2}}\cos^2 x\,dx$, $I_4 = \int_0^1 e^{-\frac{x^2}{2}}dx$ then -

(A) $I_2 > I_4 > I_1 > I_3$ (B) $I_2 < I_4 < I_1 < I_3$ (C) $I_1 < I_2 < I_3 < I_4$ (D) $I_1 > I_2 > I_3 > I_4$

8. The value of $A = \int_1^{\infty}\left[\operatorname{cosec}^{-1}x\right]dx$ {where [.] denotes greatest integer function} is equal to -

(A) cosec 1 – 1 (B) 1 (C) 1 – sin 1 (D) None

9. Value of $B = \int_1^{100}\left[\sec^{-1}x\right]dx$ {where [.] denotes greatest integer function}

(A) sec1 (B) 100 – sec1 (C) 99 – sec1 (D) None

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10. $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x\,dx}{(1+\tan x)}$

(A) $\frac{1}{2}-\frac{1}{2\sqrt{2}}\ln\left(\sqrt{2}+1\right)$ (B) $\frac{1}{2}+\frac{1}{2\sqrt{2}}\ln\left(\sqrt{2}+1\right)$

(C) $\frac{1}{2}-\frac{1}{\sqrt{2}}\ln\left(\sqrt{2}+1\right)$ (D) $\frac{1}{6}-\frac{1}{2\sqrt{2}}\ln\left(\sqrt{2}+1\right)$

11. $I=\int_2^3 \frac{\log(x-1)}{x^2+1}dx$ is equal to :

(A) $\tan^{-1}3 - \tan^{-1}2$ (B) $\frac{\tan^{-1}3-\tan^{-1}2}{2}$ (C) $\ln2\,(\tan^{-1}3 - \tan^{-1}2)$ (D) $\frac{\ln 2}{2}(\tan^{-1}3-\tan^{-1}2)$

12. If $f(n) = \frac{\int_0^n [x]dx}{\int_0^n \{x\}dx}$, then $\sum_{n=1}^{n} f(n)$ is equal to (where [*] and {*} denotes G.I.F. and fractional parts of x). $n \in N$.

(A) $n^2 - 1$ (B) $\frac{n^2-n}{2}$ (C) $\frac{n(n+1)}{2}$ (D) $n^2 + 1$

13. The value of $\int_0^1 \left(\sqrt[7]{1-x^9}-\sqrt[9]{1-x^7}\right)dx$ is equal to -

(A) $\frac{1}{2}$ (B) 1 (C) 0 (D) $\frac{3}{2}$

14. If $f(x) = \int_0^x \frac{\sin t}{t}dt$, which of the following is true :-

(A) $f(0) > f(1.1)$ (B) $f(0) < f(1.1) > f(2.1)$

(C) $f(0) < f(1.1) < f(2.1) > f(3.1)$ (D) $f(0) < f(1.1) < f(2.1) < f(3.1) > f(4.1)$

15. If $I_1 = \int_0^{\pi/2} \cos\theta f(\cos^2\theta+\sin\theta)d\theta$ and $I_2 = \int_0^{\pi/2} \sin 2\theta f(\sin\theta+\cos^2\theta)d\theta$ then $\frac{I_1}{I_2}$ is :-

(A) 1 (B) –1 (C) 2 (D) –2

16. The value of $\int_0^{\pi/2} \log(1+3\cos^2\theta)d\theta$ is equal to :

(A) $\pi\log\frac{3}{2}$ (B) $\log\frac{3}{2}$ (C) $\log\frac{2}{3}$ (D) $\pi\log\frac{2}{3}$

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17. $\int_0^1 \frac{\ln x}{1+x}dx + \int_0^1 \frac{\ln(1+x)}{x}dx$ is equal to :

(A) –e (B) e (C) 0 (D) 1

18. Let f(t) be a continous function for $t \in R$. Let $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} xf\left(x(2-x)\right)dx$ and $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f\left(x(2-x)\right)dx$,

then :-

(A) $\frac{I_1}{I_2} = 2$ (B) $\frac{I_1}{I_2} = 1$ (C) $\frac{I_1}{I_2}$ is not difined (D) $\frac{I_1}{I_2} = \frac{1}{2}$

19. If $\int_{-1/4}^{1/4}\left(\sqrt{16x^2+16x+41+40\sqrt{x^2+x+1}}+\sqrt{16x^2+16x+41-40\sqrt{x^2+x+1}}\right)\cos x\,dx$

$= K\sin\frac{1}{4}$, then the value of K is :-

(A) 10 (B) 0 (C) 20 (D) 30

20. The value of $\int_a^b (x-a)^3(b-x)^4 dx$ is $\frac{(b-a)^m}{n}$ then (m, n) is :-

(A) (6, 260) (B) (8, 280) (C) (4, 240) (D) None

21. $\sum_{k=1}^{\infty}\left(\frac{1}{3k-1}-\frac{1}{3k}\right)$ is equal to :

(A) $\log\sqrt{3}-\frac{\pi}{6\sqrt{3}}$ (B) $\log\sqrt{3}-\frac{\pi}{6}$ (C) $\log 3-\frac{\pi}{6\sqrt{3}}$ (D) $\log\sqrt{3}-\frac{\pi}{\sqrt{3}}$

22. Let I = $\int_0^{\pi/6}\frac{\cos x}{x}dx$ J = $\int_{\pi/3}^{\pi/2}\frac{\cos x}{x}dx$ which of the following is correct ?

(A) $I < \frac{\pi}{6}, J < \frac{\pi}{6}$ (B) $I > \frac{\pi}{6}, J < \frac{\pi}{6}$ (C) $I < \frac{\pi}{6}, J > \frac{\pi}{6}$ (D) $I > \frac{\pi}{6}, J > \frac{\pi}{6}$

23. Absolute value of $\int_{10}^{19}\frac{\sin x}{1+x^8}dx$, is :-

(A) more than 10^{-5} (B) more than 10^{-7} (C) less than 10^{-6} (D) more than 10^{-6}

24. If $I_n = \int_0^1 \frac{dx}{(1+x^2)^n}$, $n \in N$, then which of the following statements hold good ?

(A) $2nI_{n+1} = 2^{-n} + (2n-1)I_n$ (B) $I_2 = \frac{\pi}{16}+\frac{1}{4}$

(C) $I_2 = \frac{\pi}{8}-\frac{1}{4}$ (D) $I_3 = \frac{\pi}{16}-\frac{5}{48}$

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25. If $f(2-x)=f(2+x)$ and $f(4-x)=f(4+x)$ and $f(x)$ is a function for which $\int_0^2 f(x)dx=5$, then $\int_0^{50} f(x)dx$ is not equal to :-

(A) 120 (B) $\int_{-4}^{46} f(x)dx$ (C) 125 (D) $\int_2^{52} f(x)dx$

26. If $I=\int_0^{\pi/2} e^{-a\sin x}\,dx$, where $a \in (0, \infty)$, then which of the following doesn't hold good :-

(A) $I < \frac{\pi}{2}$ (B) $I > \frac{\pi}{2}(e^{-a}+1)$ (C) $I > \frac{\pi}{2}e^{-a}$ (D) $I > 0$

27. If $\int_0^1 \left(x^2 f(x) - x f^2(x)\right)dx$ is maximum, then $f\left(\frac{1}{2}\right)$ is :

(A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) Can not be determined

28. $\int_0^1 \frac{2x^2-2x+1}{\sqrt{x^2-2x+2}}dx$ is equal to :

(A) $\tan\frac{\pi}{8}$ (B) $\sqrt{2}\tan\frac{\pi}{8}$ (C) $\frac{1}{\sqrt{2}}\tan\frac{\pi}{8}$ (D) $2\sqrt{2}\tan\frac{\pi}{8}$

29. $\lim\limits_{n\to\infty}\frac{1+2^{\frac{1}{n}}+2^{\frac{2}{n}}+....2^{\frac{n}{n}}}{n}$ is equal to :

(A) $\log_2 e$ (B) $\log_e 2$ (C) $\log_e e$ (D) $\log_{\sqrt{e}} e$

30. In $I_n=\int \cot^n x\,dx$, then $I_0 + I_1 + 2(I_2 + I_3 + ... I_8) + I_{9} + I_{10}$ equals to :
(where $u = \cot x$)

(A) $u+\frac{u^2}{2}+....+\frac{u^9}{9}$ (B) $-\left(u+\frac{u^2}{2}+....+\frac{u^9}{9}\right)$

(C) $-\left(u+\frac{u^2}{2!}+....+\frac{u^9}{9!}\right)$ (D) $\frac{u}{2}+\frac{2u^2}{3}+....+\frac{9u^9}{10}$

31. The value of $\int_0^1 \{nx\}^{\frac{1}{3}}dx$ is - ($n \in N$ & {.} represents fractional path fn) :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) dependent on 'x'

32. The value of definite integral $\int_{-a}^{a}\frac{x^2\cos x+e^x}{e^x+1}dx$ is equal to :

(A) $2a\cos(a) - a + (a^2-2)\sin(a)$ (B) $2a\cos(a) + a + (a^2-2)\sin(a)$
(C) $2a\cos(a) - a - (a^2-2)\sin(a)$ (D) $2a\cos(a) + a + (a^2+2)\sin(a)$

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33. If $L = \lim\limits_{n\to\infty} \frac{1}{n^4}\prod\limits_{i=1}^{2n}(n^2+i^2)^{\frac{1}{n}}$, then ln L is equal to :

(A) $\ln 2 + \frac{\pi}{2} - 2$ (B) $2\tan^{-1} 2 - 4 + 2\ln 5$

(C) $2\tan^{-1} 2 + 4 + 2\ln 5$ (D) $2\ln 5 - 4 - 2\tan^{-1} 2$

34. The value of the definite integral $\int\limits_{-1}^{1} e^{-x^4}\left(1+\ln(x+\sqrt{x^2+1})+5x^3-4x^4\right)dx$ is equal to :

(A) 4e (B) $\frac{4}{e}$ (C) 2e (D) $\frac{2}{e}$

35. If $I(m,n) = \int\limits_0^1 x^m \log^n x\,dx$, then I(m,n) (where $m,n \in \mathbb{N}$) is equal to :

(A) $\frac{(-1)^n n!}{(m+1)^{n+1}}$ (B) $\frac{n!}{(m+1)^{n+1}}$ (C) $\frac{(-1)^n(n-1)!}{(m+1)^{n+1}}$ (D) $\frac{(n-1)!}{(m+1)^{n+1}}$

36. The value of definite integral $\int\limits_{-\pi}^{\pi} \frac{x^2}{1+\sin x+\sqrt{1+\sin^2 x}}dx$, is :

(A) $\frac{3x^2}{2}$ (B) $\frac{\pi^3}{3}$ (C) $\frac{2\pi^3}{3}$ (D) $\frac{\pi^3}{6}$

37. $\lim\limits_{x\to 0} \frac{\int\limits_0^x (t^2+e^{t^2})^{\frac{1}{1-\cos t}}dt}{(e^x-1)}$ is equal to :

(A) e^4 (B) e^2 (C) e^3 (D) e

38. The value of the definite integral $\int\limits_{-1}^{1} \frac{dx}{1+x^3+\sqrt{1+x^6}}$ equals :

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\sqrt{2}$

39. Let $f : R \to [4, \infty)$ be an onto quadratic function whose leading coefficient is 1, such that $f'(x) + f'(2-x) = 0$. Then the value of $\int\limits_1^3 \frac{dx}{f(x)}$, is equal to :

(A) $\frac{1}{4}$ (B) $\frac{1}{8}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{8}$

40. Let $V_n = \int\limits_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x}dx$. If $\sum\limits_{r=1}^{100} V_r = \frac{k\pi}{2}$, where $k \in N$, then k is equal to :

(A) 100 (B) 2525 (C) 5050 (D) 4950

41. $\int\limits_0^{\pi/2} \frac{\sin^3 x}{(\cos^4 x+3\cos^2 x+1)\tan^{-1}(\sec x+\cos x)}dx$ is equal to :

(A) $\frac{\pi}{2} - \tan^{-1} 2$ (B) $\ln\frac{\pi}{2} - \ln(\tan^{-1} 2)$ (C) $\ln(\tan^{-1} 2)$ (D) $\ln\frac{\pi}{2}$

42. If $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+....+\infty=\frac{\pi^2}{6}$ and $\int_0^1 \frac{\ln(1+x)}{x}dx=\frac{3\pi^2}{k}$ then k equals :

(A) 72 (B) 36 (C) 24 (D) 12

43. Let $I_1=\int_0^x e^{tx}.e^{-t^2}dt$ and $I_2=\int_0^x e^{-t^2/4}dt$ where x > 0 then the value of $\frac{I_1}{I_2}$ is :

(A) $e^{-x^2/2}$ (B) $e^{x^2/4}$ (C) $e^{-x^2/4}$ (D) $e^{x^2/2}$

44. Let $f(a)=\int_0^a \ln(1+\tan a\tan x)dx$, then $f'\left(\frac{\pi}{4}\right)$ equals :

(A) $\frac{\pi}{4}+\frac{\ln 2}{2}$ (B) $\frac{\pi}{2}+\frac{\ln 2}{2}$ (C) $\frac{\pi}{4}+\ln 2$ (D) $\frac{\pi}{2}+\ln 2$

45. If $\int_0^\infty\left(\frac{\sin x}{x}\right)^3 dx=A$ and $\int_0^\infty\left(\frac{x-\sin x}{x^3}\right)dx=\frac{aA}{b}$, where a and b are relative prime then the value of (a+b) equals :

(A) 3 (B) 4 (C) 5 (D) 6

46. The value of $\int_{-\pi}^{\pi}(1+\cos x+\cos 2x+...+\cos(2013x))(1+\sin x+\sin 2x+...+\sin(2013))dx$, is :

(A) 0 (B) π (C) 2π (D) 2013π

Multiple Correct Answer Type

47. If $I=\int_0^1\frac{dx}{1+x^{\pi/2}}$, then

(A) I > ln2 (B) I < ln2 (C) I < $\pi/4$ (D) I > $\pi/4$

48. Let p(x) be a polynomial such that $\frac{p(x+1)}{p(x)}=\frac{x^2+x+1}{x^2-x+1}$ and $p(2)=3$, then $\int_0^1 \tan^{-1}(p(x)).\tan^{-1}\sqrt{\frac{x}{1-x}}dx$ is equal to :-

(A) $\frac{\pi}{2}\ln 2$ (B) $\frac{\pi}{4}\ln 2$ (C) $\frac{\pi}{8}\ln 4$ (D) $\frac{\pi}{4}\ln 4$

49. $I_n=\int_{-\pi}^{\pi}\frac{1}{\left(1+2^{\sin(x/2)}\right)}\left[\frac{\sin\left(\frac{nx}{2}\right)}{\sin\left(\frac{x}{2}\right)}\right]^2 dx$; n = 0, 1, 2, 3,........

(A) $I_n=I_{n+1}\ \forall\ n\geq 1$ (B) $I_0, I_1, I_2, ...\ I_n$ are in A.P.

(C) $\sum_{m=0}^{9} I_{2m}=90\pi$ (D) $\sum_{m=0}^{10} I_m=55\pi$

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50. Let $S_n = \sum_{r=1}^{n}\left(\frac{n^2+nr+r^2}{n^3}\right)$ and $T_n = \sum_{r=0}^{n-1}\left(\frac{n^2+nr+r^2}{n^3}\right)$ for n = 1, 2, 3, ... then

(A) $T_n < \frac{11}{6}$ (B) $T_n > \frac{11}{6}$ (C) $S_n < \frac{11}{6}$ (D) $S_n > \frac{11}{6}$

51. Let $I_1 = \int_0^{\pi/2} \cos\left(\pi\sin^2 x\right)dx$, $I_2 = \int_0^{\pi/2} \cos\left(2\pi\sin^2 x\right)dx$ and $I_3 = \int_0^{\pi/2} \cos\left(\pi\sin x\right)dx$ then -

(A) $I_1 = 0$ (B) $I_2 + I_3 = 0$ (C) $I_1 + I_2 + I_3 = 0$ (D) $I_2 + I_3 \neq 0$

52. Let $I = \int_0^1 \sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}dx$ and $J = \int_0^1 \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}dx$ then correct statements is/are -

(A) $I + J = 4$ (B) $I - J = 2$ (C) $I = \frac{2+\pi}{2}$ (D) $J = \frac{4-\pi}{2}$

53. $\int_{-\infty}^{\infty} \frac{\ln|t|}{x^4+t^2}dt - \frac{\pi}{2}\ln 2 = 0$ (where $x \in \mathbb{R}$), then -

(A) Number of values of x satisfying the equation is 5

(B) Roots of $x^2 - \left(2+\sqrt{2}\right)x + 2\sqrt{2} = 0$, satisfy the above equation

(C) Product of values of x satisfying above equation is 8

(D) Roots of $x^3 + 31x^2 - 2x - 21 = 0$ satisfy the above equation

54. Let $f(x) = \int_0^x \frac{e^t}{t}dt\ (x > 0)$ then $e^{-a}\left[f(x+a) - f(1+a)\right]$ equals -

(A) $\int_0^x \frac{e^t}{t+a}dt$ (B) $\int_1^x \frac{e^t}{t+a}dt$ (C) $e^{-a}\int_{1+a}^{x+a} \frac{e^t}{t}dt$ (D) $\int_0^x \frac{e^{t-a}}{t-a}dt$

55. Let $S_n = \sum_{r=n}^{3n-1} \frac{n}{r^2 - 4rn}$; $S_n' = \sum_{r=n+1}^{3n} \frac{n}{r^2 - 4rn}$

(A) $S_n < \ln\frac{1}{\sqrt{3}}$ (B) $S_n > \ln\frac{1}{\sqrt{3}}$ (C) $S_n' < \ln\frac{1}{\sqrt{3}}$ (D) $S_n > S_n'$

56. $\int_4^8 \left(\sqrt{\sqrt{16x-64}+x} + \sqrt{x-\sqrt{16x-64}}\right)dx$ is equal to -

(A) $2\int_4^6 \left(\sqrt{\sqrt{16x-64}+x} + \sqrt{x-\sqrt{16x-64}}\right)dx$ (B) $2\int_8^{10} \left(\sqrt{\sqrt{16x-64}+x} - \sqrt{x-\sqrt{16x-64}}\right)dx$

(C) $2\int_7^9 \left(\sqrt{\sqrt{16x-64}+x} - \sqrt{x-\sqrt{16x-64}}\right)dx$ (D) 16

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57. If $\int \cos(2016x).\left(\sin^{2014}x\right)dx = f(x)+c, f(0)=0$ then -

(A) f(x) is odd (B) $\int_{\pi}^{2\pi} f(x)dx = 0$ (C) $\int_{0}^{\pi} f(x)dx = 0$ (D) $\lim_{x\to 0}\frac{2015f(x)}{x^{2015}} = 1$

Linked Comprehension Type

Passage for Q.58 to Q.60

Suppose f(x) and g(x) are two continuous functions defined for $0 \le x \le 2$.

Given $f(x) = \int_0^1 e^{x+t}.f(t)dt$ and $g(x) = \int_0^1 e^{x+t}.g(t)dt + x$.

58. The value of f(1) equals :

(A) 0 (B) 1 (C) e^{-1} (D) e

59. The value of g(0) – f(0) equals :

(A) $\frac{2}{3-e^2}$ (B) $\frac{3}{e^2-2}$ (C) $\frac{1}{e^2-1}$ (D) 0

60. The value of $\frac{g(0)}{g(2)}$ equals :

(A) 0 (B) $\frac{1}{3}$ (C) $\frac{1}{e^2}$ (D) $\frac{2}{e^2}$

Passage for Q.61 to Q.63

Let f and g be two real - valued differentiable functions on R satisfying $\int_0^x g(t)dt = 3x + \int_x^0 \cos^2 t\, g(t)dt$

and $f(x) = \lim_{\alpha\to 0}\frac{1}{\alpha^4}\int_0^{\alpha}\frac{\left(e^{x+t}-e^x\right)\ln^2(1+t)}{2t^3+3}dt$.

61. f(ln 2) is greater than -

(A) 0 (B) $\frac{1}{6}$ (C) $\frac{3}{20}$ (D) $\frac{4}{25}$

62. Range of g(x) is equal to -

(A) $\left[\frac{3}{2},3\right]$ (B) $\left[\frac{1}{2},3\right]$ (C) $\left[\frac{3}{2},\infty\right)$ (D) [–3, 3]

63. The value of definite integral $\int_0^{\frac{\pi}{2}} g(x)dx$ lies in the interval -

(A) $\left(\frac{2\pi}{3},\frac{4\pi}{5}\right)$ (B) $\left(\frac{\pi}{2},\frac{5\pi}{6}\right)$ (C) $\left(\frac{3\pi}{4},\frac{6\pi}{5}\right)$ (D) $\left(\pi,\frac{3\pi}{2}\right)$

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Passage for Q.64 & Q.65

Let $f_n(x) = \sum_{n=1}^{n} \frac{\sin^2 x}{\cos^2\left(\frac{x}{2}\right) - \cos^2\left(\frac{2n+1}{2}\right)x}$ and let $g_n(x) = \prod_{n=1}^{n} f_n(x)$ $\forall$ n = 1,2,3.........

64. Let $I_n = \int_0^{\pi} \frac{f_n(x)}{g_n(x)} dx$. If $\sum_{k=1}^{100} I_n = k\pi$ then the value of k is :

(A) 50 (B) 25 (C) 100 (D) 75

65. The value of $\left[\lim_{x\to 0} \int_0^x \frac{9dt}{x f_9(t) g_9(t)}\right]$ is -

(A) 50 (B) 10 (C) 100 (D) 25

Passage for Q.66 to 68

If f(x) satisfies the functional equation $f(mx + ny) = [f(x)]^m [f(y)]^n$ $\forall$ x, y, m, n $\in$ R and $f'(0) = (-\ln 2) f(0)$.

66. $\lim_{n\to\infty} \int_{-2}^{2} \frac{1}{1+(f(x))^n} dx$ is equal to -

(A) 0 (B) 2 (C) 1 (D) 4

67. Number of real solutions of $f(x) = x^2$ is-

(A) 0 (B) 2 (C) 1 (D) 3

68. $\lim_{x\to -\infty} \frac{x^2}{f(x)}$ is -

(A) 0 (B) 2 (C) 1 (D) 3

Passage for Q.69 to 70

Let f : R $\to$ R be a differentiable function $\forall$ x $\in$ R such that for all points (x, y) on the curve y = f(x) , there exists a point (1 – x, 2 – y) on the curve.

69. There the $\int_0^1 f(x) \sin \pi x \, dx$ is equal to -

(A) $\int_0^1 \sin \pi x \, dx$ (B) $\sqrt{2}\int_0^{1/2} \sin \pi\left(x + \frac{1}{4}\right)dx$

(C) $\frac{2}{\pi}$ (D) $\int_2^3 f(x-2) \sin \pi x \, dx$

70. If slope of tangent at (2, 1) on the curve is $\frac{1}{2}$, then there exist atleast one distinct tangent to the curve with slope as $\frac{1}{2}$; is given by -

(A) x – 2y + 3 = 0 (B) 2y – x – 1 = 0

(C) 2y – x + 3 = 0 (D) 2x – y + 3 = 0

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Subjective Type Questions

71. Let $f(x)=\lim\limits_{n\to\infty}\left(\dfrac{n!}{n^n}\right)^{\frac{1}{n}}$ then the value of $[e^2 f(x)]$, where [*] denotes G.I.F

72. If $S_n=\dfrac{1}{2n}+\dfrac{1}{\sqrt{4n^2-1}}+\dfrac{1}{\sqrt{4n^2-4}}+.....\dfrac{1}{\sqrt{3n^2+2n-1}}$ is $n\in N$ then $\lim\limits_{n\to\infty}S_n$ is $\dfrac{\pi}{N}$, the N is -

73. $I_1=\displaystyle\int_0^{\infty}\frac{|\ln x|}{1+3x+x^2}dx$ and $I_2=\displaystyle\int_0^{\infty}\frac{x}{e^x+e^{-x}+3}dx$, then $\dfrac{I_1}{I_2}$ is equal to -

74. Let $f(x)+f\left(x+\dfrac{1}{2}\right)=1\ \forall\ x\in R$ and $I_1=\displaystyle\int_0^3 f(x)dx$ & $I_2=\displaystyle\int_1^2 f(x)dx$ then $\dfrac{I_1}{I_2}$ is -

75. Evaluate $\displaystyle\int_{-4}^{-5}e^{(x+5)^2}dx+3\int_{1/3}^{2/3}e^{9\left(x-\frac{2}{3}\right)^2}dx$

76. If $I=\displaystyle\int_1^3\left(\tan^{-1}\left(\frac{x^2-5x+6}{x^3-6x^2+12x-7}\right)+\tan^{-1}\left(\frac{1}{x^2-4x+4}\right)dx\right)$, then find the value of [I]

where [.] denotes the greatest integer function.

77. Find the value of $\displaystyle\int_0^x[t+1]^3dt$, where [.] denotes the greatest integer function and $x\in R^+$.

78. $I_1=\displaystyle\int_{-\infty}^{\infty}\frac{x^3+1}{x^5+1}dx$ and $I_2=\displaystyle\int_{-1}^{1}\frac{x^5-x^3}{x^5+1}dx$, then I_1+2I_2 is equal to -

79. $\lim\limits_{x\to\infty}\left[x\left(\dfrac{1}{(2x+1)^2}+\dfrac{1}{(2x+3)^2}+...+\dfrac{1}{(4x-1)^2}\right)\right]=L$, then 8L equals to -

80. Le the line $y = 100x - 199$ be intersect the graph of a function $f(x) = x^3-6x^2+ax-2a+17$. $a\in R$ at three distinct points whose abscissae are x_1, x_2 and x_3 $(x_1 < x_2 < x_3)$ such that $x_3 - x_1 = 6$.

If $\displaystyle\int_{x_1}^{x_3}(f(x)-98x+198)dx=\lambda$, the n find the value of $\lambda/2$

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Matrix Match Type

81. Match the column -

Column-I	Column-II
(A) $\displaystyle\int_4^{10}\frac{[x^2]dx}{[x^2-28x+196]+[x^2]}=$	(1) $\frac{1}{100}$
(B) $\displaystyle\int_{-1}^{2}\frac{\lvert x\rvert dx}{x}$	(2) 3
(C) $\displaystyle\lim_{x\to\infty}\frac{1^{99}+2^{99}+3^{99}.....x^{99}}{x^{100}}=2A$ then value of A	(3) 1
(D) $\displaystyle 5050\int_{-1}^{1}\sqrt{x^{200}}dx=\frac{1}{\alpha}$ then α =	(4) $\frac{1}{200}$

ANSWER KEY

1. C	**2.** C	**3.** A	**4.** B	**5.** A	**6.** A	**7.** C
8. A	**9.** B	**10.** A	**11.** D	**12.** B	**13.** C	**14.** D
15. A	**16.** A	**17.** C	**18.** B	**19.** C	**20.** B	**21.** A
22. B	**23.** C	**24.** A	**25.** A	**26.** B	**27.** B	**28.** B
29. A	**30.** B	**31.** C	**32.** B	**33.** B	**34.** D	**35.** A
36. B	**37.** A	**38.** B	**39.** D	**40.** C	**41.** B	**42.** B
43. B	**44.** A	**45.** C	**46.** C	**47.** A,C	**48.** B,C	**49.** B,C,D
50. A,D	**51.** A,B,C	**52.** A,D	**53.** B,C	**54.** B,C	**55.** A,C	**56.** A,B,D
57. A,B,C,D	**58.** A	**59.** A	**60.** B	**61.** A,C,D	**62.** A	**63.** C,D
64. A	**65.** C	**66.** B	**67.** D	**68.** A	**69.** A,B,C,D	**70.** A
71. 2	**72.** 6	**73.** 2	**74.** 3	**75.** 0	**76.** 3	

77. $\left(\frac{[x]([x]+1)}{2}\right)^2+([x]+1)^3\{x\}$ **78.** 4 **79.** 1 **80.** 9

81. A-2 ; B-3 ; C-4 ; D-1

NAME : ....

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PART # 01

AREA UNDER CURVE

Straight Objective Type

1. Area bounded by curves $|y| \geq e^{-|x|} - \frac{1}{2}$ and $\frac{|x|+|y|}{2} + \left|\frac{|x|-|y|}{2}\right| \leq 2$

(A) $14 + \ell n2$ (B) $14 + \ell n4$ (C) $7 + \ell n4$ (D) $7 + \ell n\ 2$

2. Let f(x) be a continuous function given by $f(x) = \begin{cases} 2x & |x| \leq 1 \\ x^2 + ax + b & |x| > 1 \end{cases}$

The area of the region in the third quadrant bounded by the curve $x = -2y^2$ and $y = f(x)$ lying on the left of line $8x + 1 = 0$

(A) $\frac{257}{192}$ sq. unit (B) $\frac{192}{257}$ sq. unit (C) $\frac{157}{192}$ sq. unit (D) $\frac{192}{157}$ sq. unit

3. Consider a square ABCD of side length 1. Let P be set of all segments of length 1 with end points on adjacent sides of square ABCD. The mid points of segments in P enclose a region with area A, the value of A is

(A) $\frac{\pi}{4}$ (B) $1-\frac{\pi}{4}$ (C) $4-\frac{\pi}{4}$ (D) $2-\frac{\pi}{4}$

4. Area enclosed between the curves $|y| = 1 - x^2$ and $x^2 + y^2 = 1$ is

(A) $\frac{3\pi-8}{3}$ (B) $\frac{\pi-8}{3}$ (C) $\frac{2\pi-8}{3}$ (D) $\frac{4\pi-8}{3}$

5. Area bounded by curves $y^2 = 16|x|$ & $x^2 = 3|y|$ is-

(A) 48 (B) 64 (C) 48 (D) 16

6. The area bounded by the curve $\frac{[|x-1|]}{[|y-1|]} + \frac{[|y-1|]}{[|x-1|]} = 2$ is -

(where $-2 \leq x \leq 2$, $-2 \leq y \leq 2$ and [.] denotes greatest integer function)

(A) 2 (B) 3 (C) 6 (D) 9

7. The area of figure enclosed by the curve $5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) π (D) 2π

8. Let f be differentiable function such that

$(x - y)\, f(x + y) - (x + y)f(x - y) = 4xy(x^2 - y^2)$

and $f(1) = 2$. Then area enclosed by

$\frac{|f(x)-x|^{\frac{1}{3}}}{17} + \frac{|f(y)-y|^{\frac{1}{3}}}{2} \leq \frac{1}{4}$, is

(A) $\frac{3f(4)}{4}$ sq. units (B) $\frac{f(4)}{8}$ sq. units (C) $\frac{f(4)}{16}$ sq. units (D) $\frac{3f(4)}{16}$ sq. units

9. Area bounded by $y \leq 3 - |3 - x|$ and $y \geq |x - 3|$ is

(A) $\frac{5}{2}$ (B) $\frac{9}{2}$ (C) $\frac{7}{2}$ (D) 4

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10. Let A be area between co-ordinate axes, $y^2 = x - 1$, $x^2 = y - 1$ and the line which makes the shortest distance between two parabolas and A' be the area between $x = 0$, $x^2 = y - 1$ $x = y$ and shortest segment between $y^2 = x - 1$ and $x^2 = y - 1$ then

(A) $A = A'$ (B) $A = (A')^{1/2}$ (C) $A = 2A'$ (D) $2A = A'$

11. Area bounded by curves $y = [\cos A + \cos B + \cos C]$, $y = \left[7\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right]$ and curve $|x - 4| + |y| = 2$ is

(where [.] denotes greatest integer function and A, B, C are vertices of triangle)

(A) 2 (B) 3 (C) 4 (D) 6

12. Area bounded by $[x] + [y] = n$, $(n \in N)$ in the first quadrant (where [.] denotes greatest integer function) is

(A) n Sq. unit (B) (n + 1) Sq. unit (C) (n – 1) Sq. unit (D) (n – 2) Sq. unit

13. For which of the following values of m, the area of region bounded by the curve $y = x - x^2$ and the line $y = mx$ equals $\frac{9}{2}$

(A) –4 (B) –2 (C) 2 (D) 4

Multiple Correct Answer Type

14. The curve $C_1 : y = e^{-x}$ and $C_2 : y = e^{-x}\sin x$ $(x > 0)$ touch each other at infinitely many points. Let $0 < x_1 < x_2 < .... < x_n < ......$ be the abscissa of these points of contact. If A_n denotes the area bounded by two curves & ordinates $x = x_n$ & $x = x_{n+1}$, then

(A) $A_1 = \frac{1}{2}\frac{(e^{2\pi} - 1)}{e^{5\pi/2}}$ (B) $A_2 = \frac{1}{2}\frac{(e^{2\pi} - 1)}{e^{9\pi/2}}$

(C) A_1, A_2, A_3 are in A.P. (D) A_1, A_2, A_3 are in G.P.

15. If a polynomial function $y = f(x)$ satisfying the conditions $f(x) + f(y) = f(x).f(y) + f(xy)$ where $f(1) = 0$ & $f'(1) = -2$ and the area bounded by $y = f(x)$ and $y = |\cos^{-1}(\cos x) - \sin^{-1}(\sin x)|$ for $x \in \left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ is A, then

(A) function $y = f(x)$ is $1 + x^2$ (B) function $y = f(x)$ is $1 - x^2$

(C) $A = \frac{4\sqrt{2} - 3}{3}$ Sq. unit (D) $A = \frac{6 + 2\sqrt{2}}{3}$ Sq. unit

16. If $A = \left(\frac{3}{\sqrt{2}}, \sqrt{2}\right)$, $B = \left(\frac{-3}{\sqrt{2}}, \sqrt{2}\right)$, $C = \left(\frac{-3}{\sqrt{2}}, -\sqrt{2}\right)$ and $D = (3\cos\theta, 2\sin\theta)$ are four points, then value of θ for which area of quadrilateral ABCD is maximum is α, (where $\theta \in \left(\frac{3\pi}{2}, 2\pi\right)$) such that choose correct option(s).

(A) maximum area of ABCD 10 Sq. unit (B) $\alpha = \frac{7\pi}{4}$

(C) $\alpha = 2\pi - \sin^{-1}\left(\frac{3}{\sqrt{85}}\right)$ (D) Maximum area of ABCD is 12 Sq. unit

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Linked Comprehension Type

Paragraph for Question 17 to 19

Answer the following questions based on evaluation of area of figure which are represented by parametric equation

17. The area enclosed by the curve $\left(\frac{x}{a}\right)^{2/3}+\left(\frac{y}{a}\right)^{2/3}=1$ is

(A) $\frac{3}{4}a^2\pi$ (B) $\frac{3}{18}\pi a^2$ (C) $\frac{3}{8}\pi a^2$ (D) $\frac{3}{4}a\pi$

18. The area of the region bounded by an arc of the cycloid x = a(t – sint), y = a(1 – cost) and the x-axis is

(A) $6\pi a^2$ (B) $3\pi a^2$ (C) $4\pi a^2$ (D) $2\pi a^2$

19. Area of loop described as $x=\frac{t}{3}(6-t)$, $y=\frac{t^2}{8}(6-t)$ is

(A) $\frac{27}{5}$ (B) $\frac{24}{5}$ (C) $\frac{27}{6}$ (D) $\frac{21}{5}$

Paragraph for Questions 20 & 21

A curve y = f(x) passes through the point P(1, 1). The normal to the curve at P is a(y – 1) + (x – 1) = 0 and the slope of the tangent at any point on the curve is proportional to the ordinate of that point.

20. Area bounded by the curve, y-axis, and the normal to the curve at P is

(A) $2+\frac{1}{2a}+\frac{1}{ae^a}$ (B) $1-\frac{1}{2a}+\frac{1}{ae^a}$ (C) $1-\frac{1}{3a}+\frac{1}{2ae^a}$ (D) $1-\frac{1}{a}+\frac{1}{3ae^a}$

21. If A_1 is the area in the above question and A_2 is the area bounded between the curve, normal at P, x-axis and y-axis, then $\left[\frac{A_2}{A_1}\right]$, when a = 1 is - {[·] greatest integer function}

(A) 1 (B) 2 (C) 3 (D) 4

Numerical Grid

22. Area of the region bounded by the curve f(x) = max.$\{4-x^2, |x-2|, (x-2)^{1/3}\}$ for $x \in [2, 4]$ and x-axis is $\frac{p}{q}$ then value of (p + q) is equal to (where p & q are coprime)

23. The area bounded by y = f(x), y-axis and line 2y = π(x + 1)

where $f(x) = \sin^{-1}x + \cos^{-1}x + \tan^{-1}x + \tan^{-1}\frac{1}{x}$ is $\frac{\pi}{k}$ then k is equal to

24. Let $S=\left\{(x,y);\ \frac{y(3x-1)}{x(3x-2)}<0\right\}$ and S' = {(x, y) ∈ A × B; –1 ≤ A ≤ 1 and –1 ≤ B ≤ 1}. Then area of $S \cap S'$ is

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25. A 2×3 rectangular plate has vertices at (0,0), (2,0), (0,3) & (2,3). It rotates 90° clockwise about the point (2,0). It then rotates 90° clockwise about the point (5,0), then 90° clockwise about (7,0) and finally 90° clockwise about (10,0). Let S be the curve traced by the point on the plate which was initially located at (1,1) and the area bounded by S & x–axis is $\ell + \frac{m}{2}\pi$ then $m - \ell$ is equal to (where $\ell, m \in N$)

26. Let $f(x)$ be a differentiable function defined for all $x > 0$ and $\lim_{x \to 0^+} f(x) = 0$. Also $f(x)$ satisfies

(i) $f(xy) = xf(y) + yf(x) \ \forall \ x, y \in R^+$ (ii) $\int_0^x f(x)dx = \frac{f(x^2) - x^2}{4}$

If the area enclosed by $y = f(x)$ lying in the fourth quadrant is S, then $\log_{\sqrt{6}}\left(\frac{9}{S}\right)$ is equal to

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PART # 02
DIFFERENTIAL EQUATION
Straight Objective Type

1. Solution of the differential equation $xy\left(mydx + nxdy\right) = \dfrac{xdy + ydx}{x^m y^n}$, given m + n = 1 is

(A) $x^{m+1}.y^{n+1} + 1 = \dfrac{cx}{y}$ (B) $x^{m+1}.y^{n+1} - 1 = cxy$

(C) $x^{m+1}.y^{n+1} = cxy - 1$ (D) $x^m y^n + 1 = cxy$

2. If $y = f(x)$ be a curve passing through (e, e^e) and which satisfy the differential equation $(2ny + xy\log x)dx - x\log x dy = 0$, value of $\int_{1/e}^{e} g(x)dx$ (where $g(x) = \lim_{n\to\infty} f(x)$) is

(A) 0 (B) 1 (C) 2 (D) –1

3. Solution of the differential equation $x^6dy + 3x^5ydx = xdy - 2ydx$ is

(A) $x^3y = \dfrac{y}{x^2} + c$ (B) $x^3y = \dfrac{2y}{x^2} + c$ (C) $x^3y^2 = \dfrac{y}{x^2} + c$ (D) $x^3 = \dfrac{y}{x^2} + c$

4. Solution of differential equation $\dfrac{dy}{dx} = \dfrac{2x^3y + 3x^4 + y}{x - x^4}$ is

(A) $x^2y + x^3 = \dfrac{y}{x} + c$ (B) $x^2y + 2x^3 = \dfrac{y}{x} + c$ (C) $x^2y + x^3 = \dfrac{2y}{x} + c$ (D) $x^2 + y = \dfrac{y}{x} + c$

5. Solution of the differential equation $(xdy - ydx)(x + y)^2 = 4xy(x^2 + y^2)(xdx - ydy)$

(A) $\tan^{-1}\dfrac{x}{y} = \dfrac{1}{2}\log_e\dfrac{x}{y} + x^2 - y^2 + c$ (B) $\tan^{-1}\dfrac{y}{x} = \dfrac{1}{2}\log_e\left(\dfrac{x}{y}\right) + x^2 + y^2 + c$

(C) $\tan^{-1}\dfrac{y}{x} = \dfrac{1}{2}\log_e\dfrac{x}{y} + x^2 - y^2 + c$ (D) $\tan^{-1}\dfrac{y}{x} = \dfrac{1}{2}\log_e\dfrac{x}{y} + x^2 - y^2 + c$

6. Solution of differential equation $\dfrac{x + y\dfrac{dy}{dx}}{y - x\dfrac{dy}{dx}} = \dfrac{x\sin^2(x^2 + y^2)}{y^3}$

(A) $\cot(x^2 + y^2) + x^2 + c = 0$ (B) $y^2\cot(x^2 + y^2) + x^2 + c = 0$

(C) $y^2\cot(x^2 + y^2) - x^2 + c = 0$ (D) $y^2\cot(x^2 + y^2) + x^2 + cy^2 = 0$

7. The differential equation of family of ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = c$ is $\left(y' = \dfrac{dy}{dx}, y'' = \dfrac{d^2y}{dx^2}\right)$

(A) $\dfrac{y''}{y'} + \dfrac{y'}{y} - \dfrac{1}{x} = 0$ (B) $\dfrac{y''}{y'} + \dfrac{y'}{y} + \dfrac{1}{x} = 0$ (C) $\dfrac{y''}{y'} - \dfrac{y'}{y} - \dfrac{1}{x} = 0$ (D) $\dfrac{y''}{y'} - \dfrac{y'}{y} = 0$

8. Solution of differential equation given by $\dfrac{dy}{dx} = \dfrac{x^2 + y + 1}{x(1 - x\sin y)}$, where y(1) = 0, is

(A) $y + x\cos y = x^2 + x - 1$ (B) $y + x\cos y = x^2 - x + 1$

(C) $y + x\cos y = -x^2 + x + 1$ (D) $y + x\cos y = x^2 + x + 1$.

9. Let f(x, y) = 0 be the solution of the differentiable equation $(2y^2x \log x + x^2)\frac{dy}{dx} + xy \log y + y^3 + y = 0$ (where f(x, y) = 0 passes through (1, 1) and x > 0, y > 0). If x = α is the point of intersection of f(x, y) = 0 and y = 2 then

(A) $0 < \alpha < \frac{1}{2}$ (B) $\frac{1}{2} < \alpha < 1$ (C) $1 < \alpha < 2$ (D) $\alpha > 2$

10. Uranium disintegrates at a rate proportional to the amount present at any instant. If 6 and 3 grams of uranium are present at time t_1 and t_2 respectively. Then the **half life** of uranium is-

(A) $\frac{(t_2 - t_1)}{\ell n2}$ (B) $(t_2 - t_1)$ (C) $\frac{\ell n2}{t_2 - t_1}$ (D) $\frac{1}{2}(t_2 - t_1)$

11. Solution of differencial equation $\frac{dy}{dx} + \frac{1}{x}\tan y = \frac{1}{x^2}\tan y.\sin y$ is

(A) $2y = \sin y\,(1 - cx^2)$ (B) $2x = \cot y\,(1 + 2cx^2)$
(C) $2x = \sin y\,(1 + 2cx^2)$ (D) $2x \sin y = 1 - 2cx^2$

12. Given y = f(x) is a solution of differential equation xdy + xdx = ydy – ydx, also $f(0) = 0$, $f(1) = \tan\frac{3\pi}{8}$, then the area bounded by y = f(x) on x-axis & x = 0 to $x = \tan\frac{\pi}{8}$ is equal to

(A) $\frac{1}{2}(\sqrt{2} - 1)$ (B) $\frac{1}{2}(\sqrt{2} + 1)^3$ (C) $\frac{1}{2}(\sqrt{2} - 1)^3$ (D) $\frac{1}{2}(\sqrt{2} + 1)^2$

13. Suppose f(x) is a differentiable real function such that $f(x) + f'(x) \le 1$ for all x and $f(0) = 0$. The largest possible value of $f(1)$ is

(A) 1 (B) e (C) $\frac{1}{e}$ (D) $\frac{(e-1)}{e}$

14. If the solution of differential equation $x^3\frac{dy}{dx} = y^3 + y^2\sqrt{y^2 - x^2}$ is $cxy = y + k\sqrt{y^2 - x^2}$ (c is arbitrary) then

(A) k = 1 (B) k = 2 (C) $k = \frac{1}{2}$ (D) $K = \frac{3}{2}$

15. If $y^{1/m} + y^{-1/m} = 2x$ then $(x^2 - 1)\frac{d^3y}{dx^3} + 3x\frac{d^2y}{dx^2} + f(m)\frac{dy}{dx} = 0$, then f(m) is equal to

(A) $1 + m^2$ (B) $2 - m^2$ (C) $m + m^2$ (D) $1 - m^2$

16. Solution of differential equation $x^2\frac{dy}{dx} + y^2e^{\frac{x(y-x)}{y}} = 2y(x - y)$ be given by

(A) $x(x + y) = y \log (Ce^x - 1)$ (B) $x(x - y) = x \log (Ce^x - 1)$
(C) $x(x + y) = x\log (Ce^x + 1)$ (D) $x(x - y) = y \log (Ce^x - 1)$
Where C is constant of integration

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17. The function $f(x)$ satisfying $\{f(x)\}^2 + 4f(x)f'(x) + \{f'(x)\}^2 = 0$ is given by

(A) $ke^{(2+\sqrt{3})x}$ (B) $k.e^{(4\pm\sqrt{5})x}$ (C) $k.e^{(-2\pm\sqrt{3})x}$ (D) $k.\log(2+\sqrt{3})x$

18. Consider curve, $r^2 = a\sin 2\theta$ then its orthogonal trajectory is

(A) $r^2 = C$ (B) $r^2 = C\sin\theta$ (C) $r^2 = C\cos 2\theta$ (D) $r^2 = C\cos\theta$

19. If the given curves satisfies the differential equation, $e^4dx + (xe^4 + 2y)dy = 0$, and also passes through (0, 0) then the possible equation of curve can be

(A) $xe^4 + 4 = 0$ (B) $x + y^2e^y = 0$

(C) $x^2e^x + ye^4 = 1$ (D) $x.e^y = -\frac{2}{e^4}\left(ye^y - e^y + 1\right)$

20. The differential equation $\{xy^3(1 + \cos x) - y\}\,dx + xdy = 0$, represent the curve $\frac{x^2}{ay^2} = \frac{x^3}{b} + x^2\sin x + cx\cos x - d\sin x + k$. Then a + b + c + d is equal to

(A) 8 (B) 9 (C) 10 (D) 11

Multiple Correct Answer Type

21. Let $y = f(x)$ is the solution of the differential equation $y\frac{d^2y}{dx^2} - 2\left(\frac{dy}{dx}\right)^2 = 0$, $f(1) = 1$. S is area bounded by curve $y = f(x)$, $y = f^{-1}(x)$ and $xy = 0$. If $f^{-1}(2) = 0$, then which of the following statement(s) can be true ?

(A) Angle of intersection of $y = f(x)$ and $y = f^{-1}(x)$ is $\tan^{-1}\left(\frac{3}{4}\right)$

(B) Angle of intersection of $y = f(x)$ and $y = f^{-1}(x)$ is $\frac{\pi}{2}$

(C) [S] = 1

(D) [S] = 2

(where [S] denotes greatest integer less than or equal to S)

22. Radioactive substances A & B decay following the first order differential equation where rate of decay is proportional to amount of substance. Initially at t = 0, $\frac{da}{dt} : \frac{db}{dt} = 2:1$ after 1 hour this ratio is $\frac{5}{2}$ then (where a, b are amount of substances A & B). Select most appropriate option(s)

(A) After two hour ratio will be $\frac{25}{8}$

(B) Let $f(t) = \frac{a}{b}$ at time t then $f'(x)$ proportional to $f(x)$

(C) half life of A & B is different (time taken to decay to half of original amount)

(D) After two hour ratio will be $\frac{25}{4}$

23. Solution of differential equation $ydx(1 + x^2.\ell n(xy)) + xdy(1 + y^2\ell n(xy)) = 0$ is

(A) $\ell n(\ell nxy) + x^2 + y^2 + c = 0$ (B) $2\ell n(\ell nxy) + x^2 + y^2 + c = 0$

(C) $\ell n(xy) = ce^{\frac{(x^2+y^2)}{2}}$ (D) $\ell n(xy) = c.e^{-\frac{(x^2+y^2)}{2}}$

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Linked Comprehension Type

Paragraph for Question 24 to 25

If any differential equation in the form $f(f_1(x, y)).d(f_1(x, y)) + Q(f_2(x, y)).d(f_2(x, y)) + ... = 0$ then each then can be integrated separately.
Answer following question

24. The solution of the differential equation $xdy - ydx = \sqrt{x^2 - y^2}\ dx$ is

(A) $cx = e^{\sin^{-1}(y/x)}$ (B) $xe^{\sin^{-1}(y/x)} = c$ (C) $x + e^{\sin^{-1}(y/x)} = c$ (D) $x - e^{\sin^{-1}(y/x)} = c$

25. The solution of the differential equation $(xy^4 + y)dx - xdy = 0$ is

(A) $\frac{x^3}{4} + \frac{1}{2}\left(\frac{x}{y}\right)^2 = c$ (B) $\frac{x^4}{4} + \frac{1}{3}\left(\frac{x}{y}\right)^3 = c$ (C) $\frac{x^4}{4} - \frac{1}{2}\left(\frac{x}{y}\right)^2 = c$ (D) $\frac{x^3}{4} - \frac{1}{2}\left(\frac{x}{y}\right)^2 = c$

26. Solution of differential equation $(2x\cos y + y^2\cos x)dx + (2y\sin x - x^2\sin y)dy = 0$, is
(A) $x^2\cos y + y^2\sin x = c$ (B) $x\cos y - y\sin x = c$
(C) $x^2\cos^2 y + y^2\sin^2 x = c$ (D) $x\cos y + y\sin x = c$

Paragraph for Question 27 to 28

Let the curve $y = f(x)$ passes through the point (4,–2) and satisfying the differentiable equation

$y\left(x + y^3\right)dx = x\left(y^3 - x\right)dy$ and $g(x) = \int_{1/8}^{\sin^2 x} \sin^{-1}\sqrt{t}\,dt + \int_{1/8}^{\cos^2 x} \cos^{-1}\sqrt{t}\,dt$. $\left(0 \le x \le \frac{\pi}{2}\right)$

27. The area of the region bounded by curves $f(x)$, $g(x)$ and $x = 0$ is-

(A) $\frac{k^4}{8}, k = \frac{3\pi}{16}$ (B) $k^2 + 8, k = \frac{3\pi}{16}$ (C) $k^4, k = \frac{3\pi}{16}$ (D) $\frac{1}{8}\left(k^2 + 8\right), k = \frac{3\pi}{16}$

28. Number of solutions of equation $f(x) + x = 0$ are-
(A) 1 (B) 2 (C) 3 (D) 0

Numerical Grid

29. Solve the differential equation $(x^2 + 4y^2 + 4xy)dy = (2x + 4y + 1)dx$

30. The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 4} = \frac{7x^6 + 2x}{\sqrt{4 - x^2}}$ in $(-2,2)$

satisfying $f(0) = 1$. If $\int_{-\sqrt{3}}^{\sqrt{3}} f(x)dx = a\pi + b\sqrt{3}$, a,b are rational numbers, then value of $3a + 2b$ is

31. Let the function $f : [0, 2] \to [0, \infty)$ satisfies $f^3(x)f''(x) = -1$ for all $x \in (0, 2)$ and $f'(1) + f(1) = f(1) - f'(1) = 1$. If the area bounded by $f(x)$ with x-axis is S, then $\sqrt{\frac{8S}{\pi}}$ is equal to

32. The small orifice between two communicating vessels have area 1 sq. cm. The surface area of the horizontal sections of the first and the second vessel are 10 m^2 and 5 m^2 respectively. Initially the level of liquid in the first and second vessel is 15 m and 5 m from the orifice. Given that velocity of liquid flow is $v = \sigma\sqrt{2g(h_1 - h_2)}$ m/s, where h_1 and h_2 is the height of liquid in the two vessels, then the time required for the liquid to reach the same level is $\frac{10^t}{15\sigma\sqrt{20g}}$, where t is equal to

33. If the solution of differential equation $yx^2\,dy - y^2x\,dx = e^{y^{-3}}\,dy$ is $kx^2 = 2y^2 e^{y^{-3}} + Cy^2$;
(where C is integration constant) then the value of k is

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ANSWER KEY

PART # 01
AREA UNDER CURVE

1.	B	**2.**	A	**3.**	B	**4.**	A	**5.**	B
6.	A	**7.**	B	**8.**	C	**9.**	B	**10.**	C
11.	B	**12.**	B	**13.**	B	**14.**	A,B,D	**15.**	B,C
16.	B,D	**17.**	C	**18.**	B	**19.**	A	**20.**	B
21.	A	**22.**	13	**23.**	2	**24.**	2	**25.**	1
26.	4								

PART # 02
DIFFERENTIAL EQUATION

1.	C	**2.**	A	**3.**	A	**4.**	A	**5.**	C
6.	D	**7.**	A	**8.**	A	**9.**	B	**10.**	B
11.	C	**12.**	A	**13.**	D	**14.**	A	**15.**	D
16.	D	**17.**	C	**18.**	C	**19.**	D	**20.**	B
21.	A,C	**22.**	A,B,C	**23.**	B,D	**24.**	A	**25.**	B
26.	A	**27.**	A	**28.**	C				

29. $y = \log_e\left(\left(x+2y\right)^2 + 4\left(x+2y\right)+2\right) - \frac{3}{2\sqrt{2}}\log_e\left(\frac{x+2y+2-\sqrt{2}}{x+2y+2+\sqrt{2}}\right) + c$

30.	6	**31.**	2	**32.**	7	**33.**	3

Only one option is correct :

1. Let A and B are two non-singular square matrices of order n with real entries such that adjA = adjB, then which of the following is necessarily true-

(A) A = B if n is even (B) A = –B if n is even

(C) A = – B if n is odd (D) A = B if n is odd

2. Given A and B are two non-singular matrices such that $B \neq I$, $A^5 = I$ and $AB^2 = BA$, then the least value of n for which $B^n = I$ is-

(A) 63 (B) 64 (C) 31 (D) 32

3. Let p, q, r are three real numbers satisfying $[p\ \ q\ \ r]\begin{bmatrix} 2 & p & q \\ -3 & q & -p+r \\ 12 & r & -q+3r \end{bmatrix} = [5\ \ b\ \ c]$, then minimum value of (b + c) is -

(A) $\frac{25}{157}$ (B) $\frac{25}{49}$ (C) $\frac{25 \times 271}{(49)^2}$ (D) $\frac{25 \times 589}{(157)^2}$

4. AB = A and BA = B, then (here A & B are matrix of n × n) which of the following must be true -

(A) A = B (B) $A^2 = A$ (C) A = I (D) B = I

5. Let B is set of all 4 × 4 skew symmetric matrices. If there are exactly 4 0's, six 3's and six (–3)'s, then number of such matrices is-

(A) 32 (B) 64 (C) 50 (D) 0

6. Let M denotes the set of all 2 × 2 matrices. Define the relation $R = \{(A, B) \in M \times M : A + B^T$ is a symmetric matrix$\}$, then R is -

(A) symmetric, transitive but not reflexive relation

(B) reflexive, symmetric but not transitive relation

(C) reflexive, transitive but not symmetric relation

(D) equivalence relation

7. Let A is a four order diagonal matrix whose entries are complex numbers such that $A^4 = I_4$. If trace of A is zero then (here I_4 is four order unit matrix)

(A) There will be 24 distinct matrices A

(B) There will be 36 distinct matrices A

(C) Determinant value of all such matrices is 1

(D) Determinant value of all such matrices is –1

8. If X is a column matrix of order (3 × 1) and $A = XX^T$, then $A^{-1} =$

(A) A + I (B) A – I (C) 2A + I (D) does not exist

9. Let $A = \begin{bmatrix} 5 & -3 & 0 \\ -3 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ be a matrix. If 't' is a real number such that $A^2X = tAX$ for some non-zero column matrix X with non-zero elements, then the number of distinct real values of 't' is

(A) 0 (B) 1 (C) 2 (D) 3

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10. Given $A = (I - B)(I + B)^{-1}$, where B is a skew-symmetric matrix of order 'n' and $I - B$, $I + B$ are non-singular matrices. If $\det(nA) - \det(\operatorname{adj}A) = 26$, then the value of 'n' is equal to (where det(X) denotes determinant of square matrix X)-

(A) 2 (B) 3 (C) 4 (D) 5

11. If the matrices A, B, (A + B) are non-singular (where A and B are of same order), then $(A(A+B)^{-1}B)^{-1}$ is equal to -

(A) $A + B$ (B) $A^{-1} + B^{-1}$ (C) $(A + B)^{-1}$ (D) AB

12. Let $A = \begin{bmatrix} 3 & 0 & 5 & 6 & -2 \\ -2 & -5 & 0 & 1 & y \\ 5 & 2 & x & 8 & 0 \\ 0 & -3 & 2 & 3 & -5 \\ -4 & -7 & -2 & -1 & -9 \end{bmatrix}$ is a 5×5 square matrix.

Each row and column of matrix A has a value assigned to it. Every element is the sum of its row and column values. For example –9 is the sum of the value assigned to 5^{th} row and 5^{th} column. Then the value of x + y is-

(A) –7 (B) 0 (C) 7 (D) 14

13. Let $A = \begin{bmatrix} p & 13 \\ -13 & p \end{bmatrix}$ and $B = \begin{bmatrix} 4q & 85 \\ -2 & 1 \end{bmatrix}$ where $p,q \in N$. It is given that $|A| = |B|$ and $p,q \in [1,1000]$. Then total number of ordered pairs (p,q) is-

(A) 31 (B) 35 (C) 41 (D) 23

14. If A is a square matrix of order 4, then $\frac{1}{8}\det(\operatorname{adj}(-2A^2))$ is equal to (where $\det A = \sqrt{2}$)

(A) –512 (B) 4096 (C) –1024 (D) 1024

15. The matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 7 & 9 \\ 1 & 4 & 8 \end{bmatrix}$ can be decomposed uniquely into the product A = BC where

$B = \begin{bmatrix} B_{11} & 0 & 0 \\ B_{21} & B_{22} & 0 \\ B_{31} & B_{32} & B_{33} \end{bmatrix}$ and $C = \begin{bmatrix} 1 & C_{12} & C_{13} \\ 0 & 1 & C_{23} \\ 0 & 0 & 1 \end{bmatrix}$, the solution of the system $BX = [2\ \ 7\ \ 10]^T$ is -

(where A^T denotes transpose of matrix A)

(A) $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}^T$ (B) $[2\ \ 1\ \ 2]^T$ (C) $\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}^T$ (D) $[2\ \ 2\ \ 1]^T$

16. If A and B are symmetric matrices of the same order and X = AB + BA, Y = AB – BA then $(XY)^T$ is equal to :

(A) XY (B) YX (C) –YX (D) None

17. Number of non-singular square matrices A of order 3 satisfying $A + A^2 = A^T$, is (where A^T is transpose of A)

(A) 0 (B) 2 (C) 1 (D) 3

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18. Let A and B are square matrices of order 3 where $AB + B^T = B$, then choose the correct option -

(A) if B is non-singular then A is singular
(B) if B is singular then A is non-singular
(C) if B is singular then $I - A$ is non-singular
(D) if B is non-singular then $I - A$ is singular

19. Let A,B be non zero square matrices satisfying $A + BA^T = I$ & $B + AB^T = I$ and $A^4 - 2A^2 = kA$, then k is equal to-

(A) 1 (B) –1 (C) 0 (D) 2

20. The value of determinant of a matrix of order 3×3 which when pre-multiplied by $A = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 2 & 3 & 1 & 2 \\ 3 & 1 & 3 & 2 \end{bmatrix}$, interchanges first and second rows of A while third remains unchanged is

(A) –1 (B) –2 (C) 0 (D) 1

21. If A and B are non-singular symmetric matrices of the same order such that AB = BA and $P = (AB^{-1})^T$ & $Q = (A^{-1}B)^T$ then $(PQ)^{-1}$ is equal to-

(A) $AB^{-1}AB$ (B) $A^{-1}B^{-1}AB$ (C) $A^{-1}BA^{-1}B$ (D) A^2B^2

22. If A is a square matrix of order less than 4 such that $|A - A^T| \neq 0$ and $B = \text{adj}A$, then $\text{adj}(B^2A^{-1}B^{-1}A)$ is-

(A) A (B) B (C) $|A|A$ (D) $|B|B$

23. Let A & B are two non singular matrices of order 3 such that $A + B = I$ & $A^{-1} + B^{-1} = 2I$, then $|\text{adj}(4AB)|$, is (where adj(A) is adjoint of matrix A)-

(A) 4 (B) 16 (C) 64 (D) 128

24. Given A and C are involutary matrices and B is a non-singular matrix, then $(AB^{-1}C)^{-1}$ is equal to -

(A) $A^{-1}BC^{-1}$ (B) ABC (C) ABC^{-1} (D) CBA

25. If $\text{adj}B = A$, $|P| = 2$, $|Q| = \frac{1}{2}$, then $\text{adj}\left(Q^{-1}BP^{-1}\right)$ is-

(A) PQ (B) QAP (C) PAQ (D) $PA^{-1}Q$

26. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$ and $X = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$. If $P = AXB$, $Q = BX^TA$ and $\text{Tr}((PQ)^{10}) = a^{10} + b^{10}$, where $a < b$, then value of $(b - 2a)$ is

(A) 3 (B) 5 (C) 7 (D) 9

27. Let $A = \begin{bmatrix} x & 2y & x+y \\ y & 2x & x-y \\ 2 & x & y \end{bmatrix}$ and $B = \begin{bmatrix} 3xy - x^2 & 2x - y^2 - 2y & xy - 4x \\ x^2 + xy - 2y^2 & -2x + xy - 2y & 4y^2 - x^2 \\ -2x^2 - 2y^2 & 2xy + y^2 - x^2 & 2x^2 - 2y^2 \end{bmatrix}$, then which of the following option(s) is/are always correct -

(A) If $|A| = 3$, then $|B| = 27$
(B) If $|A| = 4$, then $|B| = 16$
(C) If $|B| = 50$, then $|A| = 5$
(D) If $|B| = 1000$, then $|A| = 10$

28. All the elements of a square matrix A of order 'n' are non-negative. If $a_{ij}\ \forall\ i \neq j$, is the least non-negative value of the function, $f(x) = x^2 + 2x - 1$ and $\text{tr}(A) = \lambda$, then maximum value of $|A|$ will be

(A) λ^n (B) $\left(\frac{n}{\lambda}\right)^n$ (C) $\left(\frac{\lambda}{n}\right)^n$ (D) $(\lambda N)^n$

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29. In a 4×4 matrix the sum of each row, column and both the main diagonals is α. Then sum of the four corner elements

(A) is also α (B) May not be α
(C) is never equal to α (D) Independent of α

30. Let $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then sum of elements in $\lim\limits_{n\to\infty}\left(\frac{A^n}{n}\right)$, where $\theta \in R$ $(n \in N)$

(A) 0 (B) 1 (C) 2 (D) 3

31. Let $A = \begin{pmatrix} p & 0 & 1 \\ 0 & q & 2 \\ -1 & 0 & r \end{pmatrix}$, where p, q, r are natural numbers. If tr(A) = 7, then difference between the greatest and least value of det(A) is -

(A) 9 (B) 8 (C) 7 (D) 6

[Note : tr(P) and det(P) denotes trace and determinant of matrix P respectively]

32. If A and B are 2×2 matrices with real entries such that $(AB - BA)^m = I$, where m is some fixed positive integer & I is the identity matrix, then for all A and B -

(A) $(AB - BA)^{10} = -I$ (B) $(AB - BA)^4 = I$ (C) $(AB - BA)^{10} = I$ (D) $(AB - BA)^4 = -I$

33. If $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, then value of $Tr\left(\left(A(\text{adj}A)A\right)^{-1} A\right)$ is equal to

(here adj(X) denotes adjoint of matrix X and Tr(X) denotes trace of matrix X)

(A) 2 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{8}$

34. Which of the following is **INCORRECT** -

(A) Orthogonal matrices are always invertible.
(B) Skew symmetric matrices of order $(2n + 1)$ are always non invertible $\{n \in N\}$.
(C) Idempotent matrices are always invertible.
(D) Involutary matrices are always invertible.

35. Let A be a non-singular matrix of order 3 such that det. (A) = 5 and B is also a non-singular matrix satisfying $A^{-1}B^2 + AB = \mathbf{0}$. The value of det. $(A^6 - 2A^4B + A^2B^2)$ is equal to

(A) 0 (B) 5^6 (C) $2^3 . 5^6$ (D) 10^6

36. Consider the set A of all determinants of order 3 with entries 0 or 1 only. Let B be the subset of A consisting of all determinants with value 1. Let C be the subset of A consisting of all determinants with value –1, then

(A) $C \in \phi$ (B) $A = B \cup C$
(C) $B \cap C \neq \phi$ (D) Number of elements in B and that in C are equal

37. Given P_n and Q_n are the square matrices of order 3 such that $P_n = [a_{ij}]$, $Q_n = [b_{ij}]$ where $a_{ij} = \frac{3i+j}{4^{2n}}, b_{ij} = \frac{3i-j}{2^{2n}}$ for all i and j, $1 \leq i, j \leq 3$.

If $L_1 = \lim\limits_{n\to\infty} T_r\left(4P_1 + 4^2P_2 + 4^3P_3 + \ldots.. + 4^nP_n\right)$ & $L_2 = \lim\limits_{n\to\infty} T_r\left(2Q_1 + 2^2Q_2 + 2^3Q_3 + \ldots.. + 2^nQ_n\right)$, then the value of $(L_1 + L_2)$ is (where $T_r(A)$ denotes the trace of matrix A.)

(A) 20 (B) 40 (C) 60 (D) 80

38. If A is a 3×3 matrix, then the expression $\text{Adj}((\text{Adj}(|A|A^2))^2)$ is,

(where $|A|$ = determinant of A & AdjA = Adjoint of A)

(A) $|A|^{12}A^2$ (B) $|A|^6A^{12}$ (C) $|A|^{12}A^4$ (D) $|A|^6A^4$

39. Let A and B be two 2×2 invertible matrices. Consider the statements :

(i) $AB = \mathbf{O} \Rightarrow A = \mathbf{O}$ or $B = \mathbf{O}$

(ii) $AB = I_2 \Rightarrow A = B^{-1}$

(iii) $(A + B)^2 = A^2 + 2AB + B^2$

then

(A) (i) is false, (ii) & (iii) are true (B) (i) & (ii) are false, (iii) is true

(C) (i) & (iii) are false, (ii) is true (D) (ii) & (iii) are false, (i) is true

40. Let $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}$, $a_i \in N$, $i \in \{1,2,3,.....9\}$ such that $a_1 + a_2 + a_3 = a_4 + a_5 + a_6 = a_7 + a_8 + a_9 = a_1 + a_5 + a_9 = 6$, then number of such matrices is-

(A) 147 (B) 262 (C) 219 (D) 197

41. Number of 2×2 matrices $A = \{a_{ij}\}_{2\times2}$ such that $|a_{ij}| = |a_{ji}|$, where $|a_{ij}| \in \{0, 1, 2, 3, 4, 5\}$ is -

(A) 216 (B) 1342 (C) 1341 (D) 222

42. Let $M = \begin{bmatrix} \sin^4\theta & -1-\sin^2\theta \\ 1+\cos^2\theta & \cos^4\theta \end{bmatrix} = \alpha I + \beta M^{-1}$,

where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real number, and I is the 2×2 identity matrix. If

α is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and

β is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$,

then the value of $\alpha + \beta$ is **[JEE-ADV 2019]**

(1) $-\frac{37}{16}$ (2) $-\frac{29}{16}$ (3) $-\frac{31}{16}$ (4) $-\frac{17}{16}$

<u>One or more than one correct :</u>

43. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj}M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ where a and b are real numbers. Which of the following options is/are correct ? **[JEE-ADV 2019]**

(1) $a + b = 3$ (2) $\det(\text{adj}M^2) = 81$

(3) $(\text{adj}M)^{-1} + \text{adj}M^{-1} = -M$ (4) If $M\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$

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44. Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$,

$P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and $X = \sum_{k=1}^{6} P_K \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_K^T$

where P_K^T denotes the transpose of the matrix P_K. Then which of the following options is/are correct?

[JEE-ADV 2019]

(1) $X - 30I$ is an invertible matrix
(2) The sum of diagonal entries of X is 18
(3) If $X\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$
(4) X is a symmetric matrix

45. Let $x \in \mathbb{R}$ and let $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$.

Then which of the following options is/are correct ? **[JEE-ADV 2019]**

(1) For x = 1, there exists a unit vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ for which $R\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

(2) There exists a real number x such that PQ = QP

(3) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in \mathbb{R}$

(4) For x = 0, if $R\begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6\begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then a + b = 5

46. Let A be a square matrix of order 3, that satisfies $A^3 = \mathbf{O}$, Let $B = A^2 + A + 2I_3$, $C = A^2 + 2A - 4I_3$ then-
(A) A must be singular matrix.
(B) B must be invertible matrix.
(C) C must be invertible matrix
(D) BC must be non-singular matrix.

47. Choose the correct statement(s) for same order square matrices A & B,
(A) $(I + A^{-1})^{-1} = A(A + I)^{-1}$; $|I + A^{-1}| \neq 0$, $|A + I| \neq 0$
(B) $(I + AB)^{-1}A = A(I + BA)^{-1}$; $|A| \neq 0$, $|I + AB| \neq 0$, $|I + BA| \neq 0$
(C) If B is an orthogonal matrix such that $B = (I - A)(I + A)^{-1}$ then, A must be a symmetric matrix
(D) If A is a skew-symmetric matrix, then $(I - A)(I + A)^{-1}$ is an orthogonal matrix.

48. Let A and B are commutative square matrices of order three such that A is symmetric and B is skew symmetric, if $C = (A + B)^T (A + B) (A - B)^{-1}$(where X^T and $|X|$ denotes transpose and determinant value of X respectively), then
(A) A + B – C = 0
(B) A – B – C = 0
(C) |B| = 0
(D) $2A - C = C^T$

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49. Let M, N are two non-singular matrix of order 3 with real entries such that adj(M) = 2N and adj(N) = M, then -

(A) $adj(M^2N) + adj(MN^2) = 4(M + 2N)$
(B) $M = 2N^{-1}$
(C) $MN = 4I$
(D) $adj(MN^{-1}) = 4M^{-2}$

50. If A & B are square matrices of order 2 such that $A + adj(B^T) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ & $A^T - adj(B) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then-

(A) B is symmetric matrix
(B) $A^n = A \ \forall \ n \in N$
(C) $|A + A^2 + A^3 + A^4 + A^5| = 0$
(D) $|B + B^2 + B^3 + B^4 + B^5| = 0$

51. If A and B are square matrices of order of 3 such that AB = aA + bB, where $a, b \in R - \{0\}$, then the correct statements is/are.

(A) A and B cummute
(B) A – bI is invertible
(C) B – aI is invertible
(D) A = B

52. If A is a symmetric and B skew-symmetric matrix of the same order and (A + B) is non-singular and $C = (A + B)^{-1}(A - B)$, then

(A) $(A + B)C = A - B$
(B) $C^T(A + B)C = A + B$
(C) $C^T(A - B)C = (A - B)$
(D) $C^TAC = A$

53. Let A is a matrix of order 3 and (A – 2I) (A – 4I) = 0, then-

(A) $\frac{A}{6} + \frac{4A^{-1}}{3} = I$
(B) $A^2 - 6A = -8I$
(C) $\left|\frac{A}{6} + \frac{4A^{-1}}{3}\right| = 1$
(D) $|A^2 - 6A| = 8$

54. Let $x = \alpha, y = \beta, z = \gamma$, where α,β,γ are non-zero real numbers,

satisfies $[x\sin3\theta - 1 \quad x\cos2\theta \quad 2x] - [y \quad -4y \quad -7y] + [z \quad 3z-2 \quad 7z] = [-1 \quad -2 \quad 0]$, $\theta \in [0, 4\pi]$.

If N denotes number of all possible values of θ and S is sum of all possible values of θ, then

(A) N > 8
(B) $N \leq 8$
(C) $S \leq 14\pi$
(D) $S > 14\pi$

55. If three 3×3 invertible matrices A, B, C are Idempotent, Involutary and Orthogonal matrices respectively, then -

(A) $(ABC)^{-1} = (AB^TC)^T$
(B) $|adj(2AB^{-1}C)| = 8$
(C) If $(ABC)^{-1} = (CBA)^{-1}$, then BC = CB
(D) $adj(3A^{-1}BC^{-1}) = 9adj((C(B^T)^{-1}A)^T)$

56. Let A is an invertible matrix of order 3 whose entries are complex numbers and which satisfies $2A^2 = 4A + A^3$, then

(A) $|A| = 8$
(B) $\left|adj\left(\frac{A}{2}\right)\right| = 1$
(C) $tr((A - 2I)^3) = 24$
(D) $adj(A) = A^2$

57. Let $P = [a_{ij}]$ be 3×3 invertible matrix, where $a_{ij} \in \{0, 1\}$ for $1 \leq i,j \leq 3$ and exactly four elements of P are 1. If N denotes number of possible matrices P, then which of the following is (are) true ?

(A) Number of divisors of N is even.
(B) Sum of divisors of N is 91
(C) Determinant of (adjoint P) can be –1
(D) Determinant of (adjoint P) can be 1.

58. Consider a skew symmetre matrix $A = \begin{pmatrix} a & b \\ -b & c \end{pmatrix}$ such that a, b and c are selected from the set $S = \{0,1,2,3,........, 9\}$. Then which of the following is (are) correct ?
(A) Number of possible matrix A is 10.
(B) Number of invertible matrix A is 9.
(C) If $|A|$ is divisible by 3, then number of possible matrix A is 4.
(D) If b = 1, then $|A + I| = 2$ where I is an identity matrix .

59. Let B is an invertible square matrix and B is the adjoint of matrix A such that $AB = B^T$, then-
(A) A is an identity matrix.
(B) B is a symmetric matrix.
(C) A is not an identity matrix.
(D) B is a skew symmetric matrix.

60. If X is a column matrix & A is a 3rd order square matrix, then choose the correct statements(s) :
(A) XX^T is a singular matrix
(B) If $A^TA = O$, then $A = O$ provided A is a real matrix (where O is the null matrix)
(C) If $AA^T = I$ then, $A^T A$ need not to be I
(D) If $A^{10} = O$, then $(I + A + A^2 + ... + A^9)^{-1} = I - A$ (where O is the null matrix)

61. Let A is a (4×4) order diagonal matrix whose entries are complex numbers such that $A^4 = I_4$. If trace of A is zero then (here I_4 is four order unit matrix)
(A) There will be 24 distinct matrices A
(B) There will be 36 distinct matrices A
(C) Determinant value of all such matrices can be 1 or –1
(D) Determinant value of all such matrices is –1

62. Let A and B are invertible matrices of order three such that $AB = X = BA$ and $A^T + A = X^T + X = B^T + B$ then which of the following must be true ?
(A) $A = B$
(B) $\det(A - I) = 0$
(C) $\det(B - I) = 0$
(D) $\det(X - I) = 0$

63. If M be a non-singular matrix of order 3 such that $M^{-1} = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 5 & 3 \\ 5 & 3 & 4 \end{bmatrix}$, then choose the correct option(s) -
(where |X| represent determinant of matrix X, adj(X) denotes adjoint of matrix X and Tr(X) denotes trace of matrix X)
(A) Absolute value of (15tr(adjM)) is 5
(B) $|M| = -\frac{1}{36}$
(C) $M^{-2} = \begin{bmatrix} 50 & 47 & 47 \\ 47 & 50 & 47 \\ 47 & 47 & 50 \end{bmatrix}$
(D) $-36\,\text{adj}\,M = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 5 & 3 \\ 5 & 3 & 4 \end{bmatrix}$

64. Which of the following statement(s) is/are **incorrect** ?
(A) If the matrices A, X, B, C are such that the product A X B exists, then A X B = C always implies $X = A^{-1}CB^{-1}$
(B) If A & B are square matrices, then Trace(Adj(AB)) = Trace((AdjA)(AdjB))
(C) If $f(x) = x^3 + x^2 + 5x - 7$ & $f(A) = O$, where A is 3×3 matrix & O is null matrix, then Trace(A) must be –1.
(D) If A & B are orthogonal matrices, then matrix ABA must be orthogonal.

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65. Let $A = \begin{bmatrix} a & 1 \\ b & c \end{bmatrix}$ be a 2×2 matrix with non -zero real entries such that

$A\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. If $A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$ and $A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$ such that area of triangle OBC $= \frac{1}{2}$,

where $B(x_1, y_1)$, $C(x_2, y_2)$ and O is origin, then

[**Note:** adj.M denotes the adjoint of a square matrix M.]

(A) $a + b + c = 0$ (B) det. (adj. A) = 1 (C) $a^2 + b^2 + c^2 = 26$ (D) $A^{-1} = \begin{bmatrix} 3 & -1 \\ -4 & 1 \end{bmatrix}$

66. Let us consider the matrix $f(x) = \begin{bmatrix} \cos\left(x+\frac{\pi}{3}\right) & \sin\left(x+\frac{\pi}{3}\right) & \tan\frac{\pi}{11} \\ \cos\left(x+\frac{5\pi}{3}\right) & \sin\left(x+\frac{5\pi}{3}\right) & \tan\frac{2\pi}{11} \\ \cos\left(x+\frac{9\pi}{3}\right) & \sin\left(x+\frac{9\pi}{3}\right) & \tan\frac{3\pi}{11} \end{bmatrix}$

Then which of the following is **incorrect**. (where $|x|$ denotes determinant of matrix x)

(A) f(x) is a singular matrix

(B) If $|f(2016\pi)| = a$ then $\sum_{k=1}^{2016} |f(k\pi)| = a$

(C) $|\text{adj}(f(x))| = |f(x)|$

(D) $|f(x)|$ is a positive real number whenever x is an multiple of $\frac{\pi}{11}$.

67. Let A = a_{ij} be a 3×3 matrix where $a_{ij} = \begin{cases} (i^j - j^i + ij)x & ; \text{ if } i < j, x \in R \\ 1 & ; \text{ if } i > j \\ 0 & ; \ i = j \end{cases}$, then choose the correct option-

(A) minimum value of $\det A = -\frac{1}{20}$ (B) $\int_{-2}^{2} (\det A)\,dx = \frac{80}{3}$

(C) minimum value of $\det(A) = \frac{1}{20}$ (D) $\int_{-2}^{2} (\det A)\,dx = \frac{61}{3}$

68. Let A and B are two matrices of order 3×3 such that $A^2 = 4I$ & $AB^T = \text{adj}(A)$, then which of the following option(s) must be true ($|A| < 0$)

{where adj(A) represents adjoint of matrix A and A^{-1} represents inverse of matrix A}

(A) $A = 2I$ (B) $|B| = -8$

(C) B is symmetric matrix (D) $B^{-1} = \frac{B}{4}$

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69. Let A and B be square matrices of same order which commute each other such that B is orthogonal and $A^{-1} = A$, $B^{-1} = B$. Suppose $X = \text{adj}(AB^T)^n$ then which of the following is/are correct ?
(where det(AB) > 0)

(A) X =I if n is even (where I is identity matrix)

(B) X = AB if n is even

(C) X = BA if n is odd

(D) X= A^2B^2 if n is even

70. Let A,B, X and Y are square matrices such that XB = A, YA = B then which of the following is/are always correct ? (A and B are non zero matrices)

(A) if det(X) = d then det(Y) = $\frac{1}{d}$ $(d \neq 0)$

(B) if A is invertible then B is invertible and viceversa

(C) if A & B are invertible, then XY = I (where I is identity matrix)

(D) if A is invertible then Y is invertible

71. Let A be a square matrix of order two such that $A^T - I = A^TA$, then -

(A) A is symmetric matrix
(B) tr(A) = 1
(C) $A^{-1} = I - A$
(D) $|A - I| = -1$

(where tr(X) denotes trace of a matrix X, X^T denotes transpose of X and I denotes unit matrix)

72. Identify the correct statement(s)

(A) For two non-singular matrices A, B of same order, if adj(A) = adj(B), then A = B

(B) For two non-singular matrices A, B of same order, if adj(A) = adj(B), then |A| = |B|

(C) For two non-singular matrices A, B of same order, if adj(A) = adj(B), then A + B = 0 or A – B = 0

(D) For two non-singular matrices A, B of same order, if adj(A) = adj(B), then |A + B| = 0 or |A – B| = 0

(where adj(X) denotes adjoint of matrix X and |X| denotes value of its determinant)

73. Let A,B,C be n × n order real matrices and product is pairwise commutative, also ABC = O_n, if $\lambda = \det(A^3 + B^3 + C^3).\det(A + B + C)$ then λ can be-

(A) 0
(B) 2
(C) –2
(D) 4

74. Let A and B be two square matrix of order 3, then which of the following statement(s) is/are correct ?

(A) ABA^T is a symmetric matrix

(B) AB – BA is a skew symmetric matrix

(C) If $B = |A|A^{-1}$, $|A| \neq 0$, then $\text{adj}(A^T) - B$ is a skew symmetric matrix
(where |A| is determinant of matrix A)

(D) If $B + A^T = \mathbf{O}$ and A is a skew symmetric matrix, then B^{15} is also skew symmetric matrix

75. Let A & B are square matrices of third order such that $A^T + B^2 = A$ (where A^T is transpose of A), then identify the correct statement(s) -

(A) $\det(A - A^T) = 0$
(B) $\det(A^2 + (A^T)^2) = 0$
(C) $\det(B^2 - (B^T)^2) = 0$
(D) $\det(B^2 + (B^T)^2) = 0$

76. Consider three square matrices A, B and C of order n such that $A^T = A - B$, $B^T = B - C$, then (n is odd natural number)

(A) $|C| = 2^n|B|$
(B) $|C| = 2^{n-1}|B|$
(C) $|A + B| = |A - 2B|$
(D) $|A| = 0$

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77. If A & B are two square matrices of same order which commute each other, then -
(A) A and B^n commute each other $\forall$ n $\in$N
(B) A^n and B^n commute each other $\forall$ n $\in$N
(C) $A - \lambda I$ and $B + \mu I$ commute each other $\forall\ \lambda, \mu \in R$
(D) $A + \lambda I$ and $\mu I - B$ commute each other $\forall\ \lambda, \mu \in R$

78. Let A is 3×2 matrix and B is 2×3 matrix such that det(AB) = 0, then value of det(BA) can be -
(A) 0 (B) 1 (C) 2 (D) 3

79. Let matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$, then-
(A) $(A+B)^{2005} = A^{2005} + B^{2005}$
(B) $(A-B)^{2005} = A^{2005} - B^{2005}$
(C) $(3A+7B)^n = 3^{n-1}(3^n A + 7^n B)$
(D) $(3A+7B)^n = 3^{2n} A + 3^n 7^n B$

80. Let matrix $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and A is 3 ordered square matrix such that AB = BA. If all entries of matrix A are whole numbers whose sum is 6, then-
(A) If there are exactly 3 zero's in matrix A, then det(A) = 1
(B) If traceA = 6, then det(A) = 8
(C) A is always an invertible matrix
(D) there are 4 such matrices A.

81. Let A and B are two square matrices such that AB = A, BA = B, then
(A) $A^2 = A$
(B) $B^2 = B$
(C) $(A^2 - AB + B^2) = B$
(D) $(A^2 - AB + B^2) = A$

82. A and B are two non-singular square matrices of each 3×3 such that AB = A and BA = B and $|A + B| \neq 0$ then
(A) $|A + B| = 2$ (B) $|A + B| = 8$ (C) $|A - B| = 1$ (D) $|A - B| = 0$

<u>Comprehension Type :</u>

Paragraph for Questions 83 and 84

Consider two 3×3 matrices A and B satisfying $A = \text{Adj } B - B^T$ and $B = \text{Adj } A - A^T$ (where C^T denotes transpose of matrix C)

83. If A is non singular, then $(\det A)^2 + (\det B)^2$ is equal to
(A) 16 (B) 64 (C) 128 (D) 48

84. AB is equal to (if A is non singular)
(A) I (B) 2I (C) 3I (D) 4I

Paragraph for Questions 85 and 86

Let A & B are non-singular matrices of order 3 such that det(A) = 5 & $A^{-1}B^2 + AB = \mathbf{0}$.

85. The value of $\det(A^6 - 2A^4B + A^2B^2)$ is equal to-
(A) 0 (B) 5^6 (C) $2^3.5^6$ (D) 10^6

86. $A^2\det(A^2) - \text{adj}(\text{adj } B)$ is equal to-
(A) Null matrix (B) $25A^2 - 5B$ (C) $25A^2$ (D) $50A^2$

Note : det(A) denotes determinant of matrix A & adj(A) denotes adjoint matrix of matrix A.

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Paragraph for Question 87 to 89

Consider the Fibonacci sequence 0, 1, 1, 2, 3, 5, 8, …. Whose terms are related by $t_{n+2} = t_{n+1} + t_n$; $n = 0, 1, 2, \ldots\ldots$ we can form a matrix by taking three consecutive terms as $f(x) = \begin{bmatrix} t_{n+1} & t_n \\ t_n & t_{n-1} \end{bmatrix}$; $n = 1, 2, 3 \ldots..$

Further, there exists a constant matrix A such that $f(n) = f(n-1)A$; n = 2, 3, ...

87. Trace (A) i.e. sum of diagonal elements of A is

(A) 0 (B) 1 (C) 2 (D) 3

88. $f(n) =$

(A) A^n (B) A^{n-1} (C) BA^{n-1}; $B \neq A, I$ (D) A^{n+1}

89. $\det(f(11)) =$

(A) 0 (B) 1 (C) –1 (D) 2

Paragraph for Question 90 & 91

Given $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$ and $D = \text{diag}\left(\frac{d_1^2}{3}, \frac{d_2^2}{3}, \frac{d_3^2}{2}\right)$ where $d_1 > d_2$, $d_1 > d_3$, $d_2 < d_3$ and d_1, d_2, d_3 are satisfying $|A - dI| = 0$. If there exist a matrix B such that $ABA^T = D$, then (where I is the identity matrix of order 3)

On the basis of above information, answer the following questions :

90. If adjD. adj(adjD) = kI, then the value of k is equal to-

(A) 6 (B) 2 (C) 8 (D) 4

91. If $B^T = B + \lambda I$, then the value of λ is -

(A) 0 (B) 1

(C) a rational number which is not an integer (D) none of these

Paragraph for Question no. 92 & 93

Let a, b, c be distinct real numbers and f be cubic polynomial such that

$$\begin{bmatrix} a^2 & 4a & 1 \\ b^2 & 4b & 1 \\ c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} f(-1) & 0 \\ f(1) & f(0) \\ f(2) & 1 \end{bmatrix} = \begin{bmatrix} 2a^2 + 16a + 17 & 4a + 1 \\ 2b^2 + 16b + 17 & 4b + 1 \\ 2c^2 + 16c + 17 & 4c + 1 \end{bmatrix}$$

92. Number of values of k for which the equation $f(x) = k$ has exactly two distinct solutions is -

(A) 1 (B) 2 (C) 3 (D) more than 3

93. The sum of greatest and least values of $f(x)$ in $x \in [-1, 1]$ is equal to -

(A) 0 (B) 2 (C) 4 (D) 5

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Paragraph for Question no. 94 & 95

Let $A = [a_{ij}]$ be a 3×3 matrix such that $A = \begin{bmatrix} 3 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 0 \end{bmatrix}$. Suppose x_1, x_2, x_3 are three column vectors such that $Ax_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $Ax_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $Ax_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$. B is a 3×3 matrix whose first, second and third columns are x_1, x_2 and x_3 respectively.

[**Note :** det(P) denotes the determinant of square matrix P]

94. The value of det(2B) is equal to -

(A) 1 (B) 2 (C) 4 (D) 8

95. Let $C = [C_{ij}]$ be a 3×3 matrix, where $C_{ij} = \frac{3^i}{9^{-j}} a_{ij}$ for $1 \le i, j \le 3$, then det(C) is equal to -

(A) $2(3^{14})$ (B) $2(3^{15})$ (C) $2(3^{18})$ (D) $2(3^{19})$

Paragraph for Question 96 & 97

If A and B are square matrices of same order such that $|A| = |B| = 1$ and $A(\text{adj}A + \text{adj}B) = B$

{where $|X|$, adjX, X^T, X^{-1} are respectively determinant adjoint, transpose and inverse of matrix X}

96. $|A + B|$ is equal to -

(A) 0 (B) –1 (C) 1 (D) 2

97. If B is orthogonal then AB^{-1} is-

(A) $B^{-1}A$ (B) BA^{-1} (C) A^TB^{-1} (D) B^TA^{-1}

Paragraph for Question no. 98 & 99

If A is invertible matrix of order 3 and B is another matrix of same order as of A such that

$|B| = 2$ and $A^T|A|B = A|B|B^T$, then

(where adj(A) and A^{-1} denote adjoint and inverse of matrix A)

98. $\left|AB^{-1}\text{adj}\left(A^TB\right)^{-1}\right|$ is equal to-

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{16}$ (D) 1

99. If B is symmetric matrix , then which of the following is incorrect ?

(A) adj(B) is symmetric matrix (B) B^{-1} is symmetric matrix

(C) $A^{2015}B^{2016}$ is symmetric matrix (D) AB^{-1} is symmetric matrix if AB = BA

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Paragraph for Question 100 to 102

Let $A=\begin{bmatrix}5&2\\2&1\end{bmatrix}$ and matrix B is such that $AB^TA = B$ and $BA^TB = I$ where I is unit matrix of order two.

On the basis of above information, answer the following questions :

100. Matix B^2 is-

(A) $\begin{bmatrix}5&2\\2&1\end{bmatrix}$ (B) $\begin{bmatrix}1&-2\\-2&5\end{bmatrix}$ (C) $\begin{bmatrix}-1&2\\2&-5\end{bmatrix}$ (D) $\begin{bmatrix}29&12\\12&5\end{bmatrix}$

101. If $\det(B) > 0$, then $\det(B^T - B)$ is-

(A) 32 (B) –8 (C) –32 (D) 8

102. Which of the following is **INCORRECT** ?

(A) matrix A is symmetric (B) matrix B is symmetric

(C) A and B commute each other (D) $\det(B) = 1$ or -1

Paragraph for Question 103 to 104

Let $A=\begin{bmatrix}1&-1\\1&2\\1&4\end{bmatrix}$ & B is a matrix of order (2×3) such that $AB=\begin{bmatrix}0&2&-1\\3&-7&5\\5&-13&9\end{bmatrix}$

(where tr(X) denotes trace of X)

103. tr(BA)

(A) is equal to zero (B) is equal to –3 (C) is equal to 2 (D) can't be determined

104. If X = BA, then sum of all elements in X^{2015} is-

(A) 3 (B) 2015 (C) 2016 (D) 2017

Paragraph for Question 105 to 107

If $\log_e(I+A)$ is defined as $\log_e(I+A) = A-\frac{A^2}{2}+\frac{A^3}{3}-\frac{A^4}{4}\ldots\ldots = \begin{bmatrix}f(x)&g(x)\\0&f(x)\end{bmatrix}$

where A is a square matrix and $A=\begin{bmatrix}x&1\\0&x\end{bmatrix}$ & $0<x<1$, I is an identity matrix of order 2.

On the basis of above information, answer the following questions :

105. If $\int \det(\text{adj}A)g(x)dx = \ell x^2 + mx + \ell n|x+1| + c$, then $(2\ell + m)$ is -

(A) 0 (B) 1 (C) 2 (D) 3

106. The number of solutions $f(x) = g(x)$ is -

(A) 0 (B) 1 (C) 2 (D) 3

107. If $h(x)=\begin{vmatrix}f(x)&f(x^2)&f(x^3)\\g(x)&g(x^2)&g(x^3)\\g(f(x))&f(g(x))&\ell n(g(x))\end{vmatrix}$, then $\lim\limits_{x\to 0^+}\frac{h(x)}{x}$ is -

(A) $-\ell n2$ (B) $\ell n2$ (C) 0 (D) 1

Paragraph for Question 108 to 110

A Pythagorean triple is a triplet of positive integers (a,b,c) such that $a^2 + b^2 = c^2$. Define the matrices

A,B and C by $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & -2 \\ 2 & 2 & 3 \end{bmatrix}$ & $C = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$

On the basis of above information, answer the following questions :

108. $T_r(A + B^T + 3C)$ equals -

(A) 17 (B) 18 (C) 20 (D) 21

(where $T_r(A)$ denotes trace of matrix A)

109. If Pythagorean triple (a,b,c) is written as row matrix [a b c] then which of the following matrix a product is **not** a Pythagorean triplet ?

(A) [3 4 5]A (B) [3 4 5]B (C) [3 4 5]C (D) None of these

110. Which of the following **does not** hold good ?

(A) det(3A) = 27 (B) det(2B) = 8

(C) det(3C) $\neq$ 27 (D) det(2A) = 8

Paragraph for Question 111 to 113

Let N be set of all possible 2×2 matrix 'A' of integer entries such that $AA^T = I$ (where I is an identity matrix).

On the basis of above information, answer the following questions :

111. Number of matrices in set N is-

(A) 2 (B) 4 (C) 8 (D) Infinite

112. Identify correct statement for matrices in set N -

(A) Number of symmetric matrices is equal to skew symmetric matrices

(B) All are symmetric matrices

(C) Symmetric matrices are more than skew symmetric matrices

(D) All are singular

113. Number of matrices in set N such that $|A - I| \neq 0$, is-

(A) 4 (B) 3 (C) 2 (D) 1

Paragraph for Question 114 to 115

Two square matrices A & B of same order are related by $A^2 + B^3 = I$ (where I denotes identity matrix).

114. If B is a skew symmetric matrix of odd order > 1, then which of the following is always true ?

(A) $\det(A^2 + A'^2) = 2$ (B) $\det(A^2 - A'^2) = 0$

(C) $\det(A^2 + A'^2) = 4$ (D) $\det(A^2 - A'^2) \neq 0$

115. If B is a symmetric matrix of order n, then $\det(A^2 + A'^2 + 2B^3)$ is -

(A) 2^n (B) 2^{n-1} (C) 2^{n+1} (D) $\frac{1}{2}n$

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Paragraph for Question 116 to 118

If all elements of a 3 × 3 matrix A is given by $a_{ij} = \begin{cases} \left(\frac{i+j}{2}\right) + \frac{|i-j|}{2} & \text{if } i \neq j \\ \frac{i^j - (i.j)}{i^2 + j^2} & \text{if } i = j \end{cases}$

where a_{ij} denotes element of i^{th} row & j^{th} column of matrix A.

116. Matrix A is -

(A) skew symmetric matrix
(B) singular matrix
(C) symmetric matrix
(D) a matrix, sum of whose all elements is 16

117. If $A^2 + \alpha A + \beta I = 32A^{-1}$, then $\alpha + \beta$ is equal to (I is identity matrix of order 3) -

(A) –22
(B) –20
(C) 21
(D) –23

118. If a 3 × 3 matrix B is such that $A^2 + B^2 = A + B^2A$, then $\det(\sqrt{2}BA^{-1})$ can be equal to-

(A) 1/2
(B) 1
(C) 8
(D) 16

<u>Match the column</u> :

119. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	If maximum number of distinct elements in a symmetric matrix of order n is 45, then value of n is	(1)	5
(Q)	If maximum number of distinct elements in a skew symmetric matrix of order n is 31, then value of n is	(2)	6
(R)	If minimum number of zeroes in upper triangular matrix of order n is 10, then value of n is	(3)	7
(S)	If maximum number of distinct elements in lower triangular matrix of order n is 29, then value of n is	(4)	9

Codes :

	P	Q	R	S
(A)	1	4	3	2
(B)	4	3	1	2
(C)	3	4	1	2
(D)	4	2	1	3

120. Let $A(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, X = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ and $B = AXA^T$, then (P^T denotes transpose of P and I denotes unit matrix).

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	$A^2\left(\frac{\pi}{2}\right)$ is equal to	(1)	–I
(Q)	B^2 is equal to	(2)	I
(R)	$A^{-1}BA$ is equal to	(3)	X
(S)	$B^2 - X^2$ is equal to	(4)	O (Null matrix)

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Codes :

	P	Q	R	S
(A)	1	2	2	3
(B)	1	2	3	4
(C)	4	2	3	4
(D)	2	1	3	1

121. Let A is a matrix of order 4×4 with real entries. If A

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	an idempotent matrix then number of possible values of det(A) equal to	(1)	3
(Q)	an involutary matrix then number of possible values of det(A) equal to	(2)	1
(R)	an orthogonal matrix then number of possible values of det(A) equal to	(3)	2
(S)	an invertible periodic matrix with period 3, then number of possible values of det(A) is equal to	(4)	Infinite

Codes :

	P	Q	R	S
(A)	3	3	3	2
(B)	3	3	3	3
(C)	1	2	3	4
(D)	3	3	2	2

122. Match the following

Column-I

(A) If a, b, c $\in$ R $-$ {0} such that $a \neq b \neq c$ and $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$ then $\begin{pmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{pmatrix}$

(B) If $\alpha, \beta, \gamma \in R$ then
$$\begin{bmatrix} 1 & \cos(\alpha-\beta) & \cos(\alpha-\gamma) \\ \cos(\beta-\alpha) & 1 & \cos(\beta-\gamma) \\ \cos(\gamma-\alpha) & \cos(\gamma-\beta) & 1 \end{bmatrix}$$

(C) If ω $(\neq 1)$ be cube root of unity then
$$\begin{bmatrix} 1+2\omega^{100}+\omega^{200} & \omega^2 & 1 \\ 1 & 1+\omega^{101}+2\omega^{202} & \omega \\ \omega & \omega^2 & 2+\omega^{100}+2\omega^{200} \end{bmatrix}$$

(D) If a, b, c $\in$ R {0} such that $a \neq b \neq c$
$$\begin{bmatrix} 0 & (a-b)^3 & (a-c)^3 \\ (b-a)^3 & 0 & (b-c)^3 \\ (c-a)^3 & (c-b)^3 & 0 \end{bmatrix}$$

Column-II

(P) Symmetric

(Q) Singular

(R) Non-singular

(S) invertible

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Subjective :

123. A be a square matrix of order 2 with $|A| \neq 0$ such that $|A + |A|\text{adj}A| = 0$, where adj(A) is adjoint of matrix A, then the value of $|A - |A|\text{adj}A|$ is equal to

124. Let A, B and A + B are non-singular matrices of order 3 × 3 satisfying $A^{-1} + B^{-1} = (A + B)^{-1}$ and $|AB^{-1}| \in R$ then value of $\frac{|A|}{|B|}$ is

125. If 'A' is a 3rd order square matrix whose general element $a_{ij} = \frac{2i-j}{2}$ and A = L + D + U, where; L, D & U are Lower triangular, diagonal and upper triangle matrices then, the sum of non-diagonal elements of the matrix 'L' is

126. Let A and B are non singular matrices of order three, each of them commute with matrix X, where AX = B and XB = A. If X is not an identity matrix, then $\det(X^4 + X^2 + I)$ is (where I is identity matrix of order three)

127. Let $A = \begin{bmatrix} -3 & 0 & 2 \\ 1 & x & 5 \\ -2 & 0 & x^2 \end{bmatrix}, B = \begin{bmatrix} 2 \\ b \\ -1 \end{bmatrix}$ and $C = [3\ \ 5\ \ 1]$, then the number of integral value(s) of 'b' for which $\text{Tr}(ABC) \le -18\ \forall\ x \in R$ is (are)

128. Let P be a 2 × 2 matrix such that $P\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $P^2\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. If p_1 and p_2 $(p_1 > p_2)$ are two values of p for which $\det(P - pI) = 0$, where I is an identity matrix of order 2, then $(5p_1 + 2p_2)$ is equal to
[**Note :** det(M) denotes determinant of square matrix M]

129. Let $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$ and a_{ij} denotes element of i^{th} row and j^{th} column of matrix A. If a_{ij} = cofactor of b_{ij} $\forall 1 \le i, j \le 3$ where b_{ij} is element of i^{th} row and j^{th} column of matrix B. Then $\left(\sum_{r=1}^{3} b_{3r}a_{3r}\right)^2$ is equal to

130. There exist 2^m diagonal matrices of order 4 those are involutary such that value of their determinant is positive, then value of m is

131. Given $A=[a_{ij}]_{3\times3}$ be a matrix, where

$$a_{ij} = \begin{cases} i - j & \text{if} \quad i \neq j \\ i^2 & \text{if} \quad i = j \end{cases}$$

If C_{ij} be the cofactor of a_{ij} in the matrix A and $B = [b_{ij}]_{3\times3}$ be a matrix such that $b_{ij} = \sum_{k=1}^{3} a_{ik}c_{jk}$, then the value of $\left[\frac{\sqrt[3]{\det B}}{8}\right]$ is equal to

(where [.] denotes greatest integer function and det B denotes determinant value of B)

132. Let A,B,C be three 3 × 3 matrices with real entrices. If BA + BC + AC = 1 and det(A + B)=0, then the value of det(A + B + C – BAC)

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133. Let $X=\begin{bmatrix}3 & 0\\ 3 & 3\end{bmatrix}, Y=\begin{bmatrix}\sqrt{\frac{7}{8}} & \frac{1}{2\sqrt{2}}\\ \frac{-1}{2\sqrt{2}} & \sqrt{\frac{7}{8}}\end{bmatrix}$ and $Z = Y^T X Y$. If $M = YZ^{2015}Y^T$ where $M = [m_{ij}]$ is a 2×2 matrix, then sum of digits of λ where $\lambda = \dfrac{m_{21}-m_{11}-m_{22}}{3^{2015}}$, is

134. Number of 3×3 symmetric matrices which can be formed by three '0', three '1' & three '–1' only, is

135. Let $A = [a_{ij}]$, $B = [b_{ij}]$ be two square matices of order 'n' such that a_{ij}, b_{ij} are non-negative & $\det(B) = 1$ & $\sum_{j=1}^{n}\sum_{i=1}^{n}\sum_{r=1}^{n} a_{ir}b_{rj} \le 0$, then det(A) is

136. Let $A = [a_{ij}]_{2\times 2}$ such that $a_{ij} \in \{-1,0,1\}$ $\forall$ i,j and $\text{adj}(A) = -A$, $\det(A) = -1$. If N denotes number of such matrices then $\dfrac{N}{2}$ is

137. Let A be a square matrix of order 3×3 such that $\text{adj}(A) = I$, $A \neq I$. If $A\begin{bmatrix}x\\ y\\ z\end{bmatrix}=\begin{bmatrix}-1\\ 0\\ 0\end{bmatrix}$, then $x + y + z$ is (where adj(X) denotes adjoint of X and I is unit matrix)

138. An invertible matrix A of order 3 satisfies the relation $A = A^{-1} + 2I$, (where I denotes identity matrix). The value of $|A - I|.|A + I|.|A - 2I|$ is

139. Square matrices B of second order are formed using some or all the possible values of k, when system of linear equations $kx + 4y + z = 0$, $4x + ky + 2z = 0$ & $2x + 2y + z = 0$, has infinite solution, then number of singular matrices B is

140. Let a, b, c be integers such that $a^2 + b^2 + c^2 = 2$ and $A=\begin{bmatrix}a & b & c\\ b & c & a\\ c & a & b\end{bmatrix}$. Now consider the system of simultaneous equations $A\begin{pmatrix}x\\ y\\ z\end{pmatrix}=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$, then probability that system of equation has unique solution is p/q then (p + q) is

141. Let $M=[a_{ij}]_{2\times 2}$ whose all entries are distinct and $a_{ij} \in \{1,2,5,10\}$, then number of such singular matrices M is

142. Consider a variable matrix $V=\begin{bmatrix}x & y & z\\ y & z & x\\ z & x & y\end{bmatrix}$ such that $VA=\begin{bmatrix}0\\ 0\\ 0\end{bmatrix}$.........(E), where $A=\begin{bmatrix}\log a\\ \log b\\ \log c\end{bmatrix}$ and a, b, c are distinct positive reals in G.P.. If solutions (x, y, z) of matrix equation (E) lie on line $\dfrac{x}{\alpha}=\dfrac{y}{\beta}=\dfrac{z}{\gamma}$, then $\dfrac{\alpha}{\beta}+\dfrac{\beta}{\gamma}+\dfrac{\gamma}{\alpha}$ is $(x, y, z \neq 0)$

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143. Given the system of equation

$$x + y - z = 2,$$
$$2x - y + 4z = 1 \text{ and}$$
$$x + \alpha y + z = \beta, \text{ where } \alpha,\beta \in \{0,1,2,3,4\}$$

Let A,B,C denotes the number of ordered pairs (α,β) such that the system has unique solution, no solution, infinitely many solutions respectively, then the value of $(A - 4B + C)$ is equal to

144. If $\operatorname{adj}(A)=\begin{bmatrix}1 & 1 & 2\\ 2 & 1 & 1\\ 1 & 2 & 1\end{bmatrix}$ and matrix $A^{-2} = [x_{ij}]$ where $1 \le i, j \le 3$, then $\frac{1}{4}\left(\sum_{j=1}^{3}\sum_{i=1}^{3} x_{ij}\right)$ is (where adj(A) denotes adjoint of matrix A)

145. Let A and B be square matrices of order 4 such that $B = [b_{ij}]_{4\times4}$, $b_{ij} = 2$ and $B + I = A$. If A^{-1} exist and $9A^{-1} + B = mI$, then the value of m is (where I denotes Identity matrix)

146. If $A.\operatorname{adj}(A^2)=\begin{bmatrix}1 & 0 & 2\\ 2 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}$, then absolute value of sum of elements of adj A is

(where adj(X) denotes adjoint of matrix X)

147. Let A be a 3×3 matrix whose each entry is either 0 or 1. If the probability that A is invertible is $\frac{3n}{64}$, then n is equal to

148. Let $I_n = \int_0^1 \frac{x^n}{x^{2012}-1}dx$ and $J_n = \int_0^1 \frac{x^n}{x^{2013}+1}dx$, where $n \in N$. If matrix $A = [a_{ij}]_{3\times3}$, where

$a_{ij} = \begin{cases} I_{2012+i} - I_i, & i = j \\ 0, & i \ne j \end{cases}$ and matrix $B = [b_{ij}]_{3\times3}$, where $b_{ij} = \begin{cases} J_{2016+j} + J_{j+3}, & i = j \\ 0, & i \ne j \end{cases}$, then value of

$\frac{1}{10}\det(B^{-1}) - 2\operatorname{trace}(A^{-1})$ is

149. If A and B are non singular matrices of same order satisfying $B - A^{-1}BA = I$, then $\frac{|B+I|}{|B-I|}$ is equal to

(where I represents the identity matrix and |A| represents the determinant of matrix A)

150. Let B be a skew symmetric matrix of order 3×3 with real entries. If $A = (I + B)(I - B)^{-1}$ where $I + B$, $I - B$ are non-singular matrices, then sum of values of n satisfying the equation

$\left|\sqrt[3]{n^2}.A\right| - n\left|\sqrt[3]{5}.\operatorname{adj}(\operatorname{adj}A)\right| + 6|\operatorname{adj}A|^3 = 0$ is equal to (where |A| denotes determinant value of A.)

151. Two matrices A and B have in total 6 elements (none repeated). How many different matrices A and B are possible such that product AB is defined

152. How many 3×3 skew symmetric matrices can be formed using the numbers –2,–1,1,2,3,4,0 (any number can be used any number of times but zero can be used atmost 3 times.)

153. If

$$x_1 = 3y_1 + 2y_2 - y_3, \qquad y_1 = z_1 - z_2 + z_3$$
$$x_2 = -y_1 + 4y_2 + 5y_3, \qquad y_2 = z_2 + 3z_3$$
$$x_3 = y_1 - y_2 + 3y_3, \qquad y_3 = 2z_1 + z_3$$

express x_1,x_2,x_3 in terms of z_1,z_2,z_3.

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154. Show that the equations

$$2x + 6y = -11$$
$$6x + 20y - 6z = -3$$
$$6y - 18z = -1$$

are not consistent.

155. Using matrix method to solve the following equations :

(i) $5x + 3y + 7z = 4$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$

(ii) $x + y + z = 3$, $2x - y + z = 2$, $x - 2y + 3z = 0$

(iii) $3x + 2y + 7z = 0$, $4x - 3y - 2z = 0$, $5x + 9y + 23z = 0$

(iv) $2x + 3y - z = 0$, $x - y - 2z = 0$, $3x + y + 3z = 0$

156. Determine the values of a and b for which the system

$$\begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$$

(i) has unique solution.

(ii) has no solution.

(iii) has infinitely many solutions.

157. Show that the equations $-2x + y + z = a$, $x - 2y + z = b$, $x + y - 2z = c$ have no solution unless $a + b + c = 0$ in which case they have infinitely many solutions. Find the solutions when $a = 1$, $b = 1$, $c = -2$.

158. For what values of k the set of equation $2x - 3y + 6z - 5t = 3$, $y - 4z + t = 1$, $4x - 5y + 8z - 9t = k$ has

(i) no solution (ii) infinite number of solutions.

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ANSWER KEY

1.	A	**2.**	C	**3.**	B	**4.**	B	**5.**	B
6.	D	**7.**	B	**8.**	D	**9.**	C	**10.**	B
11.	B	**12.**	B	**13.**	A	**14.**	B	**15.**	B
16.	C	**17.**	A	**18.**	A	**19.**	B	**20.**	A
21.	B	**22.**	A	**23.**	C	**24.**	D	**25.**	C
26.	A	**27.**	B	**28.**	C	**29.**	A	**30.**	A
31.	A	**32.**	B	**33.**	B	**34.**	C	**35.**	D
36.	D	**37.**	A	**38.**	C	**39.**	C	**40.**	C
41.	C	**42.**	2	**43.**	1,3,4	**44.**	2,3,4	**45.**	3,4
46.	A,B,C,D	**47.**	A,B,D	**48.**	A,C,D	**49.**	A,B,D	**50.**	A,C
51.	A,B,C	**52.**	A,B,C,D	**53.**	A,B,C	**54.**	A,D	**55.**	A,C,D
56.	B,C,D	**57.**	B,D	**58.**	A,B,C,D	**59.**	A,B	**60.**	A,B,D
61.	B,C	**62.**	B,C	**63.**	A,B,C,D	**64.**	A,B,C,D	**65.**	B,C
66.	A,B,C,D	**67.**	A,B	**68.**	B,C,D	**69.**	A,C,D	**70.**	B,C,D
71.	A,B,C	**72.**	C,D	**73.**	A,B,D	**74.**	C,D	**75.**	A,C,D
76.	A,B,C	**77.**	A,B,C,D	**78.**	A,B,C,D	**79.**	A,B,C	**80.**	A,B
81.	A,B,C	**82.**	B,D	**83.**	C	**84.**	D	**85.**	D
86.	A	**87.**	B	**88.**	A	**89.**	C	**90.**	D
91.	A	**92.**	B	**93.**	D	**94.**	D	**95.**	C
96.	C	**97.**	A	**98.**	C	**99.**	A	**100.**	B
101.	C	**102.**	B	**103.**	C	**104.**	D	**105.**	A
106.	B	**107.**	A	**108.**	A	**109.**	A	**110.**	C
111.	C	**112.**	C	**113.**	B	**114.**	B	**115.**	A
116.	C	**117.**	D	**118.**	A	**119.**	D	**120.**	B
121.	A	**122.**	(A)→(P,R,S), (B)→(P,Q), (C)→(Q), (D)→(Q)					**123.**	4
124.	1	**125.**	6	**126.**	27	**127.**	5	**128.**	8
129.	3	**130.**	3	**131.**	7	**132.**	0	**133.**	6
134.	36	**135.**	0	**136.**	6	**137.**	1	**138.**	8
139.	6	**140.**	3	**141.**	8	**142.**	3	**143.**	5
144.	3	**145.**	9	**146.**	8	**147.**	7	**148.**	3
149.	1	**150.**	5	**151.**	8.6!	**152.**	64		
153.	$x_1 = z_1 - 2z_2 + 9z_3, x_2 = 9z_1 + 10z_2 + 11z_3, x_3 = 7z_1 + z_2 - 2z_3$							**154.**	5

155. (i) $x = \dfrac{7-16k}{11}, y = \dfrac{(k+3)}{11}, z = k, k \in R$ (ii) $x = 1, y = 1, z = 1$

(iii) $x = k,\ y = 2k,\ z = -k,\ k \in R$ (iv) $x = 0,\ y = 0,\ z = 0$

156. (i) $a \neq -3$ (ii) $a = -3,\ b \neq \dfrac{1}{3}$ (iii) $a = -3, b = \dfrac{1}{3}$

157. $x = k - 1,\ y = k - 1,\ z = k,\ k \in R.$

158. (i) $k \neq 7$ (ii) $k = 7$

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ASSIGNMENT # 10 (VECTOR & 3D) MATHEMATICS

[STRAIGHT OBJECTIVE TYPE]

1. Length of intercept cut off by the line $1 - x = y - 1 = 10 - z$ between the planes $\vec{r}.\vec{a} = 3$ and $\vec{r}.\vec{a} = -3$, where $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$ is -

(A) $\sqrt{2}$ units (B) $2\sqrt{2}$ units (C) $\sqrt{3}$ units (D) $\sqrt{5}$ units

2. Shortest distance between the lines $\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} - \hat{k})$ and $\frac{x}{1} = \frac{y-1}{-1} = \frac{z+1}{1}$ is-

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{1}{2}$

3. If $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = 2\hat{i} - \hat{k}$, then the point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is-

(A) (3,–1,1) (B) (3,1,–1) (C) (–3,1,1) (D) (–3,–1,–1)

4. Consider two lines $\vec{r} = \vec{a} + \lambda\vec{b}$ & $\vec{r} = \vec{p} + \mu\vec{q}$. If $\vec{b} = 2\vec{a} - 2\vec{p}$, then $\frac{[\vec{a}\ \vec{b}\ \vec{q}]}{[\vec{p}\ \vec{b}\ \vec{q}]}$ is

(where $[\vec{p}\ \vec{b}\ \vec{q}] \neq 0$) -

(A) $\frac{1}{2}$ (B) 2 (C) 3 (D) 1

5. Consider L_1, L_2 & L_3 be three mutually perpendicular lines. Let line L_A lies in the plane of L_1 & L_2 and makes equal angles with L_1 & L_2 whereas line L_B lies in the plane of L_2 & L_3 and makes equal angles with L_2 & L_3. Then acute angle between lines L_A & L_B is-

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{5}$

6. One of the diagonals of a parallelopiped is $4\hat{j} - 8\hat{k}$. If one of its face has diagonals $6\hat{i} + 6\hat{k}$ and $4\hat{j} + 2\hat{k}$, then its volume is-

(A) 60 (B) 80 (C) 100 (D) 120

7. Intercept made by the plane $\vec{r}.(2\hat{i} + \hat{j} + 2\hat{k}) = 9$ on x-axis, is-

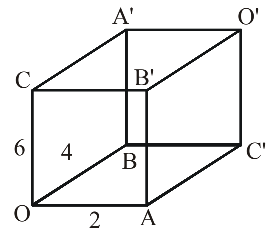
(A) 9 (B) $\frac{9}{2}$ (C) 3 (D) 18

8. Which of the following is NOT the equation of a line passing through point (1,2,3) & having direction ratios 6,2,3 -

(A) $\frac{x+5}{6} = \frac{y}{2} = \frac{z}{3}$ (B) $\frac{x-1}{6} = \frac{2-y}{-2} = \frac{z-3}{3}$

(C) $2x + 1 = 3y - 3 = z$ (D) $x = 6t - 5,\ y = 2t,\ z = 3t$ where t is parameter

9. Let OA, OB, OC are sides of a rectangular parallelopiped whose diagonals are OO', AA', BB' and CC' and D is centre of rectangle AC'O'B' and D' is centre of O'A'CB'. If sides OA,OB,OC are in the ratio 1 : 2 : 3, then angle ∠DOD' is-



(A) $\cos^{-1}\dfrac{24}{\sqrt{697}}$ (B) $\cos^{-1}\left(\dfrac{22}{\sqrt{619}}\right)$ (C) $\sin^{-1}\left(\dfrac{24}{\sqrt{697}}\right)$ (D) $\sin^{-1}\left(\dfrac{22}{\sqrt{691}}\right)$

10. Plane passing through points A(1,0,1), B(3,2,–1) and parallel to y-axis meets line $\dfrac{x-1}{1}=\dfrac{y-2}{2}=\dfrac{z-5}{3}$ at point -

(A) (0,0,2) (B) (1,2,5) (C) (0,0,0) (D) (2,4,8)

11. A plane P passes through point (1,–1,0) and is parallel to the line $\vec{r}=(1,0,1)+\lambda(2,0,3)$. The maximum distance between the plane P and the plane $\vec{r}.\left(6\hat{i}-13\hat{j}-4\hat{k}\right)-54=0$ equals-

(A) $\sqrt{\dfrac{3}{73}}$ (B) $\dfrac{35}{\sqrt{221}}$ (C) $\sqrt{\dfrac{19}{3}}$ (D) $\sqrt{\dfrac{3}{19}}$

12. If a point P whose position vector is $\vec{a}$, not lying on the plane ABC where A(2,3,0) B(3,–2,0) C(2,–1,0). Then distance of P from plane ABC is given by-

(A) $|\vec{a}.\hat{k}|$ (B) $\left|\left(\vec{a}-\left(2\hat{i}+3\hat{j}\right)\right).\left(\hat{i}-5\hat{j}\right)\right|$

(C) $\left|\left(\vec{a}-\left(2\hat{i}+3\hat{j}\right)\right).\left(\hat{j}\right)\right|$ (D) $\left|\left(\vec{a}.\left(\hat{i}-5\hat{j}\right)\right)\right|$

13. The equation of the plane passing through the line of intersection of the planes x + y + z = 1 and 2x + 3y – z + 4 = 0 and parallel to x-axis is-

(A) y – 3z – 6 = 0 (B) y – 3z + 6 = 0 (C) y – z – 1 = 0 (D) y – z + 1 = 0

14. The value of 'λ' for which the straight line 2x + y + 3z + 3 = 0 = 4x – 3y + 4z + 1 is perpendicular to the straight line 2x + y + λz – 2 = 0 = 3x – y + z – 4.

(A) $-\dfrac{12}{5}$ (B) $-\dfrac{11}{5}$ (C) $-\dfrac{9}{5}$ (D) $\dfrac{11}{5}$

15. Equation of the plane that contains the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and $\vec{r}=(\hat{i}+\hat{j})+\mu(-\hat{i}+\hat{j}-2\hat{k})$ is -

(A) $\vec{r}.(2\hat{i}+\hat{j}-3\hat{k})=-4$ (B) $\vec{r}\times(-\hat{i}+\hat{j}+\hat{k})=\vec{0}$ (C) $\vec{r}.(-\hat{i}+\hat{j}+\hat{k})=0$ (D) None of these

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16. The number of point(s) whose perpendicular distances from yz,zx and xy planes are in A.P. and whose distances from x,y,z axes are $\sqrt{25},\sqrt{20},\sqrt{13}$ respectively, is-

(A) 8 (B) 16 (C) 4 (D) infinitely many points

17. If $|\vec{a}|=|\vec{b}|=1$ and $|\vec{c}|=2$, then maximum value of $|\vec{a}-2\vec{b}|^2+|\vec{b}-2\vec{c}|^2+|\vec{c}-2\vec{a}|^2$ is-

(A) 90 (B) 50 (C) 40 (D) 42

18. A vector in the plane of $\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-\hat{j}+2\hat{k}$, perpendicular to $2\hat{i}-\hat{j}+2\hat{k}$ and having projection on $\hat{i}-\hat{j}-\hat{k}$ equal to $7\sqrt{3}$ units is -

(A) $3\hat{i}-2\hat{j}+14\hat{k}$ (B) $15\hat{i}+6\hat{j}-12\hat{k}$ (C) $15\hat{i}+6\hat{j}-18\hat{k}$ (D) $13\hat{i}+2\hat{j}-12\hat{k}$

19. If $\vec{a},\vec{b},\vec{c},\vec{a}+\vec{b},\vec{b}+\vec{c}$ and $\vec{a}+\vec{b}+\vec{c}$ are vectors of same magnitude , none of which is a null vector, then which of the following is false -

(A) Angle between $\vec{b}$ and $\vec{c}$ will always be obtuse

(B) $\vec{a}-\vec{c}$ and $\vec{b}$ are mutually perpendicular vectors

(C) $\vec{a}+\vec{c}$ will be perpendicular to $\vec{b}$.

(D) $\vec{a}$ and $\vec{c}$ are mutually perpendicular vectors

20. If projection of $\vec{x}$ along a unit vector $\vec{a}$ be 2 and $\vec{a}\times\vec{x}+\vec{b}=\vec{x}$, then vector component of $\vec{x}$ perpendicular to $\vec{a}$ is-

(A) $\dfrac{2\vec{a}+\vec{b}+\vec{a}\times\vec{b}}{2}$ (B) $\dfrac{-2\vec{a}+\vec{b}+\left(\vec{a}\times\vec{b}\right)}{2}$ (C) $\dfrac{\vec{a}-\vec{b}+\vec{a}\times\vec{b}}{2}$ (D) $\dfrac{\vec{a}-2\vec{b}+\left(\vec{a}\times\vec{b}\right)}{2}$

21. The position vectors of vertices of ΔABC are $\vec{a},\vec{b},\vec{c}$ such that $\vec{a}.\vec{a}=\vec{b}.\vec{b}=\vec{c}.\vec{c}=1$. If $\vec{a}=\lambda\vec{b}+\mu\vec{c}$ (where λ & μ are real), then position vector of orthocentre of ΔABC is-

(A) $\vec{a}+\vec{b}+\vec{c}$ (B) $\vec{b}+\vec{c}-\vec{a}$ (C) $\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$ (D) $\dfrac{\vec{a}+\vec{b}+\vec{c}}{2}$

22. Unit vectors $\vec{a},\vec{b}$ and $\vec{c}$ are coplanar. A unit vector $\vec{d}$ is perpendicular to them. If $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=\frac{1}{6}\hat{i}-\frac{1}{3}\hat{j}+\frac{1}{3}\hat{k}$ and angle between $\vec{a}$ and $\vec{b}$ is 30º, then $3\vec{c}$ is

(given that $\vec{c}.\hat{i}>0$)

(A) $\hat{i}+2\hat{j}+2\hat{k}$ (B) $\hat{i}-2\hat{j}+2\hat{k}$ (C) $\hat{i}+2\hat{j}-2\hat{k}$ (D) $2\hat{i}-\hat{j}+2\hat{k}$

23. If $\vec{a}$ and $\vec{b}$ are non-zero vectors which are linearly dependent such that $\dfrac{|\vec{a}+\vec{b}|}{|\vec{a}-\vec{b}|}=2, |\vec{b}|>|\vec{a}|$ then -

(A) $\vec{b}=3\vec{a}$ (B) $\vec{b}=-3\vec{a}$ (C) $\vec{b}=2\vec{a}$ (D) $\vec{b}=-2\vec{a}$

24. In tetrahedron OABC, if equation of altitude through O is $\vec{r}=\lambda\left(2\hat{i}+\hat{j}-2\hat{k}\right)$ and position vector of vertex A is $3\hat{i}+2\hat{j}+\hat{k}$ then length of altitude through O is (where 'O' is origin) -

(A) 6 (B) 3 (C) 2 (D) 1

25. If $\vec{a}$ and $\vec{b}$ are perpendicular unit vectors and vector $\vec{c}$ is such that $\vec{c}=\vec{a}+\vec{b}$, then $(\vec{a}\times\vec{b}).(\vec{b}\times\vec{c})+(\vec{b}\times\vec{c}).(\vec{c}\times\vec{a})+(\vec{c}\times\vec{a}).(\vec{a}\times\vec{b})$ is -

(A) 0 (B) 1 (C) –1 (D) $\vec{b}.\vec{c}+\vec{c}.\vec{a}$

26. Consider three vectors $\vec{v}_1,\vec{v}_2$ and $\vec{v}_3$ such that $\vec{v}_1=\vec{v}_2-\vec{v}_3$ where $\vec{v}_1=\left(\vec{a}\times\hat{i}\right)\times\hat{i}$, $\vec{v}_2=\left(\vec{a}\times\hat{j}\right)\times\hat{j}$ & $\vec{v}_3=\left(\vec{a}\times\hat{k}\right)\times\hat{k}$. ($\vec{a}$ is non zero vector), then -

(A) $\vec{a}.\hat{j}=0$ (B) $\vec{a}.\hat{i}=0$ (C) $\vec{a}.\hat{k}=0$ (D) $\vec{v}_1.\vec{v}_2=\left(\vec{a}.\hat{j}\right)^2$

27. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors inclined to each other at an angle of $\frac{2\pi}{3}$. If $\hat{p},\hat{q}$ & $\hat{r}$ are unit vectors along the angle bisectors of $\vec{a}\wedge\vec{b},\vec{b}\wedge\vec{c}$ & $\vec{c}\wedge\vec{a}$ respectively, then $[\hat{p}\ \ \hat{q}\ \ \hat{r}]$ is equal to-

(A) 1 (B) 0 (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

28. Let $\vec{a}\times(\vec{b}\times\vec{c})=\frac{\vec{b}}{3}+\frac{\vec{c}}{2}$ and $\vec{b}\times(\vec{c}\times\vec{a})=-\frac{\vec{c}}{2}$. If $\vec{a},\vec{b}$ and $\vec{c}$ are non-collinear pair wise unit vectors, then volume of a parallelopiped , whose coterminous edges are $\vec{a}$, $\vec{b}$ and $\vec{c}$, is-

(A) $\frac{11}{36}$ (B) $\frac{1}{2}$ (C) $\frac{\sqrt{23}}{6}$ (D) None of these

29. The scalar triple product $\left[\vec{a}+\vec{b}-\vec{c}\ \ \vec{b}+\vec{c}-\vec{a}\ \ \vec{c}+\vec{a}-\vec{b}\right]$ is equal to-

(A) 0 (B) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (C) $2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$ (D) $4\left[\vec{a}\ \vec{b}\ \vec{c}\right]$

30. Let position vectors of vertices of ΔABC be $3\hat{i}+2\hat{j}$, $-\sqrt{10}\hat{i}-\sqrt{3}\hat{j}$ and $2\hat{i}-3\hat{j}$. Let DEF be triangle obtained by joining mid-points of sides of ΔABC. If $\vec{H}$ is position vector of orthocentre of ΔDEF, then $|\vec{H}|$ is-

(A) 13 (B) 10 (C) 1 (D) 0

31. If $\vec{a},\vec{b},\vec{c}$ are non-zero vectors such that $\vec{a}\wedge\vec{b}=\vec{b}\wedge\vec{c}=\vec{c}\wedge\vec{a}=60°$, then choose the correct option(s) -

(A) If $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\left(\vec{a}\times\vec{b}\right)\times\vec{c}$ then $\vec{a}$ & $\vec{c}$ are collinear

(B) If $\vec{a}\times\left(\vec{b}\times\vec{c}\right)=\left(\vec{a}\times\vec{b}\right)\times\vec{c}$ then $\vec{a}$ & $\vec{b}$ are collinear

(C) If $\vec{a},\vec{b},\vec{c}$ are non-coplanar then $\left[(\vec{a}\times\vec{b})\times\vec{c}\ \ (\vec{b}\times\vec{c})\times\vec{a}\ \ (\vec{c}\times\vec{a})\times\vec{b}\right]=0$

(D) $|\vec{a}-\vec{b}|=|\vec{a}+\vec{b}|$

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32. If $\hat{a}, \hat{b}$ & $\hat{c}$ are unit vectors, such that $\hat{b} \wedge \hat{c} = \hat{c} \wedge \hat{a} = 60^\circ$ and the range of $\frac{5}{2}|\hat{a}+\hat{b}+\hat{c}| + 3|\hat{a}-\hat{b}+\hat{c}|$ is $\left[\alpha + \frac{5}{2}\sqrt{\beta}, \gamma\right]$ then α, β, γ are in -

(A) A.P (B) G.P (C) H.P (D) None of these

33. A vector $\vec{r}$ is equally inclined with the vector $\vec{a} = (\sin\theta)\hat{i} + (\cos\theta)\hat{j}$, $\vec{b} = (-\cos\theta)\hat{i} + (\sin\theta)\hat{j}$ and $\vec{c} = \hat{k}$ then the angle between $\vec{r}$ and $\vec{a}$ is-

(A) $\cos^{-1}\frac{1}{\sqrt{2}}$ (B) $\cos^{-1}\frac{1}{3}$ (C) $\cos^{-1}\frac{1}{\sqrt{3}}$ (D) $\frac{\pi}{2}$

34. A straight line, passing through a point P whose position vector is $\vec{p}$, is parallel to the vector which is equally inclined to two unit vectors $\vec{a}$ & $\vec{b}$, in the plane of $\vec{a}$ & $\vec{b}$, then distance of point Q with position vector $\vec{q}$ from the line is -7

(A) $\left|(q-p)\times(\vec{a}+\vec{b})\right|$ (B) $\left|(\vec{q}-\vec{p}).(\vec{a}+\vec{b})\right|$ (C) $\frac{\left|(\vec{q}-\vec{p})\times(\vec{a}+\vec{b})\right|}{|\vec{a}+\vec{b}|}$ (D) $\frac{\left|(\vec{q}-\vec{p}).(\vec{a}+\vec{b})\right|}{|\vec{a}+\vec{b}|}$

35. Distance between $\vec{r} = \hat{i} + \lambda(\hat{j}+\hat{k})$ & $\vec{r} = \hat{j} + \mu(\hat{j}+\hat{k})$ is equal to-

(A) $\frac{1}{\sqrt{2}}$ (B) 1 (C) $\frac{\sqrt{3}}{\sqrt{2}}$ (D) $\sqrt{2}$

36. Let $2\hat{a} + \vec{b} = \hat{a} \times \vec{b}$ (where $\hat{a}$ is a unit vector) then $|3\vec{b} + 4\hat{a}|$ is equal to-

(A) 0 (B) 1 (C) 2 (D) 3

[MULTIPLE OBJECTIVE TYPE]

37. Let equation of three planes are given as $\vec{r}.(a\hat{i}+\hat{j}+5\hat{k}) = 0$, $\vec{r}.(2\hat{i}+2\hat{j}+a\hat{k}) = 0$ and $\vec{r}.(5\hat{i}+6\hat{j}-\hat{k}) = 0$

If these planes intersect at a point other than origin, then b is sum of all possible values of a and c is product of all possible values of a. Then which of the following is (are) true-

(A) atleast one value of a is positive

(B) one value of a lies in (1,2)

(C) number of rational terms in expansion of $(2^b + 3^c)^{10}$ is 6

(D) sum of an infinite G.P. whose first term is c and common ratio is b is 4.

38. If sides of the triangle ABC are represented by the lines $\vec{r} = \hat{i}+\hat{j}+\lambda(\hat{j}-\hat{k})$, $\vec{r} = \hat{j}+\hat{k}+\mu(\hat{k}-\hat{i})$, $\vec{r} = \hat{k}+\hat{i}+t(\hat{i}-\hat{j})$, then (where $\lambda, \mu, t \in R$) -

(A) vertices of triangle are (1,1,0), (1,0,1) and (0,1,1)

(B) area of triangle ABC is $\frac{\sqrt{3}}{2}$ sq. units

(C) (0,0,0) lies in the plane of triangle ABC

(D) triangle ABC is an equilateral triangle

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39. If a parallelogram ABCD in which $|\overrightarrow{AB}| = a$, $|\overrightarrow{AD}| = b$, $|\overrightarrow{AC}| = c$, where a,b,c are sides of an acute angled triangle, then which of the following is (are) true-

(A) $\overrightarrow{DB}.\overrightarrow{AB}$ is positive

(B) $\overrightarrow{DB}.\overrightarrow{AB}$ is negative

(C) Angle between $\overrightarrow{DB}$ and $\overrightarrow{AB}$ is acute

(D) Angle between $\overrightarrow{DB}$ and $\overrightarrow{AB}$ obtuse.

40. Let A(1,0,0), B(1,1,0) are fixed points. If OA(where O is the origin) is rotated through O in the xz-plane by a variable angle 'θ' to reach the new position OP, then which of the following is (are) true-

(A) maximum volume of tetrahedron OAPB is $\frac{1}{6}$

(B) maximum volume of tetrahedron OAPB is $\frac{1}{2}$.

(C) equation of the plane OAP is $\vec{r}.\left(\cos\theta\hat{i} + \sin\theta\hat{k}\right) = 0$

(D) length of projection of $\overrightarrow{OP}$ on $\overrightarrow{OB}$ is $\frac{|\cos\theta|}{\sqrt{2}}$

41. From point A(–1,1,–1), the perpendicular is drawn to a plane P_1 whose foot is (1,–1,1). From point B(3,–1,–1), the perpendicular is drawn to a plane P_2 whose foot is (1,2,1). P_3 is a plane containing the line of intersection of P_1 and P_2 and passing through the mid point of the line joining A and B. Then which of the following statement(s) is/are correct ?

(A) Direction ratios of line of intersection of P_1 and P_2 are 5,4,–1

(B) Equation of plane P_3 is $16x - 19y + 4z = 12$

(C) Equation of line of intersection of P_1 and P_2 is $\frac{x-15}{5} = \frac{y-12}{4} = \frac{z-0}{-1}$

(D) The length of perpendicular from (0,0,0) to plane P_3 is $\frac{2}{\sqrt{18}}$

42. Which of the following is (are) correct ? (where $\hat{a}, \hat{b}, \hat{c}$ are unit vectors)

(A) If $|\hat{a} + \hat{b}| < 1$, then angle between $\vec{a}$ and $\vec{b}$ will be obtuse.

(B) If $|\hat{a} - \hat{b}|^2 + |\hat{b} - \hat{c}|^2 + |\hat{c} - \hat{a}|^2 = 9$, then $\hat{a} \times \hat{b} = \hat{b} \times \hat{c} = \hat{c} \times \hat{a}$.

(C) If $\vec{V}_1 = \hat{i} - 2\hat{j} + \hat{k}, \vec{V}_2 = 3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{V}_3 = \hat{i} + 2\hat{j} + \hat{k}$, then $\vec{V}_1, \vec{V}_2$ and $\vec{V}_3$ are coplanar.

(D) The plane XOY bisects the line segment joining A(1,1,2) and B(2,1,–2) (where O is origin).

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43. A line $\frac{2x-1}{1}=\frac{2y-k}{2}=\frac{2z-3}{2}$ cuts the xy- plane and xz- plane at points A and B respectively. If AB subtends a right angle at the origin, then –

(A) the value of k is 2

(B) the value of k is 3

(C) the area of ΔAOB is $\frac{\sqrt{5}}{8}$ (where O is the origin)

(D) the area of ΔAOB is $\frac{\sqrt{5}}{16}$ (where O is the origin)

44. If plane P containing lines $L_1:\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ & $L_2:\frac{x-2}{2}=\frac{y-3}{k}=\frac{z-4}{3}$, then-

(A) L_1 & L_2 can be parallel

(B) L_1 & L_2 can be intersecting

(C) equation of plane P can be $x - y + 1 = 0$

(D) equation of plane P can be $x + y - 2z + 3 = 0$

45. Consider two planes $P_1:\vec{r}.\vec{n}_1=1$ & $P_2:\vec{r}.\vec{n}_2=1$ then identify the correct statement(s) $(\vec{n}_1\neq\lambda\vec{n}_2)$

(A) P_1, P_2 have infinite points in common.

(B) There exist no plane which neither intersects P_1 nor P_2.

(C) P_1, P_2 are parallel but different.

(D) There exist a plane P_3 which intersect P_1, P_2 at exactly one point.

46. OAB is an isosceles triangle with OA = OB. If O is origin, A is (–20, –6, –5) and equation of base AB is $\frac{x+16}{2}=\frac{y}{3}=\frac{z+1}{2}$, then -

(A) If height of triangle is $3\sqrt{N}$ then N is 21.

(B) If height of triangles is $3\sqrt{N}$ then N is 3.

(C) If B has co-ordinates (a,b,c) then a + b + c is 25

(D) If B has co-ordinates (a,b,c) then a + b + c is 30

47. The plane x + y + z = 4 cuts the coordinate axes at A,B and C. Let M be the centre of tetrahedron OABC. The mid points of AM, BM and CM be D,E, & F respectively. If the shortest distance between the line passing through O & E and the line passing through D & F is $\left(\frac{a}{b\sqrt{6}}\right)$ where a & b are coprime numbers, then-

(A) a + b = 17 (B) a – b = 11 (C) a + b = 18 (D) a – b = 12

48. If $\vec{a}$ & $\vec{c}$ are unit vectors & $\vec{a}=2(\vec{b}\times\vec{c})+(\vec{c}\times\vec{a})$, then -

(A) $\vec{a},\vec{b},\vec{c}$ are coplanar vectors

(B) $\left|\vec{a}\times(\vec{b}\times\vec{c})\right|=\frac{1}{2}$

(C) $\left|\vec{b}\times\vec{c}\right|=\frac{1}{\sqrt{2}}$

(D) $\vec{a}.\vec{c}=0$

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[COMPREHENSION TYPE]

Paragraph for Question 49 to 51

Let $\vec{a} = x\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = 2\hat{i} + x\hat{j} + 4\hat{k}$ & $\vec{c} = 3\hat{i} + 3\hat{j} + x\hat{k}$.

On the basis of above information, answer the following questions :

49. The maximum possible value of volume of the tetrahedron whose co-terminous edges are $\vec{a}, \vec{b}, \vec{c}$ $(0 < x < 3)$ -

(A) $\dfrac{8\sqrt{2}}{3\sqrt{3}}$ (B) $\dfrac{16\sqrt{2}}{9\sqrt{3}}$ (C) $\dfrac{32\sqrt{3}}{3\sqrt{2}}$ (D) $\dfrac{32\sqrt{3}}{9\sqrt{2}}$

50. The value of x for which the vectors $\vec{a}, \vec{b}, \vec{c}$ are not able to make the edges of a tetrahedron $(0 < x < 3)$-

(A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) $2\sqrt{2}$

51. The vector joining the corner meeting the coterminous edges denoted by $\vec{a}, \vec{b}, \vec{c}$ of the tetrahedron to the centroid of the opposite face when the volume of tetrahedron is $\dfrac{16}{3}(x > 0)$ is

(A) $3\hat{i} + \dfrac{8}{3}\hat{j} + 2\hat{k}$ (B) $\dfrac{4}{3}\hat{i} + \dfrac{4}{3}\hat{j} + 2\hat{k}$ (C) $\dfrac{2}{3}\hat{i} + \hat{j} + \dfrac{\hat{k}}{3}$ (D) $2\hat{i} + 3\hat{j} + 5\hat{k}$

Paragraph for Question 52 and 53

P(0, 1, 2) be a point on the plane $P_1 : x + 2y + 3z = 8$. Let Q be the only point on the plane $P_2 : 2x + 2y + z = k$ $(k > 0)$ which lies at a distance 5 units from point P.

52. Co-ordinates of Q are.

(A) (4, 1, 5) (B) $\left(\dfrac{10}{3}, \dfrac{13}{3}, \dfrac{11}{3}\right)$ (C) $\left(\dfrac{7}{\sqrt{2}}, \dfrac{1}{2}, \dfrac{3}{2}\right)$ (D) (3, 1, 2)

53. Vector perpendicular to $\overrightarrow{PQ}$ which lies in the plane P_1, can be

(A) $4\hat{i} - 5\hat{j} + 2\hat{k}$ (B) $2\hat{i} + 5\hat{j} - 4\hat{k}$ (C) $3\hat{i} - 2\hat{k}$ (D) $3\hat{i} + 2\hat{j} + \hat{k}$

Paragraph for Question 54 to 56

Let a plane whose equation is $x + y + z = 50$ cuts the co-ordinate axes at points A,B & C. A tetrahedron OABC is formed where 'O' is origin.

54. If $p\hat{i} + q\hat{j} + r\hat{k}$ represents the position vector of point of concurrency of line joining the mid-point of opposite edges of tetrahedron, then $4(p + q + r)$ is-

(A) 50 (B) 100 (C) 150 (D) 200

55. Number of points with integral co-ordinates lying inside tetrahedron is (point P (a,b,c) is said to have integral co-ordinates if $a,b,c \in I$)-

(A) 19,600 (B) 18424 (C) 20,000 (D) 1960

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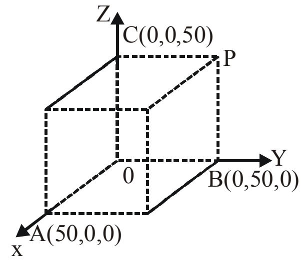
56. A cube is constructed having OA,OB and OC as coterminous edges If an insect travels from point A to point P, then shortest distance that it has to travel (journey is to be made along the faces of cube)



(A) $\sqrt{5000}$ (B) $\sqrt{10000}$

(C) $\sqrt{11000}$ (D) $\sqrt{12500}$

Paragraph for Question 57 to 59

Let A and B be two points on the lines $L_1 : \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $L_2 : \frac{x-2}{2} = \frac{y+1}{1} = \frac{z+1}{2}$ respectively, which are nearest to each other.

57. Distance between A and B equals -

(A) $\frac{1}{\sqrt{6}}$ (B) $\frac{3}{2}$ (C) $\frac{3\sqrt{2}}{2}$ (D) $3\sqrt{2}$

58. Equation of the plane OAB is (where O is origin) -

(A) $x - 34y + z = 0$ (B) $2x - 17y + 3z = 0$

(C) $x + 34y - 2z = 0$ (D) $2x + 17y - 3z = 0$

59. Volume of tetrahedron OABC is, where C(1,0,1) (O is the origin) -

(A) $\frac{1}{2}$ (B) $\frac{1}{12}$ (C) $\frac{1}{6}$ (D) 1

Paragraph for Question 60 to 61

The plane $\frac{x}{6} + \frac{y}{9} + \frac{z}{12} = 1$ cuts the coordinate axes at A, B and C. Let G be the centroid of the ΔABC. The points D, E, F and R divides OG, OA, OB and OC in the ratio of 2 : 1 respectively, where O is the origin.

On the basis of above information, answer the following questions :

60. The area of ΔODE is -

(A) $\frac{20}{9}$ (B) $\frac{20}{3}$ (C) $\frac{20}{4}$ (D) $\frac{20}{11}$

61. The ratio of volume of tetrahedron OERF to the volume of tetrahedron OABC is -

(A) $\frac{27}{8}$ (B) $\frac{8}{27}$ (C) $\frac{11}{27}$ (D) $\frac{27}{11}$

Paragraph for Question 62 to 64

A line L_1 with direction ratios –3,2,4 passes through the point A(7,6,2) and a line L_2 with direction ratios 2,1,3 passes through the point B(5,3,4). A line L_3 with direction ratios 2,–2,–1 intersects L_1 & L_2 at C & D respectively.

On the basis of above information, answer the following questions :

62. The length CD is equal to-

(A) 4 (B) 6 (C) 9 (D) 11

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63. The equation of plane parallel to line L_1 and containing line L_2 is equal to-
(A) $x + 3y + 4z = 30$ (B) $x + 2y + z = 15$ (C) $2x - y + z = 11$ (D) $2x + 17y - 7z = 33$

64. The volume of parallelopiped formed by $\overrightarrow{AB}, \overrightarrow{AC}$ & $\overrightarrow{AD}$ is equal to-
(A) 140 (B) 138 (C) 134 (D) 130

Paragraph for Question 65 to 67

A vector $\vec{r}$ satisfying the equations $\vec{r} \times (\vec{a} \times \vec{b}) = \vec{b} + 2\vec{a}$ and $\vec{a} \times \vec{r} + (2\vec{a} + \vec{b}) = (\vec{a} \times \vec{b}) \times \vec{a}$ (where $\vec{a}$ and $\vec{b}$ are non zero and non collinear vectors).

On the basis of above information, answer the following questions :

65. The vector $\vec{r}$ is equal to-

(A) $\dfrac{(\vec{a} \times \vec{b})(1 - \vec{a}^2) - \vec{a}}{\vec{a}^2}$ (B) $\dfrac{(\vec{a} \times \vec{b})(1 + \vec{a}^2) + \vec{a}}{\vec{a}^2}$

(C) $\dfrac{(\vec{a} \times \vec{b})(1 - \vec{a}^2) + \vec{a}}{\vec{a}^2}$ (D) $\dfrac{(\vec{a} \times \vec{b})(1 + \vec{a}^2) - \vec{a}}{\vec{a}^2}$

66. Identify the correct option-
(A) $\vec{a}.\vec{b} = 2\vec{a}^2$ (B) $\vec{a}.\vec{b} = -2\vec{a}^2$ (C) $\vec{a}.\vec{b} = \vec{a}^2$ (D) $\vec{a}.\vec{b} = -\vec{a}^2$

67. If angle between $\vec{a}$ and $\vec{b}$ is $\dfrac{3\pi}{4}$, then the maximum value of $\left|[\vec{r}\ \vec{a}\ \vec{b}]\right|$ is (where $|\vec{a}| < 1$) -

(A) $\dfrac{1}{4}$ (B) $\dfrac{1}{2}$ (C) 1 (D) 2

[SUBJECTIVE TYPE]

68. Let volume of a tetrahedron ABCD is $\dfrac{81}{2}$ sq. unit and volume of parallelopiped whose three coterminous edges are line segments joining centroid of any face of tetrahedron with centroids of its other three faces is 3^λ, then λ is

69. Let L_1 be the projection and L_2 be the image of the z-axis in the plane $3x - 4y + z + 1 = 0$. The distance of the point (1,2,3) from the plane containing the lines L_1 & L_2 is

70. If the plane $\vec{r}.(\hat{i} + \hat{j} + \hat{k}) = 1$, $\vec{r}.(\hat{i} + 2\lambda\hat{j} + \lambda^2\hat{k}) = 3$ and $\vec{r}.(\hat{i} + \hat{j} + \lambda\hat{k}) = 5$ intersect in a line, then the number of real values of λ is

71. If line $\dfrac{x-2}{a} = \dfrac{y-b}{1} = \dfrac{z-4}{4}$ meets the cylinder $x^2 + y^2 - 8x + 2y - 1 = 0$ at one point in xy plane, then maximum value of $(a + b)$ is

72. Let $R(3,\alpha,2)$ be a point on plane $4x - y - z = \beta$ & R is projection of point M which lies on the line $\dfrac{x+1}{2} = \dfrac{y-2}{3} = \dfrac{z+4}{5}$, then $|\alpha| + |\beta|$ is equal to

73. Let P_1 be the plane passing through (1,1,–1) and (2,6,1) and perpendicular to the plane $P_2 : x + 5y + 2z = 0$. Line of intersection of P_1 and P_2 is parallel to vector $\hat{i} - \hat{j} + 2\hat{k}$. If distance between P_1 and y-axis is d then $5d^2$ is

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74. Direction cosines ℓ,m,n of two lines satisfy $2\ell + 2m = 3n$ & $n^2 + 2m^2 = 3mn$. If θ is the angle between the lines then $9|\cos\theta|$ is equal to

75. Locus of all the points which are at a distance of 3 units from the line $\vec{r} = \lambda(\hat{i} + \hat{j} + \hat{k})$ is given by $x^2 + y^2 + z^2 - xy - yz - zx = \frac{k}{2}$ then the value of $\left(\frac{k}{3}\right)$ is equal to

76. Let $\vec{a} = 5\hat{i} + p\hat{j} + 12\hat{k}$ & $\vec{b} = \hat{i} + 13\hat{j} + 2\sqrt{q}\hat{k}$ where p,q $\in$ N. It is given that $|\vec{a}| = |\vec{b}|$ & p,q $\in$ [1,1000].

If total number of ordered pairs of (p,q) is λ, then $\frac{(\lambda+1)}{16}$ is

77. If the resultant $\vec{R}$ of two non parallel forces $\vec{P}$ and $\vec{Q}$ of magnitudes 5 units and 4 units, trisects the angle between them, then $4|R|$ is

78. Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}, \vec{b} = 2\hat{j} - \hat{k}$ and vector $\vec{c}$ satisfying the conditions (i) $\vec{c}.\vec{b} = 0$ (ii) $\vec{a}.\vec{c} = 4$, if $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, then the value of $[\alpha + \beta]$ is (where [.] denotes greatest integer function)

79. Let $\vec{a}, \vec{b}, \vec{c}$ be non coplanar unit vectors, making an angle of 60° with each other. If $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$, then the value of $2(p^2 + q^2 + r^2)$ is equal to

80. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors having magnitude 1,1,2 respectively. If $\vec{b} + (\vec{a}.\vec{c})\vec{a} = (\vec{a}.\vec{a})\vec{c}$, then cosecant of acute angle between $\vec{a}$ and $\vec{c}$ is

81. In a tetrahedron OABC, G is the centre of the tetrahedron, P, Q and R are the mid points of OA, OB and OC respectively. If the shortest distance between the line GR and PQ is $\frac{1}{2\sqrt{k}}$, then the value of k is (where O is origin, $\overrightarrow{OA} = \hat{i}$, $\overrightarrow{OB} = \hat{j}$ and $\overrightarrow{OC} = \hat{k}$)

82. In a tetrahedron OABC, the edges are of lengths, $|OA| = |BC| = a$, $|OB| = |AC| = b$, $|OC| = |AB| = c$. Let G_1 and G_2 be the centroids of the triangle ABC and AOC (where AOC is not a right angled triangle) such that $OG_1 \perp BG_2$, then the value of $\frac{a^2 + c^2}{b^2}$ is

83. If A is the area of triangle formed by the lines L_1, L_2, L_3 where $L_1 : \vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} - 2\hat{j} + \hat{k})$, $L_2 : \vec{r} = \hat{i} - \hat{j} + \mu(\hat{i} + \hat{j} + \hat{k})$ & $L_3 : \vec{r} = 2\hat{i} + \hat{k} + t\hat{j}$ then $(4A^2)$ is equal to

84. A fixed vector $\vec{d}$ is perpendicular to three non-zero vectors $\vec{a}, \vec{b}, \vec{c}$ whose magnitudes are 1,2,3 respectively. It is given that $\vec{a} \wedge \vec{b} = \alpha$, $\vec{b} \wedge \vec{c} = \beta$, $\vec{c} \wedge \vec{a} = \gamma$ then $\begin{vmatrix} 1 & 2\cos\alpha & 3\cos\gamma \\ 2\cos\alpha & 4 & 6\cos\beta \\ 3\cos\gamma & 6\cos\beta & 9 \end{vmatrix}$ is equal to

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85. Let $\hat{a}, \hat{b}, \hat{c}$ be unit vectors such that $\hat{b} \wedge \hat{c} = 60^\circ$ & $\left(\hat{a} \times \left(\hat{b} \times \hat{c}\right)\right) \times \hat{a} = \vec{0}$ $\left(\left[\hat{a}\ \hat{b}\ \hat{c}\right] \neq 0\right)$ and if volume of the parallelepiped formed by $\hat{a} + \hat{b}, \hat{b} + \hat{c}, \hat{c} + \hat{a}$ is k, then (k^2) is equal to

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ANSWER KEY

1.	C	**2.**	B	**3.**	B	**4.**	D	**5.**	A
6.	D	**7.**	B	**8.**	C	**9.**	A	**10.**	A
11.	B	**12.**	A	**13.**	B	**14.**	B	**15.**	C
16.	A	**17.**	D	**18.**	B	**19.**	C	**20.**	B
21.	A	**22.**	B	**23.**	A	**24.**	C	**25.**	C
26.	A	**27.**	B	**28.**	C	**29.**	D	**30.**	D
31.	C	**32.**	A	**33.**	C	**34.**	C	**35.**	C
36.	C	**37.**	A,B,C	**38.**	A,B,D	**39.**	A,C	**40.**	A,D
41.	A,B,C	**42.**	A,B,D	**43.**	A,D	**44.**	A,B,C,D	**45.**	A,B,D
46.	A,C	**47.**	A,B	**48.**	B,C,D	**49.**	B	**50.**	D
51.	A	**52.**	B	**53.**	A	**54.**	C	**55.**	B
56.	D	**57.**	C	**58.**	A	**59.**	B	**60.**	B
61.	B	**62.**	C	**63.**	D	**64.**	B	**65.**	A
66.	B	**67.**	C	**68.**	2	**69.**	2	**70.**	0
71.	4	**72.**	74	**73.**	9	**74.**	8	**75.**	9
76.	2	**77.**	9	**78.**	2	**79.**	3	**80.**	2
81.	6	**82.**	3	**83.**	18	**84.**	0	**85.**	3

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Straight Objective Type

1. If two events A and B are such that $P(\bar{A}) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.2$, then $P((A \cup \bar{B}) | \bar{A})$ is

(A) 0.3 (B) 0.5 (C) 0.75 (D) 0.8

2. All possible 5 lettered word formed from the letters of CASHLESS, then the probability that the word formed have "ASH" is -

(A) $\frac{13}{1370}$ (B) $\frac{50}{671}$ (C) $\frac{39}{1520}$ (D) $\frac{14}{1533}$

3. The probability of winning the U.P. election by BJP, SP and BSP are 0.4, 0.4 and 0.2 respectively. The probability that corruption is stopped if BJP, SP and BSP wins are 0.04, 0.03 & 0.01 respectively. If corruption is stopped, then probability that BJP wins, is

(A) $\frac{8}{15}$ (B) $\frac{13}{15}$ (C) $\frac{2}{3}$ (D) $\frac{2}{5}$

4. Six balls b_1, b_2, b_3, b_4, b_5 & b_6 are kept at random in six boxes B_1, B_2, B_3, B_4, B_5 & B_6 one in each box. The probability that exactly 2 balls go to the corresponding numbered boxes, is -

(A) $\frac{5}{16}$ (B) $\frac{1}{16}$ (C) $\frac{3}{16}$ (D) $\frac{3}{8}$

5. Let A and B are two events of a sample space S such that $P(A \cap B) = 0.1$ and $P(A' \cap B') = 0.7$, then the value of $P(A \cap B') \cup (A' \cap B))$ is equal to -

(A) 0.3 (B) 0.5 (C) 0.2 (D) 0.1

6. Given A= $\{a_1,a_2,a_3.......a_n\}$ & B = $\{b_1,b_2,b_3\}$ and a function $f : A \to B$ is randomly defined. Then the probability that f is onto is-

(A) $\frac{3^n - 3.2^n - 3}{3^n}$ (B) $\frac{3^n - 3.2^n + 1}{3^n}$ (C) $\frac{3^n + 3.2^n - 3}{3^n}$ (D) $\frac{3^n - 3.2^n + 3}{3^n}$

7. There are three different Urns, Urn-I, Urn-II and Urn-III containing 1 Blue, 2 Green ; 2 Blue, 1 Green; 3 Blue, 3 Green balls respectively. If two Urns are randomly selected and a ball is drawn from each Urn and if the drawn balls are of different colours then the probability that chosen Urn was Urn-I and Urn-II is-

(A) $\frac{1}{7}$ (B) $\frac{5}{13}$ (C) $\frac{5}{14}$ (D) $\frac{5}{7}$

8. A tennis match of best of 5 sets, where a player winning 3 sets first is declared as winner of the match, is played between two players A & B. The probability that a set is won by A is $\frac{1}{2}$ and if he looses the set then probability of his winning the next set is $\frac{1}{4}$, otherwise it remains same, then probability that A wins is -

(A) $\frac{5}{16}$ (B) $\frac{7}{16}$ (C) $\frac{3}{32}$ (D) $\frac{1}{8}$

9. Persons A,B,C,D & E are to be seated on a circular table. Number of ways they can be seated so that exactly two from A,B & C are adjacent is equal to -

(A) 6 (B) 8 (C) 12 (D) 24

10. Ram is sitting along with 9 other students in a line. He goes to purchase a balloon and when he comes back he finds that 4 students are not there. The probability that his neighbours are still there, is (Assume that Ram is not sitting at the ends of row).

(A) $\frac{1}{3}$ (B) $\frac{5}{9}$ (C) $\frac{5}{18}$ (D) $\frac{1}{18}$

11. The probability that Rishi's Gardner will forget to water the rose bush is $\frac{2}{3}$. The rose bush is in questionable condition. Any how if watered, the probability of its withering is $\frac{1}{2}$ and if not watered then the probability of its withering is $\frac{3}{4}$. Rishi went out of station and after returning he finds that rose bush has withered. The probability that the Gardner did not water the rose bush is-

(A) $\frac{2}{3}$ (B) $\frac{3}{5}$ (C) $\frac{3}{4}$ (D) $\frac{4}{5}$

12. There are twenty points on a circle. If four points are selected at random, then the probability that there are at least two points in between any two selected points is -

(A) $\frac{{}^{11}C_3}{{}^{20}C_4}$ (B) $\frac{{}^{5}C_1 \times {}^{11}C_3}{{}^{20}C_4}$ (C) $\frac{{}^{20}C_1 \times {}^{10}C_3}{4.{}^{20}C_4}$ (D) None of these

13. From a pack of 52 well shuffled cards, cards are drawn one by one without replacement. If 4th drawn card is found to be ace, then what is the probability, that there are no more aces left in the pack, is-

(A) $\frac{1}{{}^{48}C_3+3.{}^{49}C_2+1}$ (B) $\frac{1}{{}^{48}C_3+{}^{49}C_2+1}$ (C) $\frac{1}{3.{}^{48}C_3+{}^{49}C_2+1}$ (D) $\frac{1}{{}^{52}C_4+1}$

14. Let X,Y and Z be three events such that events X and Y are mutually exclusive, if $P(X \cup Y) = 1$, $P(X \cap Z) = \frac{1}{5}$ and $P(Z) = \frac{7}{15}$, then $P(Y \cap Z)$ is equal to-

(A) $\frac{4}{15}$ (B) $\frac{2}{3}$ (C) $\frac{7}{25}$ (D) $\frac{11}{15}$

15. Sum of digits of a 5 digit number is 41. Then the probability that such a number is divisible by 11 is

(A) $\frac{2}{15}$ (B) $\frac{3}{35}$ (C) $\frac{11}{36}$ (D) $\frac{6}{35}$

16. A box contains 3m coins, m of which are fair and 2m are biased. The probability of getting a head when a fair coin is tossed is $\frac{1}{2}$, while it is $\frac{1}{3}$ when a biased coin is tossed. A coin is drawn from the box at random and tossed which fell headwise. The probability that the same coin when tossed again will show head is -

(A) $\frac{17}{42}$ (B) $\frac{17}{36}$ (C) $\frac{13}{72}$ (D) dependent on m.

17. Initially a bag was known to contain some one rupee ('0' or more) and some fifty paisa ('0' or more) coins. In all bag was known to have 4 coins. Two coins were randomly drawn from the bag and both found to be one rupee coin. If these coins are replaced, what is the probability that next drawn coin is fifty paisa coin ?
(Assumption : Initially all number of rupee coins in the bag are equiprobable).

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{1}{8}$ (D) $\frac{1}{16}$

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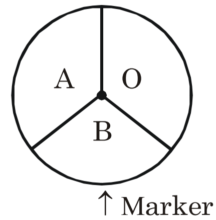
18. A circular disc with 3 sectors marked as alphabet O,A,B has circumference 1. The alphabet coming infront of marker is noted down after each spin (as shown in diagram). Assume boundary of sectors do not come infront of marker. It is spun 6 times around its centre and resulting alphabet which comes infront of marker are noted down in order. If the probability that the word BAOBAA to be formed is maximised, then (where x,y,z respectively denotes the probability of occurrence of alphabet O,A,B) -



(A) $x = \frac{1}{2}$ (B) $y = \frac{1}{2}$ (C) $y = \frac{1}{3}$ (D) $z = \frac{1}{6}$

19. A drunkard removes two randomly chosen letters of the message HAPPY HOUR that is attached on a sign board. His drunkard friend put those two letters back in a random order. Then the probability that HAPPY HOUR appears again is equal to

(A) $\frac{17}{36}$ (B) $\frac{18}{36}$ (C) $\frac{19}{36}$ (D) $\frac{25}{46}$

20. 5 married couples take part in a race. There are 5 race tracks on the ground. On each track a male & a female must have to run. Probability such that no husband & wife run on the same track, is-

(A) $\frac{44}{119}$ (B) $\frac{11}{30}$ (C) $\frac{1}{5}$ (D) $\frac{119}{120}$

Multiple Correct Answer Type

21. Ram tosses n + 1 fair coins and Shyam tosses n fair coins ($n \in N$). If P(n) is the probability that Ram gets more heads than Shyam, then -

(A) P(n) is independent of n (B) P(n) is dependent on n
(C) P(10) = 1/2 (D) P(10) > P(5)

22. There are four dice, two of which are fair and two are biased with each biased die having chance of throwing six as $\frac{1}{8}$. Two of them are randomly selected and thrown. If both the die thrown shows the face six, then-

(A) The probability that both are biased is $\frac{9}{73}$

(B) The probability that neither of them is biased is $\frac{16}{73}$

(C) The probability that both are biased is $\frac{4}{13}$

(D) The probability that neither of them is biased is $\frac{8}{13}$

23. The probabilities that a man makes a certain dangerous journey by car, motor cycle or on foot are 1/2, 1/6 and 1/3 respectively. The probabilities of an accident when he uses these means of transport are 1/5, 2/5 and 1/10 respectively. Which of the following statement(s) hold good?

(A) The probability of an accident occurring in a single journey, is $1/5$.

(B) If an accident is known to have happened, the probability that the man was travelling by car, is $1/2$.

(C) If an accident is known to have happened, the probability that the man was travelling by motor cycle, is $1/5$.

(D) If an accident is known to have happened, the probability that the man was travelling on foot, is $1/3$.

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24. If 3 numbers are chosen randomy from the set $\{1, 2, 2^2, ..., 2^n\}$ without replacement then the probability they are in increasing geometric progression is:

(A) $\frac{3}{2n}$(n odd)
(B) $\frac{4}{3n-1}$(n odd)
(C) $\frac{3n}{2(n^2-1)}$(n even)
(D) $\frac{2n}{n^2+4}$ (n even)

25. From an urn containing M white and (N – M) black balls, r balls have been lost. Then probability of drawing a white ball -

(A) Before the loss is M/N
(B) After the loss is M/N if r = 1
(C) After the loss is M/N if r = 2
(D) After the loss is M/N if r = 3

26. Suppose we have 7 coins (C_i), where i = 1, 2, ... 7 such that the probability of head on i^{th} coin is $P(H_i) = \frac{i+1}{8}$. One of the coin is randomly selected and tossed and H shows the event that it shows up the head. Then which of the following are true -

(A) $P(C_4/H) = \frac{1}{7}$
(B) $P(C_1/H) = \frac{2}{35}$
(C) $P(H) = \frac{5}{8}$
(D) $P(\bar{H}) = \frac{5}{8}$

27. If 10 balls are selected randomly from unlimited number of red, white, green and blue balls, then probability of these selections having balls of all 4 different colours is equal to -

(A) $\frac{42}{143}$

(B) $\frac{^4C_3}{^{10}C_3}$

(C) probability of obtaining a triangle if three points are joined randomly from given 13 points out of which 4 are collinear & rest are non collinear.

(D) probability of getting 3 cards which are neither Ace, nor King nor Queen nor Jack drawn randomly from pack of 52 playing cards, if it is known that drawn cards are all spade cards.

28. A, B, C in order roll a fair die indefinitely until a winner is obtained. A wins if he rolls a prime number B wins if he rolls a composite number whereras C wins if he rolls neither prime nor composite. Let P(A), P(B), P(C) denotes their respective chances of winning then identify the correct statement(s)

(A) $P(A) + P(B) = \frac{12}{13}$
(B) $P(B) - P(C) = \frac{2}{13}$
(C) P(A) = 9P(C)
(D) $P(A) - P(C) = \frac{7}{13}$

29. Two numbers are selected randomly from the set S = {1, 2, 3, 4, 5, 6} one by one without replacement. Let P_1 denotes the probability that maximum of the two numbers is greater than 4 and P_2 denotes the probability that minimum of the two numbers is 3, then

(A) $P_1 + P_2 = 1$
(B) $P_1^2 + P_2^2 = \frac{2}{5}$
(C) $P_1 > 2P_2$
(D) $2P_1 > 5P_2$

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30. In a city a person has either a Sedan car or an SUV with probability $\frac{3}{10}$ & $\frac{4}{10}$ respectively. If he has Sedan only, then he keeps a driver with probability $\frac{6}{10}$, whereas if he owns SUV only, then he keeps a driver with probability $\frac{7}{10}$, whereas if he keeps both types of cars, then his probability of keeping a driver is $\frac{9}{10}$. Now which of the following holds good ?

(Assume owning Sedan or SUV are independent events)

(A) Probability that person keeps a driver is $\frac{206}{500}$.

(B) Given that the person keeps driver, then probability that he owns SUV is $\frac{76}{103}$

(C) Probability that person keeps a driver is $\frac{403}{500}$.

(D) Given that the person keeps driver, then probability that he owns SUV is $\frac{54}{103}$

31. An experiment has 14 equally likely outcomes. Let A and B be two non-empty independent events of the experiment. If A consists of 4 outcomes, then which option(s) can be true-

(A) $P(A \cup B) = \frac{9}{14}$ (B) $P(A \cap \bar{B}) = 0$ (C) $P\left(\frac{A}{B}\right) = \frac{4}{14}$ (D) $P\left(\frac{B}{A}\right) = \frac{1}{2}$

32. The probability that a student passes in at least one of the three independent exams A,B,C is 0.75, the probability that he passes in atleast two of the exams is 0.5 and the probability that he passes exactly two of the exams is 0.4. If P_1,P_2 and P_3 denotes the probabilities of the student passing in A,B and C respectively then which of the following is/are true-

(A) P_1,P_2 and P_3 are the roots of equation $20x^3 - 27x^2 + 14x - 2 = 0$

(B) P_1,P_2 and P_3 are the roots of equation $20x^3 - 17x^2 + 54x - 2 = 0$

(C) $P_1 + P_2 + P_3 > (1 - P_1)(1 - P_2)(1 - P_3)$

(D) $\sum_{i=1}^{3} P_i^2 < \frac{1}{2}$

Linked Comprehension Type

Paragraph for Question 33 to 35

'A' is a set containing 5 distinct elements. A subset P of 'A' is chosen. The set 'A' is reconstructed by replacing the elements of P. Now a subset Q of 'A' is again chosen.

On the basis of above information, answer the following questions :

33. The number of ways of choosing P and Q so that $P \cap Q = \phi$, is -

(A) 243 (B) 242 (C) 32 (D) 31

34. The number of ways of choosing P and Q so that $P \cap Q$ contains only one element, is -

(A) 80 (B) 81 (C) 405 (D) 16

35. The number of ways of choosing P and Q so that P and Q are identical sets, is -

(A) 32 (B) 16 (C) 256 (D) 64

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Paragraph for Question 36 to 38

(i) If a pencil of length AB is having two specified mark P and Q in between AB, then probability of cutting the pencil between P and Q = $\frac{\text{length of PQ}}{\text{length of AB}}$.

(ii) If in an Archery competition, the arrow has to hit a circle made upon a rectangular wooden board, then the probability of hitting of arrow inside the circle = $\frac{\text{Area of circle}}{\text{Area of rectangular board}}$.

On the basis of above information answer the following :

36. If $k \in [0, 5]$, then the probability of the equation $x^2 + kx + \frac{1}{4}(k+2) = 0$ to have real roots is -

(A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$

37. There are two circles in the x-y plane whose equations are $x^2 + y^2 - 2y = 0$ and $x^2 + y^2 - 2y - 3 = 0$. If A point (x,y) is picked up randomly inside the larger circle then the probability of the point has been taken from the smaller circle is -

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

38. If a stick of unit length is broken into three parts each of length x, y & z respectively, then probability of forming of triangle from broken length is -

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{1}{5}$

Paragraph for Question 39 to 41

Family A has 6 members of which 4 are males and 2 are females & family B has 5 members consisting at least one male and one female. (Assume it is equiprobable for a member of unknown gender to be a male or female).

On the basis of above information, answer the following questions :

39. Probability that family B has more females than males is equal to -

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

40. Let two members are selected, one from family A & other from family B. Probability that selected members are one male and one female is equal to -

(A) $\frac{1}{3}$ (B) $\frac{23}{40}$ (C) $\frac{21}{40}$ (D) $\frac{1}{2}$

41. 2 members are selected randomly either from family A or family B. If both the members are female, then the probability that they belong to family A is equal to

(A) $\frac{1}{2}$ (B) $\frac{8}{35}$ (C) $\frac{2}{31}$ (D) $\frac{2}{9}$

Paragraph for Question 42 & 43

A slip of paper is given to A. who marks it with either a plus sign or a minus sign, the probability of his writing a plus sign is known to be $\frac{1}{3}$. He then passes the slip to B, who may either leave it alone or change the sign before passing it on to C. Next C passes the slip to D after perhaps changing the sign. Finally D passes it to an honest judge after perhaps, changing the sign. It is known that B, C and D each change the sign with probability 2/3.

42. The probability that judge see a plus sign on the slip is -

(A) $\frac{1}{81}$ (B) $\frac{17}{81}$ (C) $\frac{25}{81}$ (D) $\frac{41}{81}$

43. If the judge see a plus sign on the slip, then the probability that A originally wrote a plus sign is -

(A) $\frac{1}{41}$ (B) $\frac{12}{41}$ (C) $\frac{13}{41}$ (D) $\frac{14}{41}$

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Paragraph for Question 44 and 45

There are n urns each containing 2n balls such that the i^{th} urn contains i white and 2n – i black balls. Let E_i be the event of selecting i^{th} urn, i = 1, 2, ... n and w denotes the event of getting a white ball. P(E) denotes probability of occurence of event E.

44. If $P(E_i) \propto i^2$, where i = 1, 2, ... n, then -

(A) $P(w) = \frac{3(2n+1)}{16(n+1)}$

(B) $P(w) = \frac{3(n+1)}{4(2n+1)}$

(C) $\lim_{n\to\infty} P(w) = \frac{3}{8}$

(D) $\lim_{n\to\infty} nP(E_n) = \frac{3}{2}$

45. If $P(E_i) \propto i$, where i = 1, 2, ... n, then

(A) $P(E_1 / w) = \frac{6}{n(n+1)(2n+1)}$

(B) $P(E_i/w)$ increases as i increases from 1 to n

(C) $P(E_i/w)$ decreases as i increases from 1 to n

(D) $\lim_{n\to\infty} nP(E_n / w) = 3$

Paragraph for Question 46 & 47

There are four boxes B_1, B_2, B_3 and B_4. Box B_i has i cards and on each card a number is printed, the numbers are from 1 to i. A box is selected randomly, the probability of selecting box B_i is $\frac{i}{10}$ and a card is drawn. Let E_i represent the event that a card with number i is drawn.

46. $P(E_1)$ is equal to-

(A) $\frac{1}{10}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{2}{5}$

47. $P(B_3/E_2)$ is equal to-

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Paragraph for Question 48 & 49

A special die is constructed in the shape of a regular tetrahedron with faces numbered 1,2,3 and 4 and the score is taken from the face on which the die lands. The probabilities of throwing 1,2,3 and 4 are $\frac{1-3k}{4}, \frac{1+k}{4}, \frac{1-2k}{4}$ and $\frac{1+4k}{4}$ respectively.

48. If the probability of getting a sum equal to 4 when two such dice are thrown is $\frac{3}{16}$, then number of possible values of k is-

(A) 0 (B) 1 (C) 2 (D) Infinitely many

49. If probability of getting both numbers even given that sum of numbers is even is $\frac{1}{2}$ then-

(A) die is unbiased (B) k = 5 (C) $k = \frac{1}{2}$ (D) data in consistent

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Paragraph for Question 50 & 51

Three players A, B, C alternately throw a die in the order. The first player to throw a 6 will be the winner. A's die is fair where as B and C throw a biased die with probability of throwing a 6 being p_1 and p_2 respectively.

50. Ratio of probability of winning of B to probability of winning of C is -

(A) $\frac{1-p_1}{p_1p_2}$ (B) $\frac{p_1}{1-p_1}$ (C) $\frac{p_1}{p_2(1-p_1)}$ (D) $\frac{1-p_1}{p_1}$

51. Ratio of p_1 and p_2 so that the game becomes equally likely for all the three players is -

(A) 5 : 4 (B) 1 : 2 (C) 4 : 5 (D) 2 : 1

Paragraph for Question 52 to 53

Contents of the Urn-I,II and III are as follows.

Urn	W	B
I	3	3
II	2	3
III	4	2

A dice is thrown if a prime number appears, then a ball is drawn from Urn-I and if composite number appears, a ball is drawn from Urn-II otherwise a ball is drawn from Urn-III. Given that a white ball is drawn, then

52. What is the probability that Urn-III was selected ?

(A) $\frac{45}{89}$ (B) $\frac{20}{89}$ (C) $\frac{24}{89}$ (D) $\frac{35}{89}$

53. If another ball is drawn from the same Urn, then what is the probability that it is a black ball (Given that first drawn ball is not replaced)

(A) $\frac{53}{89}$ (B) $\frac{45}{89}$ (C) $\frac{53}{180}$ (D) None of these

Matrix Match Type

54. Consider an experiment of tossing a fair of coin six times. Let A and B be two events defined as follows

A : Number of heads greater than number of tails.

B : The third toss results in head. Now match the columns.

Column-I	**Column-II**
(A) If $P(A)=\frac{a}{b}$ where a and b are relatively prime then 'a' is	(P) an even number
(B) If $P(B)=\frac{p}{q}$, where p and q are relatively prime then 'q' is	(Q) an odd number
(C) If $P\left(\frac{A}{B}\right)=\frac{u}{v}$ where u and v are relatively prime then 'u' is	(R) a prime number
(D) If $P\left(\frac{B}{A}\right)=\frac{m}{n}$ where m and n are relatively prime then 3(m + n) is	(S) a composite number
	(T) neither prime nor composite number

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55. India and Australia play a series of 'n' one day matches and probability that India wins a match is 1/2 and losses against Australia is $\frac{1}{2}$

	Column-I		Column-II
(A)	If 'n' is not fixed and series ends when any one team completes its 4^{th} win then the probability that India wins the series.	(P)	$\frac{4}{2^7}$
(B)	If n = 7, then probability that India wins atleast three consecutive matches	(Q)	$\frac{47}{2^7}$
(C)	If n = 7, probability that India wins series through exactly four consecutive wins and three losses (all wins should be consecutive)	(R)	$\frac{1}{2}$
(D)	If n = 7 probability that India doesn't win two consecutive matches	(S)	$\frac{17}{2^6}$

56. Suppose n (>6) people are asked a question successively in a random order and exactly 3 out of 'n' people know the answer. Let P_r dentoe the probability that r^{th} person asked is the first to know the answer, then

	Column-I		Column-II
(A)	Probability that first four do not know the answer	(P)	$\frac{3(n-3)}{n(n-1)}$
(B)	P_2	(Q)	$\frac{(n-4)(n-5)(n-6)}{n(n-1)(n-2)}$
(C)	P_{n-2}	(R)	$\frac{1}{{}^nC_3}$
(D)	P_r	(S)	$\frac{3(n-r)(n-r-1)}{n(n-1)(n-2)}$

57. Let $f : A \to B$ such that $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3\}$.

	Column-I		Column-II
(A)	The probability that mapping is one-one	(P)	0
(B)	The probability that element 3 in set B has exactly two pre-image in set A	(Q)	$\frac{50}{81}$
		(R)	$\frac{1}{4}$
(C)	The probability that mapping is onto	(S)	$\frac{40}{243}$
(D)	The probability that $f(5) = 3$ is k, then k is greater than	(T)	$\frac{80}{243}$

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Subjective Type Questions

58. A certain kind of bacteria either die, split into two or split into three bacteria. All splits are exact copies. The chances of dying is $\frac{1}{4}$, the chances of splitting into two is $\frac{1}{2}$ and splitting into three is $\frac{1}{4}$. If the probability that it survives for infinite length of time is $\frac{m-\sqrt{13}}{n}$ (m,n $\in$ N), then the value of (m+n) is

59. 4 identical dice are rolled simultaneously. Probability that product of first two outcomes is more than the product of last two outcomes is $\frac{m}{n}$(m, n coprime) then value of $n - 2m - 78$ is equal to

60. 5 boys and 10 girls sit at random in a row. The probability that end seats are occupied by boys and between any two boys, there always lie an odd number of girls, is $\frac{\alpha.10!5!}{15!}$, where sum of digits of α is

61. There are two packs A and B of 52 playing cards. All the four kings are removed from the pack A whereas from the pack B, one ace, one king, one queen and one jack is removed. One of these two packs is selected randomly and two cards are drawn simultaneously from it, and found to be a pair (i.e both have same rank i.e. two 5's or two 8's etc). If $\frac{p}{q}$ denotes the probability that pack 'A' was selected where 'p' and 'q' are co-prime then $(2p - q)$ is equal to

62. Bag I contains 2 Red and 3 Black balls and another Bag II contains 4 Red and 5 Black balls. Two balls are drawn simultaneously from the first bag and put in the second bag then a ball drawn from the bag II. If the drawn ball from the II Bag is of Red colour and 'P' denotes the probability that both drawn balls from bag-I are of Red colour, then '$\frac{1}{P}$' is equal to

63. Let A and B are independent events of an experiment such that $P(A) \geq \frac{1}{2}$, $P(\bar{A} \cup B) = \frac{11}{12}$ and $P(\bar{A} \cap B) = \frac{5}{12}$. If $P(A \cap B) = \frac{m}{n}$, then value of $(n - m)$ is
(where P(X) denotes the probability of occurrence of event X and m, n are coprime)

64. In a tournament four players are participating. Each player play with every other player. Each player has 50% chance of winning any game and there are no ties. If the probability that at the end of tournament there is neither a winless nor an undefeated player is there is $\frac{a}{b}$, where a and b are relatively prime integers then $|2a - b|$ is equal to

65. A fair coin is tossed 10 times and the outcomes are listed. Let E_i be the event that the i^{th} outcome is a head and E_m be the event that the list contains exactly m heads. If E_i and E_m are independent, then 'm' is equal to

66. A five digit number is formed by the digits 1, 2, 3, 4, 5 without repetition. If the probability that the number formed is divisible by 4, is P, then 5P is

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67. A bag contains 10 white 3 black balls. Balls are drawn one by one without replacement till all black balls are drawn. The probability that the procedure of drawing balls will come to an end at the 7^{th} draw is equal to $\frac{a}{b}$, where a and b are coprime natural numbers, then the number of divisors of 'a' is

68. Humpty plays three matches against Dumpty. In any match the probabilities of Humpty getting 0, 2 and 3 points are 0.2, 0.3 and 0.5 respectively. If probability that Humpty getting atmost 6 points is $\frac{p}{200}$, then then sum of the digits of 'p' is

69. A hunter's chance of shooting an animal at a distance r is $\frac{a^2}{r^2}$ $(r > a)$. He fires when $r = 2a$ & if he misses he reloads & fires when $r = 3a, 4a, \ldots$.. If he misses at a distance '50a', the animal escapes. Find the probability that the anmial escapes.

ANSWER KEY

1.	C	2.	C	3.	A	4.	C	5.	C
6.	D	7.	C	8.	A	9.	C	10.	C
11.	C	12.	B	13.	A	14.	A	15.	D
16.	A	17.	C	18.	B	19.	C	20.	B
21.	A,C	22.	A,B	23.	A,B	24.	A,C	25.	A,B,C,D
26.	A,B,C	27.	A,D	28.	A,B,C	29.	B,C,D	30.	A,B
31.	A,B,C,D	32.	A,C,D	33.	A	34.	C	35.	A
36.	C	37.	A	38.	B	39.	A	40.	D
41.	D	42.	D	43.	C	44.	B,C	45.	A,B,D
46.	D	47.	B	48.	B	49.	A	50.	C
51.	C	52.	B	53.	A				

54. (A)→(Q,R); (B)→(P,R); (C)→(Q,T); (D)→(Q,S) **55.** A→(R), B→(Q), C→(P), D→(S)

56. A→(Q), B→(P), C→(R), D→(S) **57.** (A)→(P); (B)→(T); (C)→(Q); (D)→(P,R,S,T)

58.	7	59.	8	60.	2	61.	1	62.	8
63.	7	64.	2	65.	5	66.	1	67.	4
68.	4.00	69.	0.49						

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HOME ASSIGNMENT # 12 (COMPLEX NUMBER) MATHEMATICS

Straight Objective Type

1. Locus of a point z satisfying $|z|+\frac{1}{z}=z+\frac{1}{|z|}$ on the argand plane is -

(A) the union of two rays originating from the same point

(B) the union of a point and a ray

(C) the union of two points

(D) the union of a ray and a line originating from the same point

2. Let z is a complex number such that $az+b\bar{z}=c+id$ (where a, b, c, d $\in$ R), then z is equal to

(A) $\frac{c}{(a-b)}+\frac{id}{(a+b)}$ (B) $\frac{c}{(a+b)}+\frac{id}{(a-b)}$ (C) $\frac{c}{(a-b)}-\frac{id}{(a+b)}$ (D) $\frac{c}{(a+b)}-\frac{id}{(a-b)}$

3. A complex number z satisfies the system of equations $|z-7|=|z-7-6i|$ and $|z-2+3i|=|z-5i|$ then $\frac{1}{\text{Re } z}+\left(\frac{1}{\text{Im } z}\right)^2=$

(A) 0 (B) 2 (C) $\frac{1}{9}$ (D) $\frac{2}{9}$

4. If $A=\left\{z:\left|\frac{z-2}{z+2}\right|=3,\ z\in C\right\}$ and $z_1, z_2, z_3, z_4 \in A$ are 4 complex numbers representing points P, Q, R, S respectively on the complex plane such that $z_1-z_2=z_4-z_3$, then maximum value of area of quadrilateral PQRS is -

(A) $\frac{9}{4}$ (B) $\frac{9}{2}$ (C) 9 (D) 16

5. If $z \in C$, then area enclosed by the curve $2|z^2+|z|^2|+3|z^2-|z|^2|=6|z|,\ z\neq 0$ is -

(A) 3π sq. units (B) 3 sq. units (C) 6π sq. units (D) 6 sq. units

6. If $z_1^2+|z_1|=z_2^2+|z_2|=z_3^2+|z_3|=0$ $(z_1\neq z_2\neq z_3)$ and z_1, z_2 and z_3 are the roots of equation $ax^3+bx^2+x+c=0$, then-

(A) a = 1 (B) b = –1 (C) a = –1 (D) b = 1

7. If $x=\frac{2\pi}{1999}$, then the value of $\cos x\cos 2x\cos 3x..........\cos 999x$ is equal to-

(A) 1 (B) $\frac{1}{2^{999}}$ (C) $\frac{1}{2^{1998}}$ (D) $-\frac{1}{2^{999}}$

8. If α is a complex n^{th} root of unity and if z_1 and z_2 are any two complex numbers, then $\sum_{r=0}^{n-1}|z_1+\alpha^r z_2|^2$ is -

(A) $n^2|z_1+z_2|^2$ (B) $\left(\frac{z_1}{n}+\frac{z_2}{n}\right)^2$ (C) $n\left(|z_1|^2+|z_2|^2\right)$ (D) $n^2\left(|z_1|^2+|z_2|^2\right)$

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9. Let z be a complex number satisfying $|z|=1$ and $|\arg(z)|<\frac{\pi}{3}$ then $\arg\left(z^2+\bar{z}\right)=$

(A) $\arg(z)$ (B) $\arg(\bar{z})$ (C) $\frac{1}{2}\arg(z)$ (D) $\frac{1}{2}\arg(\bar{z})$

10. Let z_1 and z_2 be two distinct roots of the equation $z^{101} = 1$ such that $|z_1+z_2|\ge\sqrt{2+\sqrt{3}}$, then number of such ordered pairs (z_1,z_2) is

(A) 101 × 16 (B) 101 × 32 (C) 204 × 16 (D) None

11. All roots of the equation, $(1 + z)^6 + z^6 = 0$ -

(A) lie on a unit circle with centre at the origin
(B) lie on a unit circle with centre at (–1, 0)
(C) lie on the vertices of a regular polygon with centre at the origin.
(D) are collinear

12. Value of $\sum_{k=1}^{100}\left(i^{k!}+\omega^{k!}\right)$, where $i=\sqrt{-1}$ and ω is complex cube root of unity, is -

(A) $190 + \omega$ (B) $192 + \omega^2$ (C) $190 + i$ (D) $192 + i$

13. If one of the vertex of the regular hexagon, circumscribing the circle $|z-1-i|=\sqrt{3}$ is $2+(1+\sqrt{3})i$, then the complex number which is not representing any vertex of the hexagon is -

(A) $3 + i$ (B) $(1+\sqrt{3})i$ (C) $-3 + i$ (D) $2+(1-\sqrt{3})i$

14. Let C be the set of complex numbers, $z \in C$ satisfied $|z - 3| + |z + 1| \ge 8$, then minimum value of $|z - 1|$ is-

(A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) 3 (D) $2\sqrt{3}$

15. Let C_1 and C_2 be two curves on the complex plane defined as

$C_1 : z+\bar{z}=2\left(|z-1|\right)$

$C_2 : \arg\left(z+1+i\right)=\alpha$

where $\alpha\in(0,\pi)$ such that the curves C_1 and C_2 touches each other at $P(z_0)$ then the value of $|z_0|^2$ is

(A) 1 (B) 2 (C) 3 (D) 4

16. z_1 & z_2 are two distinct points in an argand plane. If $a\,|z_1| = b\,|z_2|$, (where a, b $\in$ R) then the point $\frac{a z_1}{b z_2}+\frac{b z_2}{a z_1}$ is a point on the :

(A) line segment $[-2, 2]$ of the real axis
(B) line segment $[-2, 2]$ of the imaginary axis
(C) unit circle $|z| = 1$
(D) the line with $\arg z = \tan^{-1} 2$.

Multiple Correct Answer Type

17. Consider an equation $\left(\frac{z-1}{z}\right)^{2n}+\left(\frac{z-1}{z}\right)^{n}+1=0$ (where $n \in N$), then -

(A) $\text{Re}(z)=\frac{1}{2}\ \forall\, n\in N$
(B) roots of equation are collinear
(C) roots of equation are concyclic
(D) $\text{Im}(z)=\frac{\sqrt{3}}{2}$ or $-\frac{\sqrt{3}}{2}\ \forall\, n \in N$

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18. Let z_1, z_2 and z_3 be three complex numbers such that $|z_1 - z_2| = 4$, $|z_1 - z_3| = 6$ and $|z_1| = |z_2| = |z_3| = 8$.
The minimum value of the expression $\left|z_1 - \alpha z_2 + (\alpha - 1)z_3\right|, \alpha \in R$ is less than-

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{9}{2}$

19. Let $C_1 : |z - 1 - i| = 1 : C_2\ Arg(z) = \frac{\pi}{4}$ and $C_3 : Arg(z - 1 - i) = 0$ denotes three curves in Argand plane, where z is varibale point in Argand plane, then -

(A) Minimum area bounded by all three curves is $\frac{\pi}{4}$

(B) Minimum area bounded by all three curves is $\frac{\pi}{8}$

(C) Number of complex number 'z' which lies on exactly two curves is 4

(D) Number of complex number 'z' which lies on exactly two curves is 3

20. Let z_1 and z_2 be two complex numbers such that $z_1^2 - 4z_2 = 16 + 20i$. If α and β are roots of $x^2 + z_1 x + z_2 + M = 0$ (where M is a complex number) and $|(\alpha - \beta)^2| = 28$, then (where $i = \sqrt{-1}$)-

(A) maximum value of $|M|$ is $7 + \sqrt{41}$ (B) maximum value of $|M|$ is $5 + \sqrt{41}$

(C) minimum value of $|M|$ is $7 - \sqrt{41}$ (D) minimum value of $|M|$ is $5 - \sqrt{41}$

21. If $z^4 + 1 = z^2$, then-

(A) Least positive argument of z is $\frac{\pi}{6}$

(B) Greatest positive argument of z in $[0,2\pi]$ is $\frac{11\pi}{6}$

(C) The four possible values of z forms square on argand plane

(D) The four possible values of z forms rectangle on argand plane.

22. If a and b be two complex numbers such that $|a + b| = 20$ and $|a^2 + b^2| = 16$, then

(A) least value of $|a^3 + b^3|$ is 3520 (B) least value of $|a^3 + b^3|$ is 2520

(C) greatest value of $|a^3 + b^3|$ is 3520 (D) greatest value of $|a^3 + b^3|$ is 4480

23. Let $z_1 = 16 + 6i$, $z_2 = 10 + 6i$ (where $i = \sqrt{-1}$). If z is any complex number such that amp$\left(\frac{z - z_1}{z - z_2}\right)$ is $\frac{\pi}{4}$, then which of the following is/are always correct ?

(A) Locus of z is minor arc of a circle (B) Locus of z is major arc of a circle

(C) $|z - 13 - 9i| = 3\sqrt{2}$ (D) $|z - 13 - 3i| = 3\sqrt{2}$

24. Let n is the number of complex number(s) satisfying $|z + 3| = |z - 3| + 6$ and $|z - 4| = r$, where $r \in R^+$, then identify the correct statement -

(A) if r = 1, then n = 2 (B) if r = 1, then n = 1 (C) if r = 8, then n = 2 (D) if r = 8, then n = 1

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25. If z is a complex number satisfying $|z^2 - 1| = |z| + 2$, then-

(A) Maximum value of $|z|$ is $\frac{1+\sqrt{13}}{2}$

(B) Maximum value of $|z|$ is $\frac{1+\sqrt{5}}{2}$

(C) Minimum value of $|z|$ is $\frac{1+\sqrt{5}}{2}$

(D) Minimum value of $|z|$ is $\frac{1+\sqrt{2}}{2}$

26. If $1, z_1, z_2, \ldots\ldots z_{10}$ are the roots of equation $z^{11} - 1 = 0$, then-

(where $i = \sqrt{-1}$ and $\omega\ (\neq 1)$ is complex cube root of unity)

(A) $\prod_{k=1}^{10}(i - z_k) = i$ (B) $\prod_{k=1}^{10}(i - z_k) = -i$ (C) $\prod_{k=1}^{10}(\omega + z_k) = \omega$ (D) $\prod_{k=1}^{10}(\omega + z_k) = \omega^2$

27. If $w = \frac{z-1}{z-2}, |z| = 1, z, w \in C$, then -

(A) $|w - 3|_{min} = \frac{7}{3}$

(B) $|w - 3|_{max} = 3$

(C) $(\text{Arg}(w))_{max} = \frac{\pi}{4}\ (-\pi < \text{Arg}(w) < \pi)$

(D) $(\text{Arg}(w))_{min} = 0\ (-\pi < \text{Arg}(w) < \pi)$

28. Consider a complex number z on the argand plane satisfying $\arg(z^2 - \omega) = \frac{\pi}{2} + \arg(z^2 - \omega^2)$ (where $\omega = e^{i\frac{2\pi}{3}}$), then identify the correct statement(s).

(A) Maximum value of $|z|$ is $\frac{\sqrt{3+\sqrt{3}}}{2}$

(B) Maximum value of $|z|$ is $\frac{\sqrt{2+2\sqrt{3}}}{2}$

(C) Maximum value of $|z-1-i||z+1+i|$ is equal to $\frac{\sqrt{17}+\sqrt{3}}{2}$

(D) Minimum value of $|z-1-i||z+1+i|$ is equal to $\frac{\sqrt{17}-\sqrt{3}}{2}$

29. If $|z_1| = 1$, $|z_2 - 2| = 3$ and $|z_3 - 5| = 6$, then maximum value of $|2z_1 - 3z_2 - 4z_3|$ is not less than (where z_1, z_2, z_3 are complex numbers)-

(A) 43 (B) 44 (C) 45 (D) 46

30. Consider complex numbers z and w satisfying the equation $z + w = \frac{\bar{z}}{w}$ and $\frac{1}{w} + z = w\bar{z}$, then which of the following is/are correct ?

(A) The value of $|z| + |w|$ is 2

(B) If $\arg(z) = \pi$ then, $\arg(w) = -\frac{\pi}{3}$ or $\frac{\pi}{3}$

(C) If $w = 1$, then $\text{Re}(z) = \frac{1}{2}$

(D) If $w = -1$, then $\text{Re}(z) = \frac{1}{2}$

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31. Given z_1, z_2, z_3, z_4 are four points in the complex plane such that $|z_1| < 1$, $|z_2| = 1$ and $|z_3| \le 1$ and $z_3 = \dfrac{z_2(z_1 - z_4)}{\bar{z}_1 z_4 - 1}$, then $|z_4|$ can be ($z_1 \ne z_4$) -

(A) 2 (B) $\dfrac{2}{5}$ (C) $\dfrac{1}{3}$ (D) $\dfrac{5}{2}$

32. Let θ is real. If $2^7\cos^3\theta\sin^5\theta = a\sin 8\theta + b\sin 6\theta + c\sin 4\theta + d\sin 2\theta$, then

(A) $|a + 2b| = 3$ (B) $a^2 + b^2 + c^2 + d^2 = 45$

(C) $b = 4$ (D) $\dfrac{d+a}{c+b} = -\dfrac{7}{4}$

33. All complex numbers 'z' which satisfy the relation $\left|z - |z+1|\right| = \left|z + |z-1|\right|$ on the complex plane can lie on the

(A) ellipse with foci (– 1, 0) and (1, 0)

(B) radical axis of the circles $|z - 1| = 1$ and $|z + 1| = 1$

(C) line x = 0

(D) on a line segment joining (–1, 0) to (1, 0)

34. If for any two non-zero complex numbers z_1 & z_2, if $z_1(1+|z_2|) = iz_2(1+|z_1|)$, then -

(A) $\arg\left(\dfrac{z_1}{z_2}\right) = \dfrac{\pi}{2}$ (B) $\bar{z}_1 z_2$ is purely imaginary

(C) $z_1\bar{z}_2$ is purely real (D) $|z_2|z_1 + |z_1|\bar{z}_2 = 0$

35. a,b,c are three complex numbers representing the vertices of a triangle ABC, such that $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, then which of the following **CAN BE** true -

(A) centroid of ΔABC is 0 (B) ΔABC is an equilateral triangle

(C) $|a| = |b| = |c|$ (D) $|a^3| + |b^3| + |c^3| = 3|abc|$

Linked Comprehension Type

Paragraph for Question 36 & 37

If $z_1 = a + ib$ & $z_2 = c + id$ are two complex number such that $|z_1| = |z_2| = 1$ and $\text{Re}(z_1\bar{z}_2) = 0$. Then

36. If a,b > 0 and c < 0. Then-

(A) $0 \le |z_1 - \bar{z}_2| \le 2$ (B) $0 < |z_1 - \bar{z}_2| \le \sqrt{2}$ (C) $\sqrt{2} < |z_1 - \bar{z}_2| \le 2$ (D) $\sqrt{2} < |z_1 - \bar{z}_2| < 2$

37. Let $k = |z_1 + z_2| + |a + ic|$, then value of k, is-

(A) $\sqrt{2} - 1$ (B) 1 (C) $\sqrt{2} + 1$ (D) $2\sqrt{2}$

Paragraph for Question 38 and 39

Let $z_1, z_2, z_3, \ldots\ldots\ldots, z_n$ be the complex numbers corresponding to the vertices $A_1, A_2, \ldots\ldots, A_n$ of a n-sided regular polygon $A_1A_2A_3 \ldots\ldots A_nA_1$ inscribed in a circle. Let z_0 is the centre and R is radius of the circle.

38. If n = 3, $z_0 = 2$ and $z_3 = w^2$, where $w = \dfrac{-1+\sqrt{3}i}{2}$, then z_1 can be -

(A) 3 + 2w (B) 3 – 2w (C) 3 + w (D) 3 – w

39. If n = 6 and $z_0 = 0 + i0$ and R = 2, then $(A_1A_2)(A_1A_3)(A_1A_5)$ is -

(A) 8 (B) 6 (C) 24 (D) 12

Paragraph for Question 40 to 41

Let P(z) is a point in complex plane which satisfies $|z - 1| = 2$

40. Locus of point Q(w) which satisfies $w = z + \frac{2}{z-1}$ is a conic, whose eccentricity is -

(A) 0 (B) 1 (C) $\frac{3}{\sqrt{2}}$ (D) $\frac{2\sqrt{2}}{3}$

41. Maximum value of amp(z + 3) is-

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) π

Paragraph for Question 42 to 44

Let the origin O of the argand plane represents the home town of a person. Suppose the man travels $\sqrt{2}$ units in north-east direction to reach city A. Then he travels 1 unit in east direction to reach town B. From town B he starts to move along a circular path, in anticlockwise direction with centre O and radius OB, to reach to city C such that $\angle$ BOC is right angle. Finally he returns home by the shortest path from city C. Let z_1, z_2 and z_3 be the complex numbers corresponding to the points A, B and C respectively. On the basis of above information answer the following :

42. The image of complex number z_2 under the locus $|z - z_1| = |z - z_3|$ -

(A) $\frac{16}{5} + \frac{16}{5}i$ (B) $\frac{2}{5} + \frac{2}{5}i$ (C) $\frac{16}{5} + \frac{2}{5}i$ (D) $-\frac{8}{5} + \frac{14}{5}i$

43. Let z_p & z_q be the two complex numbers satisfying the equations $|z - z_2| = 1$ & $|z - z_3| = 1$ respectively, then maximum value of $|z_p - z_q|$ is -

(A) $\sqrt{10}$ (B) $\sqrt{10} - 2$ (C) $2 + \sqrt{10}$ (D) none of these

44. The area enclosed by the path traced by the person during his complete journey is -

(A) $\frac{5\pi}{4} - \frac{1}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{5\pi}{4} - 1$ (D) $\frac{5\pi - 1}{4}$

Paragraph for Question 45 to 47

Let z_1 and z_2 be two complex numbers satisfying $(z - \omega)^2 + (z - \omega^2)^2 = 0$, where ω is non real cube root of unity. Also let 'α' be a variable point on the circle $\left|z + \frac{1}{2}\right| = \frac{\sqrt{3}}{2}$ and α_1, α_2 are values of α such that $|\alpha|$ is maximum and minimum respectively.

On the basis of above information, answer the following questions :

45. Value of $|z_1 - \alpha|^2 + |z_2 - \alpha|^2 + |\omega - \alpha|^2 + |\omega^2 - \alpha|^2$ is -

(A) 3 (B) $\sqrt{3}$ (C) 6 (D) 1

46. Which of following is incorrect ?

(A) $|\alpha_1| + |\alpha_2| = |\alpha_1 - \alpha_2|$

(B) $\alpha_1 = z_2$ and $\alpha_2 = z_1$ where $z_1 < 0$, $z_2 > 0$

(C) $|2z_1 + 1| = |2z_2 + 1|$

(D) $\text{amp}\left(\frac{z_1}{z_2}\right) = \text{amp}(z_1.z_2)$

47. Number of complex numbers α such that $|\alpha - \alpha_1| + |\alpha - \alpha_2| = \sqrt{6}$ is

(A) 4 (B) 3 (C) 2 (D) 0

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Paragraph for Question 48 & 49

For complex number z, consider sets $A = \{z \in C : |z - 2 - 2i| \leq 2\}$ and

$B = \left\{z \in C : i(z - \bar{z}) - (z + \bar{z}) + 4 \leq 0\right\}$

48. Area of the region traced by points in set $A \cap B$ (in sq. units) is-

(A) $3\pi - 2$ (B) $3\pi + 2$ (C) $\pi + 2$ (D) $\pi + 3$

49. The minimum value of $|z + 2|$ $(z \in A \cap B)$ is-

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $2\sqrt{2}$ (D) 2

Paragraph for Question 50 & 51

Let z_1,z_2,z_3,z_4 are roots of equation $z^6 + z^4 + z^3 + z^2 + 1 = 0$ such that $z_r^3 + 1 \neq 0$ for $\forall r \in \{1,2,3,4\}$

50. $z_1^{99} + z_2^{99} + z_3^{99} + z_4^{99}$ is equal to-

(A) 0 (B) –1 (C) 4 (D) 5

51. Area of quadrilateral made by joining z_1,z_2,z_3,z_4 on the argand plane is-

(A) $2\sin 18°$ (B) $2\cos 36°$ (C) $2\sin 72°$ (D) $2\cos^3 18°$

Paragraph for Question 52 to 53

Let a curve be $C_1 : |z - 1| = 1$. If $w = \frac{z^2 - z - 2}{1 - z}$, then locus of w is curve C_2.

52. Eccentricity of curve C_2 is -

(A) $\frac{8}{9}$ (B) $\frac{2\sqrt{2}}{9}$ (C) $\frac{2\sqrt{2}}{3}$ (D) $\frac{2}{3}$

53. Product of slopes of normals to curve C_1 which touch curve C_2 is -

(A) $\frac{9}{8}$ (B) $-\frac{9}{8}$ (C) –3 (D) 3

Paragraph for Question 54 to 56

Let z_1 and z_2 are two complex numbers such that $|z_1| = a$ and $|z_2| = b$ satisfying the equation $3z_1^2 - 2z_1z_2 + 2z_2^2 = 0$ and also $\text{Re}\left(\frac{z_1 - 2}{z_1 + 2}\right) = 0$. Also P_1,P_2 and O are points in complex plane corresponding to z_1,z_2 and origin respectively.

On the basis of above information, answer the following questions :

54. Eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, is-

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

55. Radius of the circle touching the parabola $y^2 = a^2x$ at the upper end of the latus rectum and passing through its focus, is-

(A) $\sqrt{2}$ (B) 2 (C) $\sqrt{3}$ (D) 3

56. Area of triangle OP_1P_2 is-

(A) $\sqrt{5}$ (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

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MATCHING LIST TYPE

57. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	$\lvert z-1+i\rvert = \lvert z+1-i\rvert$, then locus of z is	(1)	square
(Q)	$\lvert z-3\rvert + \lvert z-4i\rvert = 6$, then locus of z is	(2)	circle
(R)	$\frac{\lvert z+1-2i\rvert}{\lvert z-3i\rvert}=2$, then locus of z is	(3)	ellipse
(S)	$\lvert z+\bar{z}\rvert+\lvert z-\bar{z}\rvert=4$, then locus of z is	(4)	straight line

Codes :

	P	Q	R	S
(A)	3	4	2	1
(B)	3	4	1	2
(C)	4	3	1	2
(D)	4	3	2	1

58. The set $A = \{z : z^{18} = 1\}$, $B = \{w : w^{48} = 1\}$, $C = \{zw : z \in A \text{ and } w \in B\}$ are three sets of complex roots of unity and $D=\left\{z : 0 \le \arg(z) \le \frac{\pi}{3}\right\}$

Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	$n(A \cap B)$	(1)	4
(Q)	$\sqrt{n(C)}$	(2)	6
(R)	$n(A \cap D)$	(3)	9
(S)	$n(B \cap D)$	(4)	12

Codes :

	P	Q	R	S
(A)	2	4	2	3
(B)	2	1	1	3
(C)	3	4	1	2
(D)	2	4	1	3

59. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	Let $z_1 = 6 + i$, $z_2 = 4 - 3i$, z be complex number such that $\arg\left(\frac{z-z_1}{z_2-z}\right)=\frac{\pi}{2}$ and $\lvert z-5+i\rvert=\sqrt{p}, p \in N$. The value of p is (where $i=\sqrt{-1}$)	(1)	6
(Q)	If α is 15^{th} root of unity and $\prod_{r=1}^{14}\left(11-\alpha^r\right)=\frac{11^a-b}{c}$, then the value of a + b – c is (where $i=\sqrt{-1}$)	(2)	3
(R)	The roots of cubic equation $(z+\alpha\beta)^3=\alpha^3$, $(\alpha \ne 0), \alpha, \beta \in R$ represents the vertices of equilateral triangle of side $\sqrt{k}\lvert\alpha\rvert$, $k \in N$. The value of k is (where $i=\sqrt{-1}$)	(3)	7
(S)	If $\lvert z-i\rvert \le 2$, $z_1 = 5 + 3i$, then the maximum value of $\lvert iz + z_1\rvert$ is (where $i=\sqrt{-1}$)	(4)	5

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Codes :

	P	Q	R	S
(A)	2	1	4	3
(B)	4	2	1	3
(C)	4	1	2	3
(D)	1	3	2	4

60. If $z_1, z_2 \in C$ such that $z_1 = 5 + 12i$ and $|z_2| = 4$,
then match List-I with List-II and select the correct answer using the code given below the list.

	List-I		List-II
(P)	maximum value of $\lvert z_1 + iz_2\rvert$ is	(1)	$13-4\sqrt{2}$
(Q)	minimum value of $\lvert z_1 + (1 + i)z_2\rvert$ is	(2)	$\frac{13}{5}$
(R)	maximum value of $\left\lvert \frac{z_1}{z_2+\frac{4}{z_2}} \right\rvert$	(3)	$\frac{13}{3}$
(S)	minimum value of $\left\lvert \frac{z_1}{z_2+\frac{4}{z_2}} \right\rvert$	(4)	17

Codes :

	P	Q	R	S
(A)	4	1	2	3
(B)	1	4	3	2
(C)	4	1	3	2
(D)	1	4	2	3

Matrix Match Type

61. The complex numbers $z_1, z_2 ....z_n$, $\sum_{i=1}^{n} z_i \neq 0$, represents the vertices of a regular polygon of n sides in order, inscribed in a circle of unit radius and $z_3 + z_n = Az_1 + \bar{A}z_2$, Then the values of $[|A|]$ for prescribed values of n in column-I (where [x] be the greatest integer $\leq x$).

	Column-I		Column-II
(A)	4	(p)	0
(B)	6	(q)	1
(C)	8	(r)	2
(D)	12	(s)	3
		(t)	7

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62. **Column-I** **Column-II**

	Column-I		Column-II
(A)	If $\|z\| = 1$, then the greatest value of $\|z^2 - 3z + 1\|$ is less than	(P)	0
(B)	If $\|z\| = 1$, then least value of $\|z^3 - 3\|$ is greater than	(Q)	1
(C)	If α,β and γ are the roots of equation $x^3 - 5x^2 + 6x - 4 = 0$ then $\alpha^2(\beta + \gamma) + \beta^2(\alpha + \gamma) + \gamma^2(\alpha + \beta)$ is divisible by	(R)	3
(D)	$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}^{20} - 2\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}^{19} + \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & b \\ 0 & 0 \end{pmatrix}$, then b is divisible by	(S)	6
		(T)	18

Subjective Type Questions

63. Let a,b and c be three distinct complex numbers satisfying $|a| = |b| = |c| = 1$ and $a + b + c = 0$, then $|a^2 + b^2 + c^2|$ is equal to

64. Let α be a complex number satisfying the equations $|z - 2| = 2$ and $\bar{z} - zi = 2(1 - i)$. Then the area of triangle formed by $\alpha, \bar{\alpha}$ and 2 is $\left(i = \sqrt{-1}\right)$

65. Given $a = \cos\theta + i\sin\theta$ and the equation $az^2 + z + 1 = 0$ has a purely imaginary root and $f(x) = x^3 - 3x^2 + 3(1 + \cos\theta)x + 5$.

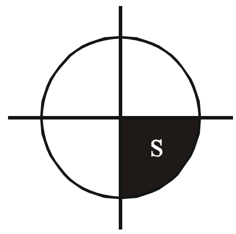
If p = Number of points of local extrema of $y = f(x)$

q = Number of points of inflection of $y = f(x)$

r = Number of points of negative real roots of equation $f(x) = 0$, then 'p + q + r' is equal to

66. Let z_n denotes the n^{th} term of a squence of complex numbers defined by

$z_n = (5 + 12i)\left(\dfrac{i}{2}\right)^n$. Let S be the intersection of the closed unit disk centered at origin and the fourth quadrant in the complex plane, then the smallest value of 'n' $\in$ N such that z_n lies in S



67. Let z_1, z_2, z_3 be complex numbers such that $|z_1| = |z_2| = |z_3| = |z_1 + z_2 + z_3| = 2$ and $|z_1 - z_2| = |z_1 - z_3|$ $(z_2 \neq z_3)$, then the value of $|z_1 + z_2|\,|z_1 + z_3|$ is

68. If area of the region in the complex plane that consist of all points z such that both $\dfrac{z}{40}$ and $\dfrac{40}{\bar{z}}$ have real and imaginary parts lies in the interval [0, 1] is A, then least integer greater than or equal to $\dfrac{A}{100}$ is

69. Let z_1 and z_2 be the points lying on the curve $|z| = 1$ and $|z-1| + |z - 3| = 4$ respectively, then minimum value of $|z_1 - z_2|$ is equal to

70. If $z_1, z_2, z_3 \in C$, satisfy the system of equations given by $|z_1| = |z_2| = |z_3| = 1$, $z_1 + z_2 + z_3 = 1$ and $z_1z_2z_3 = 1$ such that $\text{Im}(z_1) < \text{Im}(z_2) < \text{Im}(z_3)$, then the value of $\left[\left|z_1^{2017} + z_2^{2018} + z_3^{2019}\right|\right]$ is

(where [.] denotes greatest integer function)

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71. Given $\theta \in \left(0, \frac{\pi}{2}\right)$ such that $\cos\theta = \frac{1}{3}$, then the value of $\left[\lim_{n\to\infty} \sum_{k=0}^{n} \frac{\cos k\theta}{5^k}\right]$ is equal to

(where [.] denotes greatest integer function)

72. Consider the curves $C_1 : |z - 2| = 2 + \text{Re}(z)$ and $C_2 : |z| = 3$ (where $z = x + iy$, $x,y \in R$ and $i = \sqrt{-1}$). They intersect at P and Q in the first and fourth quadrants respectively. Tangents to C_1 at P and Q intersect the x-axis at R and tangents to C_2 at P and Q intersect the x-axis at S. If area of ΔPRS is $\lambda\sqrt{2}$ sq.units, then (λ^2) is

ANSWER KEY

1.	B	**2.**	B	**3.**	D	**4.**	B	**5.**	B
6.	A	**7.**	B	**8.**	C	**9.**	C	**10.**	C
11.	**D**	**12.**	**D**	**13.**	**C**	**14.**	**D**	**15.**	**B**
16.	**A**	**17.**	A,B	**18.**	B,C,D	**19.**	B,D	**20.**	A,C
21.	A,B,D	**22.**	A,D	**23.**	B,C	**24.**	A,D	**25.**	A,C
26.	A,D	**27.**	A,B	**28.**	B,C	**29.**	**A,B,C,D**	**30.**	**B,D**
31.	**B,C**	**32.**	**A,B,D**	**33.**	**C,D**	**34.**	**A,B**	**35.**	**A,B,C,D**
36.	C	**37.**	C	**38.**	B	**39.**	C	**40.**	D
41.	A	**42.**	D	**43.**	C	**44.**	A	**45.**	C
46.	B	**47.**	C	**48.**	B	**49.**	C	**50.**	B
51.	D	**52.**	**C**	**53.**	**C**	**54.**	B	**55.**	A
56.	A	**57.**	D	**58.**	D	**59.**	C	**60.**	C
61.	(A–r, B–r, C–q, D–q)			**62.**	(A)→(S,T); (B)→(P,Q); (C)→(Q,R,S,T); (D)→(Q,R,S)				
63.	0	**64.**	2	**65.**	2	**66.**	7	**67.**	8
68.	6	**69.**	0	**70.**	2	**71.**	1	**72.**	**100**